LEEBOY
FAYAT GROUP

Safety and Operations Manual

LEEBOY MODEL 8680 COMMERCIAL PAVER

Manual No. 1033428-SOP

**This manual applies to
Serial Number 365152
and above.**

Thank you for purchasing the LeeBoy 8680 Commercial Paver. We wish you many years of safe and efficient operation of your machine.

READ THIS MANUAL PRIOR TO OPERATING the unit. This is an important part of the machine and should be kept in the dedicated storage container at all times. Though you may be familiar with similar equipment, you MUST read and understand this manual before operating this machine to help prevent injury or damage to the unit. If this manual becomes lost or damaged, contact your authorized LeeBoy Dealer immediately to order a replacement (see **Contact Information** in **Section 2**).

This manual is intended as a guide for the safe and efficient use for proper operation and maintenance of the machine. This manual contains information that was available at the time of printing and are subject to change without notice.

Use this manual along with all related supplemental books, engine and transmission manuals included with your machine. Related Service Bulletins should be reviewed to provide information regarding some of the recent changes. Contact your authorized LeeBoy dealer for the latest available information.

SAFETY 1
Contains general and specific safety guidelines for product and safety label locations.

SPECIFICATIONS 2
Contains warranty, contact information, machine specification tables, and machine dimensions.

COMPONENTS 3
Contains overview of major component locations and functions.

OPERATION 4
Contains instructions for safe operation and information for optional equipment.

MAINTENANCE 5
Contains routine maintenance procedures, mechanical adjustments, component replacement and troubleshooting charts.

LeeBoy is proud to be ISO 9001 certified. The International Standards Organization (ISO) establishes guidelines to ensure that products and services are safe, reliable, and of good quality. ISO certifies companies who demonstrate compliance with all aspects of product safety, customer satisfaction, efficiency, environmental stewardship and social responsibility. Our teams work hard to deliver quality industrial machines that exceed customer expectations and we strive for continuous improvement in everything we do. The LeeBoy family of companies is committed to total quality management with a strong focus on meeting customer needs.

LeeBoy is also proud to be an accredited ANAB manufacturer, which is a certification process comprised of quality standards established by the American National Standards Institute (ANSI) and the American Society for Quality (ASQ). The ANSI-ASQ National Accreditation Board plays an important role in ensuring the safety and quality of goods and services, along with protecting the environment.

Disclaimer

All information, illustrations and specifications in this manual are based on the latest information available at the time of publishing. The illustrations used in this manual are intended as representative reference views only. Moreover, because of our continuous product improvement policy, we may modify information, illustrations and/or specifications to explain and/or exemplify a product, service or maintenance improvement. We reserve the right to make any change at any time without notice.

Title 40, Code of Federal Regulations (CFR) 1068

This product meets certified-emission requirements set by the EPA (Environmental Protection Agency), governed by Title 40 CFR 1068, which specifies actions that are prohibited by law and lists civil penalties for noncompliance. As part of those regulations, modification or rendering inoperative any emission-related component can subject you to government penalties (and void your warranty). Tampering with emission controls is in violation of federal law, and can result in monetary penalties for each day an engine or piece of equipment is operated in violation.

Please be aware that you are responsible for maintaining the machine and the certified emission engine installation. Failure to comply to comply could result in penalties as listed above and void the warranty on this engine and this machine.

For more information, visit: https://www.epa.gov/laws-regulations/regulations

California Proposition 65 Warning

Diesel engine exhaust and some of its constituents are known to the State of California to cause cancer, birth defects, and other reproductive harm.

Battery posts, terminals and other related accessories contain lead and lead compounds, chemicals known to the State of California to cause cancer and other reproductive harm. Wash hands after handling.

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Section 1

SAFETY

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NOTES

This manual provides important information to familiarize you with safe operating and maintenance procedures. Even though you may be familiar with similar equipment, read and understand this manual before operating the LeeBoy Model 8680 Commercial Paver and follow its instructions when operating the machine.

Safety is everyone's business and our top concern. Knowing the guidelines covered in this section will help ensure your safety, the safety of those around you, as well as proper paver operation.

Keep safety labels in good condition. If safety labels become missing or damaged, replace them with new matching labels. Replacement safety labels are available from your LeeBoy authorized dealer (see contact information in **Section 2**).

You can find more information about occupational health and safety in the paving industry on the internet. A few resources are listed below:

www.osha.gov

cdc.gov

www.asphaltpavement.org

www.safety.fhwa.dot.gov/

LOOK FOR THESE SYMBOLS THROUGHOUT THIS MANUAL. THESE ITEMS ARE EXTREMELY IMPORTANT FOR THE SAFETY OF YOU AND YOUR COWORKERS. READ AND UNDERSTAND THOROUGHLY. HEED THE WARNINGS AND FOLLOW THE INSTRUCTIONS.

⚠ DANGER Indicates a hazardous situation which, if not avoided, *will* result in death or serious injury.

⚠ WARNING Indicates a hazardous situation which, if not avoided, *could* result in death or serious injury.

⚠ CAUTION Indicates a hazardous situation which, if not avoided, *could* result in minor or moderate injury.

NOTICE **Indicates a situation which can cause damage to the equipment, personal property and/or the environment, or cause the machine to operate improperly.**

NOTE: Indicates a procedure, practice or condition that should be followed in order for the machine or component to function in the manner intended.

SAFETY PRECAUTIONS

⚠ CAUTION

The safety messages that follow have CAUTION level hazards.

Pre-Operation Hazard

Read and understand this Operation Manual before operating or servicing the engine to ensure that safe operating practices and maintenance procedures are followed.

- Never permit anyone to service or operate the machine without proper training.
- Safety signs and labels are additional reminders for safe operating and maintenance techniques.
- Contact LeeBoy or an authorized LeeBoy dealer for additional training.
- Make sure you are aware of all laws and regulations that are in effect for the location in which the paver is operated.
- Make sure you have all necessary licenses to operate the paver.
- Never operate a unit in need of repair.
- Ensure all covers are installed.

Poor Lighting Hazard

The work area must be well lit to ensure safe and proper operation.

- Ensure that the work area is adequately illuminated.
- Always install wire cages on portable safety lights.

Tool Hazard

Always use tools appropriate for the task at hand and use the correct size tool for loosening or tightening LeeBoy machine parts.

⚠ DANGER

The safety messages that follow have DANGER level hazards.

Power Lines Hazard

If your machine comes into contact with electric power lines, observe the following:

- Stay in the operators seat.
- Warn other workers to stay away and do not touch any control or any part of the machine.
- If contact can be broken, drive the machine away from the danger zone.
- If contact cannot be broken, stay in the operators seat until told that power is off.
- Failure to observe these directions could result in electrocution or death.

Electrocution Hazard

Disconnect the battery before welding anywhere on the machine.

Suffocation Hazard

Carbon monoxide poisoning is a serious condition that occurs as a result of improper ventilation.

- Never operate the internal combustion engine on this machine in an enclosed area with poor ventilation. Ensure proper ventilation to reduce risk of carbon monoxide poisoning or death.

Exhaust Hazard

All internal combustion engines create carbon monoxide gas during operation and special precautions are required to avoid carbon monoxide poisoning:

- Never block windows, vents or other means of ventilation.
- Always ensure that all connections are tightened to specifications after repair is made to the exhaust system.

⚠ WARNING

The safety messages that follow have WARNING level hazards.

Crush Hazard

Keep bystanders away from work area before and during operation.

Modification Hazard

Never modify the LeeBoy machine without the written consent of LeeBoy. Any modification can affect the safe operation of the paver and may cause personal injury or death.

Exposure Hazard

Operators of the paver must be aware of their work environment and the equipment needed to work safely.

- Always wear personal protective equipment, including appropriate clothing, gloves, work shoes, and protection for eyes and ears, as required by the task at hand.

Explosion Hazard

While the engine is running or the battery is charging, hydrogen gas is being produced and can be easily ignited. Keep the area around the battery well-ventilated and keep sparks, open flame, and any other form of ignition out of the area.

- Always disconnect the negative (-) battery cable before servicing the paver.
- Do not start the engine by shorting the starter circuit or any other starting method not stated in this manual. Only use the starting procedure as described in this manual to start the engine.
- Never charge a frozen battery. Always slowly warm the battery to room temperature before charging.

Fire and Explosion Hazard

- Diesel fuel is flammable and explosive under certain conditions.
- Never use a shop rag to catch fuel.
- Wipe up all spills immediately.
- Never refuel with the engine running.
- Store any containers containing fuel in a well-ventilated area, away from any combustibles or sources of ignition.

Fire Hazard

When operating machinery there is a risk for fire. Always have appropriate safety equipment available.

- Keep a charged fire extinguisher within reach when working in an environment where a fire may occur.
- Have all fire extinguishers checked periodically for proper operation and/or readiness.
- Always read and follow safety-related precautions found on containers of hazardous substances like parts cleaners, primers, sealants and sealant removers.
- Undersized wiring systems can cause electrical fires.

Entanglement/Sever Hazard

Verify there are no people, obstacles or other equipment near the machine before starting the engine. Sound the horn as a warning before starting the engine.

If the engine must be serviced while it is operating, remove all jewelry and tie back long hair before operating or servicing the machine.

- Keep hands, other body parts, and clothing away from moving/rotating parts.
- Always stop the engine before beginning service. Before maintenance, remove negative battery cable from battery post to ensure vehicle is not operated during maintenance.
- Verify that all guards and covers are properly attached before starting the engine. Do not start the engine if any guards or covers are not properly installed on the paver.

- If you must run the engine during maintenance procedures, make sure you have a helper to keep bystanders clear of the paver and make observations of moving parts as requested by the operator.
- Always turn the start switch to the OFF position after operation is complete and remove the key from the switch. Keep the key in your possession when the paver is not operating.
- Attach a “Do Not Operate” tag near the key switch while performing maintenance on the equipment.
- Never operate the engine while wearing a headset to listen to a radio or music because it will be difficult to hear the warning signals.
- Always start the engine and operate the controls while seated in the operators seat.

Alcohol and Drug Hazard

Never operate the machine while under the influence of alcohol, drugs, or when ill.

Piercing Hazard

High-pressure hydraulic fluid or fuel can penetrate your skin and result in serious injury. Avoid skin contact with high-pressure hydraulic fluid or diesel fuel spray caused by a hydraulic or fuel system leak such as a broken hydraulic hose, fuel injection or high-pressure, common-rail engine line.

If you are exposed to high-pressure hydraulic fluid or fuel spray, obtain prompt medical treatment.

- Never check for a hydraulic fluid or fuel leak with your hands. Always use a piece of wood or cardboard. Have your authorized LeeBoy dealer or distributor repair the damaged parts.

Flying Object Hazard

Always wear eye protection when cleaning the machine with compressed air or high-pressure water.

Dust, flying debris, compressed air, pressurized water or steam may cause eye injury.

Coolant Hazard

Coolant must be handled properly to ensure operator safety.

- Wear eye protection and rubber gloves when handling engine coolant.
- If contact with the eyes occurs, flush eyes with clean water for 15 minutes.
- If contact with skin occurs, wash immediately with soap and clean water.

Burn Hazard

Some of the machine’s surfaces become very hot during operation and shortly after shutdown.

- Keep hands and other body parts away from hot machine surfaces.
- Handle hot components with heat-resistant gloves.

NOTICE

The safety messages that follow have NOTICE level hazards.

- Any part that is found defective as a result of inspection or any part whose measured value does not satisfy the standard or limit must be replaced.
- Always tighten components to the specified torque. Loose parts can cause damage to the machine or cause it to operate improperly.
- Only use replacement parts approved by LeeBoy. Other replacement parts may affect warranty coverage.

Follow the guidelines of the EPA or other governmental agencies for the proper disposal of hazardous materials such as oils, diesel fuel, coolant, etc. Consult the local authorities or reclamation facility.

- Dispose of hazardous materials in accordance with all applicable laws and regulations. Never dispose of hazardous materials by dumping them into a sewer, on the ground, or into groundwater or waterways.
- Clean all accumulated dirt and debris away from the body of the paver and its components before you inspect the paver or perform preventive maintenance procedures or repairs. Operating a paver with accumulated dirt and debris will cause premature wear of paver components. Accumulated dirt and debris also hinders effective paver inspection.
- Retrieve any tools or parts that may have dropped inside of the paver to avoid improper paver operation.
- If any alert indicator illuminates during paver operation, stop the engine immediately. Determine the cause and repair the problem before continuing to operate the paver.

MACHINE PRECAUTIONS

Hot Material Precautions

- Wear protective gear for face, hands, feet, and body when operating the paver.
- Allow machine to cool before repairing or maintaining working components.
- If hot asphalt touches skin, flush area immediately with cold water. DO NOT apply ice to the affected area. DO NOT ATTEMPT TO REMOVE ASPHALT CEMENT with products containing solvents or ammonia. Natural separation will occur in about 48 to 72 hours. Get medical attention as soon as possible.
- DO NOT remove radiator cap, drain plugs, service grease fittings, or pressure taps while engine is hot. Add coolant to the radiator and perform other services only when the engine is stopped and fully cooled.

Hydraulic Systems Precautions

- Ensure all components are in good working condition. Replace any worn, cut, abraded, flattened or crimped hoses and metal lines.
- DO NOT attempt makeshift repairs using tape, clamps or cements. The hydraulic system operates under extremely high pressure and such repairs could cause serious injury.
- Wear proper hand and eye protection when checking for a high pressure leak. Use a piece of wood or cardboard as a back stop to isolate and identify leaks.

⚠WARNING **Hydraulic oil under pressure can cause serious personal injury. Check for oil leaks with a piece of cardboard. DO NOT expose hands to possible high-pressure oil. Turn off engine before attempting to tighten oil lines and fittings.**

- Escaping pressurized hydraulic fluid has force sufficient to penetrate the skin, which could cause serious personal injury. Ensure all pressure is relieved before disconnecting line, hoses or valves.
- If injury from concentrated high pressure steam or hydraulic fluid occurs, seek medical attention immediately. Injuries resulting from hydraulic fluid penetrating the skin's surface can result in serious infections or toxic reactions.

Refueling Precautions

- Do NOT overfill the fuel tank as overflow creates a fire hazard when spilled on hot components.
- DO NOT smoke when refueling and never refuel when the engine is running. Fuel is highly flammable and should be handled with care. Death or serious injury can occur due to explosion and/or fire.
- DO NOT fill tank to capacity. Allow room for expansion to reduce the risk of fuel expanding and spilling from the tank.
- Tighten fuel cap securely. Should fuel cap be lost, replace it with an original manufacturer's approved cap. Pressurization of the tank may result from use of non-approved cap.
- Prevent fires by keeping the machine clean of accumulated debris, grease, and spilled fuel.
- Use ultra-low sulfur diesel fuel (ULSD) only.

Battery Precautions

- Keep all sparks and flames away from batteries, as gas given off by electrolytes is explosive.
- Acid propelled by an explosion can cause blindness if it comes in contact with eyes. Always wear safety glasses when working near batteries.
- If you come in contact with battery electrolyte solution, wash off immediately. Chemicals can cause burns.
- Always disconnect the battery ground cable before working on the electrical system to avoid injury from spark or short circuit. Electrical shock and burns can occur.
- To avoid electrolyte loss, DO NOT tip batteries more than 45 degrees.

Starting and Stopping Precautions

- Check all around the paver to make sure there are no people working on the machine or in the path of the machine before starting. DO NOT start until area is clear. Death or serious injury can occur to bystanders from being crushed under a moving machine.
- Check brakes, steering and other control devices in accordance with instructions before starting. Be sure both steering levers (joysticks) are in neutral with the neutral lock engaged.
- DO NOT bypass the paver neutral-start system.
- Be aware of job site conditions and terrain. Adjust your speed if operating in rough or hilly conditions.

⚠WARNING **DO NOT operate the engine in an enclosed area without proper ventilation. Exhaust gasses are odorless and deadly.**

Parking Precautions

- Park paver on level ground whenever possible, set steering levers in neutral and lock. Apply the parking brake. Park the paver with wheels securely blocked on grades.
- Remove ignition key when leaving paver parked or unattended.

Operating Precautions

- Always comply with local regulations regarding moving equipment on public roads and highways.
- Know and use the hand signals required for a particular job. Know who has the responsibility for signaling.
- Make sure that all lights and reflectors comply with state and local regulations. Make sure that they are clean, in good working order, and can be seen clearly by all traffic.
- DO NOT stand between the equipment and the truck while the truck is being backed to the paver. Death or serious injury can result from being crushed between the two machines.
- DO NOT ride on attachments.
- Check all gauges and warning instruments for proper operation. If malfunctions are found, shut down the machine and report the problem for resolution. If the failure causes loss of steering control, loss of brake control, or loss of engine power, stop paver motion as quickly as possible. Apply parking brake (if equipped). Keep the machine securely parked until the failure is corrected or the machine can be safely towed.
- Drive the machine with care. Make sure speed is compatible with conditions. Use caution on rough ground, slopes, and while turning.
- Be alert for hazards and obstructions such as ditches, trees, cliffs, overhead power lines, and areas where there is danger of a slide.

- Be aware of and understand the job site traffic flow patterns.
- Obey flagmen, road signs, and signals.
- Watch for bystanders. Never allow anyone near the machine during operation. Never allow anyone to reach into the machine during use.
- Operator must know how to use signaling devices when driving the paver. Operator must also understand which circumstances require use of each signal. Use tail lights, slow moving vehicle signs, and warning beacon as needed when traveling on public roads. It is recommended that you provide an escort on the road.
- DO NOT tow the paver, except to remove from road or to load on trailer.
- The machine is shipped with two seats, one for the left side operation and one for the right side operation. It is recommended that you operate the paver from the seats. If for any reason the seats are removed, care must be taken to operate from a position with stable footing and use of the grab handles provided.
- When moving the machine, adjust speed and direction of travel for the terrain and ground conditions present. Always consider adjusting travel speed to match ground conditions.

Poor Visibility

Increasingly, asphalt maintenance equipment is used during less than ideal lighting conditions such as fog and smog or at night. These conditions present safety hazards for workers, bystanders and passing traffic.

⚠WARNING **When operating under restricted light conditions, equip the machine with special lighting to prevent serious injury.**

While the machine is equipped with a beacon and a work light, contact your authorized LeeBoy dealer for additional lighting packages if working under these conditions.

Use reflective tape on the sides of machines when working at night. Ensure all workers wear reflective safety vests. Use impact barriers (movable or stationary) to protect workers and direct the traffic flow safely away from the work site.

Storage Precautions

- Store paver in an area away from human activity.
- DO NOT permit children to play on or around the stored machine. Serious injury or death can occur from improper/unauthorized use of the machine.
- Make sure the unit is stored on a surface that is firm, level, and free of debris.
- Store the machine inside a building or cover securely with a weatherproof tarpaulin.

Maintenance Precautions

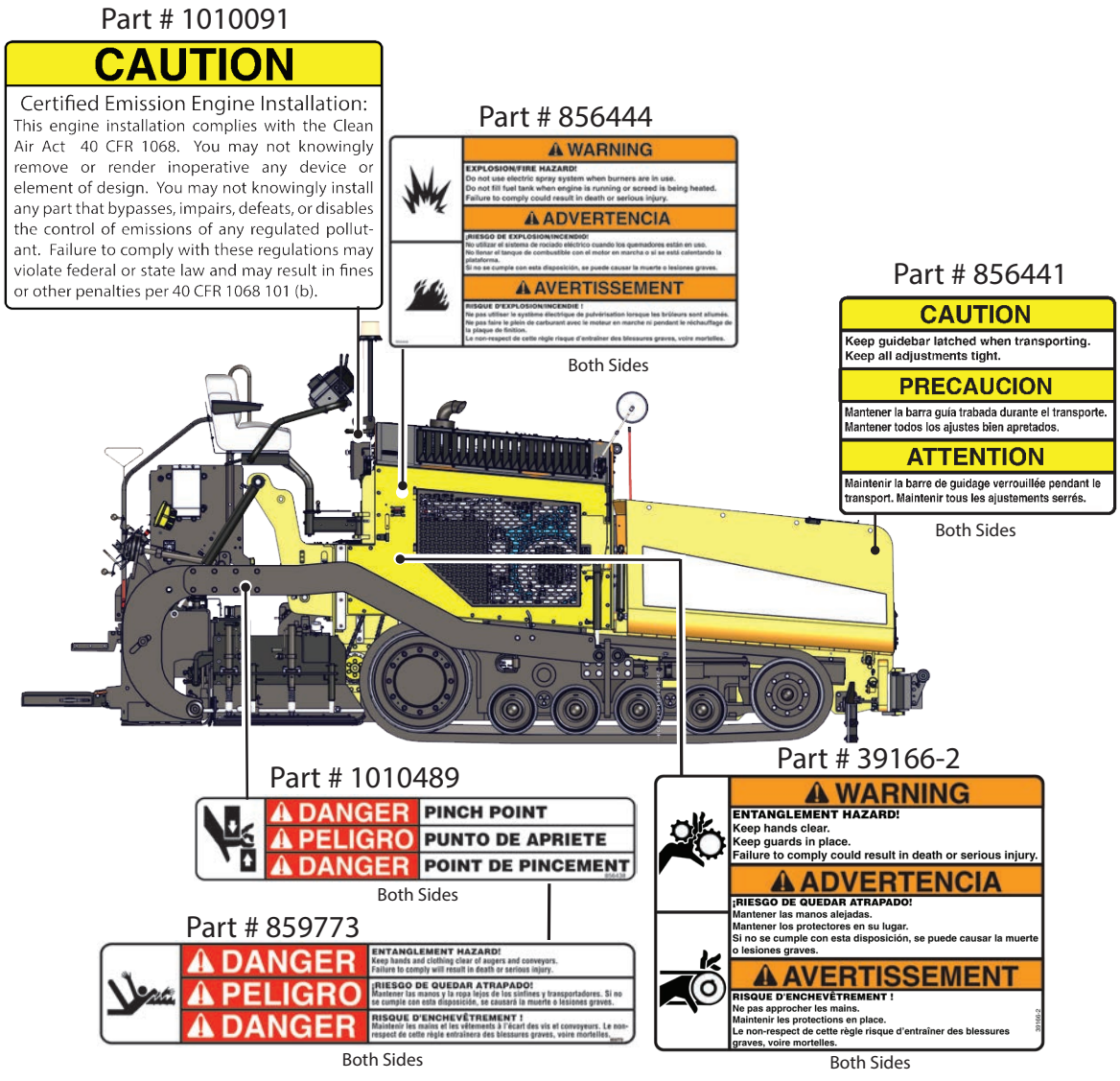
- DO NOT attempt repairs unless trained to do so. Refer to manuals and experienced repair personnel for help.
- Before working on the machine, securely block the machine and any components that may fall. Block any working components to prevent unexpected movement while repairs are being made.
- Always wear safety glasses and other required safety equipment when servicing or making repairs.
- Disconnect battery before working on the electrical system.
- Avoid lubrication or mechanical adjustments while the paver is in motion or while engine is operating.
- If lubrication or mechanical adjustment is necessary, use extreme caution and complete the following steps:
  1. Place all four joysticks in NEUTRAL (center position) on both control panels. This engages the brake.
  2. Lower screed to the ground.
  3. Shut off engine.
- Never make repairs on pressurized components such as fluid lines, the gas system, or mechanical items until the pressure has been relieved.
- When servicing or replacing hardened pins, use a brass drift or other suitable material between the hammer and pin.
- Keep brake and steering systems in good operating condition.

SAFETY DECALS

If your machine is repainted, it is extremely important that you replace all the CAUTION, WARNING and DANGER safety decals as shown in the following graphics. For additional help, refer to the parts listing in **Section 7** and contact your authorized LeeBoy dealer to order a replacement kit.

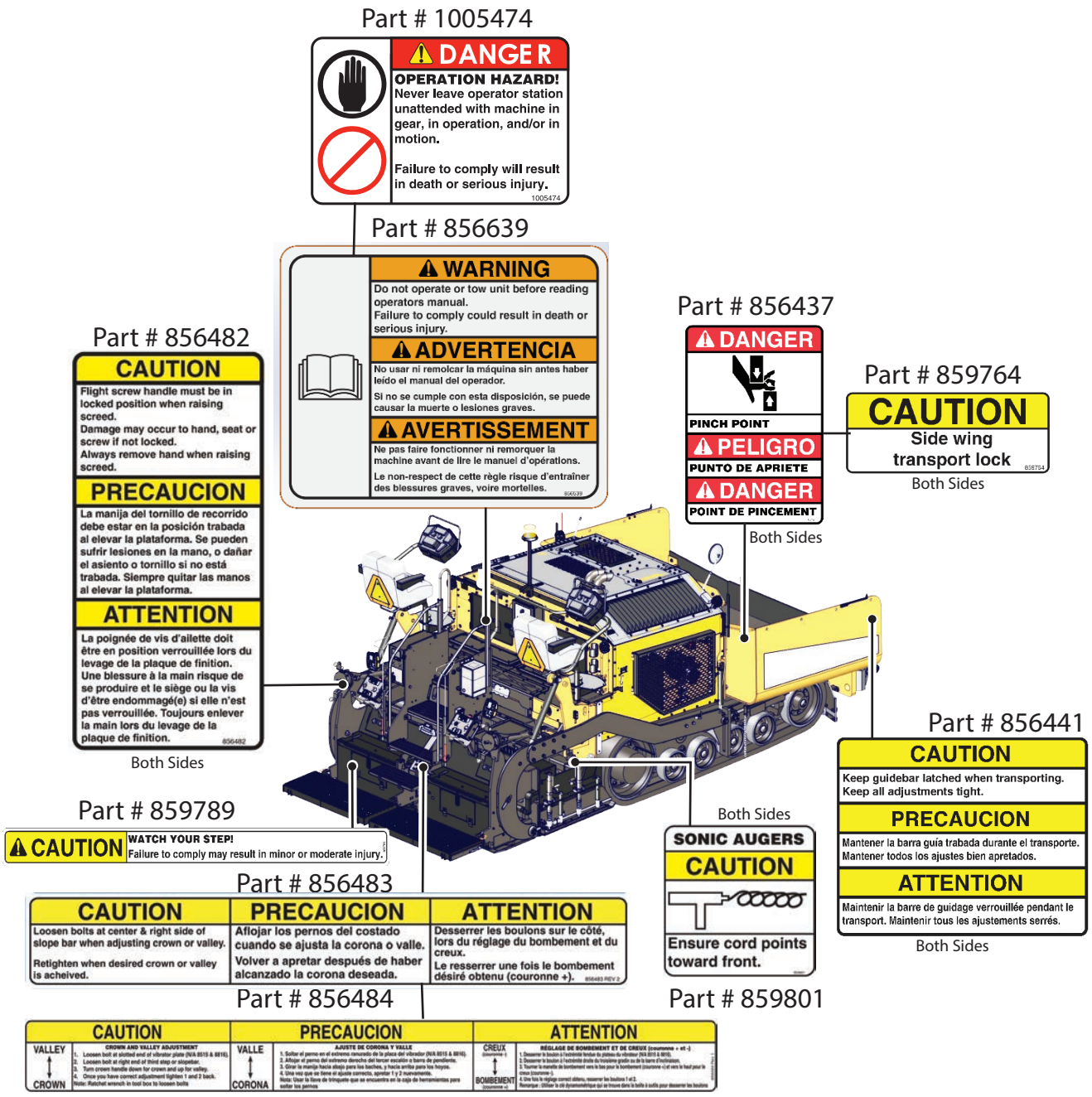
NOTE: It is the responsibility of the owner and operator to make sure that all safety labels are readable and located on the paver as designated by LeeBoy.

Safety Decal Locations



1

Safety Decal Locations



Safety Decals Care

1. Keep safety decals and signs clean and legible at all times.
2. Be familiar with the content and position of each safety decal. Decals include important information.
3. Replace decals and signs that are missing or become impossible to read.
4. When replacing parts that display a safety decal, ensure that the new part is fitted with a decal as well.
5. Obtain replacement safety decals or signs from your authorized dealer.

Decal Application

Be sure that the installation area is clean and dry. Use hot, soapy water to clean the surface where the decal will be applied.

1. Thoroughly dry the surface.
2. Measure and fit decal before removing the paper backing.
3. For decals with no top protection paper, remove the smallest split-backed paper.
4. Align decal over the specified area and carefully press exposed portion into place.
5. Slowly remove the remaining backing and carefully smooth the remaining portion of the decal into place.
6. Small air pockets can be pierced with a pin and smoothed using a piece of the decal backing.

Section 2

INFORMATION AND SPECIFICATIONS

2

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TWO YEAR LIMITED WARRANTY

Limited Warranty

If a manufacturing defect in factory supplied materials or factory workmanship is found and the authorized LeeBoy dealer ("Dealer") is notified during the warranty period, LeeBoy will be responsible for repairing or replacing any part or component of the unit or part that fails to conform to the warranty during the warranty period.

Terms and Conditions

The warranty applies for a period of two years (for unlimited hours of usage on the unit during the two years), beginning on the date that the Dealer from which the unit is purchased submits the warranty registration form to LeeBoy (typically on the date of purchase).

- The warranty is transferable.
- LeeBoy has the right to repair any component or part before replacing it with a new one.
- Replacement parts furnished by LeeBoy are covered for the remainder of the warranty period applicable to the unit or component in which such parts are installed.
- All new replacement parts purchased from a Dealer will carry a six (6) month warranty.

Exclusions and Limitations

LeeBoy has no obligation under this warranty for any defects or failures caused by:

- Misuse, misapplication, negligence, accident or failure to maintain or use in accordance with the most current operating instructions;
- Unauthorized alterations or modifications;
- Any replacement parts or attachments not manufactured by or expressly approved by LeeBoy;
- Failure to conduct normal maintenance and operating service, including without limitation, providing lubricants, coolant, fuel, tune-ups, inspections or adjustments; or
- Unreasonable delay, as determined by LeeBoy, in making the applicable units or parts available upon notification of a service notice ordered by LeeBoy.

LeeBoy is not responsible for any of the following:

- Defects or failures with respect to any engine or truck chassis;
- Defects or failures covered by a warranty provided by a third-party, including without limitation warranties with respect to engines or truck chassis;
- Costs related to travel time, mileage or overtime;
- Costs related to transporting the unit to and from the place at which warranty work is performed;
- Airfreight charges related to transporting repair parts to the place at which warranty work is performed;
- Used units or used parts of any kind;
- Repairs due to normal wear and tear, or brought about by abuse or lack of maintenance of the equipment, except for premature failures;
- Attachments not manufactured or installed by LeeBoy; or
- Miscellaneous charges.

Exclusion of Other Warranties/Limitation of Liability

OTHER THAN THE EXPRESS WARRANTY PROVIDED IN SECTION A. ABOVE, LEEBOY GIVES NO EXPRESS, STATUTORY, OR IMPLIED WARRANTY OR GUARANTEE AND HEREBY DISCLAIMS ALL SUCH WARRANTIES, INCLUDING WITHOUT LIMITATION ANY IMPLIED WARRANTY OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, SUITABILITY, DURABILITY, CONDITION, QUALITY, OR FREEDOM FROM CLAIMS OF ANY PERSON BY WAY OF INTERFERENCE, INFRINGEMENT, OR THE LIKE.

IN NO EVENT OR UNDER ANY THEORY OF RECOVERY, WHETHER AS A RESULT OF BREACH OF CONTRACT OR WARRANTY, ALLEGED NEGLIGENCE, OR LIABILITY WITHOUT FAULT, SHALL LEEBOY BE LIABLE FOR ANY SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE SALE, USE, OR LOSS OF USE OF ANY PRODUCT, INCLUDING WITHOUT LIMITATION LOST PROFITS OR REVENUES, COST OF CAPITAL, COST OF SUBSTITUTED EQUIPMENT, FACILITIES OR SERVICES, DOWNTIME COSTS, LABOR COSTS, OR DIMINUTION OF VALUE.

CONTACT INFORMATION

For information regarding parts and repairs about your LeeBoy product, contact your authorized LeeBoy dealer. If your dealer is unable to resolve the problem, contact LeeBoy directly.

Record dealer information in the space provided. For additional information about LeeBoy, please visit: www.leeboy.com.

Sales Representative: _____

Dealership Name: _____

Dealership Address: _____

Dealership Phone: _____

Record of Ownership

Please complete the following information for use if you need to contact LeeBoy for service, parts or literature.

Machine Model Number: _____

Product Serial Number: _____

Date of Purchase: _____

Nameplate

The nameplate contains the model and serial numbers used to identify the machine and its components for parts or service information. Refer to the Engine Operator's Manual for the location of the engine nameplate.

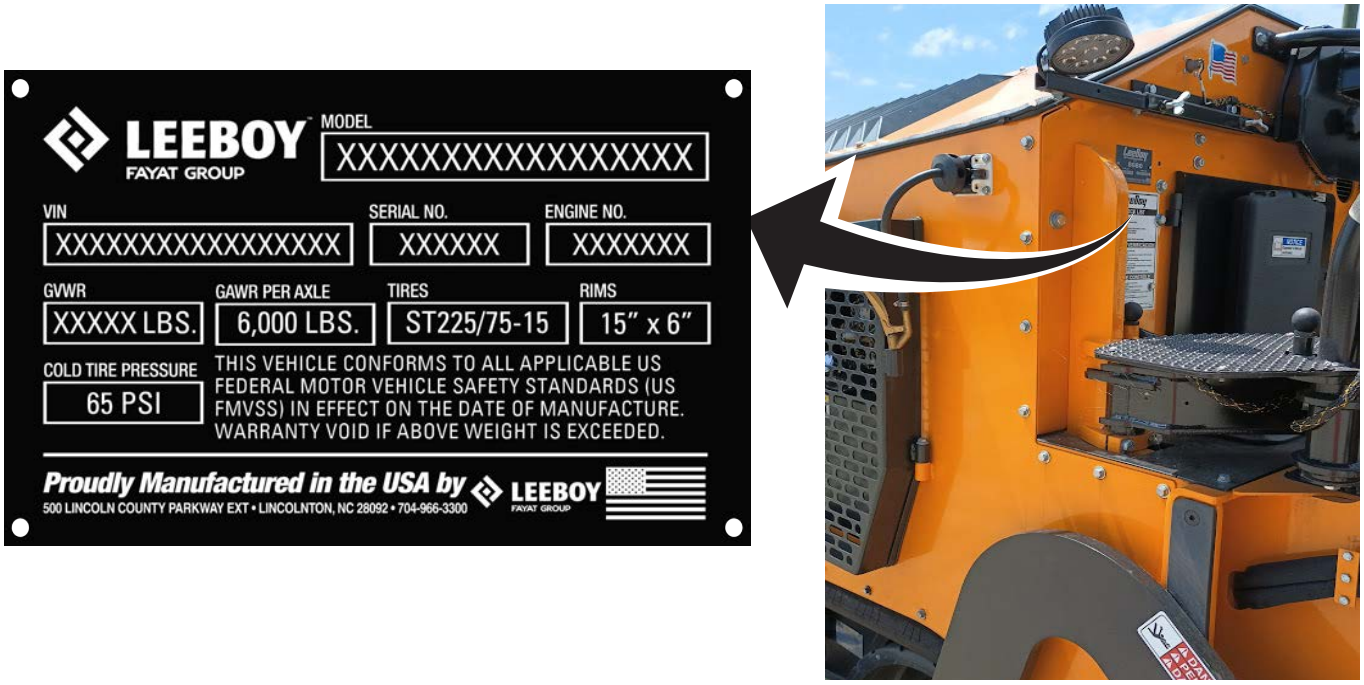

Figure 2-1. Nameplate Location

SPECIFICATION CHARTS

The specifications provided in this section include screed weights, dimensions, performance, and torque values for both metric and standard inch fastener.

⚠CAUTION Replace original equipment only with components approved by your LeeBoy dealer.

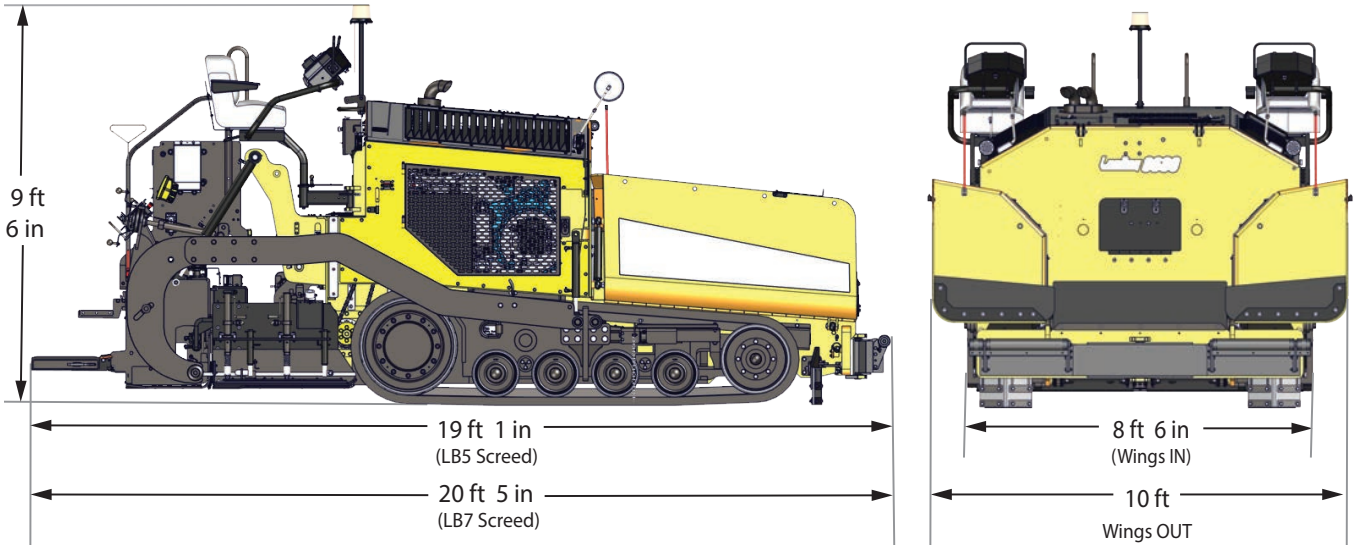

Table 2-1. Machine Dimensions

ITEM	SPECIFICATION
Overall Length	w/LB5 Screed: 19.1 ft w/LB7 Screed: 20.5 ft
Transport Length	w/LB5 Screed: 18 ft 2 in w/LB7 Screed: 19.1 ft
Overall Height	9.6 ft
Transport Height	8.9 ft
Overall Transport Width (with hopper up)	8.6 ft
Overall Width (with hopper down)	10 ft
Total Weight	With LB5 Screed: 33,600 lb With LB7 Screed: 36,000 lb
Ground Clearance	6.25 in

Table 2-2. Kubota Tier 4 Engine Specifications

ITEM	SPECIFICATION
Manufacturer	Kubota
Model	V5009-TIE5
Fuel Type	Diesel
Cylinders	4
High Idle	2200 RPMs
Max Gross Power	210 HP
Max Gross Torque	642 lb ft

Table 2-3. Performance Specifications

ITEM	SPECIFICATION
Travel Speed	0 - 6 mph
Paving Speed	0 - 3.4 mph
Maximum Paving Depth	0 - 10 in (Hydraulic tow point cylinders; includes manual flight screws on screed.)
Turning Radius	15 in (Capable of Counter-Rotation)

Figure 2-4. Lubricant Specifications

ITEM	SPECIFICATION
Engine Oil	15W-40, API, CH-4, CI-4
Hydraulic Oil	All Weather 32
Torque Hub	75W 90 Synthetic Gear Oil
Grease	Shell Avania EP Grease or Equivalent
Chain Lube	Chain Lube

Table 2-5. Machine System Capacity Specifications

ITEM	SPECIFICATION
Hydraulic Oil	37 gal
Fuel	50 gal
Engine Oil with Filter	5 gal
Coolant	7 gal
Spray Down	8 gal
Torque Hub	.88 gal

Table 2-6. Hydraulic System Specifications

ITEM	SPECIFICATION
Drives	Variable volume hydraulic pumps for each drive.
Conveyor/Augers	Load sense pump
Cylinders	Gear pump
Charge/Blower	Tandem gear pump
Primary Filter	Charge Filter (7 Microns)
Secondary Filter	Intank Return Filter (10 micron)
Tank Strainers	Suction Strainers (125 Micron)

Table 2-7. Hydraulic Pressures Specifications

ITEM	SPECIFICATION
Drive	5510 psi
Conveyors/Augers (Load Sense)	3480 psi
Cylinders	3000 psi
Track Tension Pressure	1200 psi
Track Tension Relieving	3600 psi
Standby Pressure	350 psi
Charge Pressure	500 psi

Table 2-8. Electrical Specifications

ITEM	SPECIFICATION
Battery	Two (2), Maintenance-Free
Battery Ampere per Hour Rating	1150 CCA
Battery Voltage	12 V
Alternator Type and Voltage	12 V, Negative Ground
Alternator Output Amperage	150 Amps
Alternator Fan Belt Tension	Automatic belt tension mechanism keeps serpentine belt under tension at all times.
Starter Voltage and Type	12 V, Negative Ground
Starter Rating	4.8 kW

Table 2-9. Generator Specifications

ITEM	SPECIFICATION
Model	Harrison HU34.0ALT-36N-C-5119
Power	34 KW
Voltage	480 V (277 V with 120V option)
Frequency	58 - 62 HZ

Table 2-10. Hopper Specifications

ITEM	SPECIFICATION
Material Capacity	15 Tons
Height	35.5 in
Hopper Wings	Heavy duty, 1/2 inches, with large radius corners, hydraulic lift front apron, and rear hopper wing wipers.

Table 2-11. Augers Specifications

ITEM	SPECIFICATION
System Components	2 independent, hydraulic, segmented reversible augers operate manually or automatically.
Variable Speed	1 - 150 RPMs
Diameter	16 in (or 14-inch option)

Table 2-12. Conveyors Specifications

ITEM	SPECIFICATION
System Components	2 independent, hydraulic, reversible conveyors with automatic shut-off. Heavy-duty chains. Quad bearings on front and rear conveyors.
Width	18 in

Table 2-13. Track Drive Specifications

ITEM	SPECIFICATION
Type	High-speed, smooth rubber with friction drive wheels, oscillating bogey rollers and hydraulically adjustable front idler wheel. Hydrostatically powered.
Track Length	128 in
Track Width	14 in
Number of Speeds	2-speed torque hubs with 2-speed motor consisting of 4 speeds obtained via keypad button.
Ground Contact	99 in each side
Brakes	Fail Safe
Gauge Width (outside tracks)	91.38 in

Table 2-14. LB5 and LB7 Screed Specifications

ITEM	LB5 SCREED	LB7 SCREED (Option)
Screed Weight	5200 lbs	7400 lbs
ScreeD Width	15 ft 6 in	15 ft 6 in
Electric Heat Elements	Two (2) 3000-watt S-Curve elements on main screed. One (1) 1500-watt sealed S-curve element in each screed extension.	Two (2) 3500-watt S-Curve elements on main screed. One (1) 2000-watt sealed blade element in each screed extension.
Main Screed Width	8 ft	8 ft
Screed Extension Width (2 adjustable, hydraulically-operated)	46 in	46 in
Wear Plates	Main Screed Depth: 18 in Extension Depth: 9 in	Main Screed Depth: 24 in Extension Depth: 10 in
Wear Plate Thickness	Main Screed Depth: .375 in Extension Depth: .375 in	Main Screed Depth: .50 in Extension Depth: .50 in
Bullnose Radius	Main Screed: 1.688 in Screed Extension: 1.688 in	Main Screed: 2.5 in Screed Extension: 2.5 in
Vibration	Two (2) vibration motors producing 4100 vpm	Two (2) vibration motors producing 4100 vpm
Slope @ Maximum Width	3.5 in	4 in
Crown/Valley @ 10 ft	Positive Crown: 2.5 in Negative Crown: 1.5 in	Positive Crown: 2.5 in Negative Crown: 1.5 in
End Gate	Heated, spring-loaded with vertical height adjustments and tilt.	Heated, spring-loaded with vertical height adjustments and tilt.

Figure 2-15. Push Rollers Specifications

ITEM	SPECIFICATION
Components	Two (2) 3.5 inch (diameter) adjustable rollers with sealed bearings on swivel frame and self-cleaning scrapers.
Vertical Height Adjustment	7 in

Figure 2-16. Spray Down Specifications

ITEM	SPECIFICATION
Components	Electric heavy-duty pump, with 2 hoses, sight gauge and spray wand.
Citrus Tank Capacity	8 gal

TORQUE SPECIFICATIONS

The following tables list torque values for standard hardware. This is a guide for average application involving typical stresses and machined surfaces. Values are based upon physical limitations of clean, plated and lubricated hardware. Under more extreme conditions, individual torque value should be followed.

Conversion formulas are provided below:

Conversion	Formula
ft-lb to N•m	[ft-lb]*1.3558 = [N•m)
ft-lb to in-lb	[ft-lb]*12 = [in-lb]
N•m to in-lb	[N•m]*8.8508 = [in-lb]

Standard Inch Fasteners

Table 2-17. Torque Specifications For Standard Inch Fasteners

SIZE	THREAD	CAPSCREWS: SAE GRADE 5				CAPSCREWS: SAE GRADE 8			
		TORQUE (ft lb)		TORQUE N•m		TORQUE (ft lb)		TORQUE N•m	
		Dry	Lubed	Dry	Lubed	Dry	Lubed	Dry	Lubed
1/4	20 UNC	8	6	11	8	12	9	16	12
	28 UNF	10	7	14	9	14	10	19	14
5/16	18 UNC	17	13	23	18	25	18	34	24
	24 UNF	19	15	26	20	27	20	37	27
3/8	16 UN C	31	23	42	31	44	33	60	45
	24 UNF	35	26	47	35	49	37	66	50
7/16	14 UNC	49	37	66	50	70	52	95	71
	20 UNF	55	41	75	56	78	58	106	79
1/2	13 UNC	75	57	102	77	106	80	144	108
	20 UNF	85	64	115	87	120	90	163	122
9/16	12 UNC	109	82	1 48	111	154	115	209	156
	18 UNF	121	91	164	123	171	128	232	174
5/8	11 UNC	150	113	203	153	212	159	287	216
	18 UNF	170	127	230	172	240	180	325	244
3/4	10 UNC	267	200	362	271	376	282	510	382
	16 UNF	297	223	403	302	420	315	569	427
7/8	9 UNC	429	322	582	437	606	455	822	617
	14 UNF	474	355	643	481	669	502	907	681
1	8 UNC	644	483	873	655	909	681	1232	923
	14 UNF	722	542	979	735	1020	765	1383	1037
1-1/4	7 UNC	1121	840	1520	1139	1817	1363	2464	1848
	12 UNF	1241	930	1683	1261	2012	1509	2728	2046
1-1/2	6 UNC	1950	1462	2644	1982	3162	2371	4287	3215
	12 UNF	2194	1645	2975	2230	3557	2668	4823	3617

Metric Fasteners

Table 2-18. Torque Specifications for Metric Fasteners

NOMINAL SIZE AND PITCH	CLASS 8.8 [GRADE 5 EQUIVALENT]				CLASS 10.9 [GRADE 8 EQUIVALENT]			
	TORQUE (ft lb)		TORQUE N•m		TORQUE (ft lb)		TORQUE N•m	
	Dry	Lubed	Dry	Lubed	Dry	Lubed	Dry	Lubed
M4 x 0.7	2	2	3	2	3	2	4	3
M5 x 0.8	5	3	7	4	7	5	9	7
M6 x 1	8	6	11	8	11	8	15	11
M8 x 1.25	19	14	26	19	27	20	37	27
M10 x 1.5	37	28	50	38	53	40	72	54
M12 x 1.75	65	49	88	66	93	70	126	95
M14 x 2	104	78	141	106	148	111	201	150
M16 x 2	161	121	218	164	230	173	312	235
M18 x 2.5	222	167	301	226	318	239	431	324
M20 x 2.5	314	236	426	320	449	337	609	457
M22 x 2.5	428	321	580	435	613	460	831	624
M24 x 3	543	407	736	552	777	582	1053	789
M27 x 3	796	597	1079	809	1139	854	1544	1158
M30 x 3.5	1079	809	1463	1097	1544	1158	2093	1570

Hydraulic Fittings

Tightening Flare-Type Tube Fittings

1. Check the flare and flare seat for defects that might cause leakage.
2. Align tube with fitting before tightening.
3. Lubricate connection.
4. Hand tighten swivel nut until snug.
5. To prevent twisting the tube(s), use two wrenches. Place one wrench on the connector body and tighten the swivel nut with the second to the torque shown in the following table:

NOTE: The torque values shown are based upon lubricated connections.

Table 2-19. Torque Specifications for Steel Flare Type Tube Fittings

TUBE SIZE OUTER DIAMETER	NUT SIZE ACROSS FLATS	TORQUE VALUE	
(IN)	(IN)	(LB FT)	(N•m)
3/16	7/16	8	11
1/4	9/16	12	16
5/16	5/8	16	22
3/8	11/16	23	31
1/2	7/8	38	52
5/8	1	54	73
3/4	1 1/4	75	102
7/8	1 3/8	83	113

Determining Proper Torque

The only reliable method of creating a consistently leak-free and long-lasting connection is to ensure the coupling is brought to the proper torque. Using a torque wrench with crowfoot is the best method, but the flats method can be used if a torque wrench is not available.

The most straightforward method of determining the correct torque setting is to multiply the desired torque by the length of the wrench from the center of the handle to the center of the drive (L); divided by the length of the wrench from the center of the handle to the crowfoot center (LA) as shown below:

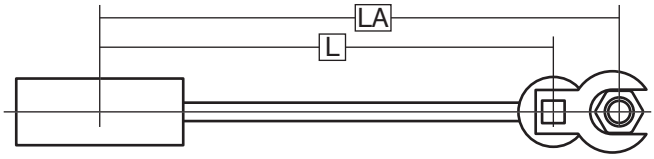

Figure 2-2. Torque Wrench - Crowfoot

NOTICE **The minimum torque values are adequate for sealing most applications. Maximum torque values should never be exceeded.**

There are several methods of determining the correct setting on the torque wrench when using a crowfoot. All of the methods involve making the setting proportional to the effective change in length of the wrench multiplied by the desired final torque. The equations and illustration below describe proper measurements.

Equations

- Torque setting if the crowfoot is placed in line with respect to the wrench:

TS = TD * L / LA

or

TS = TD * L / (L+E)

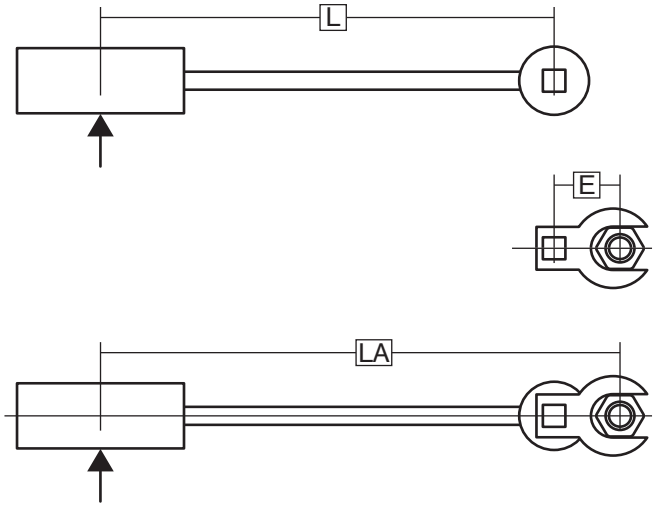

Figure 2-3. Measurements Needed

LEGEND

L = Distance from center of torque wrench handle to the center of socket drive

E = Distance from center of socket drive to the center of crowfoot

LA = Distance from center of torque wrench handle to the center of crowfoot

TD = Desired torque at the fitting

TS = Torque setting indicated on wrench

2

Coupling Installation

Use the following steps for proper coupling installation:

1. Determine the correct torque value for your coupling.

NOTE: Only use the torque values specified from the manufacturer. DO NOT use SAE torque recommendations.

2. Ensure the seal face and threads are clean and in good condition. O-Rings should be lubricated with light oil, but threads should be completely dry unless making pipe thread connections (interference seal).

NOTE: Attach the male end of the hose onto the equipment first since it may be necessary to rotate the entire hose assembly to tighten the male threads. Then route the hose into position while avoiding twisting the hose.

3. Tighten the connection (by hand), bringing the seal face into contact and rotating the nut until it stops.
4. Mark a line across the coupling nut and backup hex for the flats method verification of coupling torque.
5. Apply a wrench to the backup hex to prevent the coupling and hose from moving while tightening the nut with a torque wrench.

NOTICE **Failure to retain the backup hex during installation will also result in additional clamp load force that could cause damage to the seal face.**

NOTE: The coupling nut must be in motion for an accurate torque reading. If the nut is stopped before final torque value is achieved, it must be loosened and retightened until the torque is attained while the nut is in motion.

If a torque wrench cannot fit into the coupling area or if it is unavailable, the flats method may be used to ensure that the coupling is properly tightened, as shown in the following figure.

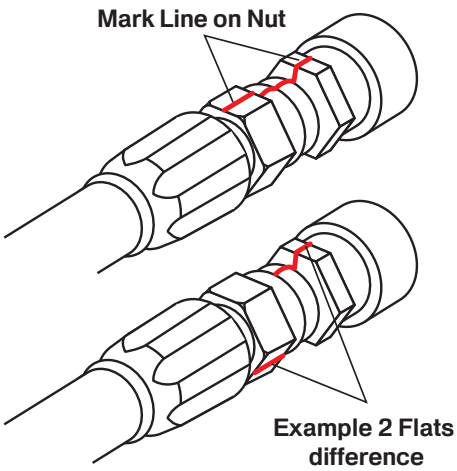

Figure 2-20.

Figure 2-21. Flats Method Tightening

NOTE: The mark placed on the nut and backup hex after tightening by hand should rotate during final tightening according to the table below. The nut and backup hex can then be marked to indicate if the coupling loosens over time.

Table 2-22. Flats Method Values for Selected Terminations

FLATS METHOD VALUES		
Termination Type	**Dash Size**	**Flats**
JIC	-4	1.5 - 1.75
JIC	-6	1.0 - 1.5
JIC	-8	1.5 - 1.75
JIC	-10	1.0 - 1.5
JIC	-12	1.0 - 1.5
JIC	-16	.75 - 1.0
JIC	-20	.75 - 1.0
JIC	-24	.75 - 1.0
JIC	-32	.75 - 1.0
JIS	-4	.5 - 1.5

1. Seal faces must be in contact with the fitting fully tightened by hand before marking flats.
2. The flats method is most accurate for the first assembly cycle. For multiple disassembly and assembly cycles, torque values are more reliable.
3. Tightening two (2) flats or more may damage seal faces.

Table 2-23. Torque Specifications For US Style Coupling Terminations

JIC, SAE 45°, ORFS, O-RING BOSS, GATES ADAPTERLESS AND MEGASEAL										
DASH SIZE	JIC 37°, SAE 45° & Mega-Seal (Steel)		JIC 37°, SAE 45° & Mega-Seal (Brass)		Flat Face O-Ring Seal (Steel)		SAE O-Ring Boss (Steel) & Gates Adapterless ≤ 4000 PSI		SAE O-Ring Boss (Steel) & Gates Adapterless > 4000 PSI	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
-3									8	10
-4	10	11	5	6	10	12	14	16	14	16
-5	13	15	7	9					18	20
-6	17	19	12	15	18	20	24	26	24	26
-8	34	38	20	24	32	40	37	44	50	60
-10	50	56	34	40	46	56	50	60	72	80
-12	70	78	53	60	65	80	75	83	125	135
-14					65	80			160	180
-16	94	104	74	82	92	105	111	125	200	220
-20	124	138	75	83	125	140	133	152	210	280
-24	156	173	79	87	150	180	156	184	270	360
-32	219	243	158	175						

Table 2-24. Torque Specifications for DIN 24, DIN 60, and Inverted Cone Style Coupling Terminations

DIN 24, DIN 60, AND INVERTED CONE			
Size (mm)		Torque (lb ft)	
Light Series Tube OD	Heavy Series Tube OD	Min	Max
6		7	15
8		15	26
10	8	18	30
12	10	22	33
14	12	26	37
15	14	30	52
	16	30	52
18	20	44	74
22	25	59	89
28	30	74	111
	38	74	162
35		133	184
42		148	221

Table 2-25. Torque Specifications for 4-Bolt Flange Connections

Figure 2-26. 4-BOLT FLANGES		
Dash Size	Bolt Size (in)	Torque (lb ft)
-8	0.31	17
-12	0.38	26
-16	0.44	43
-20	0.50	65
-24	0.63	130
-32	0.75	220

1. Align faces and tighten bolts (by hand) before applying final torque in a pattern. The seal faces must be parallel with an even bolt tension to seal properly.
2. Torque values apply to bolts that are plated or coated in light engine oil.
3. Before assembly, lubricate O-Ring with light oil (SAE 10W or 20W).

Table 2-27. Torque Specifications for NPTF Dry Seal Pipe Threads

Figure 2-28. NPTF	
Dash Size	**Max Torque (ft-lb)**
-2	20
-4	25
-6	35
-8	45
-12	55
-16	65
-20	80
-24	95
-32	120

1. The torque values obtained from tightening pipe threads can vary considerably depending upon thread condition. Adequate sealing can occur at values much lower than the maximum values listed above. Only enough torque to achieve adequate sealing should be used.
2. When using a male tapered pipe thread with a female straight or parallel pipe thread, maximum values are 50% of those listed in the table above.
3. If thread sealant is used, maximum values shown should be decreased by 25%.

Table 2-29. Torque Specifications for BSP 30° Inverted Cone and JIS Coupling Terminations

BSP 30° INVERTED CONE AND JIS		
Dash Size	**Torque (ft-lb)**	
	Min	**Max**
-2	7	9
-4	11	18
-6	19	28
-8	30	36
-10	37	44
-12	50	60
-16	79	95
-20	127	152
-24	167	190
-32	262	314

Section 3
COMPONENT LOCATION

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NOTES

COMPONENTS OVERVIEW

The LeeBoy 8680 Commercial Paver delivers higher horsepower and production capacity for a larger footprint in the heavy commercial class market. It's designed and built for producing superior mat quality in a variety of road and other pavement construction projects.

Dual main operator control platforms with ergonomic control keypads include auto-throttle, auto-idle and pause functions. Adjustable operator platforms that swing out beyond the frame to provide optimum visibility with the 8680 commercial paver. The beacon light can also be lowered as needed for overhead clearance.

The rugged 8680 commercial paver features high-speed, smooth rubber tracks with oscillating bogies on a four-speed drive system. This paver can travel up to 6 mph with a paving speed of 300 feet per minute (3.4 mph) due to the rubber tracks and a 210 HP engine. Continuous and flexible, the rubber track provides the traction and flotation of a tracked paver, but with better mobility and a smoother ride.

The 8680 features the LB5 screed or the optional LB7 screed with the ability for variable paving widths from 8 to 15.5 feet. Independent control of one or both hopper wings in its 9-ton hopper allows paving alongside buildings or other obstructions with ease. Narrow conveyors and a power tunnel shields the height-adjustable augers (14 or 16-inch) are just part of LeeBoy innovation and added versatility. Proportional control of augers and conveyors enhance asphalt material delivery.

Separate pumps for each function charge, (augers, blower and conveyors), cylinders and propulsion, with proportional valves for extensions and tow cylinders to optimize hydraulic performance. Hydraulic tow point cylinders allows 10 inches of travel for quick and easy height adjustments, and integrates with sonic grade and slope automation. Sloping extensions include 2.5-inch chrome rods with multiple adjustments for improved stability.

Operator Platforms

Dual operator control platforms slide out from either side of the paver for better visibility and feature the latest in technology. Each fully-contained control platform features modern amenities such as a comfortable seat, foot rest, and an ergonomic digital control panel. The stylish control platform mobility combined with its low-profile hood line gives operators unlimited visibility with no line-of-sight disruptions while paving.

The controls are always directly in front of the operator as the entire platform can be adjusted to multiple height positions. The air cleaner and emissions system is mounted below the low-profile hood line to reduce overall height and enhance the operator's ability to see and communicate with the material truck driver. The beacon light can easily be lowered as needed for overhead clearance.

Its sleek, robust design and innovative control panel include push-button keypad controls for paving functions. Operators can choose paving functions at a glance of symbols on the control panel keypad buttons, which also illuminate to show function status. The Plus+ 1® CAN bus system is the brains behind paver controls. The main control panel features the DM430 digital display controller, work function switches, and a vandalism cover for security and protection. The display's wide-angle, anti-glare screen automatically adjusts brightness for sunlight or darker settings. The operator can set the desired paving speed in the display controller and the automated system will pave consistently throughout the day.

High-Speed Rubber Tracks

Rubber tracks offer a smooth, less noisy ride at faster speeds with maximum flotation and more stability, uniformity and consistency. This rubber tracked paver is extremely helpful when paving close to buildings or other tight areas as the turning radius is tighter--you can turn it around with itself on counter-rotation.

Rubber tracks also have an advantage when paving wide and/or deep lifts as there is no variation in machine movements when pushing trucks. The 8680 paver rubber tracks combines a stronger tractive effort than steel tracks, with the smoother ride of a wheeled paver.

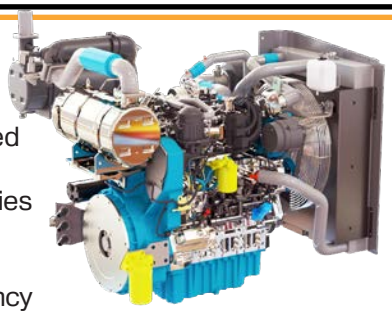
Hydraulic System

Separate pumps for the various functions enhance the paver's hydraulic power. The hydraulic system consists of a tandem drive pump, a tandem blower and charge pump, cylinder pump; and a load-sense pump for the augers and conveyors. Hydraulic motors and cylinders that work together for optimum machine component control. Valves regulate the directional, pressure and fluid flow throughout the hydraulic system. Manifolds are set up to allow for all options. Screed function manifolds are mounted onto the screed. Dual 220,000 BTU oil coolers improve reliability. Hydraulic pressure transducers sense hydraulic pressure levels and send output signals to the control panel digital display unit for ease and convenience when checking and troubleshooting main functions.

Electrical System

The two batteries produces 12 volts DC and maintains 950 cold cranking amperes (CCA) each. An engine-mounted alternator with 150 amperes charging capacity keeps the batteries charged during normal operation.

Engine



A powerful, turbo-charged 210 HP engine equipped with the latest technologies and advanced emission systems allows optimum performance and efficiency while operating functions simultaneously. Cooling systems increase water pump flow while robust lubrication systems produce more flow with higher pressure.

This powerful engine features a remotely-mounted DOC+DPF+SCR after-treatment package with more flow, high pressure, enhanced cooling system and heat reduction. A heat shield for the exhaust manifold reduces heat radiation to the engine. Baffles and louvers direct hot air up and away from the operator. Six rib serpentine belt and pulleys with P-Max fan adds more air flow through the engine cooling system.

Advanced emission-control devices and low-sulfur fuel requirements--only Ultra-Low Sulfur Diesel (ULSD) can be used in these engines--combine with after-treatment Diesel Particulate Filter (DPF) to optimize combustion. This paver is also equipped with a DEF (Diesel Exhaust Fluid) tank. Engine optimization runs all functions at 1800 RPMs for fuel-efficient operation.

The airflow design draws ambient air across the engine compartment, allowing the exhaust air to exit from the side of the machine to provide a cooler engine compartment and operator platform. This also reduces engine power demand, maximizes fuel efficiency, minimizes emissions and enhances quiet operation.

Accessing the radiator and other engine components is as simple as pushing a switch to raise and lower the engine hood. The engine is also accessible through several access doors, with a front service door where fuel filter components are located for easy access.

Refer to the Engine Operator's Manual accompanying your paver for additional engine information.

Screed

The 8680 Commercial Paver features the new LB5 screed that is packed with standard features along with other custom options. Or choose the monumental LB7 screed option, packed with easy-to-use highway paving features especially for screed operators. Both provide variable paving widths from 8 to 15.5 feet, horizontal depth screws, automatic temperature control, and spring-loaded end gates for an unblemished asphalt mat.

Multiple adjustment mechanisms are located on top of the screed, extensions and end gate along with adjustable strike-offs. Electric dual vibrators mounted on the main screed frame increase paving material compaction. The screed extensions also include slope and berm capability. The LB7 screed option also includes electronic vertical height and angle-of-attack (AOA) adjustments, along with hydraulic crown/invert adjustment. An energy chain on each screed extension protects the hard wiring as the screed extension moves back and forth for added protection and less maintenance. S-curve heating elements in the main screed with straight heating elements in the extensions cycle during heat-up to save horsepower and fuel consumption while paving.

Augers

Right and left augers rotate to move material from the conveyors to the screed. The 14 or 16-inch augers can be adjusted up or down, and operated independently of each other.

The augers are controlled from the operator platform control panel or by the screed operator using the screed control box on each side of the paver. Operators can control the augers manually or automatically using the sonic sensor mounted on either screed end gate, where it monitors and controls the pile height of material at the discharge end of the augers.

The dual augers are hardened, 3/4-inch thick, 14 or 16-inch round, HRC 55-60 augers, that can operate in reverse. The modular auger assembly, in one-foot sections, allows for a simple quick change of the assembly.

Hopper and Conveyors

Material in the hopper is moved toward the back of the paver to the screed by conveyors.

The narrow, automatic conveyors with a wide hopper design delivers asphalt more evenly and efficiently to produce a smooth asphalt mat. The conveyors are also automatic, adjustable, and reversible through the screed control boxes.

Hydraulic power raises and lowers the hopper wings, which can also be operated individually while paving close to buildings, barriers and other obstacles. The hopper's rounded design reduces cold material build-up while rear hopper wing wipers help prevent spillage. The hydraulic front apron can be raised and lowered to form a snug fit with the material truck, which also helps clean asphalt off the apron and front corners of the hopper. The hopper wings can also be locked into place (in the closed position) for transport.

MACHINE OVERVIEW

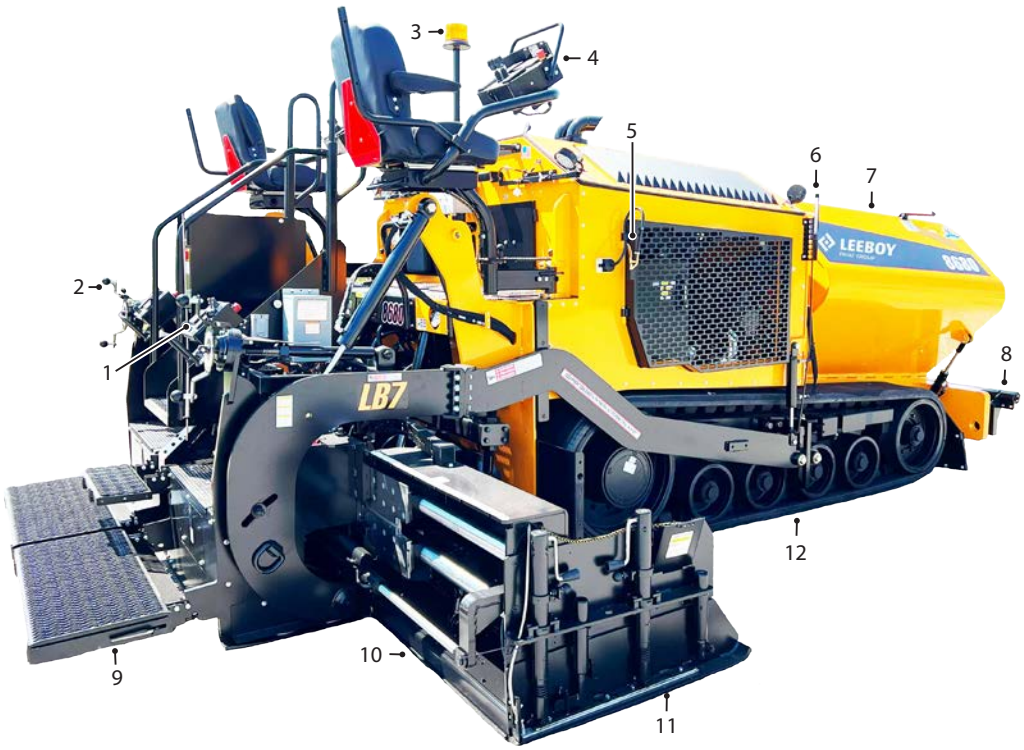


Figure 3-1. Machine Overview

ITEM NO.	CONTROL NAME	FUNCTION
1	Screed Control Panel	Located on each side of the screed. Features numerous paving controls.
2	Horizontal Depth Screw	Controls depth of the asphalt (one on each side).
3	Beacon Light	Revolving amber safety light to alert traffic and others at the work site.
4	Main Control Panel	Main operator control panel with joysticks, optional steering wheel or optional steering knob.
5	Spray Down Wand	Spray wands located on both sides for daily cleaning after paving.
6	Screed Elevation Guide	This gauge provides the operator with a visual reference for the tow point's position (one on each side).
7	Hopper	Hopper wings fold out to hold asphalt material.
8	Push Rollers or Hydraulic Truck Hitch Option	Push rollers adjusts height (manually) for roll-up to the asphalt material truck. The optional truck hitch is hydraulically driven for roll-up, but also hooks around the inside of the asphalt truck wheel, hitching the paver to the truck. Controlled from the main control panel.
9	Foldable Walkboard	Walkboard provides added convenience for screed operators. Can be folded up when not in use.
10	Screed Extension	Extends paving width from 8 to 15.5 feet (one extension on each side).
11	End Gate	Adjusts for a paving width to match joints and leave a clean edge.
12	High-Speed Rubber Tracks	Offers a smoother and more quiet ride with better traction than steel tracks.

MAIN CONTROL PANEL OVERVIEW

Left Control Panel Shown

Figure 3-2. Main Control Panel Overview

Table 3-1. Main Control Panel Overview

ITEM NO.	CONTROL NAME	FUNCTION
1	Lower Left Keypad	The lower left keypad features 8 buttons and contains the following paver functions: Left and Right Auger/Conveyor On/Off; Left and Right Cut-Off Open/Closed; and Left Screed Extension Extend/Retract.
2	Upper Left Keypad	The upper left keypad features 12 buttons and contains the following paver functions: Start/Stop (On/Off); Beacon/Work Lights; Grade/Slope; Truck Hitch In/Out; Spray Down On/Off; Slope/Berm Select; Slope/Berm Up/Down; Front Apron Up/Down and Vibrator On/Off.
3	DM430 Digital Display	Displays engine operating information, status and system faults.
4	Upper Right Keypad	The upper right keypad features 12 buttons and contains the following paver functions: Work High/Low Speed; Travel High-Low Speed; Horn; Left and Right Tow Up/Down; Auger Lift Up/Down; Pause; Apron Up/Down; Pad Builder; and Generator On/Off.
5	E-Stop Button	Immediately stops the paver engine and all electronic functions. Turn clockwise to release E-Stop button.
6	Lower Right Keypad	The lower right keypad features 8 buttons and contains the following paver functions: Hopper Wing Select; Active Console; Hopper Wing(s) Open/Close; Screed Raise; Screed Float; and Right Screed Extension Extend/Retract.
7	Joysticks	Joysticks control speed and direction forward or reverse, and also turns the paver right or left. The farther pushed in either direction, the faster the speed. The locking collar locks the joystick in NEUTRAL. You must pull up on the locking collar to move joysticks forward or reverse.
8	Steering Knob Option	If equipped with this option, the steering knob assembly replaces the joystick on the right side.

CONTROL PANEL WITH STEERING WHEEL OPTION

The control icons and keypads are the same, but arranged slightly different if equipped with the steering wheel option.

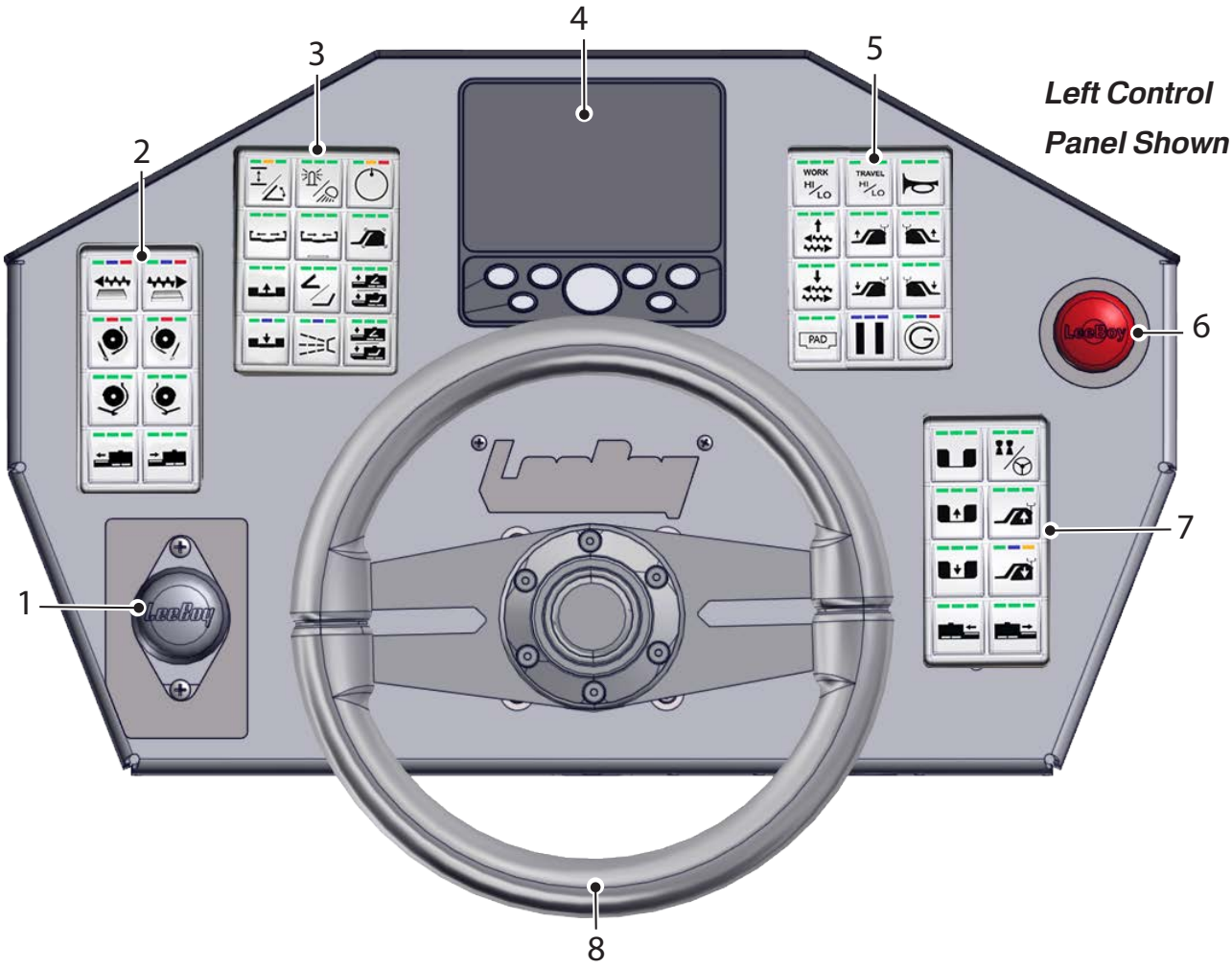


Figure 3-3. Main Control Panel With Steering Wheel Overview

Table 3-1. Main Control Panel Overview

ITEM NO.	CONTROL NAME	FUNCTION
1	Forward/Neutral/ Reverse (FNR) Joystick	Joysticks control speed and direction by pushing the joystick forward or reverse. The center position is neutral. The farther pushed in either direction, the faster the speed. The locking collar locks the joystick in NEUTRAL. You must pull up on the locking collar to move joysticks forward or reverse.
2	Lower Left Keypad	The lower left keypad features 8 buttons and contains the following paver functions: Left and Right Auger/Conveyor On/Off; Left and Right Cut-Off Open/Closed; and Left Screed Extension Extend/Retract.
3	Upper Left Keypad	The upper left keypad features 12 buttons and contains the following paver functions: Start/Stop (On/Off); Beacon/Work Lights; Grade/Slope; Truck Hitch In/Out; Spray Down On/Off; Slope/Berm Select; Slope/Berm Up/Down; Front Apron Up/Down and Vibrator On/Off.
4	DM430 Digital Display	Displays engine operating information, status and system faults.
5	Upper Right Keypad	The upper right keypad features 12 buttons and contains the following paver functions: Work High/Low Speed; Travel High-Low Speed; Horn; Left and Right Tow Up/Down; Auger Lift Up/Down; Pause; Apron Up/Down; Pad Builder; and Generator On/Off.
6	E-Stop Button	Immediately stops the paver engine and all electronic functions. Turn clockwise to release E-Stop button.
7	Lower Right Keypad	The lower right keypad features 8 buttons and contains the following paver functions: Hopper Wing Select; Active Console; Hopper Wing(s) Open/Close; Screed Raise; Screed Float; and Right Screed Extension Extend/Retract.
8	Steering Wheel	Steers right and left.

Upper Left Keypad

(Left Operator Platform Control Panel Shown)

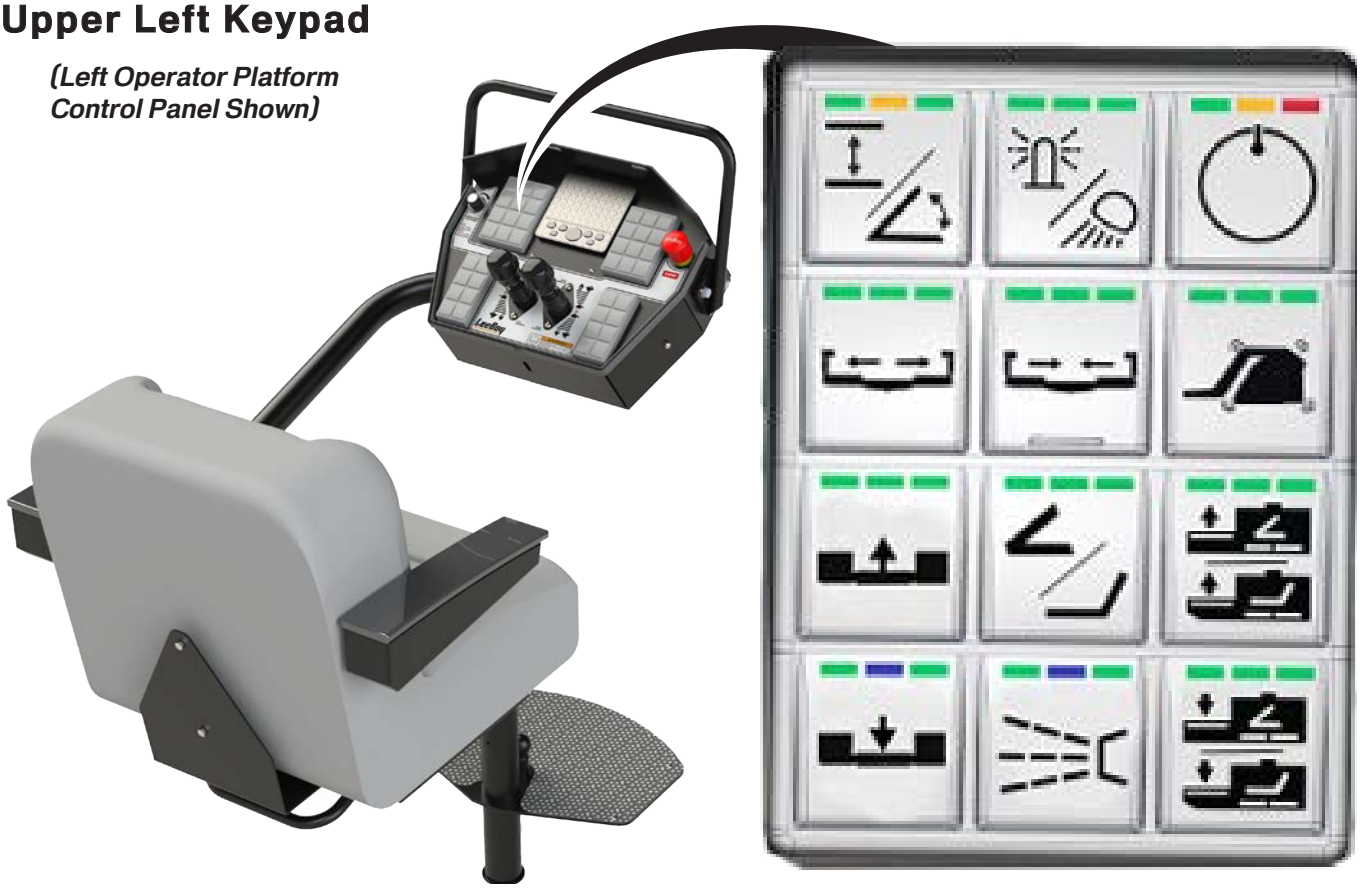
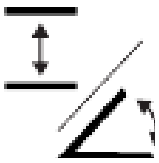
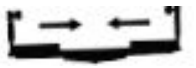
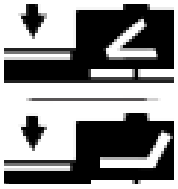
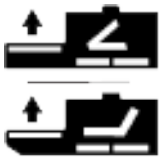


Table 3-2. Control Panel - Upper Left Keypad

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Grade/Slope	You can choose to control tow points manually or automatically via the TopCon® sonic automatic grade/slope control. The default setting is MANUAL. The left LED light illuminates GREEN in this mode on both consoles. Press and hold the button to set up the TopCon® sensor system and operate in AUTO mode. The right LED light illuminates GREEN in this mode on both consoles. **NOTE: The Grade/Slope function is disabled when joysticks are in neutral. The center light will illuminate YELLOW.**
	Beacon/Work Lights	Upon start-up, the lights are OFF (default) and can be operated from the active and inactive consoles. LED status lights also illuminate on both consoles. Press the button once to activate the beacon light (the left LED light illuminates GREEN). Press the button again to activate the work lights (the center LED light illuminates GREEN). Press the button a third time to activate the beacon and work lights (all three LED lights illuminate GREEN). Pressing the button a fourth time turns all lights OFF.
	Start/Stop (On/Off)	This button allows the operator to start and stop the engine from the active or inactive console. With ignition key in the ON position and all four joysticks in neutral, press this button and hold one second to start the engine. The center light illuminates amber during “Wait to Start” stage. The left LED lights illuminate GREEN on both consoles when the engine is running. Push the button again to stop the engine. The right LED light illuminates RED. **NOTE: All joysticks on BOTH operator platform control panels *must* be in NEUTRAL and the E-Stop button engaged to start the engine.**

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Truck Hitch OUT (OPTION)	Attaches truck hitch to the material truck if equipped with this option. Can be operated from active and inactive consoles (LED status lights Illuminate GREEN while engaged on both consoles). Press and hold button to move truck hitch arms outward.
	Truck Hitch IN (OPTION)	Attaches truck hitch to the material truck if equipped with this option. Can be operated from active and inactive consoles (LED status lights Illuminate GREEN while engaged on both consoles). Press and hold button to move truck hitch arms inward.
	Vibrator	This button turns the screed vibrator ON and OFF and can be operated from active or inactive consoles. The screed vibrator helps compact material while paving. Press the button once to turn ON. All three LED lights will illuminate green on both consoles when engaged. The vibrator automatically turns OFF if paver is placed in reverse or neutral.
	Apron UP	Use this button to raise the hopper's front apron to help clean asphalt off the apron and front corners of the hopper. Press and hold button to desired angle. The LED status lights Illuminate GREEN while engaged on both consoles.
	Apron Down	Use this button to lower the hopper's front apron of the hopper to help clean asphalt off the apron and front corners of the hopper. Press and hold button to desired angle. The LED status lights Illuminate GREEN while engaged on both consoles.
	Slope/Berm Select	Use this button to select SLOPE or BERM mode. Press the button once to select SLOPE. The left LED light illuminates GREEN in this mode on both consoles and the left or right screed keypad (corresponding to the console activated). Push the button twice to select BERM mode. The right LED light illuminates GREEN on both consoles and the corresponding screed keypad.
	LEFT Slope/Berm DOWN	Pressing this button enables the Slope or Berm DOWN function, depending upon which mode is selected on the Slope/Berm Select button. When activated by pressing the button, all three LED lights illuminate GREEN on both consoles and the left or right screed keypad (corresponding to the console activated).
	LEFT Slope/Berm UP	Pressing this button enables the Slope or Berm UP function, depending upon which mode is selected on the Slope/Berm Select button. When activated by pressing the button, all three LED lights illuminate GREEN on both consoles and the left or right screed keypad (corresponding to the console activated).
	Spray Down	Engages the spray down system for spraying release agent to clean, lubricate and prevent asphalt from hardening on machine surfaces. Press the button once to turn ON. All three LED lights will illuminate GREEN. Press the button again to turn OFF. The spray down function can be engaged on both active and inactive consoles.

Lower Left Keypad

(Left Operator Platform Control Panel Shown)

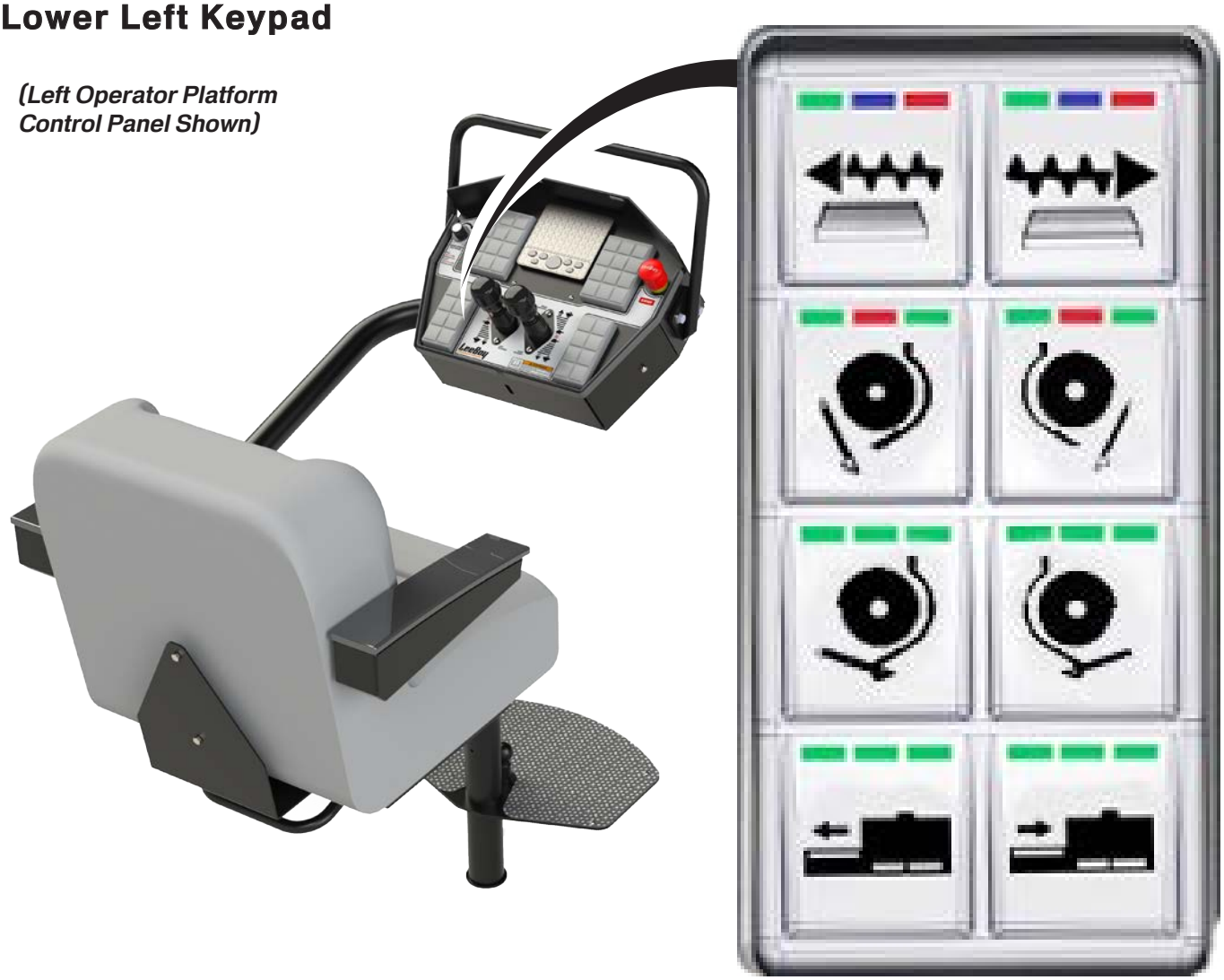
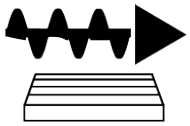


Table 3-3. Control Panel - Lower Left Keypad

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Auger/Conveyor LEFT	Turns the LEFT auger and conveyor on and off. The left LED light illuminates GREEN while conveyor is ON; the right RED light illuminates when OFF, and BLUE if paver is paused while conveyors are operating (or the material has shut off the sensor). Press the button once to turn ON the LEFT auger and conveyor (AUTO mode). Press the button and hold for MANUAL override. Press button again to turn off the LEFT auger and conveyor. **NOTE: Moving the joysticks into neutral pauses operation in AUTO mode.**

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Conveyor RIGHT	Turns the RIGHT auger and conveyor on and off. The right LED light illuminates GREEN while auger/conveyor is ON; the left RED light illuminates when OFF, and BLUE if paver is paused while the right auger/conveyor are operating (or the material has shut off the sensor). Press the button once to turn ON the RIGHT auger and conveyor (AUTO mode). Press the button and hold for MANUAL override. Press button again to turn off the RIGHT auger and conveyor. **NOTE: Moving the joysticks into neutral pauses operation in AUTO mode.**
	LEFT Cut-Off CLOSED	Press and hold to close the LEFT Under Auger Material Cut-Off. When closed, the center LED light illuminates RED on both operator consoles and the LEFT screed control keypad. If paver is put into reverse, the cut-offs will automatically close and the center LED light illuminates RED if operating in AUTO mode.
	RIGHT Cut-Off CLOSED	Press and hold to close the RIGHT Under Auger Material Cut-Off. When closed, the center LED light illuminates RED on both operator consoles and the LEFT screed keypad. If paver is put into reverse, the cut-offs will automatically close and the center LED light illuminates RED if operating in AUTO mode.
	LEFT Cut-Off OPEN	Press and hold to open the LEFT Under Auger Material Cut-Off. All three LED lights illuminate GREEN on both operator consoles and the LEFT screed keypad while open.
	RIGHT Cut-Off OPEN	Press and hold to open the RIGHT Under Auger Material Cut-Off. All three LED lights illuminate GREEN on both operator consoles and the LEFT screed keypad while open.
	LEFT Extension EXTEND	Extends the LEFT screed extension. Press and hold button to extend the left screed extension to desired width. All three LED lights illuminate GREEN on both consoles and the LEFT screed keypad while engaged.
	LEFT Extension RETRACT	Retracts the LEFT screed extension. Press and hold button to retract the left screed extension to desired width. All three LED lights illuminate GREEN on both consoles and the LEFT screed keypad while engaged.

Upper Right Keypad

(Left Operator Platform Control Panel Shown)

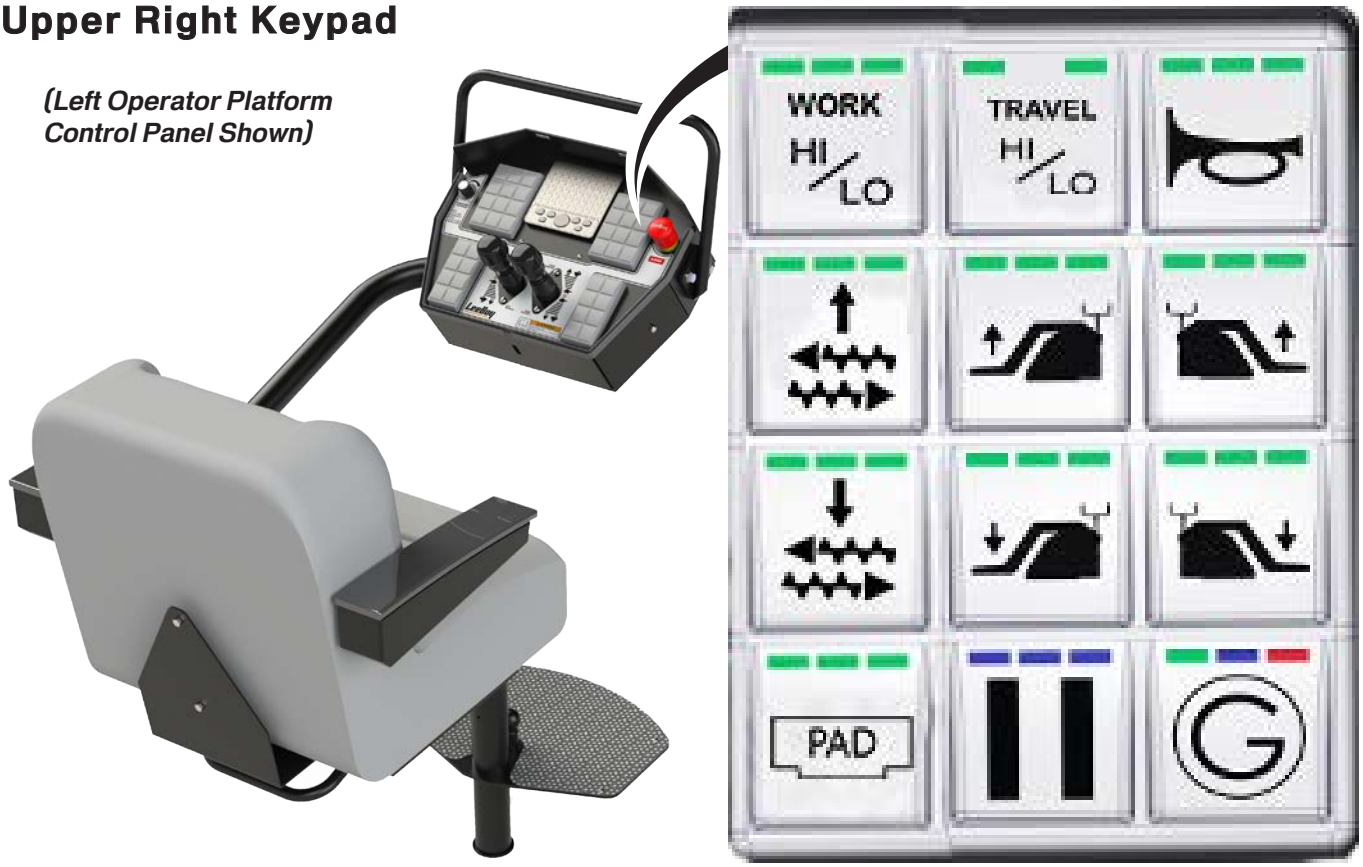
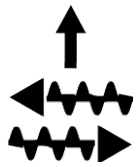
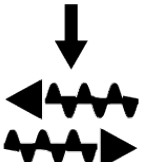


Figure 3-4. Control Panel - Upper Right Keypad

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
WORK HI/LO	Work High/Low Speed	This button is used to change paver low and high speed in the working mode. The right LED light illuminates GREEN on both consoles at low speed. Press the button to shift to high work speed. The left LED light illuminates GREEN on both consoles at high speed. The work speed can be shifted between the two speeds as needed. **NOTE: Only pave in work speed.**
TRAVEL HI/LO	Travel High/Low Speed	This button is used to change paver low and high speed in travel mode. The right LED light illuminates GREEN on both consoles at low speed. Press the button to shift to high travel speed. The left LED light illuminates GREEN on both consoles at high speed. The work speed can be changed by stopping, then shifting, between the two speeds as needed. **NOTE: Only pave in work speed.**
	Horn	Pressing this button sounds the horn, and functions on both consoles along with the screed keypads. All three LED lights will illuminate GREEN when engaged.

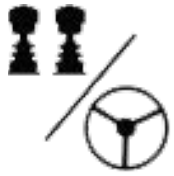
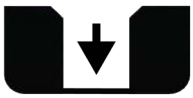
KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Auger Lift	Raises the augers. • Press and hold button to raise both the left and right auger. • Releasing the button will hold the augers at that position. **NOTE: The Under Auger Material Cut-Offs will automatically open when raising the augers.**
	Auger Lower	Lowers the augers. • Press and hold button to lower both the left and right auger. • Releasing the button will hold the augers at that position.
	LEFT Tow UP	Sets the asphalt mat thicker and controls the hydraulic tow cylinders. Press and hold button to raise LEFT tow. All three LED lights illuminate GREEN while engaged on both consoles and the LEFT screed keypad. **NOTE: Does not function when the grade control is set to AUTO.**
	RIGHT Tow UP	Sets the asphalt mat thicker and controls the hydraulic tow cylinders. Press and hold button to raise RIGHT tow. All three LED lights illuminate GREEN while engaged on both consoles and the RIGHT screed keypad. **NOTE: Does not function when the grade control is set to AUTO.**
	LEFT Tow DOWN	Sets the asphalt mat thinner and controls the hydraulic tow cylinders. Press and hold button to lower LEFT tow. All three LED lights illuminate GREEN while engaged on both consoles and the LEFT screed keypad. **NOTE: Does not function when the grade control is set to AUTO.**
	RIGHT Tow DOWN	Sets the asphalt mat thinner and controls the hydraulic tow cylinders. Press and hold button to lower RIGHT tow. All three LED lights illuminate GREEN while engaged on both consoles and the RIGHT screed keypad. **NOTE: Does not function when the grade control is set to AUTO.**
	Pad Builder	Pressing this button turns on the augers and conveyors, along with opening the Under Auger Material Cut-Offs. • Press and hold this button to manually overide both conveyors.
	Pause	Pauses propel (travel), conveyors and augers functions. Press the button once to pause these functions. The three LED lights illuminate BLUE on both consoles. Pressing the button again resumes these functions. **NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)**
	Generator	The generator can be activated in AUTO or MANUAL mode from the active or inactive console. The default is Auto. • Press the button once for AUTO operation (the temperature setting controls operation). The left LED lights illuminate GREEN. • Press the button again to turn off (the right LED light illuminates RED).

Lower Right Keypad

(Left Operator Platform Control Panel Shown)

Figure 3-5. Control Panel - Lower Right Keypad

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Hopper Wing Select	Used to select the right, left or both hopper wings, and can be operated on both active and inactive consoles. Upon start-up, the operational mode is set by default to operating both hopper wings. The LED lights illuminate GREEN. • Pressing the button once enables operating only the LEFT hopper wing. The left LED light illuminates GREEN on both consoles. • Pressing the button again operates only the RIGHT hopper wing. The right LED light illuminates GREEN on both consoles. • Pressing the button again returns to the default operational control of both hopper wings.

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Active/Inactive Console	Both consoles are inactive upon start-up. All four joysticks must be in the NEUTRAL position to activate either console. Press this button on the desired operator platform console once to activate its control panel console. The three LED lights will illuminate GREEN on the active console. **NOTE: All joysticks on both operator platform control panels *must* be in NEUTRAL to deactivate the console in use and activate the other console.** **NOTE: The opposite console can be activated by selecting multiple times.** **NOTE: If an active console becomes inactive while the paver is moving, the paver will automatically slow down to a stop and apply park brake.**
	Hopper Wing(s) OPEN	Opens the right, left or both hopper wings, depending upon hopper mode selected, and can be operated from active and inactive consoles. Press and hold the button to open the hopper wing(s). The center LED lights GREEN while engaged on both consoles.
	Screed Raise	Press and hold button to raise the screed. All three LED lights illuminate GREEN while engaged on both consoles. This function can only be operated from the active console.
	Hopper Wing(s) CLOSE	Closes the right, left or both hopper wings, depending upon hopper mode selected, and can be operated from active and inactive consoles. Press and hold the button to close the hopper wing(s). All three LED lights GREEN while engaged on both consoles.
	LEFT Screed Float	This button controls floating the screed, and can only be operated from the active console. Placing the joysticks on the active console into neutral moves into screed hold, and float resumes when joysticks are returned to the forward position. The center LED light illuminates GREEN when screed float is active, and BLUE when in screed hold, on both consoles.
	RIGHT Screed Extension RETRACT	Retracts the RIGHT screed extension. Press and hold button to retract the right screed extension to desired width. All three LED lights illuminate GREEN on both consoles and the RIGHT screed keypad while engaged.
	RIGHT Screed Extension EXTEND	Extends the RIGHT screed extension. Press and hold button to extend the right screed extension to desired width. All three LED lights illuminate GREEN on both consoles and the RIGHT screed keypad while engaged.

JUNCTION BOX CONTROLS

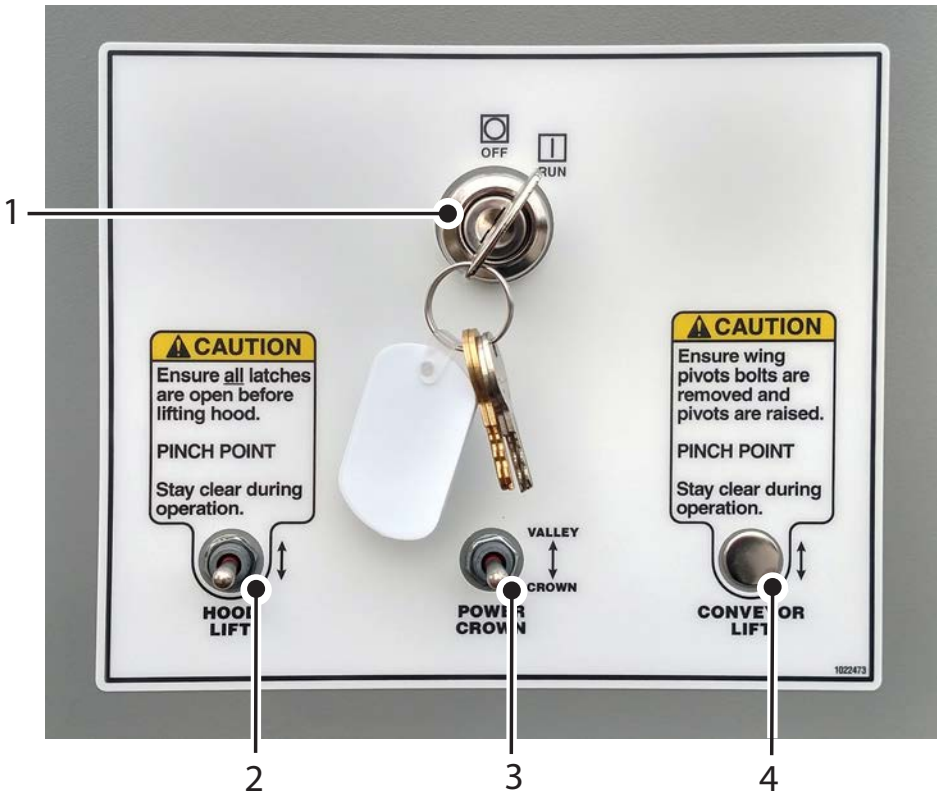


Figure 3-6. Junction Box Controls

ITEM NO.	CONTROL NAME	FUNCTION
1	Ignition	Insert key into the ignition switch and turn clockwise to the RUN position, press the Start/Stop button on the main control panel, and activate the desired control panel to start the paver. ***(Page 4-5)*** **NOTE: The engine will not start unless *both* joysticks on both operator platforms (4 total) are in the NEUTRAL position.**
2	Hood Lift Switch	Raises and lowers the engine hood. ***(Page 5-5)*** • Push switch UP and hold to raise the hood. • Press switch DOWN and hold to lower the hood. ⚠CAUTION **Ensure all hood latches (5) are unlatched before raising the hood. Ensure all latches are securely latched before operating.** ***(Page 5-6)***
3	Power Crown Switch (OPTION)	Automatically adjusts crown and valley. • Push switch UP for positive crown adjustment. • Push switch DOWN for negative valley adjustment. **NOTE: The switch does not function if not equipped with this option.**
4	Nonfunctional	Nonfunctional on this model.

SCREED CONTROLS

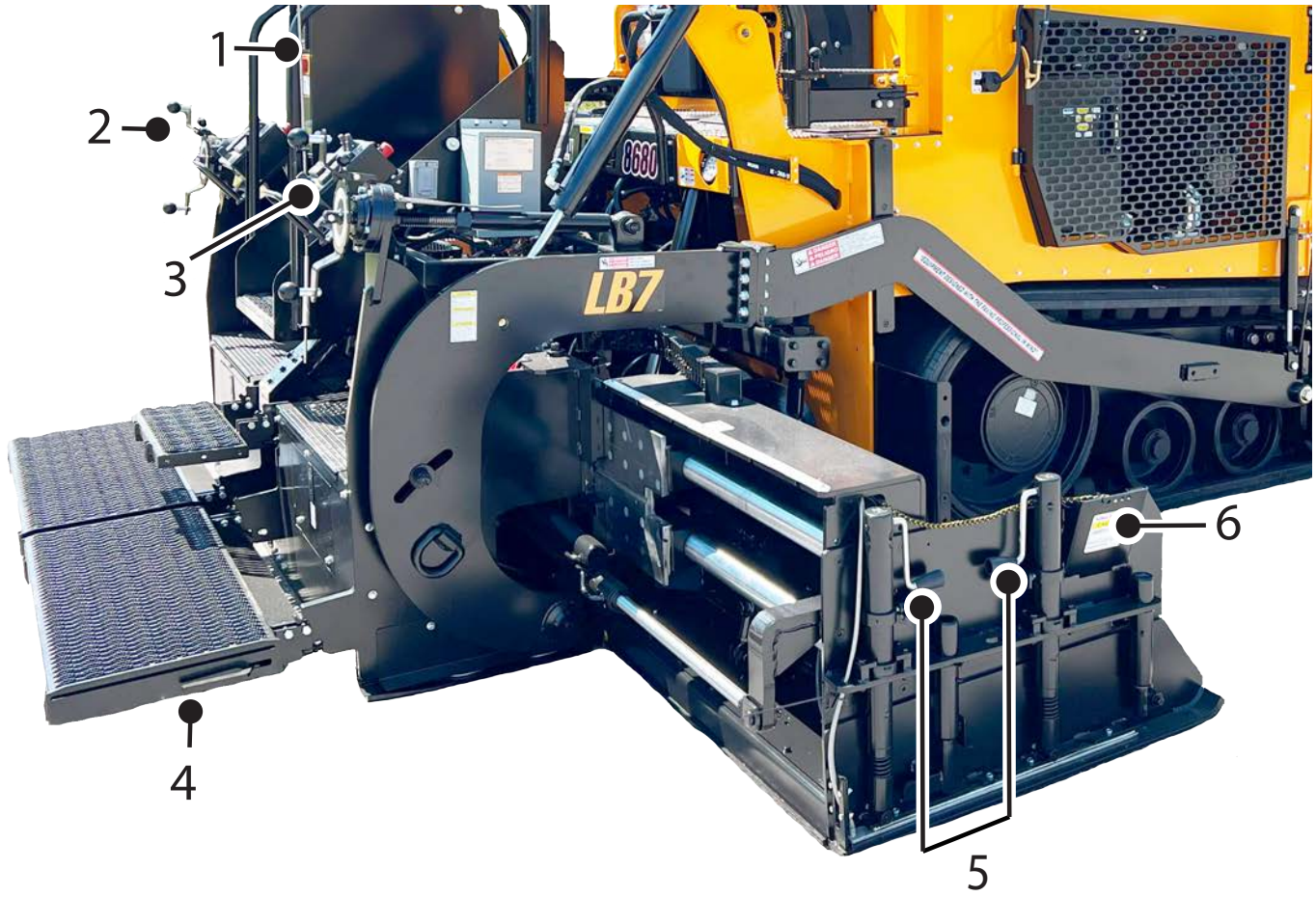


Figure 3-7. Screed Controls

ITEM NO.	CONTROL NAME	FUNCTION
1	Electric Heat Box	Control box for the electric screed heat and heated end gate.
2	Horizontal Depth Screw	Controls depth of the asphalt (one on each side).
3	Screed Control Panel	Located on each side of the screed. Features numerous paving controls.
4	Foldable Walkboard	Walkboard provides added convenience for screed operators. Can be folded up when not in use.
5	End Gate Adjusters	Raises, lowers and adjusts angle of end gates.
6	Sonic Auger Sensor Mount	Mount for the Sonic Auger Sensor, which senses and automatically gauges the amount of material in the extensions.

LEFT Screed Control Box - Left Keypad

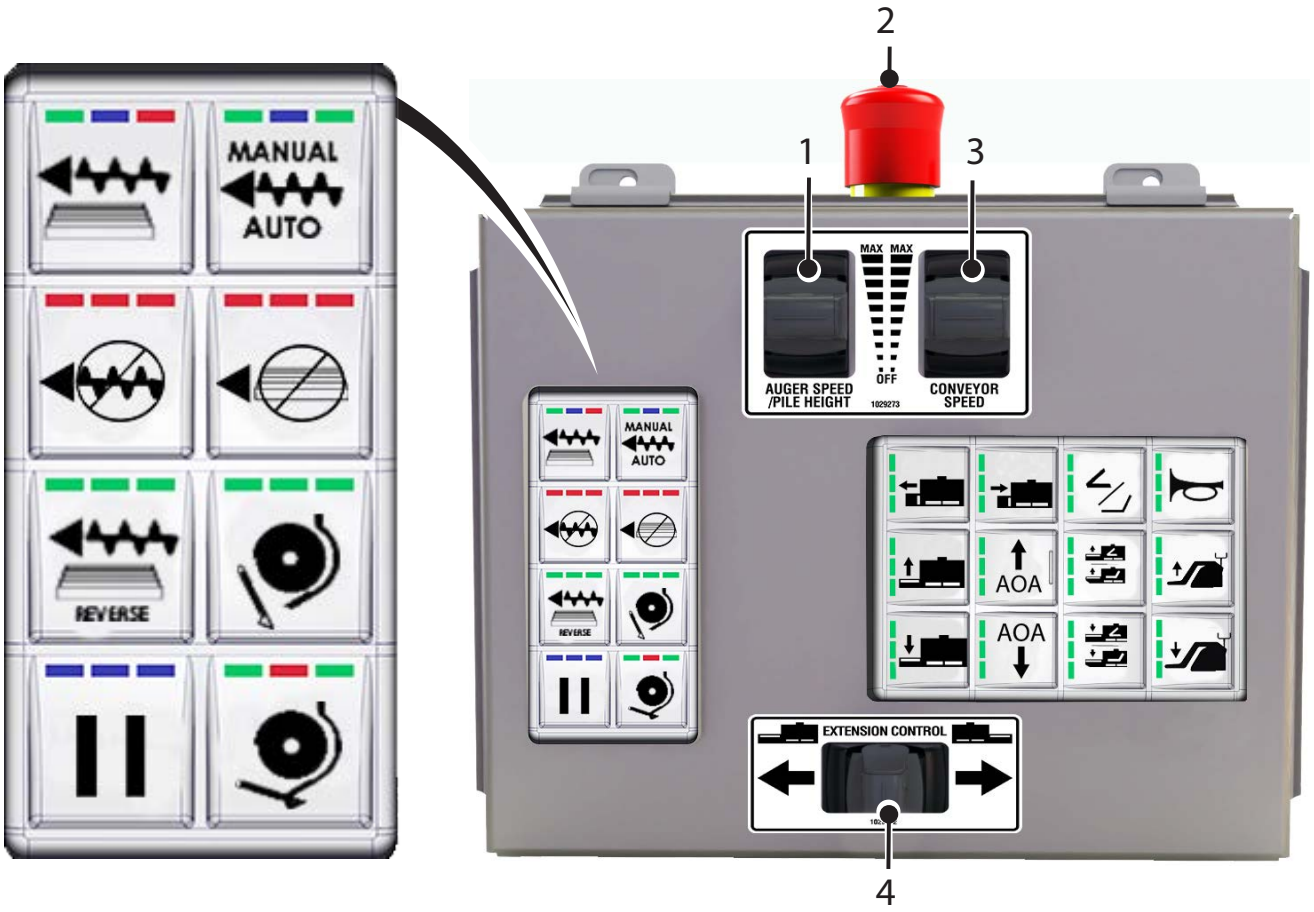
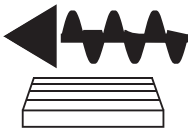
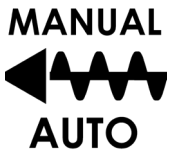
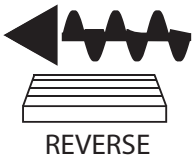


Figure 3-8. LEFT Screed Controls (Left Keypad)

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
1	Auger Pile Speed	• Set switch up/down to increase/decrease the auger speed when auger is set to MANUAL mode. • Set switch up/down to increase/decrease the pile height when auger is set to AUTO mode.
2	Emergency Stop (E-Stop) Button	Immediately stops the paver engine and all electronic functions. Turn clockwise to release E-Stop button.
3	Conveyor Speed	Set switch up/down to increase/decrease the conveyor speed when operating in MANUAL mode.
4	Extension IN/OUT	• Push and hold this switch to extend (OUT) or retract (IN) the screed extension on the corresponding control box side of operation.

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Auger/Conveyor ON/OFF	Press this button once to turn ON the auger and conveyor. • The left LED light illuminates GREEN when the auger and conveyor is on. • The default setting is AUTO mode, which is controlled by the Auger MANUAL/AUTO button. • Press the button again to turn OFF the auger and conveyor. • You can also use the auger OR conveyor bypass button to pause operation of either function separately.
	Auger MANUAL/ AUTO Select	This button only functions if the Auger/Conveyor ON/OFF button is turned ON. Independent right and left augers can be operated in MANUAL or AUTO (automatic) mode. AUTO mode is the default. • Press button twice to operate auger in MANUAL mode. The left LED light illuminates GREEN. In MANUAL mode, the operator controls auger speed using the auger pile speed switch above the keypad. • Press the button once to operate in AUTO mode. The right LED light illuminates GREEN. The center blue light illuminates when material reaches the set material height. In AUTO mode, the auger dial controls material height. **The auger sonic sensor must be connected to operate in AUTO mode.**
	Auger/Conveyor Reverse	Press the Auger/Conveyor Reverse button for reversing the direction of both functions to help clear material as needed. All three LED lights illuminate while the auger/conveyor reverse is active.
	LEFT Cut-Off OPEN	Press and hold to open the LEFT Under Auger Material Cut-Off. All three LED lights illuminate GREEN on both operator consoles and the LEFT screed keypad while open.
	Pause	Pauses propel (travel), conveyors and augers functions. Press the button once to pause these functions. The three LED lights illuminate blue. • Pressing the button again resumes these functions. **NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)**
	LEFT Cut-Off CLOSED	Press and hold to close the LEFT Under Auger Material Cut-Off. When closed, the center LED light illuminates RED on both operator consoles and the LEFT screed control keypad. If paver is put into reverse, the cut-offs will automatically close and the center LED light illuminates RED when operating in AUTO mode.

LEFT Screed Control Box - Right Keypad

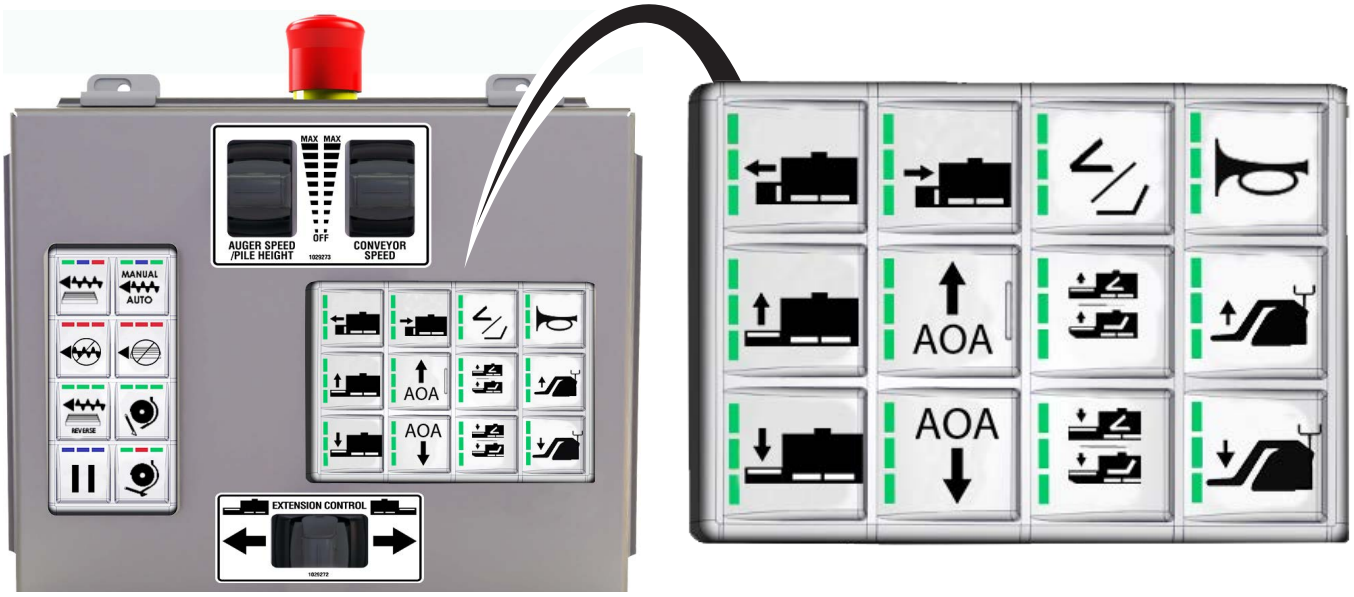
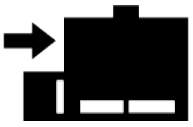
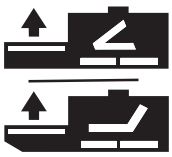
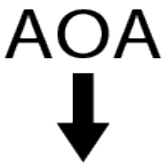
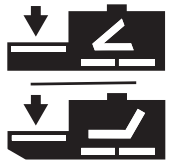


Figure 3-9. LEFT Screed Controls (Right Keypad)

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Tunnel Extension OUT	The tunnel extension allows material to flow while paving at wide widths. Press and hold the button to extend the auger. The left LED light illuminates green in MANUAL mode.
	Tunnel Extension IN	The tunnel extension allows material to flow while paving at wide widths. Press and hold the button to retract the auger. The right LED light illuminates green in MANUAL mode.
	Slope/Berm SELECT	Use this button to select SLOPE or BERM mode. • The default is set to OFF. • Press the button once to select SLOPE. The LEFT LED light illuminates GREEN on the keypad (and both consoles) in this mode. • Push the button twice to select BERM mode. The right LED light illuminates GREEN in this mode.
	Horn	Pressing this button sounds the horn whether the console is active or not. All three LED lights will illuminate GREEN when engaged.
	Vertical Up	Automatically increase the screed extension vertical height by pressing and holding the vertical UP. All three LED lights illuminate while in operation.

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	AOA Up	Press and hold this button to make adjustments to the Angle-of-Attack (AOA). Adjust the mat texture using the Angle-of-Attack (AOA) adjuster to produce a smooth or coarse mat texture.
	Slope/Berm UP	Press and hold this button to enable the Slope or Berm UP function, depending upon which mode is selected on the Slope/Berm Select button. When activated by pressing the button, all three LED lights illuminate GREEN on both main consoles and the left or right screed control box (corresponding to the active console).
	Tow UP	Sets the asphalt mat thicker and controls the hydraulic tow cylinder. Press and hold button to raise LEFT tow. All three LED lights illuminate GREEN on the keypad (and both consoles) while engaged. **NOTE: Only functions with joysticks in neutral when the grade control is set to AUTO.**
	Vertical Down	Automatically decrease the screed extension vertical height by pressing and holding the vertical DOWN. All three LED lights illuminate while in operation.
	AOA Down	Press and hold this button to make adjustments to the Angle-of-Attack (AOA). Adjust the mat texture using the Angle-of-Attack (AOA) adjuster to produce a smooth or coarse mat texture.
	Slope Berm DOWN	Press and hold this button to enable the Slope or Berm DOWN function, depending upon which mode is selected on the Slope/Berm Select button. When activated by pressing the button, all three LED lights illuminate GREEN on both main consoles and the left or right screed control box (corresponding to the active console).
	Tow DOWN	Sets the asphalt mat thicker and controls the hydraulic tow cylinder. Press and hold button to lower LEFT tow. All three LED lights illuminate GREEN on the keypad (and both consoles) while engaged. **NOTE: Only functions with joysticks in neutral when the grade control is set to AUTO.**

RIGHT Screed Control Box - Left Keypad

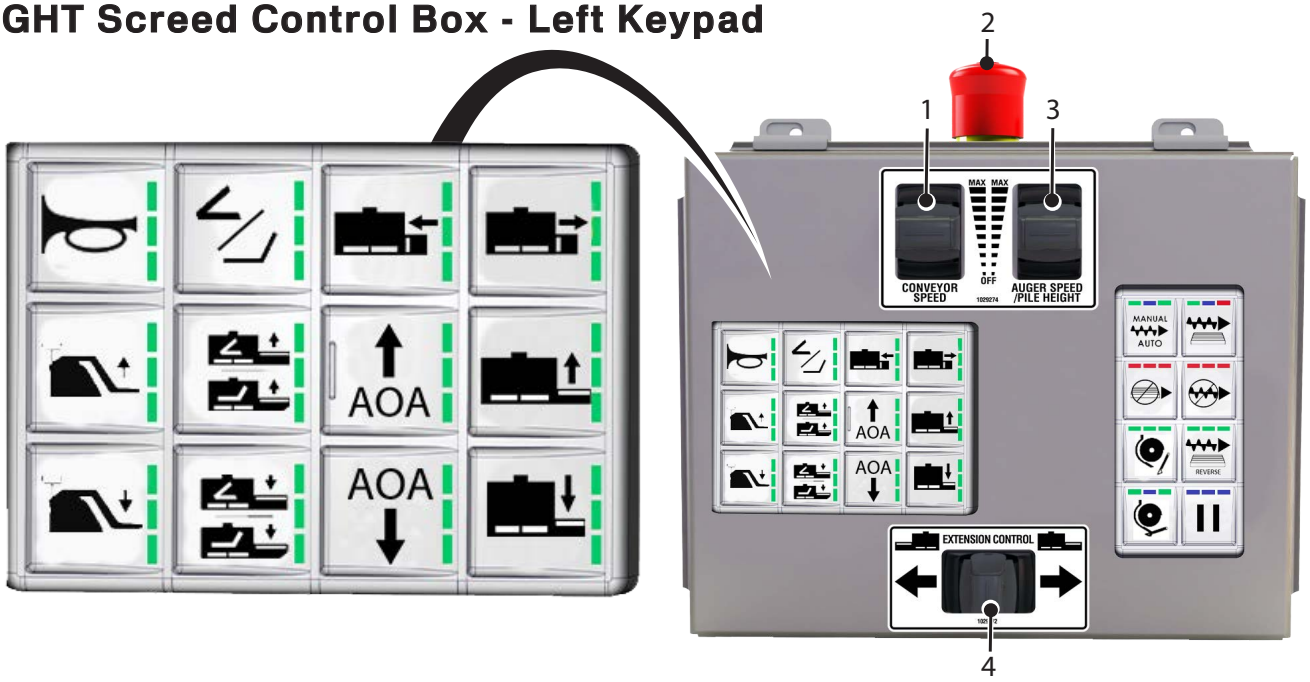
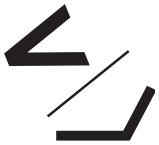
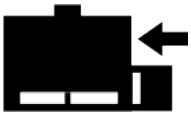
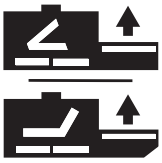
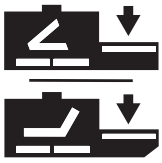
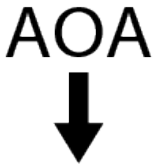


Figure 3-10. RIGHT Screed Controls (Left Keypad)

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
1	Conveyor Speed	• Set this switch up/down to increase/decrease the conveyor speed when operating in MANUAL mode.
2	Emergency Stop (E-Stop) Button	Immediately stops the paver engine and all electronic functions. Turn clockwise to release E-Stop button.
3	Auger Pile Speed	• Set switch up/down to increase/decrease the auger speed when auger is set to MANUAL mode. • Set switch up/down to increase/decrease the pile height when auger is set to AUTO mode.
4	Extension IN/OUT	• Push and hold this switch to extend (OUT) or retract (IN) the screed extension on the corresponding control box side of operation.
	Horn	Pressing this button sounds the horn whether the console is active or not. All three LED lights will illuminate GREEN when engaged.
	Slope/Berm SELECT	Use this button to select SLOPE or BERM mode. • The default is set to OFF. • Press the button once to select SLOPE. The LEFT LED light illuminates GREEN on the keypad (and both consoles) in this mode. • Push the button twice to select BERM mode. The right LED light illuminates GREEN in this mode.
	Tunnel Extension IN	The tunnel extension allows material to flow while paving at wide widths. Press and hold the button to retract the auger. The left LED light illuminates green in MANUAL mode.

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Tunnel Extension OUT	The tunnel extension allows material to flow while paving at wide widths. Press and hold the button to extend the auger. The right LED light illuminates green in MANUAL mode.
	Tow UP	Sets the asphalt mat thicker and controls the hydraulic tow cylinder. Press and hold button to raise RIGHT tow. All three LED lights illuminate GREEN on the keypad (and both consoles) while engaged. **NOTE: Only functions with joysticks in neutral when the grade control is set to AUTO.**
	Slope/Berm UP	Press and hold this button to enable the Slope or Berm UP function, depending upon which mode is selected on the Slope/Berm Select button. When activated by pressing the button, all three LED lights illuminate GREEN on both main consoles and the left or right screed control box (corresponding to the active console).
	AOA Up	Press and hold this button to make adjustments to the Angle-of-Attack (AOA). Adjust the mat texture using the Angle-of-Attack (AOA) adjuster to produce a smooth or coarse mat texture.
	Vertical UP	Automatically increase the screed extension vertical height by pressing and holding the vertical UP. All three LED lights illuminate while in operation.
	Tow DOWN	Sets the asphalt mat thicker and controls the hydraulic tow cylinder. Press and hold button to lower RIGHT tow. All three LED lights illuminate GREEN on the keypad (and both consoles) while engaged. **NOTE: Only functions with joysticks in neutral when the grade control is set to AUTO.**
	Slope Berm DOWN	Press and hold this button to enable the Slope or Berm DOWN function, depending upon which mode is selected on the Slope/Berm Select button. When activated by pressing the button, all three LED lights illuminate GREEN on both main consoles and the left or right screed control box (corresponding to the active console).
	AOA Down	Press and hold this button to make adjustments to the Angle-of-Attack (AOA). Adjust the mat texture using the Angle-of-Attack (AOA) adjuster to produce a smooth or coarse mat texture.
	Vertical Down	Automatically decrease the screed extension vertical height by pressing and holding the vertical DOWN. All three LED lights illuminate while in operation.

RIGHT Screed Control Box - Right Keypad

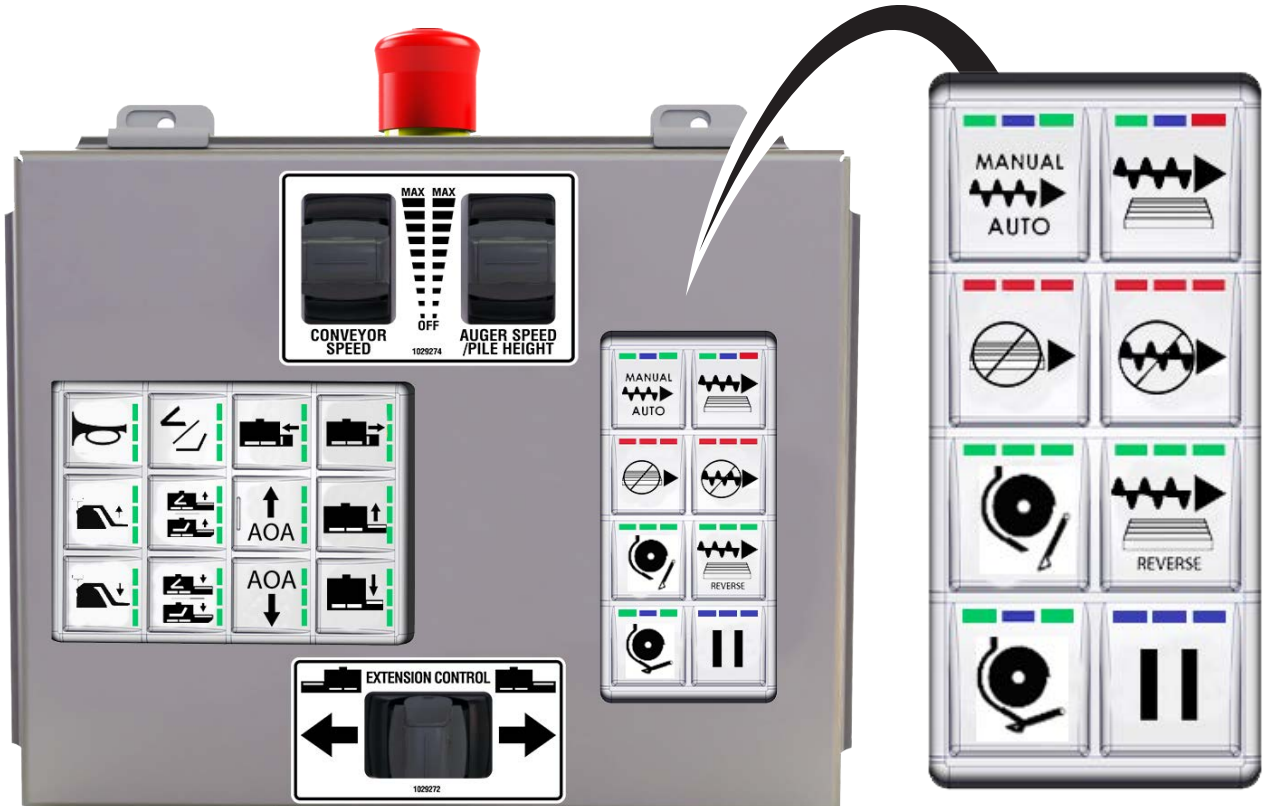
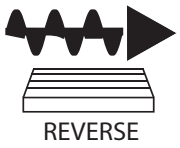


Figure 3-11. RIGHT Screed Controls (Right Keypad)

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
MANUAL / AUTO	Auger MANUAL/ AUTO Select	This button only functions if the Auger/Conveyor ON/OFF button is turned ON. Independent right and left augers can be operated in MANUAL or AUTO (automatic) mode. AUTO mode is the default. • Press button twice to operate auger in MANUAL mode. The left LED light illuminates GREEN. In MANUAL mode, the operator controls auger speed using the auger pile speed switch above the keypad. • Press the button once to operate in AUTO mode. The right LED light illuminates GREEN. The center blue light illuminates when material reaches the set material height. In AUTO mode, the auger dial controls material height. **The auger sonic sensor must be connected to operate in AUTO mode.**
	Auger/Conveyor ON/OFF	Press this button once to turn ON the auger and conveyor. • The left LED light illuminates GREEN when the auger and conveyor is on. • The default setting is AUTO mode, which is controlled by the Auger MANUAL/AUTO button. • Press the button again to turn OFF the auger and conveyor. • You can also use the auger OR conveyor bypass button to pause operation of either function separately.

KEYPAD SYMBOL	CONTROL NAME	FUNCTION
	Auger/Conveyor Reverse	Press the Auger/Conveyor Reverse button for reversing the direction of both functions to help clear material as needed. All three LED lights illuminate while the auger/conveyor reverse is active.
	RIGHT Cut-Off OPEN	Press and hold to open the LEFT Under Auger Material Cut-Off. All three LED lights illuminate GREEN on both operator consoles and the LEFT screed keypad while open.
	Pause	Pauses propel (travel), conveyors and augers functions. Press the button once to pause these functions. The three LED lights illuminate blue. • Pressing the button again resumes these functions. **NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)**
	RIGHT Cut-Off CLOSED	Press and hold to close the LEFT Under Auger Material Cut-Off. When closed, the center LED light illuminates RED on both operator consoles and the LEFT screed control keypad. If paver is put into reverse, the cut-offs will automatically close and the center LED light illuminates RED in AUTO mode.

LB5 Electric Heat Controls

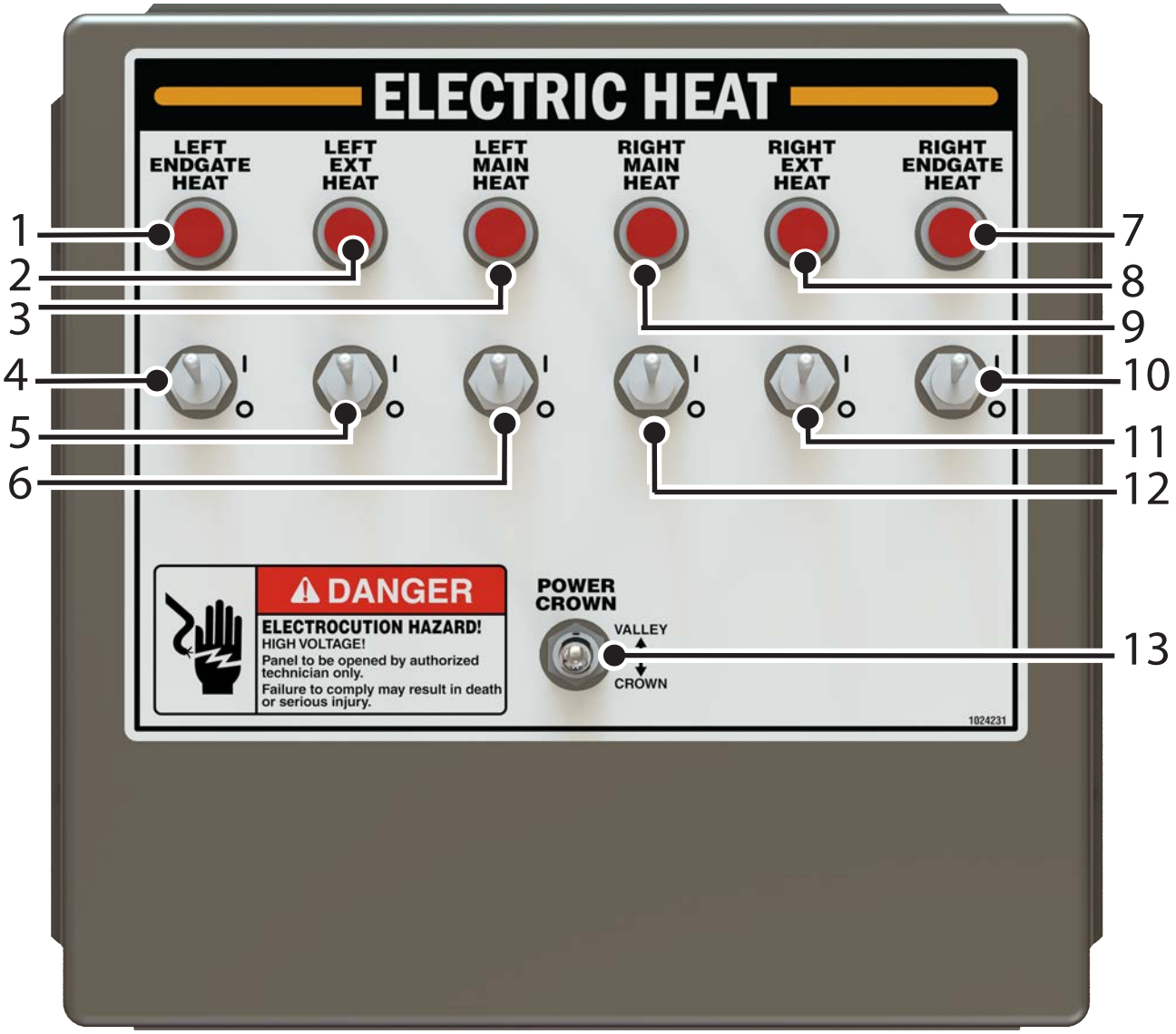


Figure 3-12. Electric Heat Controls

ITEM NO.	CONTROL NAME	FUNCTION
1	Left End Gate Heat ON Light	Illuminates when the LEFT end gate heat switch is ON to notify operator that the left end gate heating is engaged.
2	Left Screed Extension Heat ON Light	Illuminates and stays on as long as the screed extension heating elements are heating the screed plates.
3	Left Main Screed Heat Light	Illuminates and stays on as long as the main screed heating elements are heating the screed plates.
4	Right End Gate Heat ON Light	Illuminates when the RIGHT end gate switch is turned ON to notify operator that the right end gate heat is engaged.

ITEM NO.	CONTROL NAME	FUNCTION
5	Right End Gate ON/OFF Switch	Turns the RIGHT End Gate electric heat system on and off. • Push switch UP to turn ON the end gate heat. P • Push switch DOWN to turn OFF.
6	Right End Gate Heat ON Light	Illuminates when the RIGHT end gate switch is turned ON to notify operator that the right end gate heat is engaged.
7	Left End Gate ON/OFF Switch	Turns the LEFT end gate electric heat system on and off. • Push switch UP to turn ON the end gate heat. • Push switch DOWN to turn OFF.
8	Left Screed Extension ON/OFF Switch	Turns the LEFT screed extension heat system on and off. • Push switch UP to turn ON the left screed extension heat. • Push switch DOWN to turn OFF.
9	Left Main Screed ON/ OFF Switch	Turns the LEFT main screed heat system on and off. • Push switch UP to turn ON the left main screed heat. • Push switch DOWN to turn OFF.
10	Right Main Screed ON/ OFF Switch	Turns the RIGHT main screed heat system on and off. • Push switch UP to turn ON the right main screed heat. • Push switch DOWN to turn OFF.
11	Right Screed Extension ON/OFF Switch	Turns the RIGHT screed extension heat system on and off. • Push switch UP to turn ON the right screed extension heat. • Push switch DOWN to turn OFF.
12	Left End Gate ON/OFF Switch	Turns the RIGHT screed extension heat system on and off. • Push switch UP to turn ON the right screed extension heat. • Push switch DOWN to turn OFF.
13	Power Crown Switch [Option]	Use this switch to increase/decrease crown and valley. • Push and hold switch UP to increase valley. • Push and hold switch DOWN to increase crown.

LB7 Electric Heat Controls [Option]

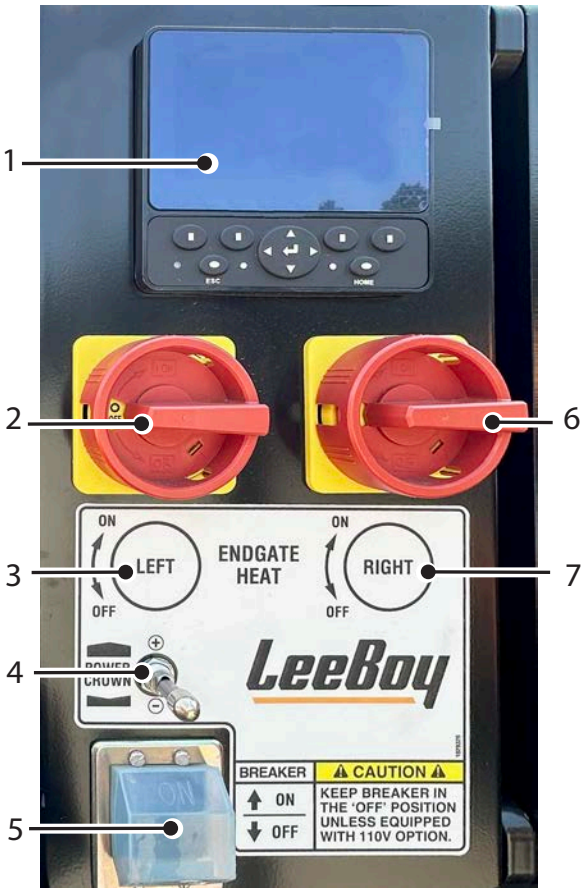


Figure 3-13. Electric Heat Controls

ITEM NO.	CONTROL NAME	FUNCTION
1	Heat Box Digital Controls	Use the digital display to set the screed heat temperature and other screed heat functions.
2	LEFT Screed Heat Dial	• Turn dial clockwise to turn ON the LEFT screed heat system. • Turn dial counterclock-wise to turn OFF the LEFT screed heat system.
3	LEFT End Gate ON/ OFF Switch [Option]	Turns the LEFT End Gate electric heat system on and off. • Push switch UP to turn ON the end gate heat. • Push switch DOWN to turn OFF.
4	Power Crown Switch [Option]	Use this switch to increase/ decrease crown and valley. • Push and hold switch UP to increase valley. • Push and hold switch DOWN to increase crown.
5	Breaker Switch	Circuits are protected from damage by overloads or short circuits. If the system detects an abnormal condition, it “breaks” the current flow to prevent overheating, equipment damage, and potential fire hazards.
6	RIGHT Screed Heat Dial	• Turn dial clockwise to turn ON the RIGHT screed heat system. • Turn dial counterclockwise to turn OFF the RIGHT screed heat system.
7	Right End Gate ON/ OFF Switch [Option]	Turns the RIGHT End Gate electric heat system on and off. • Push switch UP to turn ON the end gate heat. P • Push switch DOWN to turn OFF.

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OPERATION

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RECEIVING THE MACHINE

The LeeBoy 8680 Commercial Paver is the most dependable and productive commercial asphalt paver on the market. Featuring the new LB5 screed (or optional LB7 screed), this commercial paver is designed and built for producing superior mat quality in a variety of road and other pavement construction projects.

Its heavy-duty, rubber tracks provide a smooth ride, better traction, and higher speed while paving. Independent hopper wings in its 15-ton hopper with wide conveyors, height-adjustable and reversible augers that can work independently, hydraulic tow point cylinders and Under Auger Material Cut-Offs that automatically close are just part of LeeBoy innovation and the 8680 paver's versatility.

Dual main operator control platforms with ergonomic control keypads swing out from either side of the paver, and its low-profile hood line gives operators unlimited visibility with no line-of-sight disruptions while paving. The controls are always directly in front of the operator as the entire platform can be adjusted to multiple height positions.

Electric dual vibrators mounted on the main screed frame increase paving material compaction. The screed extensions also include slope and berm capability. The screed control box features push-button keypads with every control a screed operator needs for commercial road construction jobs.

⚠CAUTION **Read this manual carefully before operating the machine. Only authorized personnel who are properly trained should operate the paver.**

Initial and Daily Inspection

Although the machine has been checked thoroughly by the manufacturer, road hazards or other factors during transport may result in damage. Inspect machine for any damage during transport. Contact your authorized dealer immediately if any damage has occurred. **(Table 4-1)**

The following inspection is essential and should be performed before the initial start-up and daily prior to operating. Visually inspect the unit to check its general condition and for familiarization. Continue with a check of the systems and components shown in **Table 4-1** below:

Table 4-1. Initial Inspection

INSPECT	PROCEDURE
Engine Oil Level	Maintain oil level between ADD and FULL marks on dipstick.
DEF Tank Level	Check DEF tank level. Refill daily.
Fuel Tank Level	Check for adequate fuel supply. Fill if needed. Always fill tank at end of the day to prevent condensation in the tank.
Hydraulic Oil Level	Check the hydraulic tank for oil level and hoses for leaks.
Radiator	Check coolant level. Add more if needed.
Battery	Ensure all cables are tight and clean. Check for corrosion on the battery terminals.
Air Cleaner	Check the air filter element and hose connections. Air cleaner has both a primary and secondary filter.
Drain Plugs	Make sure plugs are inserted and tight.
Engine Belt	Check for proper belt tension.
Generator Belt	Check for proper belt tension.
Grease Fittings	Ensure fittings are greased and in good working order.

Screed Locks

This machine is provided with screed lock mechanisms as added safety and stability during transport. A screed lock knob is located on each side of the screed on the paver frame. **(Figure 4-1)**

- Pull the knob outward to release the screed lock.
- Push the knob in and down to lock the screed in place.

⚠CAUTION Ensure locks are in the unlocked position before raising or lowering the screed.

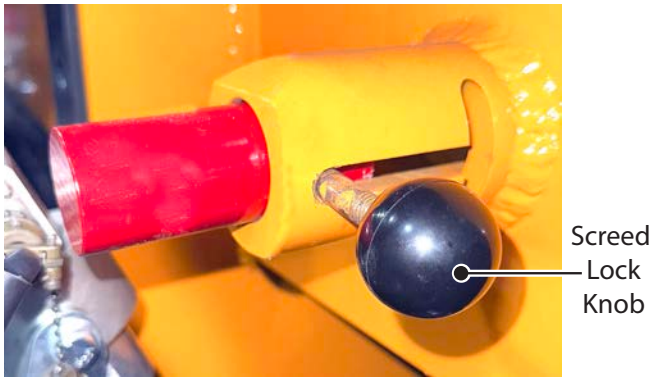


Figure 4-1. Screed Lock (Left Side Shown)

Hopper Lock Valves

The hopper cylinder valves can be used to lock the hopper wings when they are open or closed. **(Figure 4-2)** Be sure to unlock the hopper valves before operation!

- Turn the valve counterclockwise to unlock the hopper wings.
- Turn the valve clockwise to lock the hopper wings.

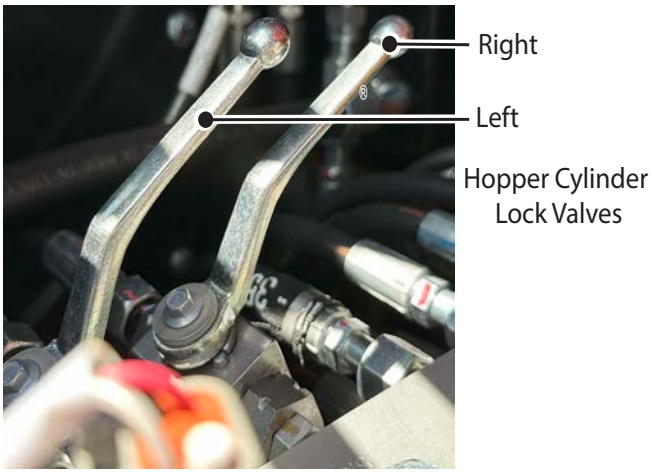


Figure 4-2. Hopper Cylinder Lock Valve

⚠CAUTION Ensure the hopper cylinder is unlocked before extending or retracting the hopper wings.

OPERATOR PLATFORM FEATURES

Figure 4-3. Control Panel Vandalism Cover

Lockable Vandalism Covers

The 8680 paver features a heavy-duty vandalism cover for both protecting the control panel keypads and switches along with security when the paver is not in use. **(Figure 4-3)** The cover has hooking tabs at the top, with lock slots on the bottom. Simply remove the lock and lift up from the bottom to remove. Align the top tabs to close and reinstall lock to reattach.

After removing the vandalism lock cover(s), attach it to the paver frame using the locking knobs as shown in **Figure 4-3.**

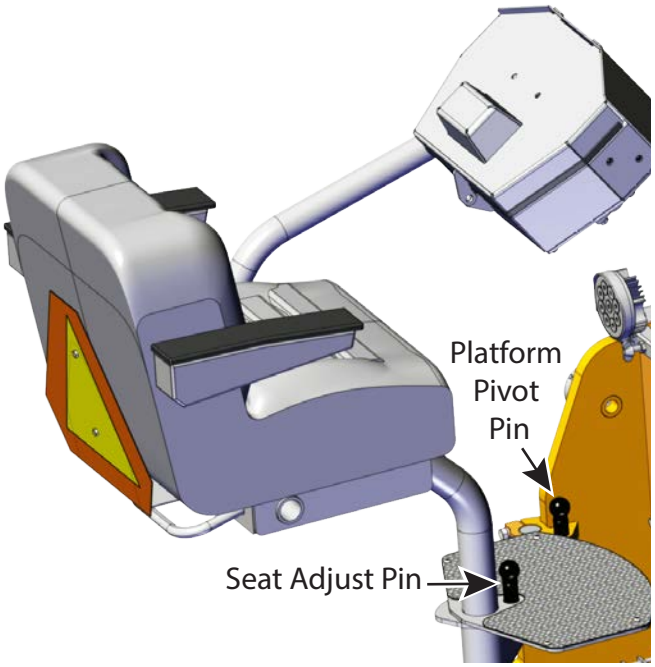


Figure 4-4. Seat Adjust and Platform Pivot Pins

Seat Adjustment and Operator Platform Pivot

You can adjust the operator seat position and swing out the entire operator platform for better visibility and comfort. **(Figure 4-4)**

There are multiple positions for the operator seats. Simply pull the seat adjust pin toward the seat to swivel the seat.

Use the platform pivot pin to swing out the operator platform to the side. Pull the platform pivot pin to move the platform to the desired position.

Ensure the platform locks into place in the desired position. It is recommended that you stand beside the paver to make these adjustments.

⚠CAUTION Verify the seat and platform are locked into position BEFORE operating the paver.

START-UP PROCEDURE

⚠ DANGER DO NOT leave the operator station unattended while the paver is in gear or in motion. Operator must return joysticks to the neutral position and press the Activate Console button to deactivate the console before leaving the operator platform.

⚠ CAUTION Verify there are no people, obstacles or other equipment in the machine's path before starting the engine.

NOTICE **Using starting additives, such as ether, is not recommended as severe engine damage can occur.**

The ignition switch is located on the junction box. **(Figure 4-5)**

Figure 4-5. *Ignition Switch on Junction Box*

When the key is turned on, the DM430 digital display controller on both operator control panels automatically come on, with an LED status light illuminating amber to indicate the system is in the "Wait to Start" mode. **The engine will not start unless both joysticks on both operator platforms (4 total) are in the NEUTRAL position**. The "Wait to Start" light will illuminate amber if one or more joysticks are not in neutral, and turn to amber when the joystick(s) has been moved into the neutral position. The amber light will turn off once the paver is ready to start, ***provided all four joysticks are in neutral.***

NOTE: An icon displaying a pair of joysticks will illuminate in red on the DM430 digital controller when a joystick(s) is out of neutral.

To start the engine:

1. Insert key into the ignition and turn clockwise to the RUN position.
2. With ignition key in the RUN position and all four joysticks in neutral, press the Start/Stop button on the main control panel. **(Figures 4-6 and 4-7 on *Page 4-8)***
   - The center light illuminates amber during the "Wait to Start" stage.
   - The left LED lights illuminate GREEN on both consoles when the engine is running.

NOTICE **DO NOT hold the starter longer than 10 - 15 seconds. If the engine does not start, allow the starter to cool two or three minutes.**

Figure 4-6. Start-Stop and Activate Console Keypad Buttons (Left Panel Shown)

3. You must activate the control console for desired driving side by pressing the Activate Console button on the lower right keypad. **(Figures 4-6 and 4-7 on *Page 4-8)*** The three LED lights illuminate green when a console is activate.
4. Allow the engine to warm before moving paver for more efficient operation. In cold weather, allow the hydraulic oil to warm to 60°F before moving.

The engine's low idle is set at 1300 RPMs, with a maximum engine speed of 2200 RPMs. The engine will remain at the operator-set RPMs after the joysticks initially leave the neutral position. If the engine is at low idle and the operator engages either of the following functions, the engine will automatically idle up to the preset RPMs:

- Generator: 1800 RPMs
- Auger: 1800 RPMs
- Conveyor: 1800 RPMs

Once the joysticks return to the neutral position, the engine will idle slow down to 1300 RPMs after eight (8) seconds, provided the generator, auger or conveyor functions are not engaged.

NOTE: The engine speed may be manually increased to a maximum of 2200 RPMs if desired.

Stopping the Engine

1. Move joysticks to the NEUTRAL position.
2. Press the Idle button on the engine screen on the digital display.
3. Press the Start/Stop button. The right LED light illuminates RED to indicate engine shutdown.
4. Turn ignition key counterclockwise to the OFF position and remove key.

NOTE: If for any reason the engine does not shut down when key is turned off, push in the red E-Stop button on the upper right side of the control panel.

DIESEL EXHAUST FLUID (DEF)

DEF is a mixture of pure urea and de-ionized water that is injected into the exhaust stream by the SCR system. DEF is a vital element in engine operation and performance. It is ***extremely*** important to keep the DEF tank filled, and DEF fluid stored properly, to prevent costly repair. ***(See Page 5-20 for detailed information.)*** The DEF tank is located inside the right engine access panel. Keeping the DEF tank full on a daily basis will prevent costly repairs and operational down time.

DEF Purge: The SCR system will automatically purge DEF after shutting down the engine. You will hear the pumping noise as the system circulates DEF to cool the DEF injector, then the pump reverses to return DEF back to the tank. **Always allow this purging process to finish** to prevent damage to the injector and other SCR components.

⚠CAUTION **NEVER disconnect battery until the purge cycle has completed to prevent serious damage to components in the SCR system.**

If the DEF tank level runs low (below 15 percent) during operation, DEF warning messages will appear on the DM430 digital display screens (described in detail beginning on ***Page 4-42).*** If the level drops to 5 percent (empty), urgent warning messages pop up. Engine torque will be reduced 50 percent and engine speed will be reduced 40 percent after 45 minutes.

If the operator continues to ignore the warning and the DEF tank is still not refilled, another urgent warning message pops up. At this point, there will be no available torque--the engine will only idle.

In either of these conditions, refilling the DEF tank will restore engine performance.

If the DEF tank level has fallen to 0 percent, there is no ability for DEF injection into the exhaust system (nor can the system perform the parked regeneration process described on ***Pages 4-42 to 4-43.*** **The DEF injector may now be damaged and require replacement from your authorized dealer.**

DEF Thaw Control Mode: Sensors on the DEF tank also detect the ambient air temperature within seconds of starting the engine. If the DEF is frozen, the SCR system automatically starts the Thaw Control Mode and turns on the DEF pump, tank and tube heaters. Engine maximum torque may be limited to protect the DEF injector.

MAIN CONTROL PANEL KEYPAD FUNCTIONS

The control panel keypads feature paving symbols that are used for the majority of paver functions. **(Figure 4-7)** LED lights illuminate to indicate the function status. Some buttons may only control the side of the machine in which they are located, or may only function on the active console. The following pages provide detailed information for each function symbol and status lights.

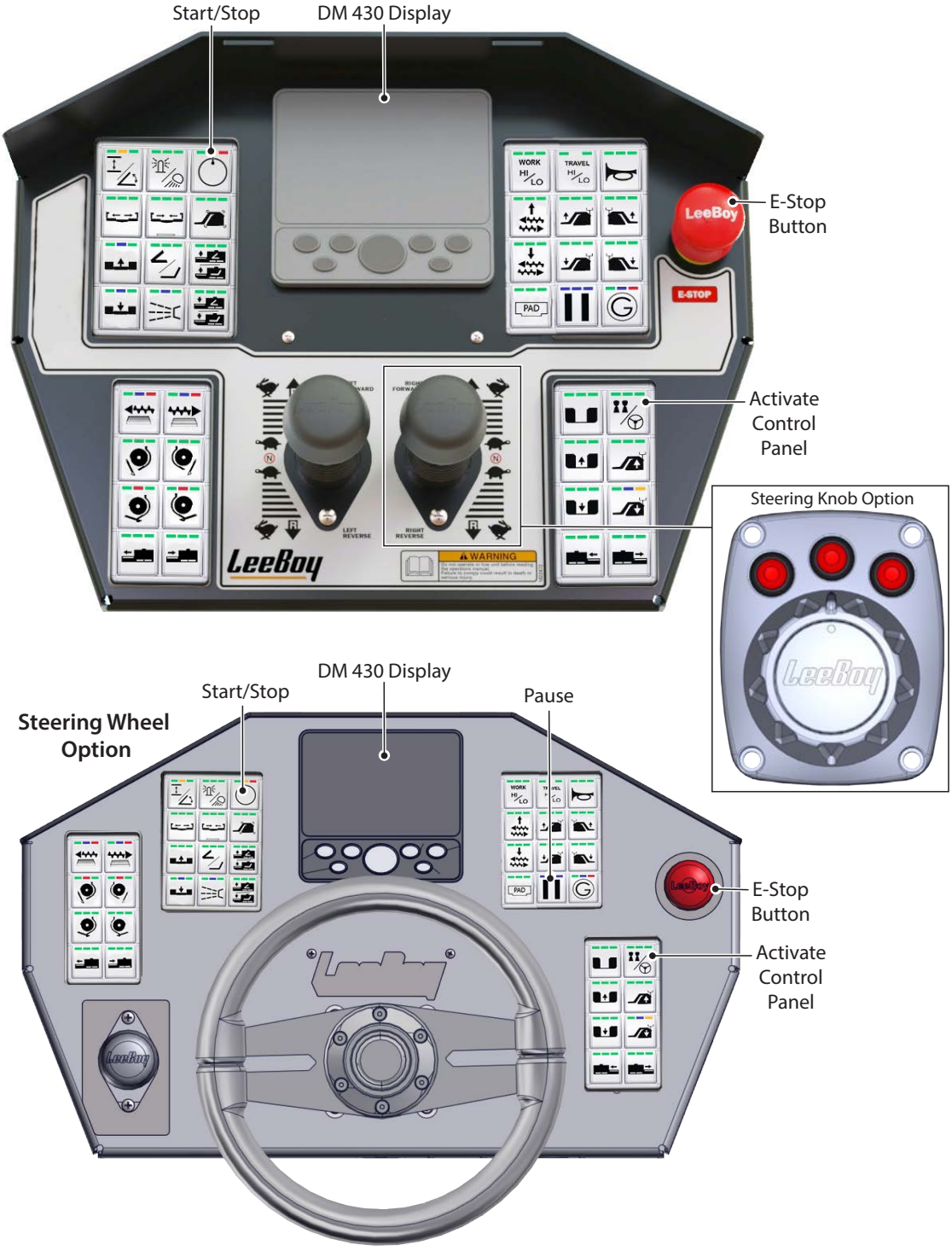


Figure 4-7. Main Control Panel

Upper Left Keypad

The upper left keypad contains push buttons for beacon/work lights, grade/slope controls, apron up/down, start, spray down, truck hitch in/out and vibrator. LED lights illuminate to indicate the function status as described below:

GRADE/SLOPE (LEFT)

The left and right tow points can be controlled automatically using the TopCon sonic automatic grade/slope control system ***(Page 4-59)*** or can be controlled manually using the keypad buttons.

- The default setting is MANUAL. The left LED light illuminates green in manual mode on both consoles. Use the Tow Up (thicker asphalt mat) and Tow Down (thinner asphalt mat) buttons to control the hydraulic tow cylinders.

NOTE: Does not function when the grade control is set to AUTO, however, the up and down buttons function when joysticks are in neutral.

- Pressing the Grade/Slope button changes to AUTO mode. The right LED light illuminates green on both consoles. The TopCon sonic control system must be connected to operate in AUTO mode.

NOTE: When in automatic mode, the Tow Up and Down buttons do not function. The tow points are paused when the machine is not traveling forward. In this situation, the center LED light will flash.

WORK/BEACON LIGHTS

Pressing this button toggles through 4 states:

- OFF (no LED lights illuminate).
- Beacon light ON (left LED light illuminates).
- Work lights ON (center LED light illuminates).
- Both beacon and work lights On (all LED lights illuminate).

START/STOP

This button is used to start and stop the engine. There are seven states that the engine start/ stop state machine can enter:

- **Boot**: Engine cannot be started while the computer system boots, but power is provided to the engine on output. The yellow and red LEDs are solid in this state. After a slight delay, the engine will leave this state and go to Ready. If the engine RPM indicates the engine is already running, then another stop cycle will be attempted (interlocks short cycling the main key switch).
- **Ready**: The yellow LED will remain on during the engine Preheat stage or if the operator attempts to start the engine while the auger, conveyor, or generator are turned on, or if one or more joysticks is not in neutral.
- **Crank**: The system will stay in the cranking state until the engine RPM indicates it is running, or the button is released. The red and green LED lights illuminate (and yellow light if still preheating). (While cranking, both the engine on and engine start outputs are active.) If the button is released before the engine is running, the system reverts back to Ready.
- **Running**: When the engine is running, only the green LED light illuminates. The engine can be stopped by pressing the Start/Stop button again.
- **Stop**: When in the stop state, the red LED light will flash for a short period of time. (The engine on output will be off in this state.) It can be held in the stop state longer if the operator presses and holds the button. The system will boot when leaving stop.
- **Shutdown**: In the case of excessively low oil pressure or high coolant temperature, a message will be sent to the engine to shut down. The red LED will flash in this state and the engine on output will remain ON. The only way to exit this state is with a power cycle.
- **Engine Offline**: When communication with the engine is lost, the system will enter an offline state. Only the yellow LED will be flashing. If the button is pressed in this state, the system will go to stop.

NOTE: All joysticks on BOTH operator main control panels *must* be in NEUTRAL to start the engine.

TRUCK HITCH or EXTENDABLE PUSH ROLLER OPTION

The truck hitch (or extendable push roller) IN/OUT function can be activated from either main control panel. All three LED lights illuminate green when pressed.

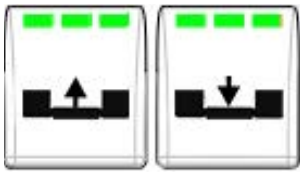
- Press and hold either button to move truck hitch (or extendable push roller) in or out as shown by arrows on the symbol.

VIBRATOR

This button turns the screed vibrator ON and OFF, and can be operated from active or inactive consoles.

- Press the button once to turn on the vibrator. All LED lights illuminate green when the vibrator is engaged.
- Press the button again to turn off the vibrator.

NOTE: The vibrator only functions when traveling forward and will automatically turn off if paver is put into reverse.



APRON UP / DOWN

Press and hold the UP or DOWN buttons to pivot the front apron to assist in cleaning out the hopper. All three LED lights illuminate while in operation.

SLOPE/BERM SELECT

Use the slope/berm select button to choose either function:

- Press button once to select the Slope function.
- Press button twice to select the Berm function.
- This function only works on the activated panel as the selection is side specific.
- The left LED illuminates in slope control mode and the right LED light illuminates in berm control mode.

SLOPE/BERM UP or DOWN

After selecting slope or berm, press and hold the UP or DOWN buttons to increase/ decrease slope or berm.

- The left LED illuminates in slope control mode and the right LED light illuminates in berm control mode.
- When the button is pressed, all three LED light illuminate green on both consoles (and the corresponding screed keypad).

SPRAY DOWN

The spray down pump can be turned on and off from either main control panel whether active or inactive.

- Press the button once to turn on the spray down system. All LED lights illuminate green when the spray down system is on.

Upper Right Keypad

The upper right keypad contains push buttons for high/low gear in Work or Travel mode, horn, tow up and down functions (on each side of paver), auger lift/lower, pad builder, pause, and generator on/off. LED lights illuminate to indicate the function status as described below:

WORK HI/LO

This button controls the two speed motor shift in the WORK mode.

- Low speed is the default. The right LED light illuminates green at low speed.
- Press the button to shift to high gear. The left LED light illuminates green at high speed.
- Pressing the button again shifts back to low gear.
- To reduce downshift harshness, the operator cannot downshift at maximum speed. If a downshift is attempted at full speed, the right LED light will flash and a message banner appears on the DM430 digital display directing operator to reduce speed.

NOTE: Pave only in Travel LO (Low) mode.

NOTE: The paver will automatically move into low gear when machine operation is paused.

TRAVEL HI/LO

This button controls the two speed torque hub shift in the TRAVEL mode.

NOTE: You must come to a complete stop before shifting from LO to HI or from HI to LO gears.

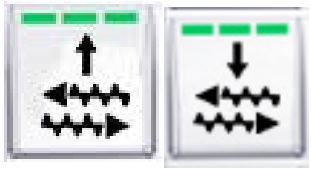
- Low speed is the default. The right LED light illuminates green at low speed.
- Stop the machine and press the button to shift to high gear. The left LED light illuminates green at high speed.
- Stopping and pressing the button again shifts back to low gear.
- To reduce downshift harshness, the operator cannot downshift at maximum speed. If a downshift is attempted at full speed, the right LED light will flash and a message banner appears on the DM430 digital display directing operator to reduce speed.

NOTE: DO NOT pave in Travel HI mode.

NOTE: The paver will automatically move into low gear when machine operation is paused.

HORN

- Pressing this button sounds the horn. All three LED lights will illuminate GREEN when engaged.
- You hold the button for longer sound.
- The horn functions whether the console is active or not.



AUGER LIFT/LOWER

These buttons allow you to lift and lower both augers.

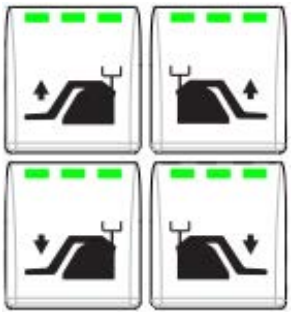
To raise the augers:

- Press and hold the auger lift button (⇧) to raise both the left and right auger.
- Releasing the button will hold the augers at that position.

To lower the augers:

- Press and hold the auger lower button (⇩) to lower both the left and right auger.
- Releasing the button will hold the augers at that position.

NOTE: ALWAYS raise the augers when loading the paver onto transport vehicles.



TOW POINTS UP or DOWN (both sides)

The left and right tow points can be controlled automatically using the TopCon sonic automatic grade/slope control system or can be controlled manually using the keypad buttons.

- The default setting is MANUAL. The left LED light illuminates green in manual mode on both consoles. Use the Tow Up (thicker asphalt mat) and Tow Down (thinner asphalt mat) buttons to control the hydraulic tow cylinders.

NOTE: Does not function when the grade control is set to AUTO, however, the up and down buttons function when joysticks are in neutral.

- Pressing the Grade/Slope button changes to AUTO mode. The right LED light illuminates green on both consoles. The TopCon sonic control system must be connected to operate in AUTO mode. (The Tow Up and Down buttons do not function in AUTO mode).

NOTE: When in automatic mode, the tow points are paused when the machine is not traveling forward. In this situation, the center LED light will flash.

PAD BUILDER

This function allows you to build a starter pad.

- Press the button once to engage the augers and conveyors. Both Under Auger Material Cut-Offs will open.
- Press the button again and the augers/conveyors will turn off.
- Press and hold button to manually engage both conveyors only.

PAUSE

When paused, the propel, augers, and conveyors will slow to a stop. All three LED lights illuminate blue when paused.

- Press the button once to pause operation. Press the button again to resume.
- If the screed temperature is above the set point, the engine will auto idle after the normal delay time.
- If the generator needs to run, the engine RPM will increase as necessary while paused.

NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)

DANGER The Pause button should NEVER be used as though it's a park brake--it is not. DO NOT leave the operator platform when paused!

GENERATOR

The generator can be activated in AUTO or MANUAL mode from the active or inactive console. The default setting is AUTO mode.

- Press the button once for AUTO operation (the temperature set point controls operation). The left LED lights illuminate green.

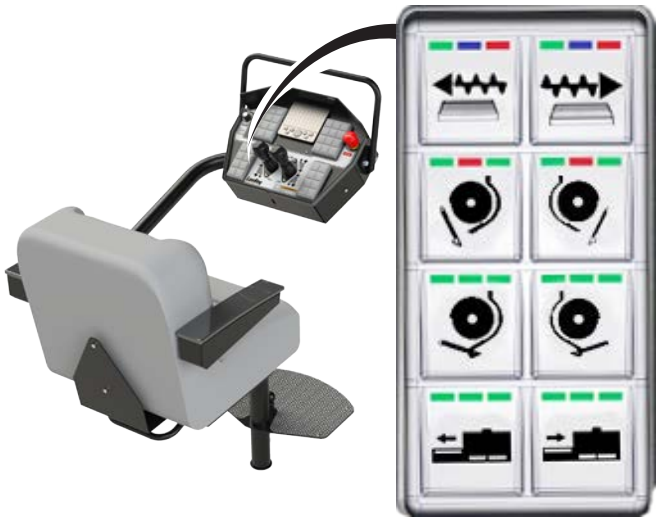
NOTE: In auto mode, if the temperature is above the set point, the center LED is solid blue. If is below the set point, the LED will flash.

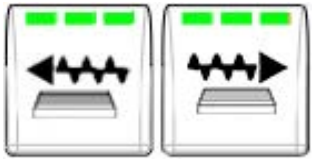
- Press the button again to turn off (the right LED light illuminates red).
- In manual mode the center LED never illuminates.

NOTE: Operator selects AUTO or manual operation in the Operator Settings screen on the DM430 display control unit. ***(Page 4-39)***

Lower Left Keypad

The lower left keypad contains push buttons for left and right auger/conveyor function, Under Auger Material Cut-off open and close, and extend or retract the left screed extension. LED lights illuminate to indicate the function status as described below:



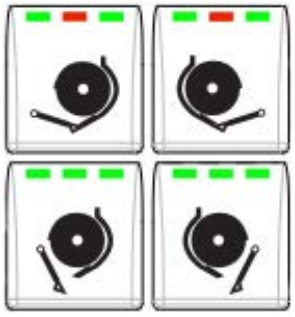


AUGERS/CONVEYORS

The augers and conveyors (both sides) can be turned on and off from either control panel. The default setting for both functions is the AUTO mode of operation.

- Press the button once to turn on the auger and conveyor on the side corresponding with the arrow (left or right side).
- The left LED illuminates green when that auger/conveyor is on; the right LED illuminates red when that auger/conveyor is off; and the center LED light illuminates blue if paver is paused while auger(s)/conveyor(s) are on (or the asphalt material has shut off the sensor).
- Press the button again to turn off the left or right conveyor.
- Pressing and holding either button is a manual override for the augers and conveyors as long as you hold the button.

NOTE: Moving the joysticks into neutral pauses auger/conveyor operation if this function is enabled.



CUT-OFFS

Both control panels and the screed keypad has buttons for controlling the left and right side Under Auger Material Cut-Offs.

- Press and hold to open or close the corresponding left or right Under Auger Material Cut-Offs.
- All three LED lights illuminate GREEN on both operator consoles and the LEFT screed keypad while open.
- If paver is put into reverse, the cut-offs will automatically close and the center LED light on both operator consoles (and the corresponding screed control keypad) illuminates RED.

NOTE: User can turn off this feature if desired.

SCREED EXTENSION (LEFT)

- Press and hold button to extend or retract the LEFT screed extension to desired width.
- Release button when desired width is achieved.
- All LED lights illuminate green when the button is pressed on the main control panel and on the corresponding left or right screed keypad.

Lower Right Keypad

The lower right keypad contains push buttons to activate the control panel consoles, operate the hopper wings, raise and float the screed, and extend or retract the right screed extension. LED lights illuminate to indicate the function status as described below:

HOPPER WING SELECT

Used to select the right, left or both hopper wings. Upon start-up, the operational mode is set by default to operating both hopper wings (the LED lights illuminate green.

- Pressing the button once enables operating only the LEFT hopper wing. The left LED light illuminates green.
- Pressing the button again operates only the RIGHT hopper wing. The right LED light illuminates green.
- Pressing the button again returns to the default operational control of both hopper wings. All three LED lights illuminate green.
- This button functions whether the console is active or not.

ACTIVATE CONSOLE

Both main control panels are inactive upon start-up. You must activate the right or left control panel to operate on the desired side of the paver. All LED lights are illuminated green on the side that is active.

- All four joysticks must be in the NEUTRAL position before either control panel can be activated.

NOTE: All joysticks on both operator platform control panels *must* be in NEUTRAL to deactivate the console in use and activate the other console.

- Press the button on the desired main control panel once to activate that control panel.
- Press the button again to deactivate an active control panel.

NOTE: If one control panel is active and the other control panel's joystick(s) is moved out of neutral, the active console will automatically deactivate.

NOTE: If an active console becomes inactive while the paver is moving, the paver will automatically slow down to a stop and apply park brake.

- A control panel cannot be activated if a parked regeneration is active. ***(Page 4-44)***

HOPPER WING OPEN

After selecting the left, right or both hopper wings on the hopper wing select button, use this button to open the selected hopper wing(s).

- Press and hold the button to open the hopper wing(s). The center LED light illuminates green while engaged.
- This button functions whether the console is active or not.

HOPPER WING CLOSE

After selecting the left, right or both hopper wings on the hopper wing select button, use this button to open the selected hopper wing(s).

- Press and hold the button to close the hopper wing(s). The center LED light illuminates green while engaged.
- This button functions whether the console is active or not.

SCREED RAISE

This function can only be operated from the active console.

- Press and hold button to raise the screed. All three LED lights illuminate GREEN while engaged (on both control panels).

SCREED FLOAT

This button controls floating the screed.

- Press the button to activate screed float (the center LED light illuminates green).
- Placing the joysticks into neutral while screed float is active moves into screed hold, and float resumes when joysticks are returned to the forward position. The center LED light illuminates blue when in screed hold.

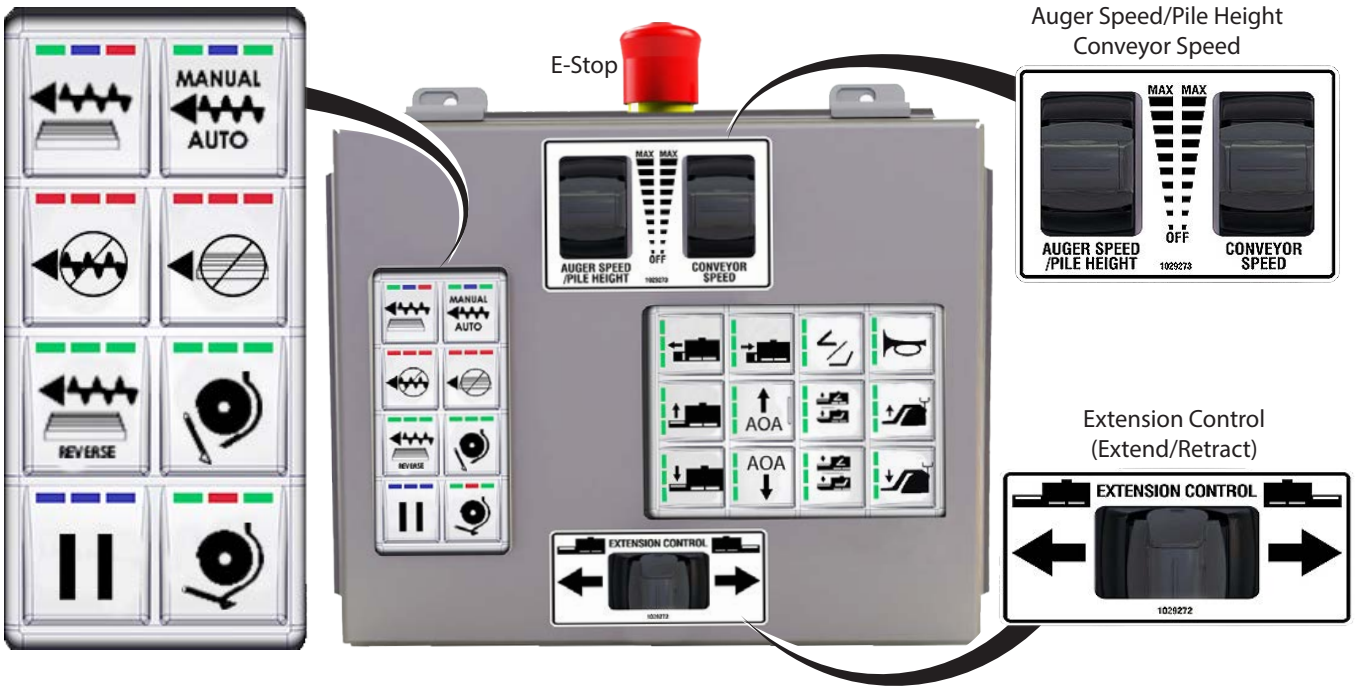
SCREED EXTENSION (RIGHT)

- Press and hold button to extend or retract the RIGHT screed extension to desired width.
- Release button when desired width is achieved.
- All LED lights illuminate green when the button is pressed on the main control panel and on the corresponding left or right screed keypad.

SCREED CONTROL BOX

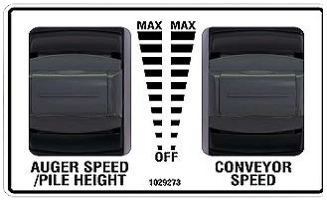
(Left Screed Control Box Shown)

Left Keypad and Control Buttons



A screed control box is located on each side of the screed, featuring keypad button controls for screed functions, auger and conveyor speed buttons, extension control buttons (extend/retract) and an emergency stop (E-Stop) button, described on the following pages.

E-Stop: Pull up on the E-Stop knob to disengage; push knob down to engage *emergency* **stopping only.**



AUGER SPEED/PILE HEIGHT

- When operating in AUTO mode, press and hold button in the desired increase/decrease direction to set the pile height at the end gate.
- When operating in MANUAL mode, press and hold button in the desired increase/decrease direction to set the auger speed.

CONVEYOR SPEED

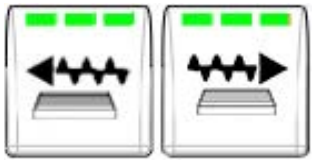
- Press and hold the button in the desired increase/decrease direction to set conveyor speed. You can also fine tune the conveyor speed in the digital display. ***(See Motor Settings on Page 4-40)***

EXTENSION CONTROL

- Press and hold the button in the OUT direction to extend the screed extension.
- Press and hold the button in the IN direction to retract the screed extension.

AUGERS/CONVEYORS

The augers and conveyors (both sides) can be turned on and off from either control panel. The default setting for both functions is the AUTO mode of operation.

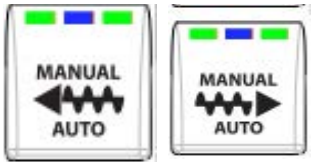


- Press the button once to turn on the auger and conveyor on the side corresponding with the arrow (left or right side).
- The left LED illuminates green when that auger/conveyor is on; the right LED illuminates red when that auger/conveyor is off; and the center LED light illuminates blue if paver is paused while auger(s)/conveyor(s) are on (or the asphalt material has shut off the sensor).
- Press the button again to turn off the left or right conveyor.

NOTE: Moving the joysticks into neutral pauses auger/conveyor operation if this function is enabled.

AUGERS MANUAL/AUTO

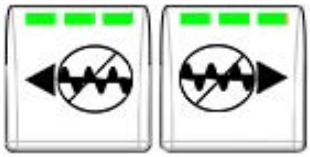
Independent right and left augers can be operated in MANUAL or AUTO (automatic) mode. Arrow direction on the symbols indicates left or right auger. The default is set for AUTO operation.



- Turn on the left or right auger by pressing the corresponding Manual/Auger button once to enable the auger on that side to operate in AUTO mode. The left LED light on the button illuminates green.
- Press and hold button 1/2 second to operate in AUTO mode. The right LED light on the button illuminates green.

NOTE: The auger sonic sensor must be connected to operate in AUTO mode. ***(Pages 4-53 to 4-54)***

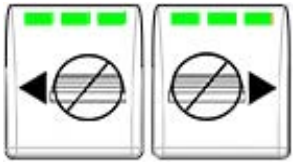
- The center blue light illuminates when material reaches the set material height or if paused.



AUGER BYPASS

If the Auger/Conveyor is on, but the augers are not needed while paving, the screed operator can press the corresponding left or right Auger Bypass button(s) to cease auger operation. All three LED illuminate to when auger operation is paused.

- Press the button again to resume auger operation.



CONVEYOR BYPASS

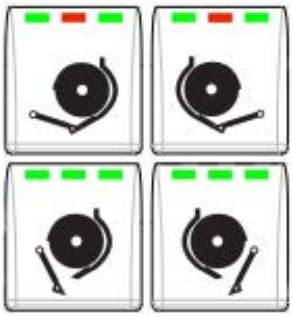
If the Auger/Conveyor button is on, but the conveyors are not needed while paving, the screed operator can press the corresponding left or right conveyor bypass button to cease operation. All three LED illuminate to when the conveyor operation is paused.

- Press the button again to resume conveyor operation.

AUGER/CONVEYOR REVERSE

Screed operators can reverse the augers and/or conveyors to help clear material as needed while paving. The default is reversing both functions. All three LED lights illuminate green in this mode.

- Press the button once to reverse only the auger(s). The left LED light will illuminate green.
- Press the button again to return to reversing both functions.



CUT-OFFS

Both control panels and the screed keypad has buttons for controlling the left and right side Under Auger Material Cut-Offs.

- Press and hold to open or close the corresponding left or right Under Auger Material Cut-Offs.
- All three LED lights illuminate GREEN on both operator consoles and the LEFT screed keypad while open.
- If paver is put into reverse, the cut-offs will automatically close and the center LED light on both operator consoles (and the corresponding screed control keypad) illuminates RED.

NOTE: User can turn off this feature if desired.

PAUSE

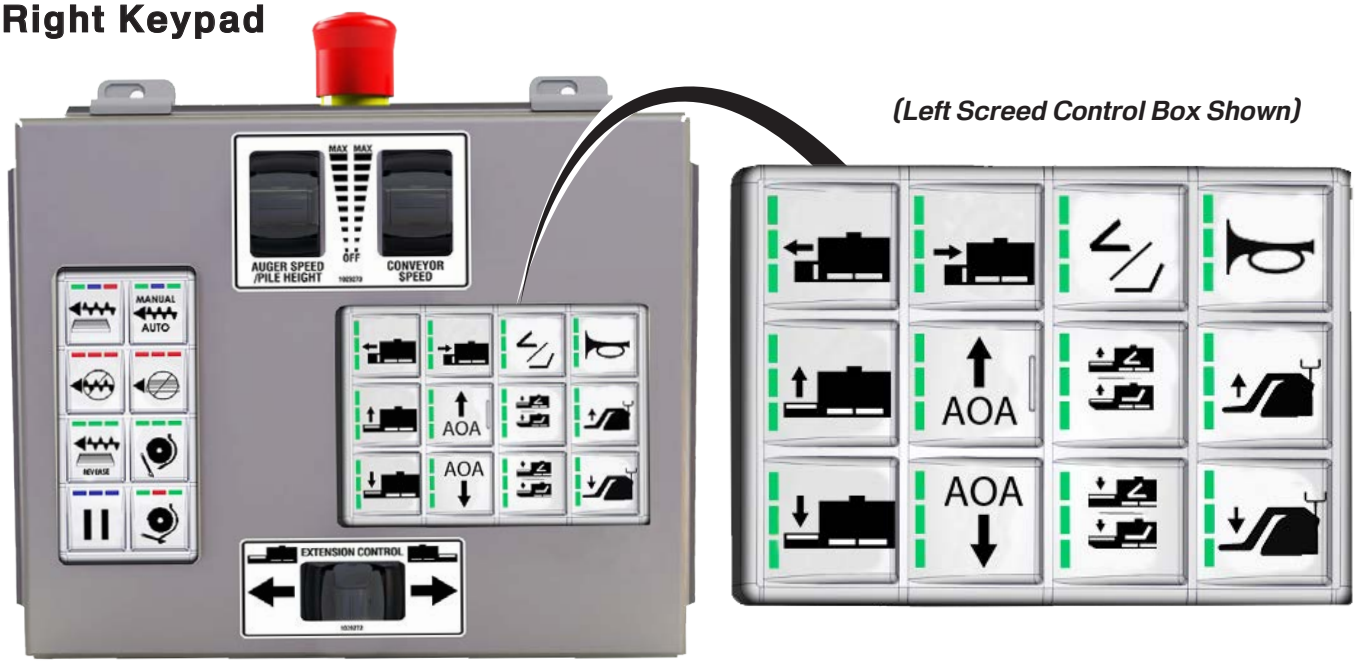
When paused, the propel, augers, and conveyors will slow to a stop. All three LED lights illuminate blue when paused.

- Press the button once to pause operation. Press the button again to resume.
- If the generator needs to run, the engine RPM will increase as necessary while paused.

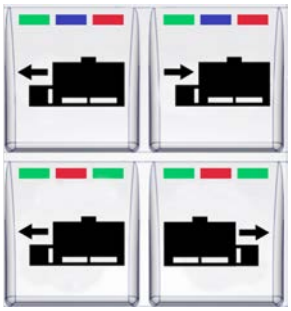
NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)

DANGER The Pause button should NEVER be used as though it's a park brake--it is not. DO NOT leave the operator platform when paused!

Right Keypad



(Left Screed Control Box Shown)



TUNNEL IN/OUT

Extend and retract the tunnel when paving at wider lengths.

If operating in MANUAL mode:

- Press and hold the Tunnel Out button once to extend the tunnel. All LED lights illuminate green.
- Press and hold the Tunnel In button to retract the tunnel. All LED lights illuminate green.

If operating in AUTO mode:

- ***Simultaneously*** press and hold the extension control switch in the extend position and the Tunnel Out button to extend the tunnel.
- ***Simultaneously*** press and hold the extension control switch in the extend position and the Tunnel In button to retract the tunnel.
- To lock the tunnel in place, press and hold the Tunnel In and Tunnel Out buttons simultaneously until the LED light flashes green.
    - To unlock, press and hold both buttons again.

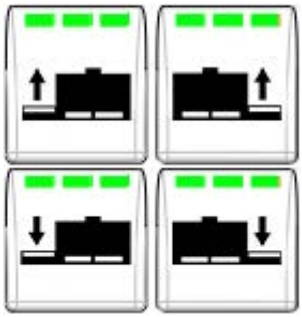
SLOPE/BERM SELECT

Use the slope/berm select button to choose either function. This function only works on the activated panel as the selection is side specific. The default is set at the neutral position.

- Press the button once to select the Slope function.
- Press the button twice to select the Berm function.
- The left LED illuminates in slope control mode and the right LED light illuminates in berm control mode.

HORN

- Pressing this button sounds the horn. All three LED lights will illuminate GREEN when engaged.
- You hold the button for longer sound.
- The horn functions whether the console is active or not.



VERTICAL UP/DOWN

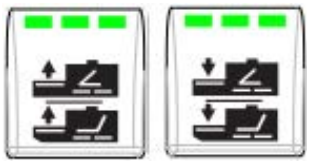
Screed operators can use the vertical UP and DOWN buttons on the corresponding left and/or right screed control box button to raise and lower the screed extension vertical height.

- Press and hold the corresponding button to raise or lower vertical height. All three LED illuminate while in use.

AOA UP/DOWN

The screed operator can use the AOA (Angle-of-Attack) buttons for AOA adjustments while paving.

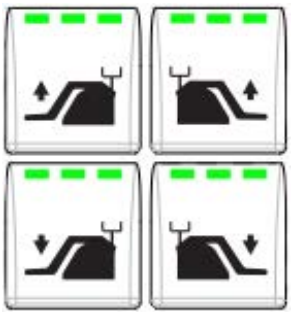
- Press and hold the UP or DOWN button to increase or decrease AOA while paving. All three LED lights illuminate while active.



SLOPE/BERM UP or DOWN

After selecting slope or berm, press and hold the UP or DOWN button to increase/ decrease slope or berm.

- The left LED illuminates in slope control mode and the right LED light illuminates in berm control mode.
- When the button is pressed, all three LED light illuminate green on both consoles (and the corresponding screed keypad).



TOW POINTS UP or DOWN (both sides)

The left and right tow points can be controlled automatically using the TopCon sonic automatic grade/slope control system or can be controlled manually using the keypad buttons.

- The default setting is MANUAL. The left LED light illuminates green in manual mode on both consoles. Use the Tow Up (thicker asphalt mat) and Tow Down (thinner asphalt mat) buttons to control the hydraulic tow cylinders.

NOTE: Does not function when the grade control is set to AUTO, however, the up and down buttons function when joysticks are in neutral.

- Pressing the Grade/Slope button changes to AUTO mode. The right LED light illuminates green on both consoles. The TopCon sonic control system must be connected to operate in AUTO mode. (The Tow Up and Down buttons do not function in AUTO mode).

NOTE: When in automatic mode, the tow points are paused when the machine is not traveling forward. In this situation, the center LED light will flash.

DPF REGENERATION (REGEN)

Before operating the paver, it is important that the operator understand the Diesel Particulate Filter (DPF) regeneration process described throughout the following pages. ***(See Section 5)*** DPF regeneration is the cleaning process where the exhaust temperature is raised to safely burn off any soot and undesired combustion gas. When the DPF filter reaches a soot saturation level between 25 - 45 percent, the onboard engine ECU (Engine Control Unit) will activate a regeneration cycle. The DPF icon on the DM430 digital controller indicates the level of soot build-up on a scale of Level 0 - 5. Regeneration will occur automatically at Level 1 - 2. DPF is further explained on the following pages.

During normal operation, the DPF is cleaned automatically. Under certain conditions, the operator may have to instruct the control system to perform the cleaning process. **It is important to perform a parked, manual regeneration BEFORE the engine reaches as high as Level 4. If the DPF reaches Level 4 - 5, non-warranty factory assistance from the engine manufacturer is required.**

Once the regeneration cycle has been activated, it is important not to shut off the engine as the regeneration cycle will fail and could possibly damage the engine. The system may attempt a second regeneration, but if it is unsuccessful, the DPF filter is blocked more than 80 percent and needs to be cleaned or replaced.

The engine speed will increase during regeneration and there may be a noticeable sound difference. Once regeneration is complete, the engine will return to normal idle speed.

NOTICE **Don't idle machine any longer than necessary as too much idling can cause soot to build up on the DPF, requiring more frequent regeneration.**

There are several factors that affect how often your DPF regenerates and the life expectancy of your DPF filter:

- Grade of engine oil.
- Grade of diesel fuel.
- Use of additives.
- Mileage.
- Wear of other components.

DM430 DIGITAL DISPLAY

The DM430 digital display features a high-resolution, anti-glare color display screen optimized with wide viewing angles in a variety of lighting conditions. There are three DM430 displays on the 8680, one on each main control panel, and the screed heat control panel. Each display is configured for the left and right operator platform controls. When starting the engine, the "boot" screen automatically displays on the DM430 digital display unit, followed by the Engine screen. .

NOTICE **Keep the controller screen clean by wiping with a soft, damp cloth. Use a mild dish washing detergent if needed. To avoid scratching or discoloring the display, DO NOT use abrasive pads, scouring powders or solvents (such as alcohol, benzene or paint thinner).**

Pressing "soft key" buttons on the control unit navigates to other screens, scrolls through information, toggles between screens and allows users to make selections as described on the following pages. A banner along the bottom of some screens correspond with the four function selection buttons (■), and may change accordingly as described on the following pages. Symbols and pop-up alert messages may also appear on the engine (and pave) screens as described throughout this section. (**Table 4-2** beginning on ***Page 4-25)***

Figure 4-8. DM 430 Button Controls

1 - Button #1

2 - ESC (Engine / Engine Diagnostics Screens)

3 - Button #2

4 - Screen Navigation

5 - Button #3

6 - Home (Pave / Pressures Screens)

7 - Button #4

ESC (Engine / Engine Diagnostics) Button: Pressing this button toggles between engine and engine diagnostics. ***(Page 4-27)***

Screen Navigation Button: Pressing this button navigates as shown in **Figure 4-9**:

- Pressing left or right navigates between all other screens (settings 1, settings 2, engine faults, and system faults).
- Pressing up or down is used on a particular screen scrolls through screen information (where prompted).
- The center of the button (↵) is used to select an item (where prompted).

Home (Pave / Pressures Screens) Button: Pressing this button toggles between the pave and pressure gauge screens. ***(Pages 4-33 and 4-41)***

If an engine fault is present, the Check Engine icon illuminates solid or flashing yellow or red, depending upon the severity of the problem.

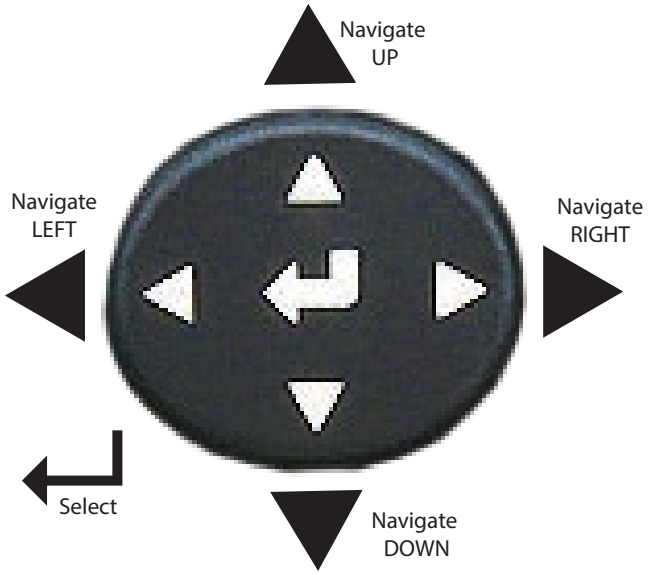


Figure 4-9. Screen Navigation Button

Engine Screen

The Engine screen is the default screen and displays various gauges. It also displays status information and icons (fuel level, DPF level, voltage, oil pressure, coolant temperature, hydraulic temperature, etc). **(Figure 4-10)**

Figure 4-10. Engine Screen

Several gauges appear on the engine screen as shown in **Figure 4-11:**

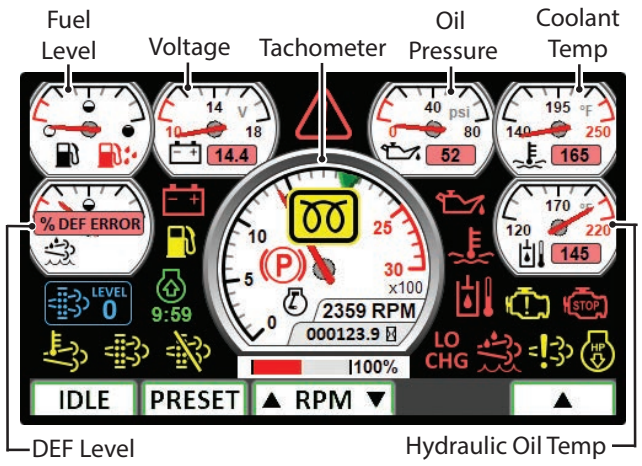


Figure 4-11. Engine Screen Gauge

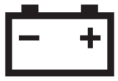
Warning symbols may also illuminate or turn red on the screen, indicating low fuel level, high exhaust temperature, low voltage, check engine, low DEF level, regeneration warnings, etc. Additional icons are shown on the screen, and will change if a fault condition is present. ***(Table 4-2 shows alert icons that may appear with descriptions.)***

- The fuel and DEF level gauges show icons to indicate fault conditions (water in fuel, improper DEF concentration).
- If the background turns red on gauges that are displayed as a digital readout, a fault exists or the computer system has lost communication.
- Engine RPM is shown in the center gauge (tachometer). The red needle is the actual RPM while the green is the RPM set point. RPMs and engine hours are also shown on the tachometer. Engine load is displayed as a red bar graph below engine hours.

DM430 Engine Icons

The following table shows icons that may appear on the engine (and other screens) to notify the operator a problem exists or component status:

Table 4-2. DM430 Engine Icons

Icon	Description
	Park Brake. Illuminates when the park brake is engaged.
	Transmission Neutral. The neutral icon displays when the transmission is in neutral. **NOTE: To perform a Parked REGEN, the "P" and "N" lamps must both be illuminated**
	**Voltage.** Illuminates if voltage level is low.
	Oil Pressure. Illuminates if oil pressure is too low.
	Check Engine. If the check engine icon illuminates, an engine problem exists that needs immediate attention. The engine may continue to operate, however, it is unable to perform DPF cleaning either automatically or manually. ⚠WARNING **Take action immediately to correct the fault.**
	Engine Stop/Protect. This icon illuminates when an existing engine fault has been allowed to progress to such a critical level that the engine will shut down automatically to protect itself.
	Fuel. Illuminates when the fuel level is low.
	Coolant Temperature. Illuminates when the coolant temperature is too high.
	Hydraulic Oil Temperature. Illuminates when the hydraulic oil temperature is too high.
	Plus +1 System Fault. If this icon illuminates, a fault exists in the engine ECU (Plus +1).

Icon	Description
LO CHG	**Low Hydrostatic Charge Pressure.** If this icon illuminates, the hydraulic charge pressure is low.
	Engine Power De-Rate Restored Timer.
	Engine Power De-Rate due to Active Fault.
	DEF Level/Quality. Illuminates yellow if the DEF tank level or DEF concentration level is becoming low (below 15 percent). This icon will illuminate red if the DEF tank level or DEF concentration becomes critically low.
	High Exhaust Temperature. Illuminates to notify operator that the exhaust temperature is high. The icon will turn off when normal operating temperatures are reached after the REGEN cycle. ⚠WARNING **Be sure engine exhaust is away from any potentially combustible materials when this is illuminated.**
	DPF Regeneration. Illuminates when the system is performing regeneration. Any time this lamp begins flashing, the operator should increase the loading on the engine so regeneration is possible. ⚠WARNING **If increased load does not cause an automatic REGEN to occur, the operator should immediately perform a Parked, Manual REGEN.**
	The user may choose to inhibit the REGEN if conditions are too hazardous for high exhaust temperatures. When this lamp is illuminated, a REGEN cannot be performed and soot levels will continue to rise.
	⚠WARNING **Unless hazardous conditions exist, the REGEN Inhibit switch and this lamp should remain off.**
	Engine Emission Fault. Illuminates when a fault exists in the emission system. Icon will be yellow at the caution level and turn red when the fault has become critical.
	Engine Preheat Active. Illuminates before starting while the engine is preheating.

You can set engine RPMs, idle the engine, or use preset RPM set points in the Engine screen using the display buttons that correspond with the function banner along the bottom of the engine screen. **(Figure 4-12)**

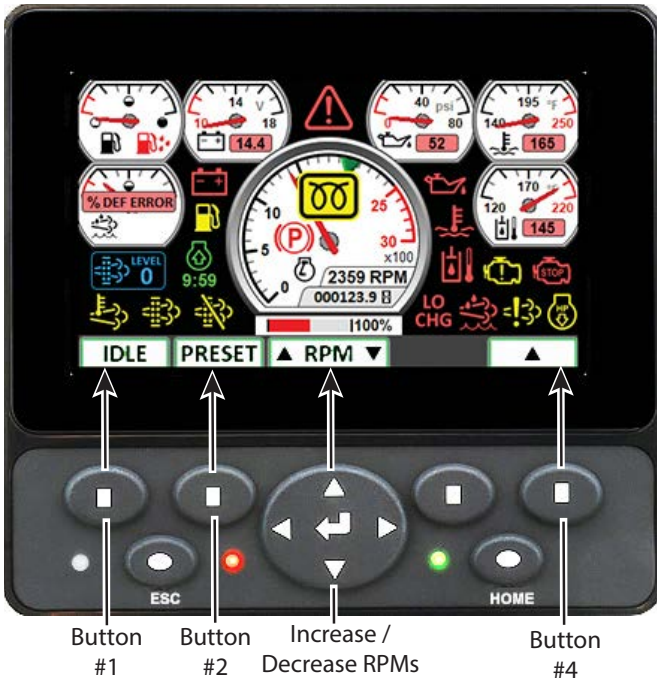


Figure 4-12. Engine Screen Navigation

- Adjust the Engine RPMs using the up (increase RPMs) and down (decrease RPMs) arrows on the center navigation button. The red needle in the tachometer gauge displays the actual RPMs.
- You can also use the preset RPM set points by pressing Button #2, which will cycle through these possible selections: Pave Low (1800 RPMs); Pave High (2100 RPMs; or Travel (2200 RPMs). The green arrow in the tachometer gauge displays set points.
- Press Button #1 to idle the engine.

NOTE: The engine will "auto idle" after a period of inactivity, and return to the previous setting when functions are activated.

The machine should be maintained in an operating condition so the DPF can automatically regenerate. When automatic regeneration occurs, the DPF regeneration icon appears on the screen. However, the operator can choose to activate the parked regeneration process, depending upon the DPF level (described on ***Pages 4-42 to 4-43).***

When you press **Button 4** in the engine screen, a pop-up banner appears with regeneration selections that correspond with the controller buttons. **(Figure 4-13)**

NOTE: If the icon background is gray, the function is unavailable (or unneeded) at the moment.

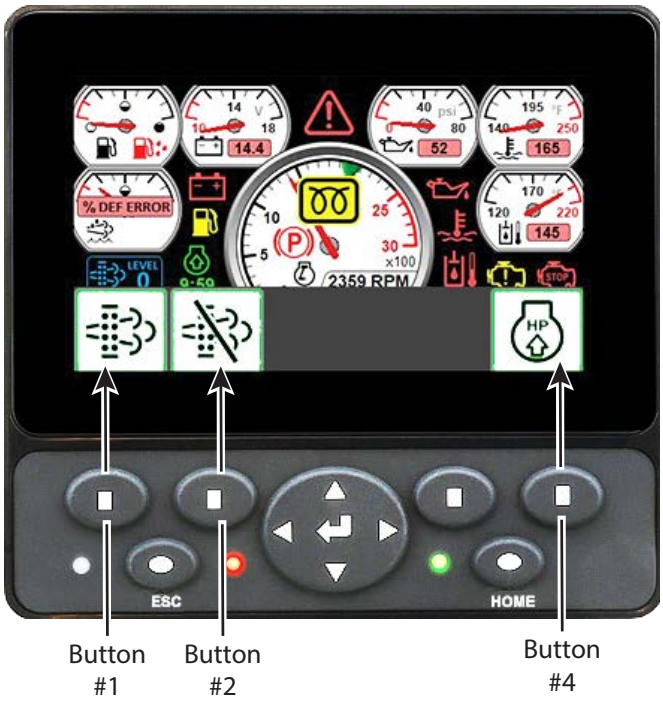


Figure 4-13. Engine Screen Pop-Up Banner

- Pressing Button #1 starts the parked regeneration process described on ***Page 4-44.***
- Pressing Button #2 inhibits the regeneration process.
- Pressing the ESC (escape) button exits the pop-up to the engine screen.

NOTE: The DEF concentration, DEF quality and DEF temperature will be shown in red when out of range.

Pressing the ESC (escape) button navigates to the Engine Diagnostics screen, which displays engine status and important information as shown in **Figure 4-14.** You can view active and logged engine fault screens *(shown on the following page)* by pressing the screen navigation button left and right. ***A complete listing of engine fault codes is shown in Trouble-shooting, Section 5.***

Figure 4-14. Engine Diagnostics Screen

- Press Button #1 to view active engine faults.
- Press Button #2 to view logged engine faults.
- Use Buttons #3 and #4 (PREV [previous] and NEXT) to scroll between fault screens.

The Active Diagnostics screen displays active warnings or faults with the appropriate Suspect Parameter Number (SPN) and Failure Mode Indicator (FMI). It may also display text description and ID/Name of the device that transmitted the message. **(Figure 4-15)**

Figure 4-15. Active Engine Faults Screen

The Logged Diagnostics screen displays nonactive warnings or faults with the appropriate Suspect Parameter Number (SPN) and Failure Mode Indicator (FMI) that are stored in the system. **(Figure 4-16)**

Figure 4-16. Logged Engine Faults Screen

Engine Alert Messages

Message banners may pop up on the engine (and pave) screen due to faults, or to alert the operator of issues. Some can be dismissed by pressing the ESC [escape] button and will periodically return to remind the operator of the issue. If the initial alert message is ignored, red alert messages will appear denoting an issue has reached critical level, and the engine output will be limited or will shut down. To prevent costly repairs, it is important to resolve a problem when it first arises.

If for some reason the DM430 controller has lost communication with the Plus +1 system or machine drive controller, the following alert message banner appears: **(Figure 4-17)**

Figure 4-17. Main Controller Offline Alert Screen

If the coolant temperature becomes too high, a red alert message banner will appear on the engine (and pave) screens to notify operator the engine will shut down. **(Figure 4-18)**

If this alert message pops up on the screen:

1. Turn off the engine (if it has not already shut down).
2. Remove key from the ignition.
3. Allow the machine to cool down several minutes, then check the coolant level. Add more coolant if needed. ***(Page 5-22)***
4. Start the engine.
5. If issues are still present or the engine will not restart, contact your authorized dealer for assistance.

Figure 4-18. High Coolant Temperature Alert Message

The following alert message appears if the engine oil pressure becomes so low that the engine will shut down. **(Figure 4-19)**

If this alert message pops up on the screen:

1. Turn off the engine (if it has not already shut down).
2. Remove key from the ignition.
3. Check the engine oil level. Add more oil if needed. ***(Page 5-17)***
4. Start the engine.
5. If issues are still present or the engine will not restart, contact your authorized dealer for assistance if needed.

Figure 4-19. Low Oil Pressure Alert Message

If the DEF is frozen, the following alert banner appears to notify operator that engine output is limited while the system attempts to thaw the DEF. You may continue operation in this state. **(Figure 4-20)**

Figure 4-20. Frozen DEF Alert Message

NOTE: Ambient temperature and humidity adversely affect DEF quality and consumption, with hot and dry climates resulting in greater DEF consumption. Temperatures above 86°F causes DEF to break down, which releases ammonia into the onboard DEF tank or storage container. DEF freezes at 12°F.

If the DEF tank temperature exceeds 120° F, the alert banner below will appear. **(Figure 4-21)** Add more DEF at your earliest opportunity.

Figure 4-21. High DEF Temperature Alert Message

If the previous warning is ignored and the DEF temperature is allowed to exceed 140° F, DEF sensor damage may occur that can adversely impact engine performance. The following URGENT message will appear: **(Figure 4-22)**

Figure 4-22. URGENT High DEF Temperature Alert Message

Other urgent alert banner messages may appear if active engine faults are ignored (or the emissions system has been tampered with), therefore becoming so severe that engine output is now limited and requires authorized dealer service. **(Figures 4-23 to 4-28)**

NOTICE **Modification or rendering inoperative any emission-related component can subject you to government penalties (and will void your warranty). Tampering with emission controls is in violation of federal law, and can result in civil penalties of up to $3,750 each day an engine or piece of equipment is operated in violation.**

Figure 4-23. URGENT Engine Output Limited Alert

Figure 4-24. Tampering Detected Alert

Figure 4-25. URGENT Engine Severely Limited Alert

If an engine fault has been resolved, the following instructional message may appear: **(Figure 4-26)**

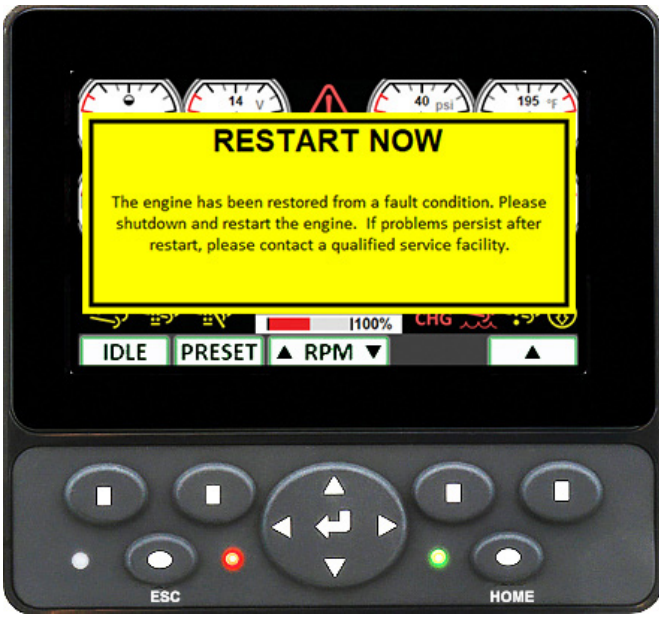


Figure 4-26. Restart Alert Message

Depending upon the engine fault, the engine output may be temporarily restored. These URGENT messages instruct the operator to move to a safe location, shut down the machine, and contact your authorized dealer for service. **(Figures 4-27 and 28)**

Figure 4-27. First Output Temporarily Restored Alert Message

If the DEF level is too low, the engine will de-rate. The restore function allows you to restore engine to full power for 10 minutes, then the engine will de-rate again if the DEF tank has not been filled. There are only two restores available per low-DEF level.

To restore engine after this event, press Button #4, which accesses the Engine screen pop-up banner.

- The engine symbol

will be functional (not gray) in this mode. Pressing Button #4 restores.

NOTE: Restores can be used during any Level 1 torque de-rate (50 percent reduction). This can be DEF level, DEF quality, or SCR system tampering. Tampering includes any sensor or controller errors of the SCR system.

NOTICE **When the DEF level is too low, the engine will de-rate. The restore function allows you to restore engine to full power for 10 minutes, then the engine will de-rate again if the DEF tank has not been filled. There are only two restores available per low-DEF level.**

Figure 4-28. Final Output Temporarily Restored Alert Message

Pave Screen

Pressing the Home Button navigates to the Pave Screen where paving function status is displayed, along with a navigation banner for making speed adjustments. **(Figure 4-29)**

Figure 4-29. Pave Screen

Engine status icons appear across the top of the screen, with auger, conveyor, DPF level, generator, and screed temperature shown in the row beneath.

NOTE: If "fault" appears instead of screed temperature, there is an issue with the sensor.

The auger, conveyor and generator icons are the same symbols shown on the main control panel and screed keypad (generator is not shown on the screed keypad). Conveyor speed is shown just below its icon. The auger and conveyor icons display in three color states:

Red: Off

Green: On

Blue: Paused.

When you turn on the generator to heat the screed, the generator icon changes from red (OFF) to green (ON). If operating in MANUAL mode, the actual temperature is shown in black. If operating generator in AUTO mode, the actual temperature is shown in black.

NOTE: The IDLE throttle speed button ceases to function when the generator is on. When the generator turns off, the IDLE functionality returns.

NOTE: The generator is set at the factory to default at 1800 RPMs.

Important gauges are shown beneath the row of paving status icons. **(Figure 4-30)**

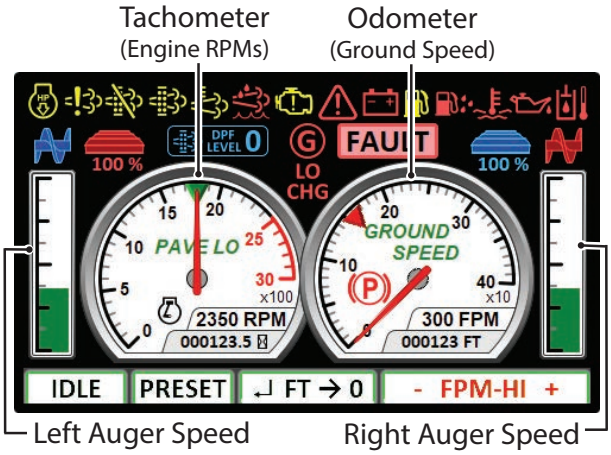


Figure 4-30. Pave Screen Gauges

- Bar graphs on the left and right sides of the screen reflect auger speed (set by the operator on the auger dial if operating in the MANUAL mode).
- The tachometer shows engine RPMs (same gauge as on the engine screen). The red needle indicates the actual speed. The set point triangle indicates speed set points. "Pave Lo" or "Pave Hi" appears in the tachometer denoting whether operating in low or high RPMs.
- The set point triangle automatically changes color when shifting between low gear [green] or high gear [red]. "FPM-HI" [high gear] or "FPM" [low gear] also displays on the FPM (feet per minute) navigation banner above Button #4.

- The ground speed gauge displays a digital value for actual speed and odometer that accumulates distance in feet. The red needle indicates the actual ground speed. The green triangle indicates speed set points.
- The maximum ground speed can be adjusted using Buttons #3 (-) [decrease] and #4 (+) [increase]. **(Figure 4-31)**
- Pressing and holding the center of the navigation button (↵) resets the odometer to zero feet.

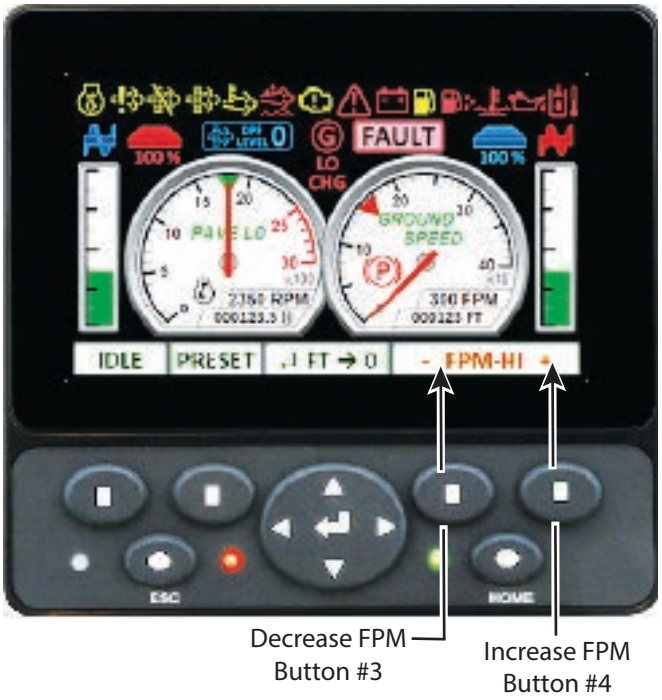


Figure 4-31. Ground Speed Adjustment

Pave Screen Alert Messages

If a fault is present, a red warning icon ⚠ will flash (on both the engine and pave screens). If the icon is solid, the main controller is offline and cannot communicate with the display, therefore no fault information can be shown. A list of paving-related fault codes is shown on the following pages.

You can view active **(Figure 4-32)** and logged **(Figure 33)** system faults by pressing the screen navigation left or right from any screen.

Figure 4-32. Active Paving Function Faults Screen

- Press Button #1 to view active paving function faults.
- Press Button #2 to view logged paving function faults.
- Use Buttons #3 and #4 (PREV [previous] and NEXT) to scroll between fault screens.

Figure 4-33. Logged Paving Function Faults Screen

Pop-Up message banners may also appear on the pave (and engine) screen to alert the operator of an issue, or if the operator attempts to do something under the wrong conditions. **(Figures 4-34 to 4-42)**

The following banner appears if the operator attempts to crank the engine with the auger, conveyor or generator are engaged. **(Figure 4-34)** This banner notifies the operator that the engine will not start unless these components are turned off.

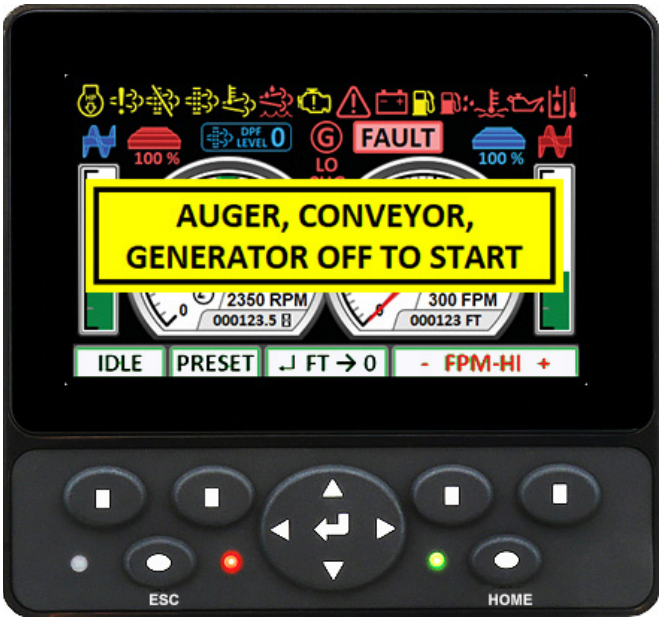


Figure 4-34. Component Engine Start Alert

The following pop-up messages appears if the operator attempts to activate a control panel while parked regeneration is in process. **(Figure 4-35)**

Figure 4-35. Parked Regen Active Alert

If the operator attempts to crank the engine or activate a control panel console while one or more joystick(s) is out of neutral, an alert message appears. **(Figure 4-36)**

NOTE: The joystick in the wrong position will be shown in red.

Figure 4-36. Joysticks Must Be in Neutral Alert

The following message alert appears if the paver in high gear with the conveyors or augers on. **(Figure 4-37)**

Figure 4-37. High Gear Pave Alert

If the operator attempts to shift from high to low gear while traveling too fast, the following message pops-up on the screen. **(Figure 4-38)**

NOTE: The downshift will automatically occur if the operator slows down.

Figure 4-38. Downshift Inhibited Alert

If the operator is attempting to use the joystick(s) on an inactive console, the following message appears that notifies operator to activate the control panel console. **(Figure 4-39)**

Figure 4-39. Activate Console Alert

If the control panel or a joystick is disabled or faulty, the following message appears: **(Figure 4-40)**

Figure 4-40. Joysticks Must Be in Neutral Alert

The following message alert appears if the operator sets the RPMs above the Pave Hi (high) preset. **(Figure 4-41)**

Figure 4-41. High Engine RPM Alert

The following messages appears if the RPM is set above idle while turbo protection is active. **(Figure 4-42)**

NOTE: The turbo protection feature takes control of the throttle for the first 1 - 30 seconds based on ambient temperatures. It does not allow the engine RPM to increase to ensure proper lubrication to the turbo and cylinders before increasing RPMs.

Figure 4-42. Throttle Up Inhibited Alert

Pave Function Fault Codes

Fault Code	Fault Description
1	Left Display Time Out
2	Left Keypad (Lower Left) Time Out
3	Left Keypad (Lower Right) Time Out
4	Left Keypad (Upper Left) Time Out
5	Left Keypad Time Out
6	Left Keypad (Upper Right) Time Out
7	Left Console, Left Joystick Fault
8	Left Console, Left Neutral Fault
9	Left Console, Right Joystick/Steer Fault
10	Left Console, Right Neutral Fault
11	Left Console, Auger Potentiometer Fault
12	Left Screed, Auger Potentiometer Fault
17	Right Display Time Out
18	Right Keypad (Upper Left) Time Out
19	Right Keypad (Lower Left) Time Out
20	Right Keypad (Upper Right) Time Out
21	Right Keypad (Lower Right) Time Out
22	Right Keypad Time Out
23	Right Console Left Joystick Fault
24	Right Console Left Neutral Fault
25	Right Console Right Joystick/Steer Fault
26	Right Console Right Neutral Fault
27	Right Console Auger Potentiometer Fault
28	Right Screed Auger Potentiometer Fault
33	Safety Controller Error Detected
34	Engine J1939 Time out
35	Engine ACU J939 Time Out
36	Engine ACU2 J1939 Time Out
37	Input Fault, Fuel Level Sender
38	Press Sensor 1 Fault, Auger/Work Pump
39	Press Sensor 2 Fault, Conveyor Manifold
40	Press Sensor 3 Fault, Cylinder Manifold
41	Press Sensor 4 Fault, Propel Left Forward
42	Press Sensor 5 Fault, Propel Right Forward
43	Sensor Power Fault
44	Input Fault, Left Auger Sonic
45	Input Fault, Right Auger Sonic
46	Input Fault, Left Propel Speed Sensor

Fault Code	Fault Description
47	Input Fault, Right Propel Speed Sensor
48	Input Fault, Hydraulic Temperature Sensor
49	Input Fault, Screed Temperature Sensor
50	Cylinder Output Module Time Out
51	Motor Output Module Time Out
52	Screed Output Module Time Out
53	Output Fault, Left Auger Proportional
54	Output Fault, Left Auger Reverser
55	Output Fault, Right Auger Proportional
56	Output Fault, Right Auger Reverser
57	Output Fault, Beacon Light
58	Output Fault, Left Berm Down
59	Output Fault, Left Berm Up
60	Output Fault, Right Berm Down
61	Output Fault, Right Berm Up
62	Output Fault, Brake Release
63	Output Fault, Backup Alarm
64	Output Fault, Left Conveyor Proportional
65	Output Fault, Left Conveyor Reverser
66	Output Fault, Right Conveyor Proportional
67	Output Fault, Right Conveyor Reverser
68	Output Fault, Left Cut-Off Close
69	Output Fault, Left Cut-Off Open
70	Output Fault, Right Cut-Off Close
71	Output Fault, Right Cut-Off Open
72	Output Fault, Engine On
73	Output Fault, Engine Start
74	Output Fault, Left Extension
75	Output Fault, Right Extension
76	Output Fault, Generator Neutral Signal
77	Output Fault, Generator On
78	Output Fault, Truck Hitch In
79	Output Fault, Truck Hitch Out
80	Output Fault, Left Hopper Close
81	Output Fault, Left Hopper Open

Fault Code	Fault Description
82	Output Fault, Right Hopper Close
83	Output Fault, Right Hopper Open
84	Output Fault, Horn
85	Output Fault, Lights
86	Output Fault, Propel High Gear
87	Output Fault, Left Propel Forward
88	Output Fault, Left Propel Reverse
89	Output Fault, Right Propel Forward
90	Output Fault, Right Propel Reverse
91	Output Fault, Screed Down
92	Output Fault, Screed Up
93	Output Fault, Left Slope Down
94	Output Fault, Left Slope Up
95	Output Fault, Right Slope Down
96	Output Fault, Right Slope Up
97	Output Fault, Left Sonic Enable
98	Output Fault, Right Sonic Enable
99	Output Fault, Spray Down Pump
100	Output Fault, Left Tow Down
101	Output Fault, Left Tow Up
102	Output Fault, Right Tow Down
103	Output Fault, Right Tow Up
104	Output Fault, Vibrator
105	Engine Shut Down, Low Oil Pressure
106	Engine Shut Down, High Coolant Temperature
107	Propel Pump Charge Pressure Low
108	Engine Oil Pressure Low
109	Water in Fuel
110	Engine Coolant High Temperature
111	Hydraulic Oil Temperature
112	Primary/Secondary Processor Mismatch
113	No Valid Parameter File Loaded
114	Output Fault, Track Tension Unloader

Operator Settings Screen

Navigate to the Operator Settings screens to set paving controls by pressing left or right on the screen navigation button. **(Figures 4-43 and 4-44)** Use this screen to select manual or AUTO mode for operating the under auger material cut-offs, conveyor and generator. You can set the screed temperature for automatic temperature control or choose to operate the screed heat manually using the keypad. You can also disable one or both screed control panels.

- Use the up and down arrow buttons to scroll to each setting window. Pressing the left/right arrow (right) to the end will navigate to the second operations settings screen.
- Press the center of the navigation button (↵) to select desired setting when highlighted in the window.

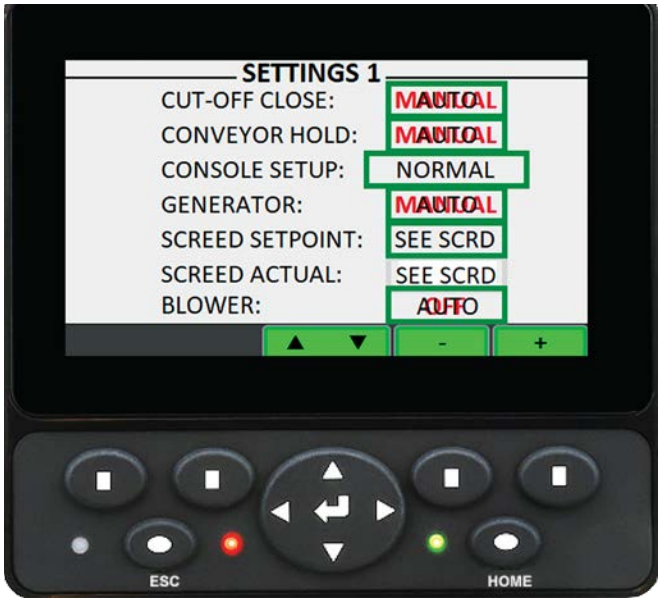


Figure 4-43. Operator Settings Initial Screen

Cut-Off Close: There are two settings for the under auger material cut-offs: Manual and AUTO.

- In AUTO mode, the cut-off will automatically close when putting the paver in reverse.

Conveyor Hold: There are two settings for the conveyor hold: Manual and AUTO.

- In AUTO mode, the conveyors will stop when the paver stops.

Console Setup: You can disable the left, right or both main control panels in this window in the event of fault conditions. For example, if joysticks are malfunctioning on one console, disable that console and use the other side.

Generator: Set the generator to operate in manual or AUTO mode in this window.

- In auto, the generator will turn on and off automatically to maintain the temperature set point.
- In manual mode, the generator will turn on and off after an elapsed timer.

NOTE: If the temperature sensor is faulted, manual mode will be forced.

Screed Set Point: Use this window to adjust the screed set point temperature.

- Use the plus and minus buttons for setting the screed set point temperature.
- If in manual mode, N/A will be shown in the window. This is the default setting.

Screed Actual: This is the actual measured screed temperature.

NOTE: If a fault exists in the sensor, FAULT will be shown rather than the temperature.

Blower: The blower will automatically turn on when either conveyor is turned on in AUTO mode, and shut off when conveyors are both off.

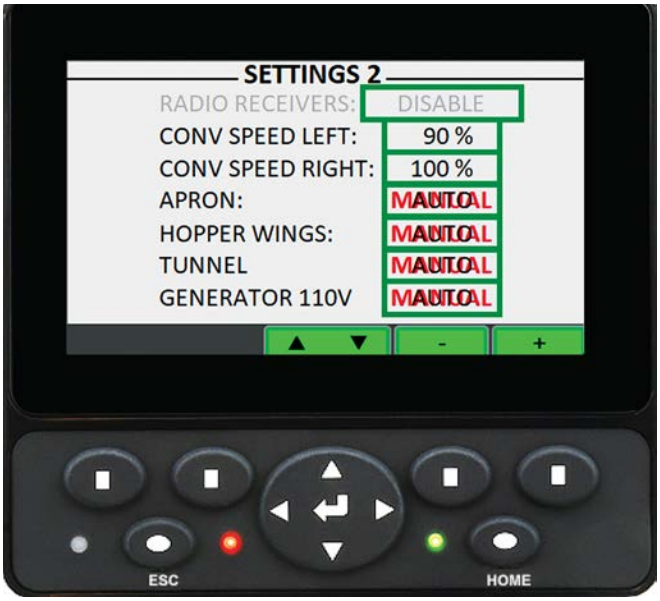


Figure 4-44. Operator Settings Secondary Screen

Conveyor Speed Left/Right: Sets the maximum speed allowed by the conveyor paddles.

Apron: When set to AUTO, the system will automatically raise/lower the apron when the hopper is opened and closed.

Hopper Wings: When set to AUTO, the system will automatically run with the hopper open when the operator presses the Hopper Open button.

Tunnel: When set to AUTO, the tunnel will automatically function while operating the screed extension.

Generator: In AUTO mode, the generator turns on and off automatically to maintain the temperature set point. In MANUAL mode, the generator turns on and off on a timer basis.

NOTE: If the temperature sensor fails, manual mode will be forced.

Motor Settings Screen

Navigate to the Motor Settings screen by the pressing left or right on the screen navigation button. **(Figure 4-45)**

Use this screen to set the left/right augers and conveyor for operating in forward or reverse. You can also set the left and right conveyor speed on this screen.

- Use the up and down arrow buttons to scroll to each setting window.
- Press the center of the navigation button (↵) to select desired setting when highlighted in the window.
- Use the plus and minus buttons for setting the conveyor speed.

Figure 4-45. Motor Settings Screen

Pressure Gauge Screen

Use the Home button to cycle between the Pave screen and the Pressure Gauge screen. The pressure gauge screen displays hydraulic pressure information for five paver functions: auger, conveyor, cylinders, propel left and propel right. **(Figure 4-46)**

NOTE: If a fault exists, the LO CHG icon will illuminate, and the pressure digital read-out background for that component will turn red.

- Press Button #1 to idle the engine.
- Press Button #4 to use preset RPMs, or adjust engine RPMs:
  - Press Button #2 to decrease RPMs.
  - Press Button #3 to increase RPMs.

NOTE: The engine RPMs is shown in the center of the navigation banner.

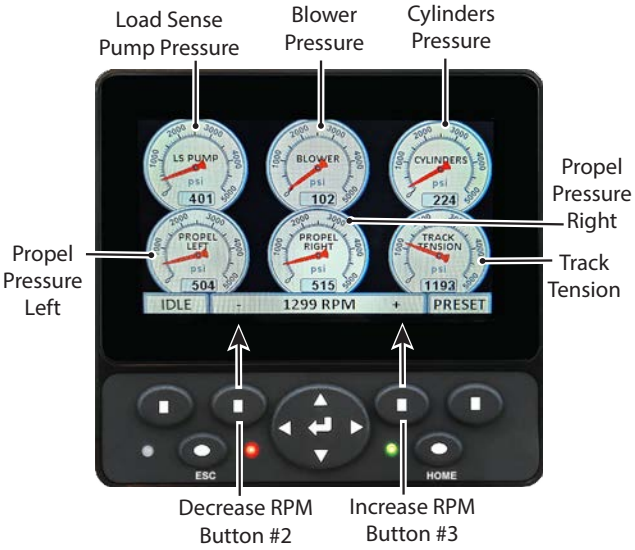


Figure 4-46. Pressure Gauge Screen

DPF and Regeneration

It cannot be emphasized enough that the paver should be maintained in good operating condition so the emission system can automatically perform the regeneration (regen) process.

⚠CAUTION If engine faults are present that prevent the ability to perform regeneration, DPF levels will continue to rise. It is important to resolve any engine faults immediately and perform the needed regeneration to prevent costly DPF technician service if DPF level warnings or engine faults are ignored.

If regeneration begins while operating, immediately move the paver to a safe location to allow regeneration to occur. During regeneration, there may be a noticeable sound difference. The high exhaust temperature icon will illuminate on engine and pave screens will appear during the process.

- If unable to move to a safe location for the engine to perform a regeneration automatically, the operator may choose to ***temporarily*** inhibit (disable) the regeneration process.
  - Press Button #4 while on the Engine screen, then use Button #2 to inhibit regeneration, described on ***Page 4-27***, until you can move to a safe location and complete the parked regeneration process. ***(Page 4-44)***

NOTE: The ability to inhibit (disable) regeneration is only possible at DPF levels 1 and 2, described on the following page.

NOTE: Regeneration will remain disabled until the engine is turned off or the operator navigates back to the screen to enable regeneration.

⚠CAUTION EXHAUST TEMPERATURES WILL BE EXTREMELY HIGH. BE SURE EXHAUST WILL NOT COME INTO CONTACT WITH ANY COMBUSTIBLE MATERIALS.

⚠CAUTION Monitor the surrounding area during regeneration. If unsafe conditions develop, shut the unit off immediately.

DPF Levels

Pay attention to the DPF level icon on the engine and pave screens. Automatic regeneration can only occur at DPF Level 1 and 2.

DO NOT allow soot levels to reach high levels to prevent damage to the engine and/or DPF components that will require costly repairs. DPF levels are described below.

NOTE: If the operator has not inhibited regeneration and there are no engine faults, but automatic regeneration is not occurring at DPF Level 1 or 2, increase engine RPMs and/or load to allow the DPF to automatically regenerate.

NOTICE **It is important to perform a parked regeneration BEFORE the DPF level reaches Level 4. If the regeneration process is not performed and the soot level reaches Level 4, non-warranty factory assistance from the engine manufacturer is required.**

If an engine fault is present at DPF Levels 1 and 2, a message banner may appear on the engine and pave screen notifying operator regeneration is not possible until the engine fault is resolved. DPF levels will continue to rise. **(Figure 4-47)**

Figure 4-47. Active Engine Code DPF Alert

DPF Level 1: The DPF is loaded with soot and needs to be regenerated. When safe, please uninhibit automatic active regeneration.

DPF Level 2: The soot level has continued to accumulate. Increase engine RPMs (if at idle) and engine load to prompt automatic regeneration.

NOTE: If the operator is inhibiting automatic regeneration and does not perform parked regeneration, the soot level will continue to rise.

DPF Level 3: An orange message alert banner may appear to warn the operator that soot levels are extremely high. **(Figure 4-48)** If you have allowed the exhaust soot level to reach Level 3, automatic regeneration is not possible. If this message appears, the operator must perform parked regeneration at the earliest opportunity. When safe, complete a parked regeneration as described on the following pages.

NOTICE **Continued operation without completing a parked regeneration can cause engine damage and will void engine warranty. The operator should perform parked regeneration immediately!**

Figure 4-48. Parked Regeneration Required Alert

DPF Level 4: If the operator continues ignoring DPF level warnings and the soot level reaches Level 4, a red message alert banner appears on the engine and pave screens. **(Figure 4-49)**

Engine output is limited and the operator can no longer perform regeneration. Regeneration is required IMMEDIATELY by a qualified engine or OEM service technician using specialized equipment. Move the paver to a safe location, shut down, and contact your authorized dealer for service.

NOTICE **If the soot level reaches Level 4 or 5, specialized equipment is needed to perform regeneration and the machine must be serviced by your authorized dealer. The DPF system may become damaged and require replacement. Therefore, performing regeneration BEFORE reaching Level 4 is imperative for preventing costly repair.**

Figure 4-49. DPF Exceeds Normal Limits Alert

Regen Level 5: If the DPF level reaches Level 5, a red message alert banner appears on the engine and pave screens notifying operator the DPF must be replaced. **(Figure 4-50)** The DPF must be removed and cleaned by a qualified cleaning facility, and the ECU must be reset by a qualified engine or OEM service technician. The engine will derate to idle with no torque available. IMMEDIATELY contact your authorized dealer.

Figure 4-50. DPF Must Be Replaced Alert

4

Parked Regeneration

If the DPF is at Level 2 or 3, and no engine faults are present, perform parked regeneration using the following procedure:

1. Park the paver in an open area away from people, equipment, trees/shrubbery, debris, and combustible materials as exhaust temperatures will increase considerably during regeneration.
2. Place the paver in NEUTRAL and engage the park brake.
   - Ensure the coolant temperature is the operating range of 155° to 160° F.
3. Set the engine to low idle.

⚠CAUTION DO NOT ATTEMPT TO OPERATE THE UNIT, CHANGE ENGINE RPM OR MOVE FROM PARK/NEUTRAL WHILE REGEN IS OCCURRING. This will abort the Regeneration process and require you to start the process over.

4. From the engine screen, press Button #4 to access DPF commands. A pop-up banner appears with regeneration selections corresponding to the buttons. **(Figure 4-51)**

NOTE: If the icon background is gray, the function is unavailable (or unneeded) at the moment.

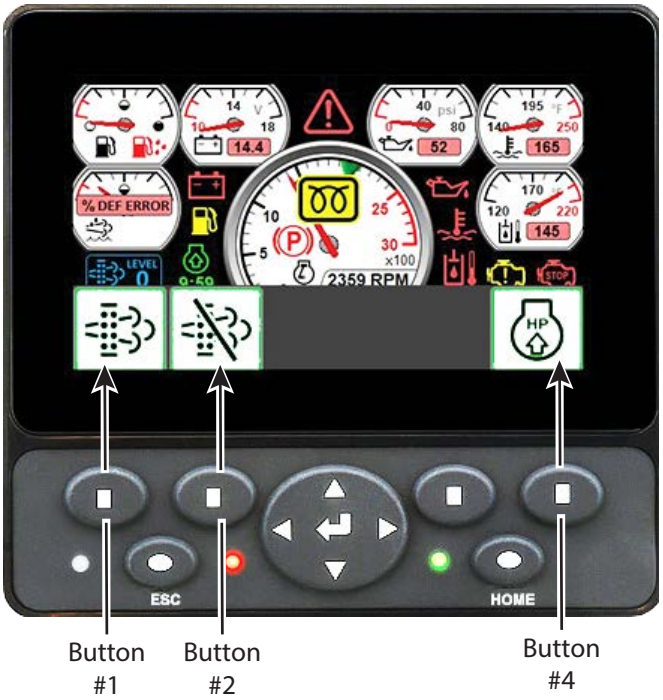


Figure 4-51. Engine Screen Pop-Up Banner

5. Press Button #1 to start the regeneration process.

During this process:

- DO NOT move machine out of park or neutral.
- DO NOT touch the accelerator pedal/lever.
- DO NOT turn the engine off or stop the regeneration process.

- The engine speed will increase and there may be a noticeable sound difference. This is normal. The "Parked DPF Regen Active" message alert will appear **(Figure 4-52)**.

NOTE: The High Exhaust icon will illuminate when the exhaust temperature reaches >842° F Both alerts will stay on during the regeneration process.

Figure 4-52. Parked Regen Active Alert

6. Monitor your surroundings while the system is regenerating, ensuring others stay away from the machine. If any unsafe condition arises, shut off the engine immediately and restart the process when the area is clear.

When the parked regeneration process is complete, the engine will automatically return to normal idle speed. The Parked DPF Regen Active message will disappear and the high exhaust icon light will turn off.

PLUS +1® CONTROLLERS

The Plus +1® system contains several controllers and OX-modules operating in the background that monitors paver control systems and alerts the operator in the event of a fault. This is the brains behind paver and engine systems. When it senses a fault, the Plus +1® sends a message to the DM430 display to alert the operator. The Plus +1 controller and OX-modules are located throughout the engine compartment and on the screed. (A complete list of fault codes with corrective action messages are shown in **Section 5,** *Troubleshooting*.**)**

The Plus One system will not allow the engine to start if:

- Steering control box not detected.
- Joystick(s) not in neutral at start-up.
- Pump control coil fault.
- Brake valve/pump neutral bypass coil fault.

DRIVING THE 8680 PAVER

The dual operator platforms allow easy and convenient access for controls of most paver functions. The paver is usually operated from the left operator station, but can also be operated from the right side. Dual joysticks propel forward and reverse, and steer the paver. Ramping for acceleration and deceleration are proportional to the movement of the joysticks. The amount of steering is controlled by the amount of joystick movement and the speed of the paver.

⚠ DANGER NEVER leave the operator platform unattended while the paver is in gear or in motion. Operator must return joystick(s) to the neutral position, engage the park brake, press the Start/Stop button, and remove the ignition key before leaving the operator platform.

⚠ WARNING Ensure there are no people, equipment or obstacles in the paver's before moving in either direction.

Make seat and operator platform adjustments described on ***Page 4-5*** before operating the paver. Follow the instructions on ***Page 4-6*** to start the engine and activate the desired operator main control panel.

Keep these items in mind for operating with rubber tracks and maximizing track life:

- Avoid detracking episodes by:
  - Maneuvering at lowest speed possible;
  - Avoiding sharp turns on hot asphalt (move to a cold area for turns);
  - Making gradual turns in as large an arc as possible, repeating zigzag motions gradually instead of pivoting;
  - Using forward instead of reverse when possible;
  - Preventing large foreign objects from being entangled in the undercarriage/tracks.
- Properly trained operators who practice good operating techniques will make a big difference.
- Avoid spot turns, spinning, constant operation on slopes and contact with the side of tracks on curbs, hard walls and other objects. Limit use around large, sharp and rocky surfaces such as manholes and drainage covers.
- Clean, inspect and check alignment on the rubber tracks at least once per day. Replace worn parts immediately if needed.
- Avoid track contact with abrasive/corrosive materials. If oil or similar products get on the tracks, clean it off immediately. Over time, oil degrades rubber quality.
- Remove asphalt buildup on the undercarriage components and rubber tracks. Buildup causes misalignment between the drive wheel and guide lugs, poor tracking performance and premature undercarriage components/rubber track wear. Periodically check the undercarriage components for wear (i.e., drive wheel, rollers and idler) and make repairs if needed.

Steering and Speed Control

Use the dual joysticks for steering and speed control. Move the joysticks SLOWLY. Lift up on the joystick collar and move joystick either FORWARD or REVERSE. The joystick collar will lock back into place in the NEUTRAL position. **(Figure 4-53)**

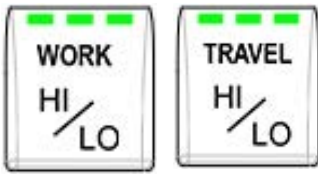
⚠ DANGER Rapid counter-rotation in either direction can result in the operator being thrown from the machine and suffer personal injury or death. Counter-rotation occurs if one joystick is in the extreme forward position and the other is in the extreme rear position while the machine is in gear and moving. Make only SLOW adjustments to the joysticks to turn the paver right or left.

⚠ CAUTION Be aware of the terrain, traffic and other driving conditions. Adjust speed accordingly.

⚠ CAUTION Avoid low limbs, power lines, and other objects that can endanger crew or paver.

NOTICE **ALWAYS pave in Work mode. Use Travel mode only for traveling.**

To drive the 8680 Commercial Paver:



Select WORK Hi/Lo or TRAVEL Hi/Lo mode on the main control panel, then use joysticks to move forward and reverse.

Figure 4-53. Joysticks

- To slow the paver, move joysticks closer to the neutral position.
- To stop the paver, pull joysticks back to the neutral position.
- To drive the paver forward, lift up on the joystick collar and push joystick forward to reach the desired speed.
- To move in reverse, lift up on the joystick collar and move joystick backwards to reach desired speed.
- Return joysticks to the neutral position to stop paver.
- To steer the unit to the left, SLOWLY push the right joystick farther forward than the left joystick. The farther the joystick is pushed, the more the paver turns. Slow and easy adjustments are required for safe operation.
- To steer the unit to the right, SLOWLY push the left joystick farther forward than the right joystick. The farther the joystick is pushed, the more the paver turns. Slow and easy adjustments are required for safe operation.

If you need to stop the paver:

1. Place joysticks into the neutral position.
2. Throttle back to idle using settings on the DM430 digital display screen.
3. Turn ignition key on the junction box counterclockwise to the OFF position and remove key.

NOTE: If the engine does not shut down when the key is turned to OFF, press the E-Stop button on the control panel to terminate power.

You can also use the Pause button once to pause propel, conveyors and augers functions. Pressing the button again resumes these functions.

NOTE: The engine idle will remain at high but automatically go into low idle if paused for a preset period of time (unless the generator is enabled.)

If equipped with the optional steering knob, it is used for steering and replaces the right joystick on the control panel. **(Figure 4-54)** Use the left joystick for forward and reverse speed control and steer right or left using the steering knob.

1 - Press button for incremental steering to the left.

2 - Press button to center paver.

3 - Press button for incremental steering to the right.

4 - Turn knob left to steer left; Turn knob right to steer right.

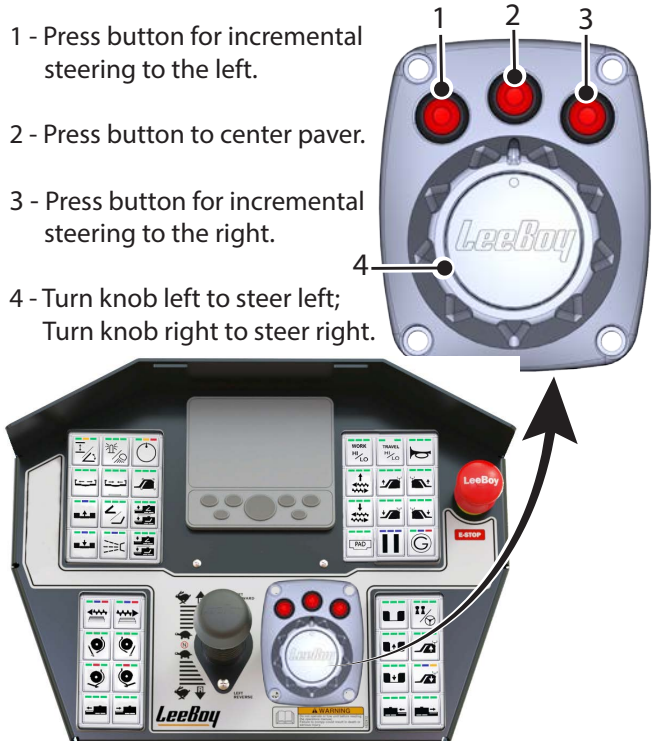


Figure 4-54. Steering Knob

If equipped with the optional steering wheel, it is used for steering and replaces the right joystick on the control panel. **(Figure 4-55)**

The joystick is used for speed control and propelling forward and in reverse. The center position is neutral.

- To drive the paver forward, lift up on the joystick collar and push joystick forward to reach the desired speed.
- To move in reverse, lift up on the joystick collar and move joystick backwards to reach desired speed.
- Return joystick to the neutral position to stop paver.

Figure 4-55. Steering Wheel Option

PAVER OPERATION

Spray Down

Always spray down the machine to clean it before and after paving. Spray solvent or release agent on any part of the paver that comes into contact with asphalt to prevent buildup that will damage the machine over time.

The citrus tank with filler cap is located under the right floorboard beside of the DEF tank. **(Figure 4-56)**

Figure 4-56. DEF and Citrus Fill Caps

⚠CAUTION Ensure the area is clear of others and flammable materials before spraying. Allow the screed to cool at least 10 minutes before spraying to prevent possible burns from solvent spray on a hot screed. The screed should be warm, but not hot.

⚠WARNING Never spray cleaning solvent or release agent near the screed heating element or near open flame. These agents are highly flammable and can ignite, causing serious personal injury.

NOTICE **When using solvents and release agents, DO NOT allow cleaning solvent to run onto the ground. Only spray down the machine in a designated area away from the paving site.**

There is a spray down system on both sides of the Leeboy 8680 paver. A spray wand is conveniently located on each side of the operator platform. **(Figure 4-57)** The hose reels are located inside the access panel on each side of the paver.

Figure 4-57. Spray Down Wand Location (Each Side)

To operate the spray down system:

1. Push the Spray Down button on the main control panel to turn system ON.
   - Press the button again to turn OFF.
2. Extend hose from the hose reel.
3. Squeeze the wand handle to spray desired areas.

Start at the front of the machine and work rearward, using a scraper if needed, paying particular attention to:

- Push Rollers: Rotate the rollers while cleaning and soak bearing ends to maintain free rotation.
- Rubber Tracks and Undercarriage: The importance of keeping the heavy-duty rubber tracks cannot be emphasized enough to prevent premature wear. Don't operate in corrosive, abrasive, salt or other contaminant materials that cause rubber to deteriorate. Look for oil and grease that may drip from the machine onto the tracks and rinse off as soon as possible. Mud can also harden and adversely affect track operation along with other problems. You can use a pressure washer to remove the mud. Cleaning is especially important in colder temperatures.

NOTE: DO NOT operate with frozen tracks. If you try to use power to force the tracks to move, you could damage them or your final drive.

- Hopper and Conveyors: Thoroughly clean surfaces and any material accumulating in chains, drive sprockets, idler shaft, bearings and tunnel areas. AFTER cleaning the machine, operate the conveyors briefly to carry solvent into return areas where it can help dissolve hidden asphalt material.
- Augers: Scrape away any accumulated material around bearings and clean contact surfaces down to the metal to avoid quick build-up while paving.
- Screed: Face and bottom plates should be clean and smooth on the main and screed extensions to achieve a smooth paving mat.
- Operator Platforms: Be sure to wash away solvent residue so walk boards and operator platforms do not become slippery.

Hopper Wings

The hopper wings are hydraulically opened so the hopper can be filled with paving material. The 8680 paver features a 9-ton, rounded design to reduce cold material build-up while rear hopper wing wipers help prevent spillage. Material in the hopper is moved toward the back of the paver to the screed by conveyors.

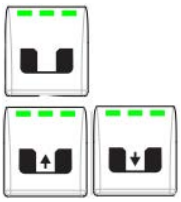
The operator can also extend and retract each hopper wing individually. This feature gives you the ability to pave very close to walls, fences, road barriers, buildings or other obstacles with ease.

⚠WARNING **DO NOT retract the hopper wings until the material truck has moved completely away from the paver or if the hopper is full of asphalt. NEVER raise the conveyor bed with asphalt in the hopper.**

⚠WARNING **Ensure no people, equipment or other obstacles are in or near the hopper when raising or lowering the hopper wings to prevent injury or damage to the paver.**

⚠CAUTION **Ensure the hopper cylinder is unlocked before extending or retracting the hopper wings.**

The hopper wings are operated using hopper buttons on the lower right control panel keypad:

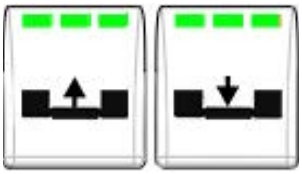


1. Use the hopper wing select button to choose hopper wing function. If operating both hopper wings, there is no need to press the hopper select button as operating both wings is the default operational mode.

- Press the hopper wing select button once to operate the LEFT hopper wing.
- Press the wing selection button again to operate the RIGHT hopper wing.
- Pressing the button again returns to the default operation control for both hopper wings.

⚠CAUTION **Be sure the hopper wings are completely opened before receiving paving material from the truck.**

2. After selecting the left, right or both wings on the hopper wing select button, use the hopper UP and Down buttons to open and close the hopper wings:
   - Press and hold the hopper wing up (Close) or down (Open) buttons to close or open one or both hopper wings based upon the hopper wing selection made.



Press and hold the Apron UP / DOWN button to pivot the front apron to assist in cleaning out the hopper. All three LED lights illuminate while in operation.

Hopper Lock Valves

The hopper cylinder valves can be used to lock each hopper wing when they are open or closed. The lock valves are located inside the left engine panel. **(Figure 4-58)** Be sure to unlock the hopper valves before operation!

- Turn the valve counterclockwise to unlock the hopper wings.
- Turn the valve clockwise to lock the hopper wings.

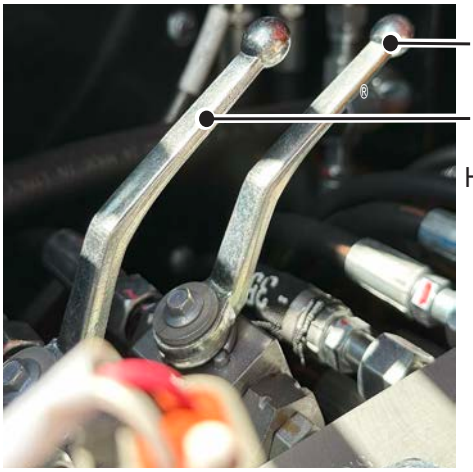


Figure 4-58. Hopper Cylinder Lock Valve

Conveyors

The conveyor is a very important part of the paver that moves the asphalt mix from the hopper to the auger, which controls the amount of material delivered to the screed. Pay close attention to integrating conveyor operation into the total operation of the paver. The left and right conveyors can be operated in conjunction with the augers from either control panel, and both can be operated in reverse.

⚠CAUTION NEVER leave the paver unattended while the conveyor is operating. DO NOT leave paver with empty conveyors running for any length of time to prevent excessive wear.

When operating the conveyor:

- Spray down the conveyor drive chains with cleaning solvent or release agent several times during the work day for optimum performance. ***(See* Section 5 *Lubrication Chart.)***

NOTICE Asphalt in the conveyors can fill tracks and cause damage to the machine. To prevent flight chains from sticking inside the conveyor pans, lubricate them daily for maximum operational efficiency.

- If equipped with the Under Auger Material Cut-Offs option, some spillage can occur while the conveyors are operating. This is normal. To help prevent this, operate conveyors manually when loading hopper and not paving.

- Irregular movement of the conveyor indicates that the conveyor chains need adjustment. The conveyor chains are easily adjusted on the 8680 paver. Fine tuning conveyor chain tension is described in detail in **Section 5.**

NOTE: Check conveyor chain adjustment every 100 hours.

NOTE: Moving the joysticks into neutral will pause conveyor operation. (This feature can be turned off.)

To operate the conveyor(s) from the main control panel:

- Press one or both auger conveyor buttons on either control panel to turn on the conveyor(s) and auger(s).
- AUTO mode is the default. The conveyor automatically stops when proximity sensors detect asphalt height at the appropriate level for pausing the augers conveyors.
- Press and hold the button(s) again for manual operation.

Screed operators can also increase or decrease conveyor speed (and auger speed/pile height) using buttons on the screed control box. **(Figure 4-59)**

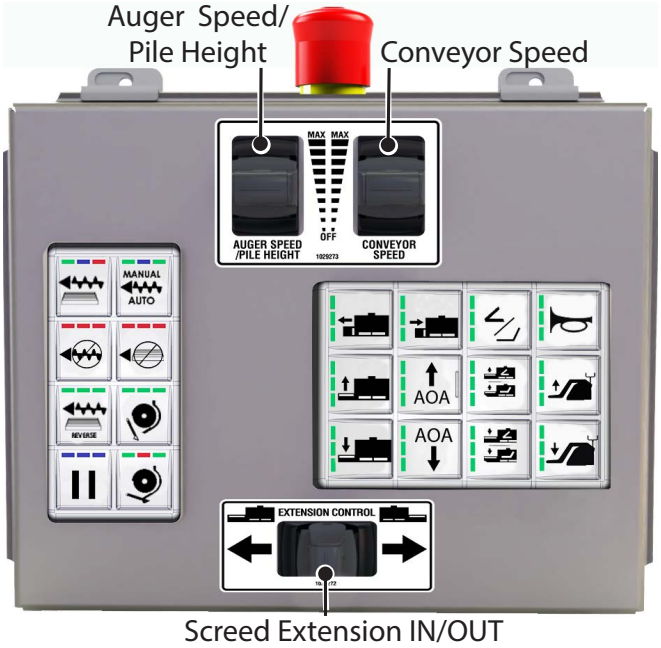
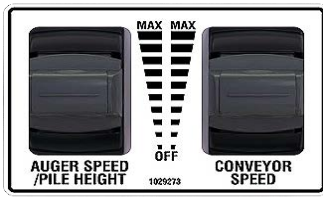
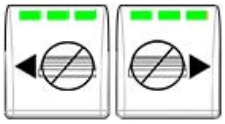


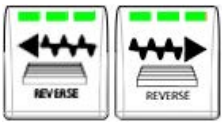
Figure 4-59. Screed Control Box (Left Box Shown)



- Press and hold the button in the desired increase/decrease direction to set conveyor speed. You can also fine tune the conveyor speed in the digital display. ***(See Motor Settings on Page 4-40)***



The screed operator can press the Conveyor Bypass button on the screed control box to pause conveyor operation while paving.



The screed operator can press the Auger/Conveyor Reverse button on the screed control box for reversing the direction of both functions to help clear material as needed.

NOTE: The conveyor will only reverse in the manual mode, and conveyor speed is reduced.

PRO TIP: Use the Pad Builder function on the main control panel to build a starter pad:

- Press the button once to engage the augers and conveyors. Both Under Auger Material Cut-Offs will open.
- Press the button again and the augers/conveyors will turn off.
- Press and hold button to manually engage both conveyors only.

Augers and Sonic Auger Sensor

Independent right and left augers can be operated in manual or AUTO (automatic) mode. The left and right augers are operated in conjunction with the conveyors from either control panel, and both can be operated in reverse.

Press one or both auger/ conveyor button on either the main control panel or the screed control box to turn on the conveyor(s) and auger(s).

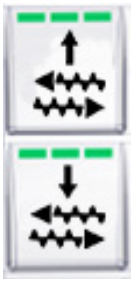
- The screed operator can also select manual or AUTO mode after the main operator has turned on the auger/ conveyor using the Auger Manual AUTO button on the screed control box. (AUTO operation is the default.)

AUTO Mode: The TopCon sonic auger sensor detects the amount of material present and controls its flow to the end gates, gauging the amount of paving material in the extensions.

The sensor sits in its mount on the end gate(s) and connects to its cable on the screed. **The sonic auger sensor system must be connected to operate in AUTO mode. (Figure 4-60)**

- When the material height moves into the sonic range, the system slows the auger speed and turns it off as needed. It will turn back on when the material level drops and continue monitoring pile height.

Figure 4-60. Sonic Auger Sensor



To raise the augers:

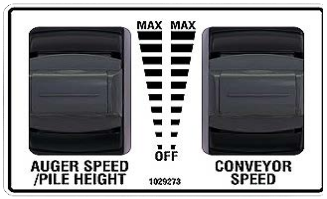
- Press and hold the auger lift button (⇧) to raise both the left and right auger.

To lower the augers:

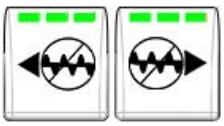
- Press and hold the auger lower button (⇩) to lower both the left and right auger.
- Releasing the button will hold the augers at that position.

NOTE: ALWAYS raise the augers when loading the paver onto transport vehicles.

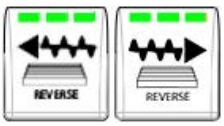
The screed operator can use the Auger Speed / Pile Height button during operation.



- When operating in AUTO mode, press and hold button in the desired increase/decrease direction to set the pile height at the end gate.
- When operating in MANUAL mode, press and hold button in the desired increase/decrease direction to set the auger speed.

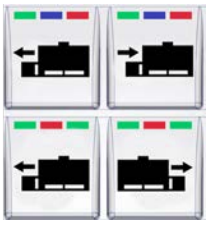


The screed operator can press the Auger Bypass button on the screed control box to pause auger operation while paving.



The screed operator can press the Auger/Conveyor Reverse button on the screed control box for reversing the direction of both functions to help clear material as needed.

NOTE: The auger/conveyor reverse function will only work when operating in the manual mode.



The screed operator can press and hold the Tunnel In or Out buttons to extend or retract the auger on the screed control boxes when paving wider lengths.

If operating in MANUAL mode:

- Press and hold the Tunnel Out button once to extend the tunnel. All LED lights illuminate green.
- Press and hold the Tunnel In button to retract the tunnel. All LED lights illuminate green.

If operating in AUTO mode:

- ***Simultaneously*** press and hold the extension control switch in the extend position and the Tunnel Out button to extend the tunnel.
- ***Simultaneously*** press and hold the extension control switch in the extend position and the Tunnel In button to retract the tunnel.
- To lock the tunnel in place, press and hold the Tunnel In and Tunnel Out buttons simultaneously until the LED light flashes green.
- To unlock, press and hold both buttons again.

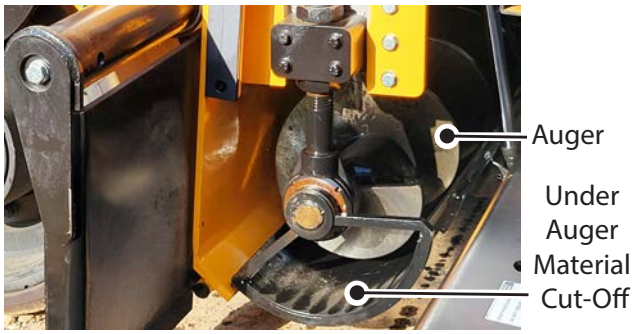


Figure 4-61. Under Auger Material Cut-Offs

Under Auger Material Cut-Offs

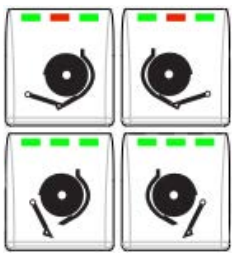
The asphalt material moves from the auger to the Under Auger Material Cut-Offs, which control asphalt flow to the screed and extensions. **(Figure 4-61)** Cut-offs open to increase asphalt flow and close to decrease flow when making narrow passes at the beginning and end of each pass or pull. Proper use of cut-offs prevents spillage and minimizes clean-up.

NOTE: To prevent excessive handwork, turn off the conveyors and close the cut-offs about three feet from the end of a pull to prevent excessive handwork.

⚠CAUTION NEVER back up with the cut-offs open to prevent machine damage.

The cut-offs can be configured to automatically close when moving paver in reverse by navigating to the Operator Settings screen on the DM430 digital controller. ***(Page 4-39)***

The Under Auger Material Cut-Offs can be operated via the main control panel and the corresponding active side on the screed control box keypad.



- Press and hold the left or right Close Cut-Off button to close.
- Press and hold the left or right Open Cut-Off button to open.

Heating the Screed, Extensions and End Gates

Heating the left and right main screed, extensions and end gates helps prevent hot mix from sticking to the cold screed plates to produce a smooth, tight mat surface. Heat the screed before paving and when the paver is idle long enough for the screed to cool between loads using the electric heat control box above the main screed.

Both the standard LB5 and optional LB7 screed heat systems are heated by a hydraulically-driven generator that powers electric heating elements. Blade heating elements in the main screed and extensions cycle during heat-up to save horsepower and fuel consumption while paving.

Both sides of the main screed elements heat when the paver is in neutral. When you start to pave, the right side heat turns off for 128 seconds while the left remains on, then turns back on, then the left sides turns off for 128 seconds. This “flip flop” system helps take some of the load off the generator for increased efficiency.

The standard LB5 Screed electric heat box features On/Off control switches for the main screed, screed extensions, heated end gates (if equipped with this option), and optional power crown control (if equipped with this option). **(Figure 4-62)** ***(See Pages 3-30 to 3-31)***

Figure 4-62. Standard LB5 Screed Electric Heat Control Box

The optional LB7 Screed electric heat box also features On/Off control switches for the main screed, screed extensions, heated end gates and [optional] power crown control. It is also equipped with a breaker switch to protect circuits from damage by overloads or short circuits. **(Figure 4-63)** ***(See Page 3-32)***

Figure 4-63. Optional LB7 Screed Electric Heat Control Box

NOTE: Lower the screed two inches off the ground while heating to prevent cold air and wind from affecting the heating process.

NOTE: Allow the screed components to heat 15 minutes before beginning to pave.

NOTE: You can set the screed directly onto a fresh mat of hot asphalt while running the heating system, allowing the heat of the asphalt to help heat the screed plate.

NOTE: Engine speed automatically increases to 1800 RPMs when the heat cycle is active.

NOTE: The engine speed automatically decreases to idle after the heat cycle completes.

NOTICE **Overheating the screed can warp the screed plate and possibly cause damage. If screed extensions lock, allow the machine to cool before attempting to extend or retract. Replace a warped screed plate immediately.**

Heat cycles for both screeds can be controlled by automatic time and temperature settings via the DM430 digital display on the main control panel or operated manually as described below.

To operate the screed heating system:

1. Start the paver and idle until the engine reaches normal operating temperature.
2. Raise the screed plate two inches off the ground.
3. Turn the desired components for heating on by moving the electric heat box switches up.

4. Press the Generator button on the control panel to activate the heating process.
5. Navigate to the Pave screen in the DM430 digital controller to set desired heating temperature and duration if operating in AUTO mode.

Once the heating function has been enabled, the electric heat box will apply electrical power to the heating elements and the heating cycle will begin. The heating cycle is timed to optimize the heat generated at the screed plates.

NOTE: The idle throttle speed ceases to function when the generator is on. When the generator turns off, the idle functionality returns.

NOTE: The generator is set at the factory to default at 1800 RPMs.

NOTE: If equipped with the power crown option, push and hold the power crown switch UP for increased crown or DOWN for increased valley.

LB5 and LB7 Screeds

The 8680 Commercial Paver features the new LB5 screed, or choose the optional LB7 screed for your paver. Both provide variable paving widths from 8 to 15.6 feet, and include berm/slope capability, horizontal depth screws, and spring-loaded end gates for matching pavement joints and seams.

Multiple adjustment mechanisms are located on top of the screed, extensions and end gate to ensure a quality mat. Electric dual vibrators mounted on the main screed frame increase paving material compaction. The optional LB7 screed includes electronic vertical height and angle-of-attack (AOA) adjustments, along with hydraulic crown/invert adjustment.

Screed operators can manage numerous functions using the screed control boxes on each side of the main screed. The screed operator can control augers/conveyors, under auger material cut-offs, tow points, and slope/grade/berm; increase/decrease Angle-of-Attack (AOA) and extension vertical; and extend/retract the screed extensions.

Screed Locks

This machine is provided with screed lock mechanisms as added safety and stability during transport. A locking lever is located on each side of the screed on the paver frame. **(Figure 4-64)**

- Push the locking lever UP to lock screed.
- Push the locking lever DOWN to unlock screed.

⚠CAUTION Ensure locks are in the unlocked position before extending or retracting the screed.

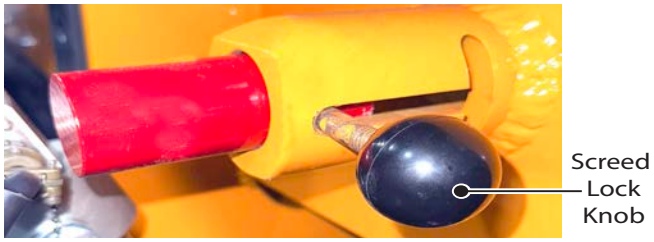


Figure 4-64. Screed Lock (Left Side Shown)

Use the procedures on the following pages for operating the screed and screed extensions:

- Always heat the screed extension(s) and end gate(s) described on the previous page before making any adjustments.
- Set the screed extension(s) and adjustment components when paving widths exceeding eight (8) feet. When adjusted correctly, pressure on the rear edge of the extended screed is the same as the rear edge of the main screed, resulting in a smooth mat the length of the screed.

NOTE: To set the asphalt depth, you can place small starter blocks under the main screed while you make adjustments. (Figure 4-65)

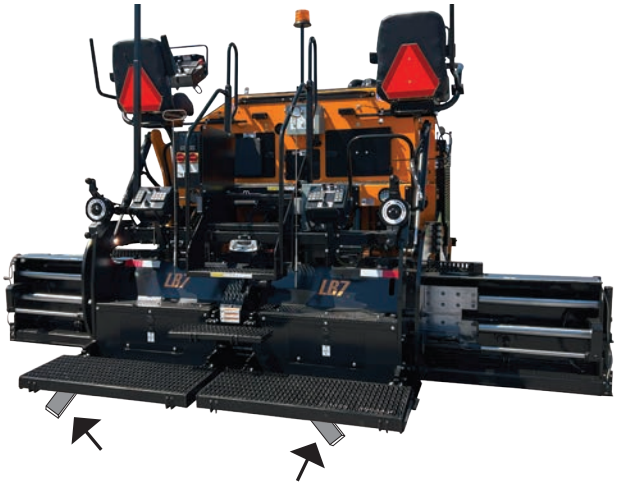


Figure 4-65. Starter Blocks Location

Use the end gate height adjusters to raise and lower the end gate height as needed. **(Figure 4-66)**

NOTE: Never allow the end gate to carry the weight of the screed to prevent uneven compaction.

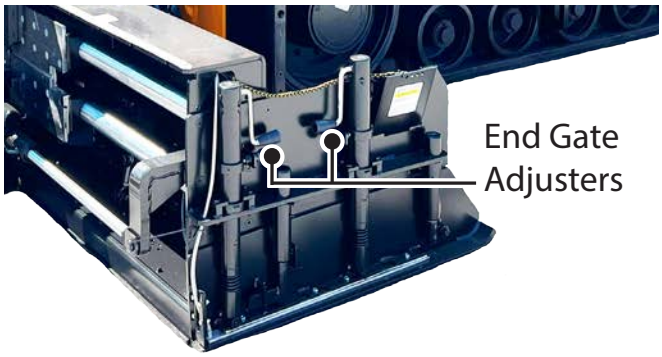


Figure 4-66. End Gate Height Adjusters

Raise/Lower Screed

Raising and lowering the screed can only be accomplished from the activated main control panel.

- Press and hold the corresponding button to raise (⇧) or lower (⇩) the screed.
- The screed lower (⇩) button also controls floating the screed:
  - Placing the joysticks into neutral (activated main control panel only) moves into screed hold, and float resumes when joysticks are returned to the forward position.
  - If the paver is moving forward, then stops, screed float is paused (the center LED light illuminates when paused. Screed float resumes when joysticks are returned to the forward position.

4

Extend/Retract Screed Extension

The screed operator can extend or retract screed extensions on either or both sides of the paver using the corresponding extend or retract buttons on the screed control box. Arrows on each symbol point the direction the screed mechanism moves.

- Press and hold the corresponding button to extend or retract the screed to desired width.
- Release button when desired width is achieved.

Adjust the screed while paving using the following procedures:

1. Move paver to the starting position for paving.
2. Extend the screed extensions to the desired width.
3. Level the screed using the tow points. ***(Page 4-59)***
   - Refer to the elevation gauge. Raise or lower until the rod end of the cable is flush with “0” on the gauge.
   - While paving, refer to both gauges and make minor adjustments if needed.
4. Null the screed on starting surface (or starter blocks) using flight screws. Make adjustments as needed while paving.
5. On the first pass, adjust the end gate so the front tilts down slightly when the screed is lifted. This will allow the end gate to set itself to the grade.
   - If the end gate digs into the asphalt at the front, adjust it until the end gate tilts back more.
6. On the first pass, leave approximately six to eight inches of unrolled asphalt where the joint will be made.
7. When making a joint, the end gate must be set to where it fits flush with the bottom of the screed.
   - If the joint looks too high or too low, adjust the flight screw on the screed one turn at a time and allow four to five feet of travel to correct itself.

NOTE: Too much adjustment up or down may cause uneven pavement.

NOTE: Keep runners clean. When making a joint, spray solvent on the runners.

Horizontal Depth Screws

The screed operator can also use the horizontal depth screws for making minor adjustments to mat thickness for the main screed as needed while paving. The depth screws also have a locking lever mechanism to lock the screw in place as needed. **(Figure 4-67)**

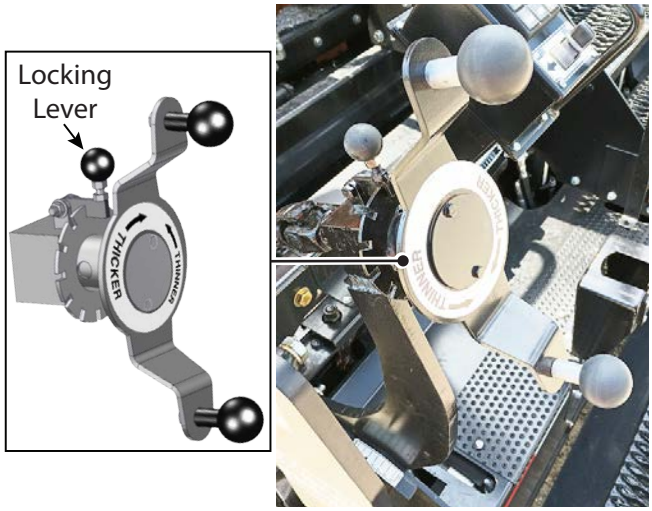


Figure 4-67. Horizontal Depth Screw

You must be operating tow point cylinders in the manual mode when using the depth screw

1. Before paving, refer to the screed elevation gauge. **(Figure 4-68)**
2. Raise or lower the screed until the cable rod end is flush with “0”on the indicator decal.
3. Slowly move both joysticks forward.
4. Use the Thicker/Thinner depth screw on either or both sides of the screed to adjust mat thickness as you pave. You can lock the depth screw when satisfied with mat depth. **(Figure 4-67)**
5. **C**heck the mat thickness regularly, making minor adjustments if needed.

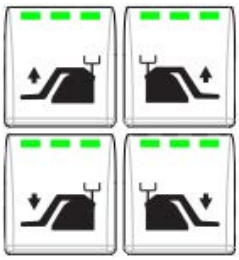
Hydraulic Tow Point Cylinders and Grade/Slope

The tow point cylinders are an added convenience to the operator for adjusting the depth of the screed. The same height between the tow point and the screed pivot point above grade creates a parallel line of pull, making it possible to lay down an asphalt mat of uniformly desired thickness. This also helps the screed maintain a stable floating condition.

The 8680 paver features hydraulic tow point cylinders with linkage that allows eight inches of travel for quick and easy height adjustments, and integrates with the sonic grade and slope automation system described on ***Pages 4-53 to 4-54.*** The tow points can be operated manually or in AUTO mode.

A screed elevation gauge is located on both sides of the paver that are connected to the hydraulic tow points. **(Figure 4-68)** This gauge provides the operator with a visual reference for the tow point's position.

Figure 4-68. Screed Elevation Gauge



The left and right hydraulic tow point cylinders can be operated in manual mode from the main control panel or the screed control box. The default setting is manual. Use the Tow Up (thicker asphalt mat) and Tow Down (thinner asphalt mat) buttons to control the tow cylinders in manual mode.

Operating in Manual mode:

1. Center the tow point by referring to the screed elevation gauge. Raise or lower the screed until the rod end is flush with the "0" on the indicator decal. **(Figure 4-68)**
2. While paving, make minor adjustments to the screed using the tow point adjustment (or flight screw described in the following section).

Operating in AUTO mode:

Pressing the Grade/Slope button changes to AUTO mode. The TopCon sonic automatic grade/slope control system (included with your paver) must be connected to operate in AUTO mode. ***(Page 4-53)***

- The Tow Up and Down buttons do not function in AUTO mode unless joysticks are in neutral.
- When operating in AUTO mode, the tow points are paused when paver is not traveling forward.

Slope/Berm

The sloping extensions provides the means of matching curb and gutter to reduce handwork while paving. The hydraulic slope extensions provide 0 - 8 percent slope. The entire extension can be angled down to 2.5 inches.

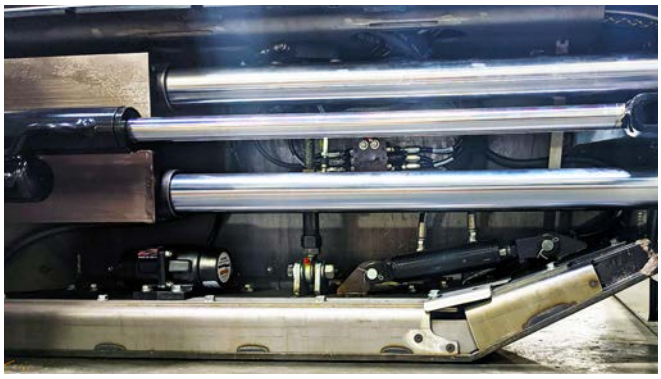
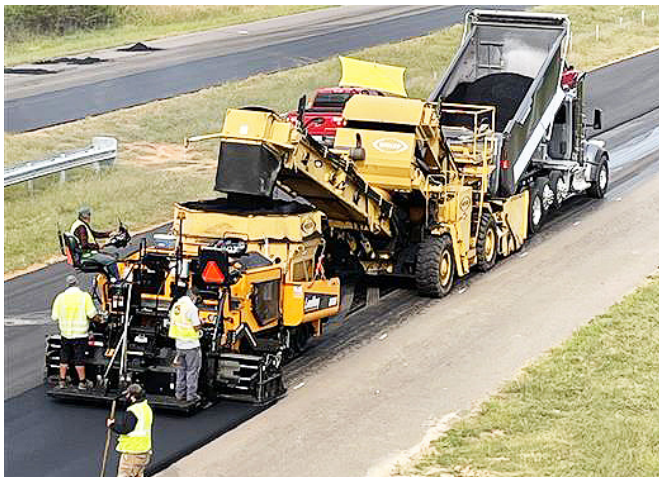


Figure 4-69. Screed Extension Berm Setting

The berm function allows you to adjust the last one foot of each extension to pave up to a six-inch roll-up curb. **(Figure 4-68)** You can raise the height to easily allow more material alongside buildings or other objects. The last foot (12 inches) of the extension can be angled up to six inches.

Use the slope/berm select button on the main control panel to choose either function, then use the slope/berm up and down buttons to increase or decrease either function.

- The default is Slope, press button for Berm. The left LED light illuminates when in slope mode and right LED light illuminates in berm mode.



STARTING TO PAVE

Inspect your paver and components to look for any issues that could affect operation or safety before work begins. Prior to startup, check all fluid levels. Start up the paver and then check all gauges.

This paver is capable of placing a mat thickness of six inches for:

- Bituminous base.
- Binder or surface course.
- Lime or Portland cement-stabilized sub-base.
- Graded aggregate materials.

Operators should be trained to operate the various machine components along with proper paving techniques before paving.

Keep these items in mind:

- Plan the project so you pave the most narrow passes first (the width of the paver), leaving the widest passes until last.
- Always pave in low gear.
- Always clean the paver after use. Remove dirt and debris from the rubber tracks and undercarriage as often as possible during the day as cleanliness directly impacts the rubber track wear rate.
- The two most important factors for laying a quality mat are all about consistency: consistent head of material in front of the screed and consistent paving speed. An increase in the head of material increases mat thickness and can translate into a wavy surface, and a decrease reduces the mat thickness and causes major flaws.

- Avoid excessive reverse operation and operating on the sides of hills/slopes as this wears tracks more quickly. Climb slopes straight up and down.
- Avoid abrupt and aggressive turns, especially at higher speeds, which puts unnecessary stress on the tracks and undercarriage. Continuous turning on the same side can also cause uneven wear on the rubber tracks.
- It is the operator's job to guide the material truck back to the hopper and signal the driver when to stop dumping asphalt into the hopper. The driver must maintain a light pressure on the brakes to prevent the truck from dumping asphalt onto the ground.
- Adjust the push rollers to the appropriate height needed for the asphalt truck. **(Figure 4-70)**
    - There are five (5) set heights on the truck push rollers. A scraper behind each one scrapes asphalt from the push rollers as you pave to help prevent asphalt build-up.
    - Remove bolts on both sides of each push roller, align with either of the five slots on the frame at desired height, and reinstall bolts.
    - Both push rollers should be at the same height to prevent damage.

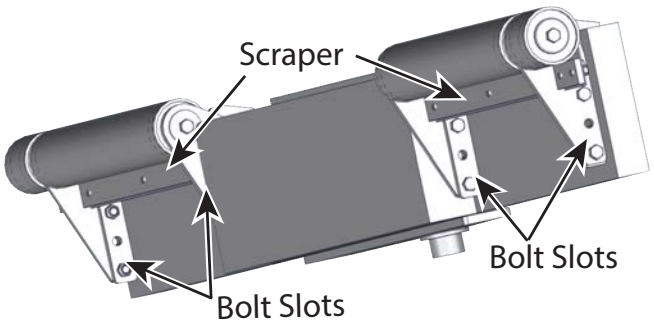
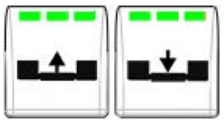


Figure 4-70. Adjustable Push Rollers

Press and hold the truck hitch IN and OUT buttons to move the truck hitch in or out as shown by arrows on the keypad symbol.

NOTE: These buttons operate the extendable [push roller if equipped with that option. ***(Page 4-67)***



Press and hold the apron UP and DOWN buttons to move the front apron up or down for additional hopper fit to the asphalt truck and reduce spillage.

- Screed operators can use the vibrator function to help with material compaction as needed while paving.

Press the button once to turn on the vibrator. Press button again to turn it off.

- It is very important that the first pass be straight as it will serve as your guideline for the following passes. Use a reference guide such as a curb, gutter, adjacent mat or string line. You can also use the screed guide at the center of the tracks on either side to further assist you in paving a straight line.
- Maintaining a precise rolling pattern is especially crucial for a quality job. Make the most number of passes while the mix is still HOT. An efficient rolling pattern that covers as much of the paved area in as short amount of time as possible is generally the best approach to achieving a uniform mat.
- If there is a line from the edge of the main screed, the screed operator needs to adjust the vertical height.
    - Lines on the inside of the main screed means the vertical is too low.

NOTE: Lines on the outside of the main screed means the vertical is too high

Automatically adjust vertical height by pressing and holding the vertical UP and DOWN buttons on the screed control box.

Basic Paving

1. Spray down the machine. ***(Pages 4-48 to 4-49)***
2. Heat screed and end gates. ***(Pages 4-55 to 5-56)***
3. Position paver at the starting point for the mat.
4. Open the Under Auger Material Cut-Offs (if equipped with this option. ***(Page 4-54)***

⚠CAUTION Never back up with cut-offs open.

5. Adjust screed as needed. ***(Pages 4-56 to 4-57)***
6. Adjust push rollers to the appropriate height needed for the asphalt truck. **(Figure 4-61)**
    - Or use extendable push roller controls if equipped with this option. ***(Page 4-67)***
7. Open hopper wings to accept asphalt from the material truck. ***(Pages 4-50 to 4-51)***

NOTE: Allow only a partial load of asphalt to enter the hopper when first starting to pave.

8. Turn the augers and conveyors on. It is recommended to operate both components in AUTO mode (default). ***(Pages 4-51 to 4-54)***
    - Ensure the sonic auger sensor is connected for AUTO operation.
    - When the material starts moving into the sonic range, the sonic auger will automatically regulate auger operation.

NOTE: Bypass auger operation when paving a basic eight-foot pull as auguring is not necessary at this width.

9. When the material begins to discharge from under the screed, press the Screed Down button to float the screed.
10. Begin paving, moving slowly at first, adjusting the screed if needed.
    - Use the hydraulic tow points, grade/slope functions, and/or depth screws as needed while paving. ***(Page 4-59)***
    - The procedures for setting the crown and making AOA (Angle of Attack) adjustments for asphalt mat texture are described on the following pages.
11. Make a paving pass until the asphalt level in the hopper is low.
12. Return paver back to starting position to begin the next pull.

Setting the Crown

The screed can also be set to control the pavement valley, or crown. A pavement crown is the elevation of the middle in relation to the edges, and insures proper drainage of the pavement.

NOTE: Maximum crown is two inches.

To set the crown:

1. Loosen the lock-down bolts in slotted bars before adjusting crown and valley mechanism. **(Figure 4-71)**

Figure 4-71. Loosen Lock-Down Bolts

2. Use the crown handle adjuster for setting positive or negative crown:
   - For increased positive crown push down on adjuster.
   - For increased negative crown pull up on adjuster.
   - To put leading edge crown in the screed, you must make the adjustment with the crown handle. Once leading edge is set where you want, remove the four locking bolts, then set the trailing edge flat and reinstall the locking bolts.
3. Use the gauge at the center of the screed above the screed platform as a visual reference.
   - If the needle is above the zero, you will have positive crown.
   - If the needle is below the zero, you will have negative crown.
4. To get exact crown or valley, measure the distance between a flat level surface to the center bottom portion of screed. Make adjustments with crown and valley control.
   - Positive crown is when the middle of the mat is raised to permit water to drain to each side. **(Figure 4-72)**
   - Negative crown is the lower in the center of the screed plate, often used in an alley where drainage is necessary.

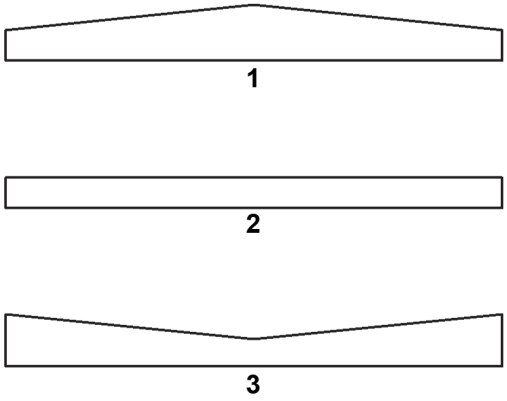


Figure 4-72. Crown Settings

1 - Positive (+)

2 - Zero (0)

3 - Negative (-)

NOTE: Crown may be placed in the leading edge and/or the trailing edge of the screed plate. Crown in the leading edge aids material flow under the screed plate only. Trailing edge crown puts a crown in the mat.

Example: The trailing edge crown is 0 and the leading edge crown is 1/8 inch. There will not be any crown in the mat, but the material flow under the screed plate will be improved.

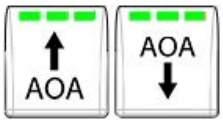
Mat Texture Adjustment

The Angle-of-Attack (AOA) is the angle between the bottom of the screed plate and the grade surface. Adjusting the AOA produces a smooth or coarse mat texture.

- The screed extension should be hot before making any adjustments.
- Spray the bottom of the screed with cleaning solvent BEFORE heating the screed.

⚠CAUTION NEVER spray cleaning solvent or release agent on or near a hot screed heating or near open flame. Cleaning solvent and release agents are flammable and could cause serious personal injury.

NOTE: Do not allow release agent chemicals to run onto the ground.



The screed operator can make AOA adjustments using the AOA Up and Down buttons on the screed control box while paving.

1. Spray the heat screed extension with release agent or solvent if needed.
2. Press and hold the AOA Up and Down buttons to increase or decrease the AOA:
   - Pressing the AOA Up button increases the pressure on the back of the extension, which gives a smoother, slicker finish.
   - Pressing the AOA Down button decreases the pressure on the back of the extension, resulting in a coarser finish.

NOTE: Too much pressure on the back of the screed extension will take the weight off the screed wear plate and cause poor material compaction, resulting in a poor finish.

UNLOADING AND LOADING

Ensure the transport trailer has the capacity to carry the paver's weight. Park the trailer in a clear, level area when loading or unloading the machine.

⚠CAUTION ALWAYS position an assistant to be a ground guide when loading and unloading the paver.

⚠CAUTION Work slowly and carefully to avoid accidents. Keep the loading area clear.

⚠CAUTION ALWAYS raise the augers when loading the paver onto transport vehicles.

⚠WARNING ALWAYS back the paver up loading ramp with the screed end first. ALWAYS be alert to uneven surfaces and the potential reactions when transferring the paver from surface to surface. Serious injury or even death can result.

⚠WARNING Use extra caution steering when loading and unloading the paver as slight steering changes on ramps and trailers can cause the machine to fall off the ramp or transport trailer. Serious injury or even death can result.

⚠CAUTION To prevent damage to the undercarriage and throughout the paver, reduce traveling speed before the paver tracks come into contact with loading ramps or any abrupt change in the surface. The track drive sprocket or other components can be damaged by excessive shock.

⚠CAUTION DO NOT allow the screed to strike the ramp as this can damage bearings on the thickness control screws or welds on the leveling arms. A longer ramp to reduce the loading angle may be needed.

NOTICE Never move the paver in reverse while the cut-offs are open.

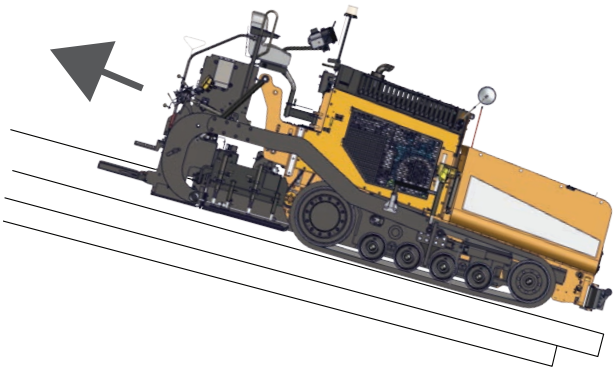


Figure 4-73. Correct Unloading and Loading Position

Unloading

1. Remove tie-down equipment.
2. Start and warm up engine.
3. Set throttle to Pave Low (1800) RPMs.
4. Set steering control lever so paver moves very slowly.
5. BEFORE moving the machine, ensure:
   - The screed position is UP.
   - Auger extensions are removed.
   - The screed extensions are IN.
   - The Under Auger Material Cut-Offs below the augers are CLOSED.
6. Move paver forward down the ramp.

Loading

⚠CAUTION **The paver must be loaded screed-end first to prevent damage.**

1. Back up paver to base of the ramp, aligning tracks with the ramp. **(Figure 4-73)**
2. BEFORE moving machine up the ramp, ensure:
   - The screed position is UP.
   - Auger extensions are removed.
   - The screed extensions are IN.
   - The Under Auger Material Cut-Offs are CLOSED.
3. Load the paver screed-end first. Set throttle to Pave Low (1800) RPMs.
4. Set steering control lever so paver moves very slowly onto the ramp.
5. Using the steering control joysticks Or steering knob if equipped), SLOWLY guide paver up the ramp.
6. Position paver in center of trailer.
7. Lower the screed.
8. Shut down the engine.
9. Attach tie-down chains to the front weight block and D-Rings in front of the torque hubs. .
10. Place chocks at tracks.
11. Ensure that all chains are tightly secured before transporting the paver.

TOWING

⚠WARNING **Ensure the paver is on level ground and blocked to prevent machine from moving while preparing for towing. Be especially careful after disconnecting the torque hub as the paver could roll.**

While it is strongly recommended that the paver be transported rather than towed, there are situations where you may need to tow the paver off the road until a transport vehicle arrives. **DO NOT tow paver long distances.**

You must manually disconnect the brakes using a hydraulic pump to prepare the machine for towing. The brake release mechanisms are located inside the right engine access door. **(Figure 4-74)**

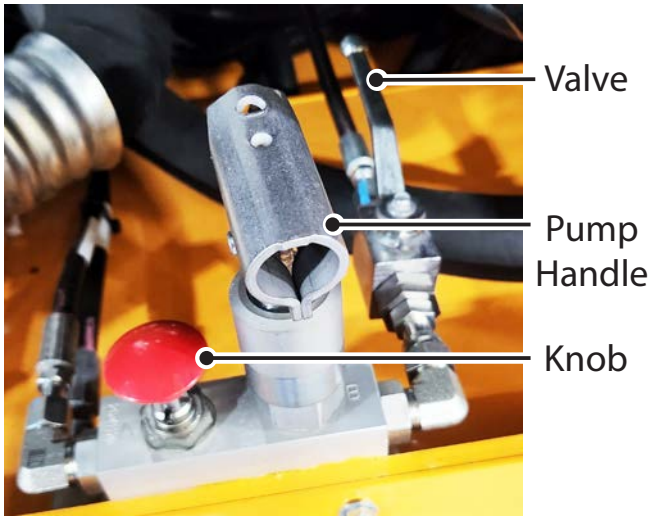


Figure 4-74. Brake Release Pump Valve

To release the brake for towing:

1. Push the red knob into the lower position.
2. Pull the valve handle 1/4 turn counterclockwise.
3. Using the pump handle, pump the brake release until tight. Brakes are now released for towing.
4. When the paver has been towed to its destination, open the valve by turning 1/4 turn clockwise to re-engage brakes.

OPTIONAL COMPONENTS

LB7 Screed

The monumental LB7 screed is packed with easy-to-use, highway class features for screed operators. **(Figure 4-75)** This versatile screed with front mounted extensions can pave up to 15.6 feet. A 34kW generator provides power to the S-curve heating elements on the screed extensions. With four independently user-adjustable heat zones, the operator is given precise and automatic heat to the zone as needed.

The extensions are equipped with electronic, on-demand vertical height and angle-of-attack (AOA) adjustments with the press of a button. This screed is equipped with spring-loaded end gates, giving the operator the downward pressure needed when matching joints or against curbs and gutters.

Additional options for the LB7 screed include:

- 20-foot Paving Kit (also available for the LB5 Screed)
- 18x16-inch Berm Extensions
- 12x6-inch Berm Extensions

Figure 4-75. Optional LB7 Screed

Umbrella Option

The umbrella kit option provides shelter from sun and other weather conditions for the operator. The umbrella kit assembly includes the mounting bracket and umbrella frame assembly that can be mounted on either side of the screed handle.

Screed Extension Vibration

Screed vibration helps compact material. If equipped with this option, use the Vibrators button on the main control panel to turn vibration on and off.

Screed Central-Point Lube System

The screed central-point lube system combines all screed lubrication points into one. When you pump grease into the central point, valves distribute the grease evenly to all the screed grease points.

Single Point Conveyor Lube Kit

The Conveyor Lube kit combines the 10 conveyor bearing grease points and the two front wheel spindle bearings. This kit combines these 12 individual grease points into one lubrication point.

Steel Poly Tracks

Steel poly tracks are available in lieu of the rubber tracks. Choose this option at purchase of the 8680 paver.

Steering Knob

The steering knob option is used for steering and replaces the right joystick on the control panel. **(Figure 4-76)**

Figure 4-76. Steering Knob Option

Steering Wheel

A steering wheel option is also available in lieu of joystick steering. **(Figure 4-77)** The Run/Stop switch on the steering panel controls running and stopping the machine, and applies parking brake when set to the STOP position. The joystick controls forward and reverse travel along with speed.

Figure 4-77. Steering Wheel Option

- To drive the paver forward, push joystick forward to reach the desired speed.
- To move in reverse, pull joystick(s) and move backwards to reach desired speed. A back-up alarm will automatically turn on when in reverse.
- To slow the paver, move joysticks closer to the neutral position.
- Return joystick to the neutral position to stop paver.

NOTE: If the joystick steering function is not being used, the Run/Stop switch must be in the center OFF position.

NOTE: The engine will not start unless the E-STOP button is disengaged and the joystick must be in the NEUTRAL (center) position.

20-Foot Paving Kit

The 20-foot paving kit includes all components for enabling your paver to lay asphalt mats up to 20-feet wide. This option is available for both the LB5 and LB7 screeds, and must be selected when ordering the 8680 Commercial Paver.

Berm Extensions

Berm extensions extend the screed's functionality to forming a sloped asphalt berm or shoulder, which can be crucial for drainage and meeting project specifications. Available only with the LB7 screed option, the berm extensions are available at 18" x 6" or 12" x 6" dimensions.

High-Lift End Gate

The high-lift end gate option is a specialized feature that helps create a defined, high-quality pavement edge, even when the screed is extended to its maximum width or encountering obstructions. The high-lift end gate can be extended to ride on top of the curb, ensuring a smooth and consistent transition between the paved surface and the curb. They are especially useful when working on projects with different paving widths, winding roads, or situations where the end gate needs to be extended.

Extendable Push Roller

The extendable, oscillating push roller option offers even more flexibility for maintaining contact with an asphalt material truck. **(Figure 4-78)** Extending the rollers various lengths as needed allows better contact for material trucks of all sizes, decreasing the load on the paver so it's easier to push the truck. Use the same procedures in the previous section for operation.

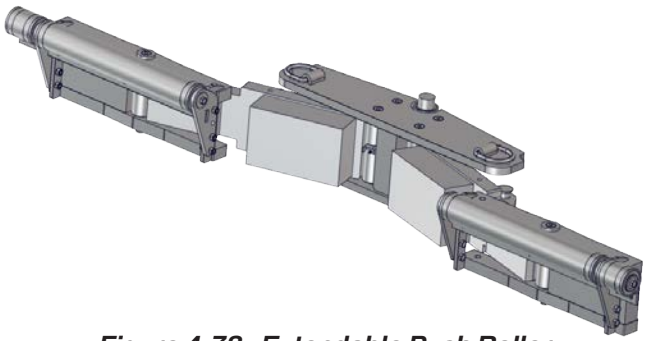


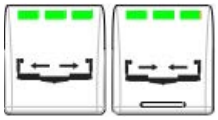
Figure 4-78. Extendable Push Roller

Truck Hitch

The truck hitch is an optional attachment that prevents excessive and uneven braking from the asphalt material truck. **(Figure 4-79)**

Figure 4-79. Truck Hitch

To engage the hitch with the rear wheels of the asphalt truck:



1. Use the Truck Hitch button on the main control panel to move truck hitch into the desired position.
2. Extend the arm extensions of the truck hitch by pushing and holding the Truck Hitch button in the OUT position.
3. The paving material truck then SLOWLY backs up to the paver until the hitch makes contact with the rear tires of the truck.
4. Retract the arm extension by pushing and holding the Truck Hitch button in the IN position until both guide rollers are fully locked onto the truck wheel rims.

NOTE: It may be necessary to adjust the roller guides to the inside of the wheel rims.

Topcon® Sonic and Dual Grade Control

Topcon® sonic grade control, dual grade control and slope options offer paving professionals screed control for improving smoothness along with assuring accurate slope and material thickness. Sonic trackers continuously measure the elevation over the surface, automatically averaging and adjusting the tow point cylinder on the paver. These controllers send signals to the screed to precisely maintain mat thickness and slope as you pave. Most states require this technology for state road paving jobs to ensure smoothness and continuity on roadways.

Choose from one of three options: Sonic Grade Control (one side); Dual Grade Control or Dual Grade Control and Slope. The dual grade controls automatically maintain a set depth on both ends of the screed. The Dual Grade and Slope controls maintain a set depth on one end of the screed and a set slope on the other end.

The paver must be pre-wired for TopCon technology. If equipped with this option, be sure to unplug and store the system after use. Always recap the cables to protect the connectors when not in use. **(Figure 4-80)**

Figure 4-80. TopCon Sonic Grade Control Box

To initiate the TopCon system:

1. Start the engine and idle a few minutes to warm hydraulic oil throughout the machine.
2. Connect the plug into the connector the main control panel column. (The plug will only fit into its correct connector.)
3. Push the power button to turn the system ON.
4. Turn the Mode Select Knob clockwise (or counterclockwise depending on last setting used) until the number 14 appears in the Mode Select display window.
5. Press and hold the Set button for three seconds and release.
6. Press the Set button again.
7. Turn the Mode Select Knob until EXT (external) appears in the Mode Select display window.
8. Press Set button again, then press Menu button.
9. Turn the Mode Select Knob until the number 50 appears in the Mode Select display window.
10. Repeat Steps 5 and 6 to set.
11. Turn the Mode Select Knob until the number 16 appears in the Mode Select display window. If the tow arm cylinder does not begin to move, continue turning the knob until it does. **DO NOT raise or lower the screed.**

NOTE: This is the valve offset setting, which varies from paver to paver. Make a note of this number on the Notes page provided in this manual for future reference.

12. Repeat Steps 5 and 6 to set.
13. Turn the Mode Select Knob until the number 14 appears in the Mode Select display window. If the tow arm cylinder does not begin to move, continue turning the knob until it does.

NOTE: This is the elevation setting, which varies from paver to paver. Make a note of this number on the Notes page provided in this manual for future reference.

NOTICE **Setting this value too high will result in overshooting the grade.**

14. Repeat Steps 5 and 6 to set.
15. Turn the Mode Select Knob until the number 15 appears in the Mode Select display window.
16. Press the Set button. You are now ready to pave.

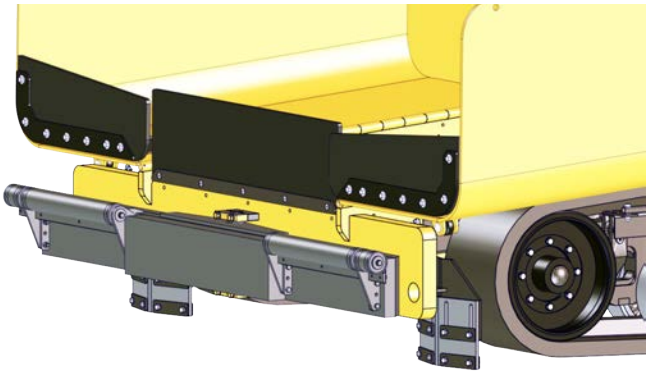


Figure 4-81. Track Plows

Track Plows

Track plows mounted on the frame in front of the tracks shield tracks from asphalt, dirt and debris. **(Figure 4-81)** See your authorized LeeBoy dealer for purchase.

Power Crown Option

The Power Crown option adjusts the positive or negative crown as needed. The power crown switch is located on the junction box. **(Figure 4-82)** (It is included with the LB7 screed option.)

- Push the switch UP for valley.
- Push the switch DOWN for crown.

Power Crown Switch

Figure 4-82. Power Crown Switch

Slope Meter

The slope meter is attached to the screed (beside the crown adjuster lever). The slope meter is used for slope and crown road construction, and continuously indicates the slope of the ground, or "percent of grade." **(Figure 4-83)**

Figure 4-83. Slope Meter

The slope meter is calibrated to read from .00 to .12 (0 - 12 percent) and works similar to a level tool. Just like a level tool, it contains a ball bearing suspended in oil that gravity always keeps at the lowest point of the meter, indicating the slope as you pave. Fluid in the vial is heavy and helps keep the indicator ball steady even with vibration, yet it still reacts quick enough to keep you accurate. While the slope meter is sturdy and functions regardless of weather conditions, take care not to break the glass tube. Keep the slope meter clean for optimum viewing at a glance.

If you selected the slope meter option when your ordered your paver, it is already mounted onto the screed at our manufacturing facility. Otherwise, you will need to mount the slope meter onto the screed.

NOTE: The slope meter is a precision instrument and must be installed correctly for accurate readings. Install the slope meter onto the metal block (if equipped), or a solid, straight, flat area on the main screed frame that will not twist or strain during operation. It cannot be emphasized enough that the paver needs to be on level ground for correct installation and operation.

To install the slope meter:

1. Park the paver on a level surface. Use a level tool to determine that the paver is level before installing.
2. The slope meter comes bolted to an aluminum mounting plate (two bolts). Remove the bolts and the slope meter from its mounting plate.
   - If your paver is equipped with a metal block welded onto the screed frame, discard the mounting plate as you will be attaching the slope meter directly to the metal block surface.

NOTE: Pay attention to the raised rims around the bolt holes. The slope meter must be remounted on the same side to prevent damage during installation.

3. Align top edge with the frame edge. Use a level to ensure it is perfectly level.
   - If using the aluminum plate, align it onto the mounting surface carefully, with the top edge of the plate aligned with the frame edge.
   - If installing onto the metal block, use a level tool to ensure
4. Reinstall the two bolts securely, ensuring the raised rims are in their original position.

NOTE: If using the aluminum plate, the lock washers provided MUST be used or the plate can shake loose.

Figure 4-84. Balloon Light

Balloon Light

The bslloon light is mounted on the operator platform to illuminate the work area when paving at night. **(Figure 4-84)** This LED light provides strong illumination in all directions around the paver. The light can also be adjusted for height, then easily retracted for storage. See your authorized dealer for purchase

Magnetic Cup Holder

Magnetic cup holders can be purchased for use with your machine (and the rest of your fleet). **(Figure 4-85)**

See your authorized dealer for details. Ask for **Part Number 1020521.**

Figure 4-85. Magnetic Cup Holder

NOTES

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MAINTENANCE

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NOTES

MAINTENANCE SCHEDULE

Before performing any maintenance procedures on the LeeBoy Model 8680 Commercial Paver, review the safety information in **Section 1**. Refer to the Engine Operator's Manual accompanying your paver for more detailed engine service information.

⚠WARNING Always use the appropriate and correct sized tools for the task at hand to prevent damage or possible injury.

⚠CAUTION When performing any routine maintenance, always include the previous routine maintenance hours in the higher hourly schedule.

Table 5-1. Periodic Maintenance Schedule

SYSTEM	ITEM	Every 10 Hours Daily	Every 50 Hours Weekly	Every 100 Hours Monthly	Every 250 Hours Quarterly	Every 500 Hours Semi-Annually	Every 1000 Hours Annually
Paver	Lubricate paver.	X					
Engine	Check oil level.	X					
	Check battery for corrosion and clean if needed.	X					
	Check coolant level.	X					
	Drain and replace coolant.					X	
	Drain fuel water separator if needed.	X					
	Clean fuel water separator.					X	
	Inspect/clean air filter.				X		
	Change air filters.						X
	Replace fuel filter.				X		
	Change oil and oil filter. **(Initial 50 Hours)**	X			X		
	Inspect/adjust engine belt.				X		
	Replace engine belt.					X	
	Check DEF tank fluid level. Fill if needed.	X					
	Check/adjust generator belt.			X			
Hydraulic	Check hydraulic oil level.	X					
	Change hydraulic oil and filter. **(Initial 50 Hours)**	X			X		X
	Check torque hub oil. Refill if needed.		X				
	Change torque hub oil.					X	
	Drain hydraulic tank. Replace strainer.						X
Mechanical	Check/adjust conveyor drive chains.			X			
	Check/adjust conveyor flight chains.			X			
	Check/adjust auger chains.			X			
	Check/adjust screed extension top guide.			X			
	Contact dealer to check valve clearance.						X
Every 1500 Hours	Replace oil separator element. Contact your authorized dealer to check engine injector tip, PCV (Positive Crankcase Ventilation) valve, and EGR cooler.						
Every 3000 Hours	Contact your authorized dealer to check EGR system and turbo-charger.						

5

ACCESSING COMPONENTS

Raise and Lower Engine Hood

Accessing the radiator and other engine components is simple and convenient on the 8680 commercial paver. The engine hood pivots forward for convenient access to the radiator and other engine service points. **(Figure 5-1)** Access doors on each side of the paver also provide easy access for servicing various components.

To raise the hood:

1. Push the center of the hood latches (5) across the **(Figure 5-1)** to unlatch. (One side of the latch mechanism pops up.)

⚠CAUTION Ensure all hood latches (5) are unlatched before raising the hood.

Figure 5-1. Engine Hood Latches

2. Turn the ignition switch to the RUN position. **(Figure 5-2)**
3. Use the Hood Lift switch on the junction box to raise and lower the hood:
   - Push the Hood Lift switch UP and hold to raise the hood.
   - Push the Hood Lift switch DOWN to close the hood.
4. After lowering the hood completely, push the latches back into place to secure the hood.

⚠CAUTION Ensure all latches are securely latched before operating.

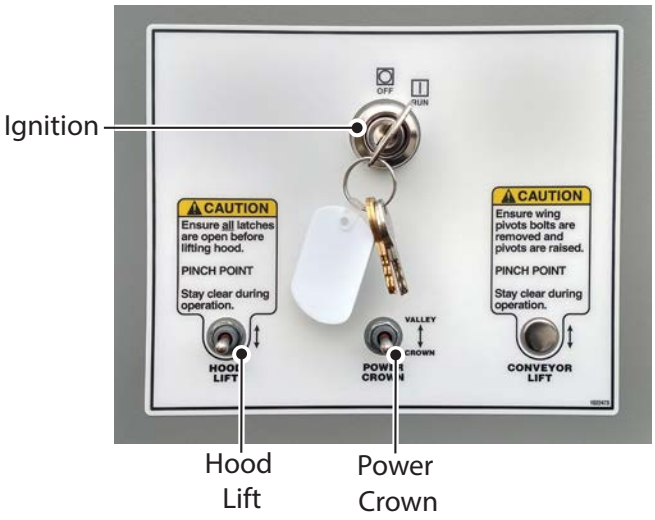


Figure 5-2. Junction Box Controls

NOTE: The conveyor lift switch is nonfunctional on this model.

ROUTINE MAINTENANCE

Following the maintenance schedules and procedures will maintain the paver in top operating condition and provide years of trouble-free operation. Refer to the Engine Operator's Manual accompanying your paver for more detailed engine service information.

NOTICE **Changing oil and cleaning the paver should only be done in a designated area where the oil and chemicals can be contained. Discard these by-products in accordance with environmental regulations.**

When performing any routine maintenance, always include the previous routine maintenance hours in the higher hourly schedule.

Prepare Paver for Maintenance

⚠WARNING **If any maintenance must be performed while the engine is running, DO NOT leave paver unattended.**

1. Park the paver on a flat surface.
2. Lower all attachments to ground level.
3. Place joysticks in neutral.
4. Run engine at 1/2 speed (RPMs) without load for three to five minutes, then reduce to slow idle.
5. Place ignition switch in the OFF position.

NOTICE **If asphalt material is allowed to remain on the paver overnight, possible damage and increased maintenance costs can result upon start-up the next day. Spray down the paver daily (or more often if needed).**

Every 10 Hours or Daily Routine Maintenance

1. Clean off paving material and spray cleaning solvent or release agent on the wear plate, extensions, and any other area that has been in contact with the paving material. Cleaning should be performed while the paver is hot.

⚠WARNING **NEVER spray cleaning solvent on hot surfaces as serious injury can result.**

2. Lubricate the required parts. ***(Pages 5-9 to 5-13)***
3. Check battery for corrosion and clean if needed. Check cables and connections. ***(Page 5-26)***
4. Check coolant level. Fill if needed. ***(Page 5-22)***

⚠WARNING **Allow the radiator to cool off completely (30 minutes after engine operation) before checking coolant level. NEVER remove the radiator cap when hot to prevent serious burns.**

5. Check DEF tank fluid level. Fill if needed. ***(Page 5-20)***
6. Check for loose, worn or damaged hoses, tubes clamps. Tighten, repair or replace promptly.
7. Check fuel level. Keep the fuel tank full to prevent condensation. Always fill at the end of the day. ***(Page 5-23)***

⚠WARNING **ALWAYS stop the engine when refilling fuel and be careful not to spill to prevent a fire. DO NOT smoke while refueling. Only add fuel in well-ventilated area or outdoors. DO NOT overfill.**

NOTICE **Use only ultra low sulfur diesel fuel.**

8. Check engine oil level. ***(Page 5-17)***
9. Check hydraulic oil level. ***(Page 5-24)***
10. Drain fuel water separator if needed. ***(Page 5-24)***
11. Perform any other engine preventive maintenance described in the Engine Operator's Manual.

NOTE: In cold weather, keep conveyor flight chains properly oiled with cleaning solvent or release agent. This will prevent conveyor bars from sticking. Neglect could result in conveyor bar damage or drive chain failure.

Every 50 Hours or Weekly Routine Maintenance

1. The FIRST 50 HOURS, change engine oil and filters, then every 250 hours thereafter. ***(Page 5-17)***
1. The FIRST 50 HOURS, change hydraulic oil and filters, then every 250 hours thereafter. ***(Page 5-25)***
2. Check torque hub oil level. Add more oil if needed. ***(Page 5-13)***
3. Lubricate the flight screws, extension cylinders, end gate adjusters and crown/valley mechanism. Grease nuts on extension screws. ***(Pages 5-10 to 5-12)***
4. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 100 Hours or Monthly Maintenance

1. Check generator belt. Adjust if needed. ***(Page 5-27)***
2. Check and adjust conveyor drive chains. ***(Page 5-15)***
3. Check and adjust conveyor flight chains. ***(Page 5-14)***
4. Check auger drive chains, lubricate and adjust.
5. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 250 Hours or Quarterly Routine Maintenance

1. Inspect/clean air filter element. ***(Page 5-18)***
2. Inspect/adjust engine belt. ***(Page 5-19)***
3. Change engine oil and filter. ***(Page 5-17)***
4. Change hydraulic oil and filter. ***(Page 5-25)***
5. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 500 Hours or Semi-Annual Routine Maintenance

1. Drain radiator and replace coolant. ***(Page 5-22)***
2. Change torque hub oil. ***(Page 5-13)***
3. Replace engine belt. ***(Page 5-18)***
4. Replace air filter. ***(Page 5-19)***
5. Clean fuel water separator. ***(Page 5-24)***
6. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 1000 Hours or Annual Routine Maintenance

1. Drain hydraulic tank and refill with fresh hydraulic oil. Change strainer. ***(Page 5-25)***
2. Replace air filter. ***(Page 5-18)***
3. Contact your dealer to check valve clearance.
4. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 1500 Hours

1. Contact your authorized dealer to check and service (if needed), the engine injector tip, PCV (Positive Crankcase Ventilation) valve, and EGR cooler.
2. Replace oil separator element. ***(Page 5-19)***
3. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 3000 Hours

1. Contact your authorized dealer to check and service (if needed) the EGR system and turbo-charger.
2. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

LUBRICATION

Proper lubrication is extremely important and necessary to maintain the paver and its components at top efficiency. Lubrication reduces wear on moving parts, seals and protects components, absorbs shock, minimizes corrosion and reduces operating temperatures.

NOTICE **Lubrication service should only be done in a designated area where the oil and chemicals are contained. These by-products should be cleaned up and discarded in accordance with environmental regulations.**

Lubrication Points Chart

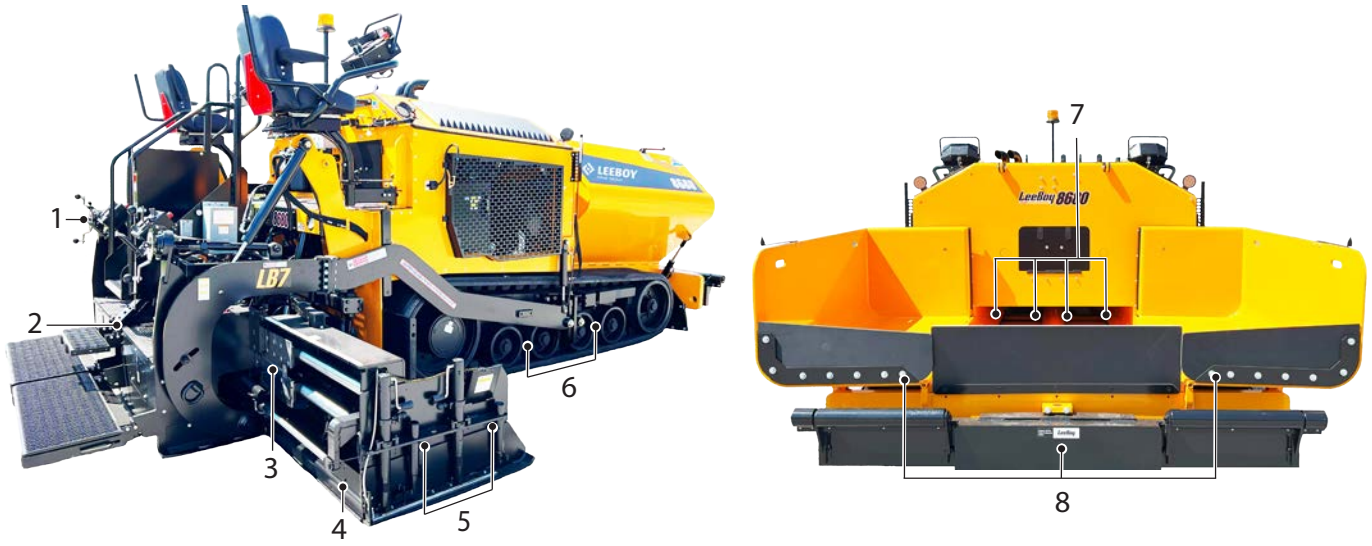


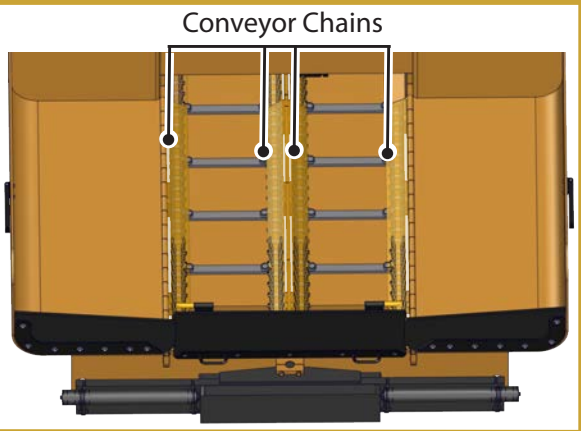
Table 5-1. Lubrication Points Chart

ITEM NO.	TYPE LUBE	DESCRIPTION AND LOCATION	INTERVAL
	A	Grease With Shell Avania EP Grease 2 or Equivalent	
	B	Spray with Release or Chain Lube Agent	
	C	Spray with Copper-Based Anti-Seize Agent	
1	A	Flight screw (both sides). Grease first in locked position, unlock, and turn 180° to grease. Also lube U-Joints.	Weekly
2	A	Crown and Valley.	Weekly
3	C	Screed extension slide plates (both sides).	Weekly
4	A	All screws and bearings on screed extensions (both sides).	Weekly
5	A	End gate adjusters (both sides).	Weekly
6	A	Track pivots (both sides).	Weekly
7	B	Conveyor chains (one on each side of each conveyor).	Daily
8	B	Spray all parts of paver that come into contact with asphalt.	Daily

Daily Lubrication

Spray these components with release agent daily:

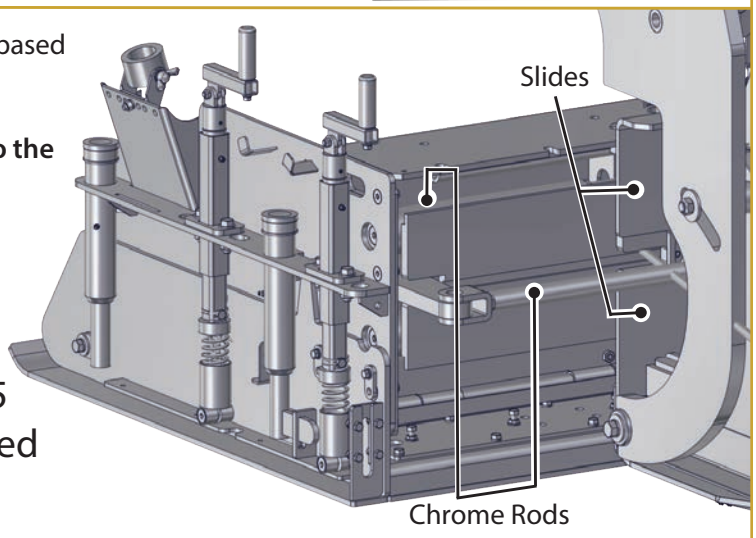
- Conveyor chains (one on each side of each conveyor).
- Any part of the paver that comes into contact with asphalt.

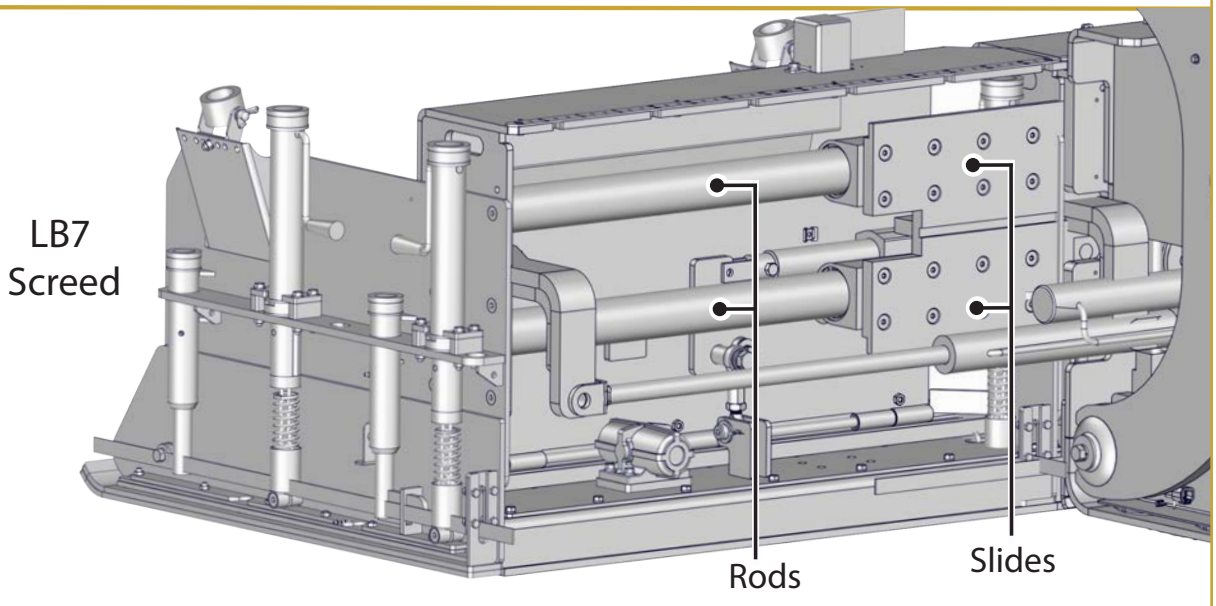


Spray both screed extension slides with copper-based anti-seize agent ONLY (LB5 and LB7 screeds).

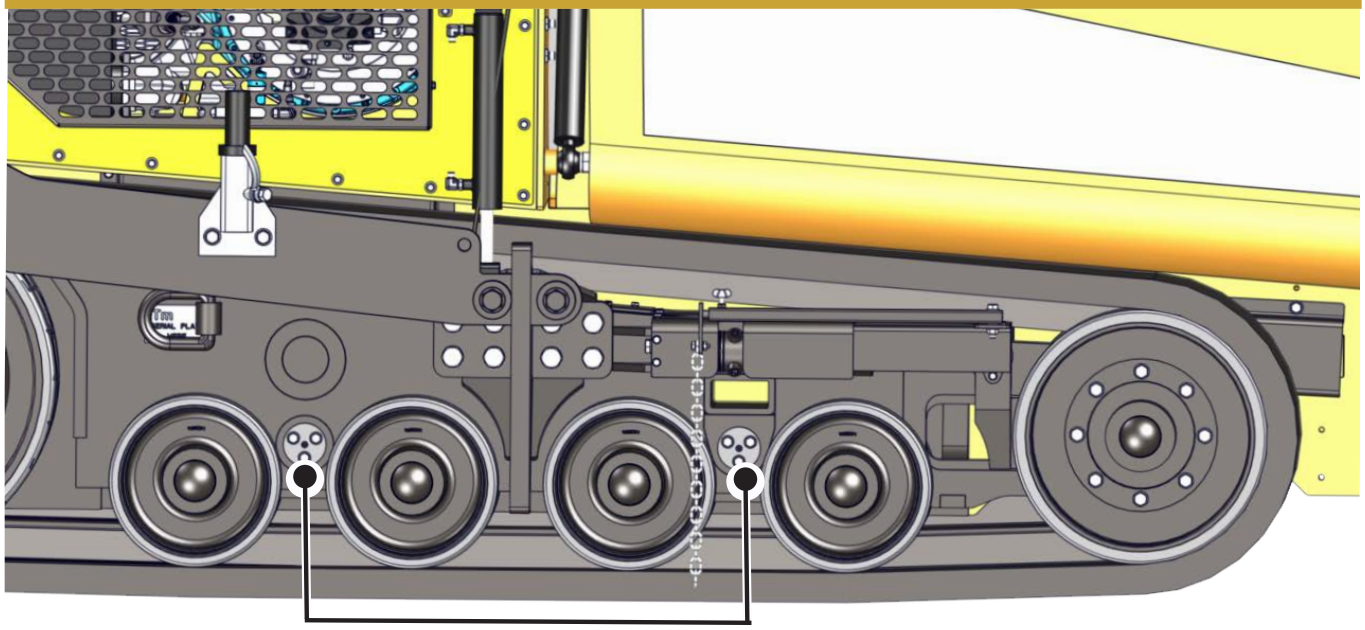
DO NOT spray release or chain lube agent onto the screed extension rods and slides.

NOTE: Keep the chrome rods clean, but DO NOT lubricate.





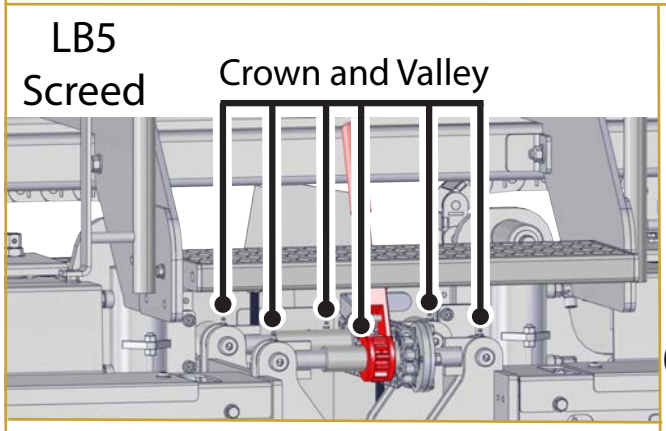
Daily Lubrication

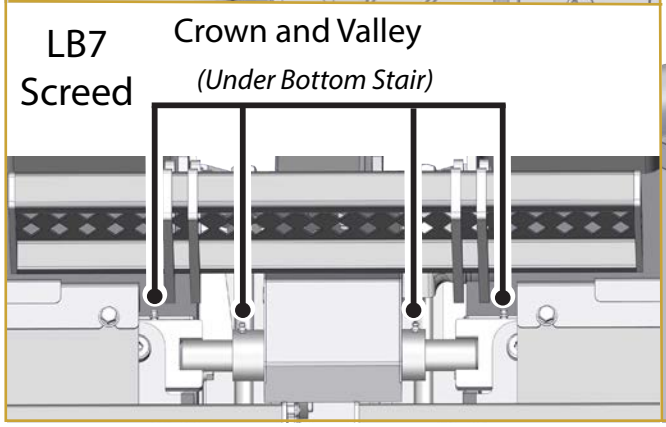


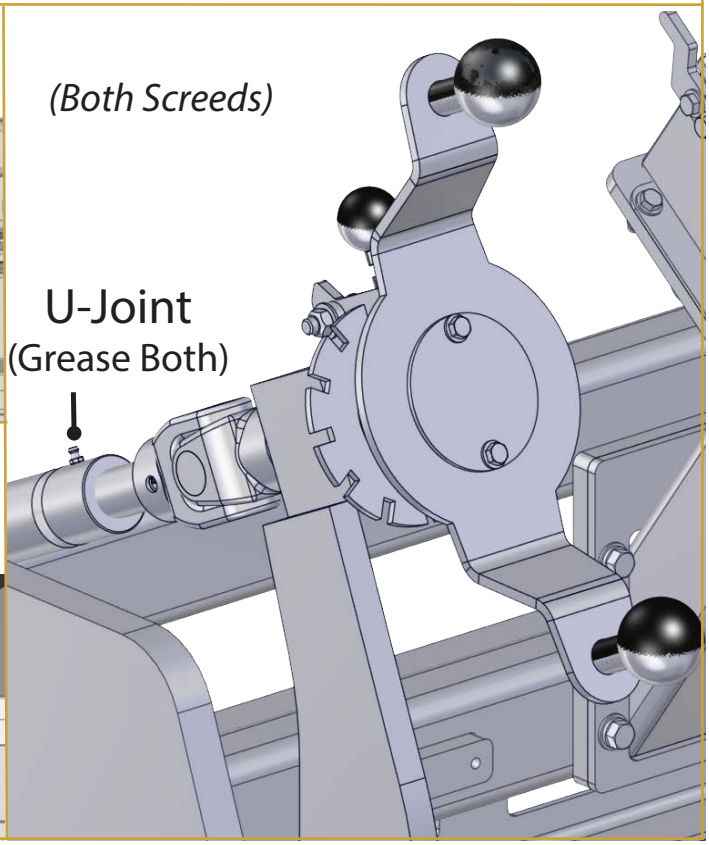
Track Pivots

Weekly Lubrication

Lubricate the screed components with Shell Avania EP Grease 2 or equivalent weekly.

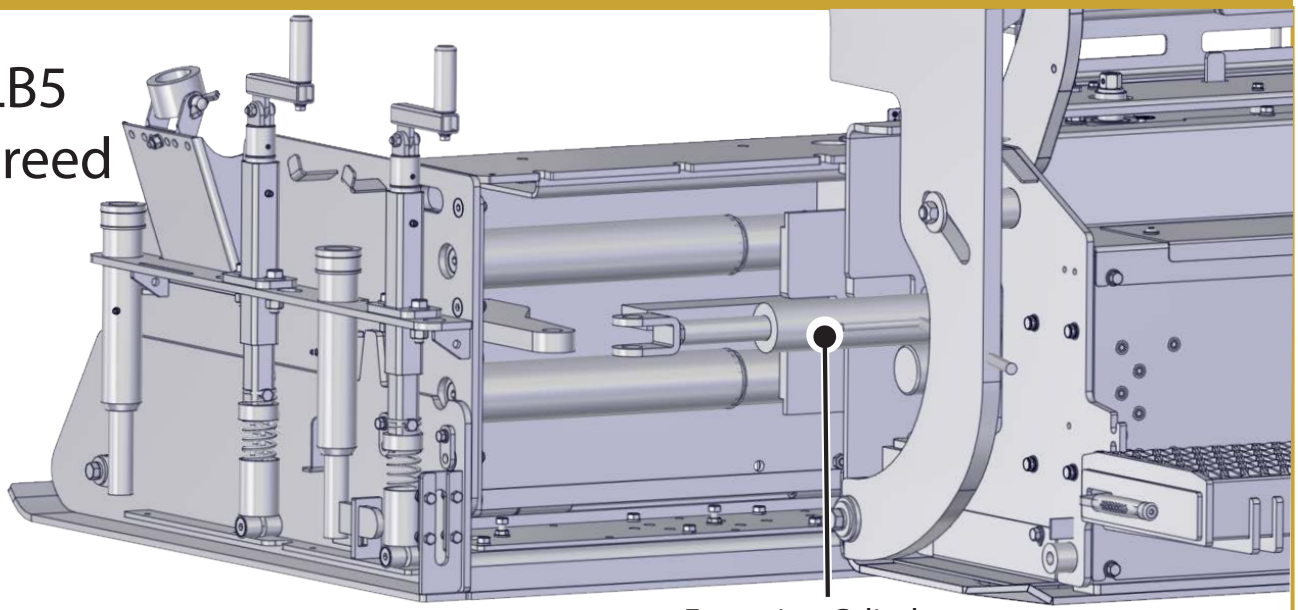






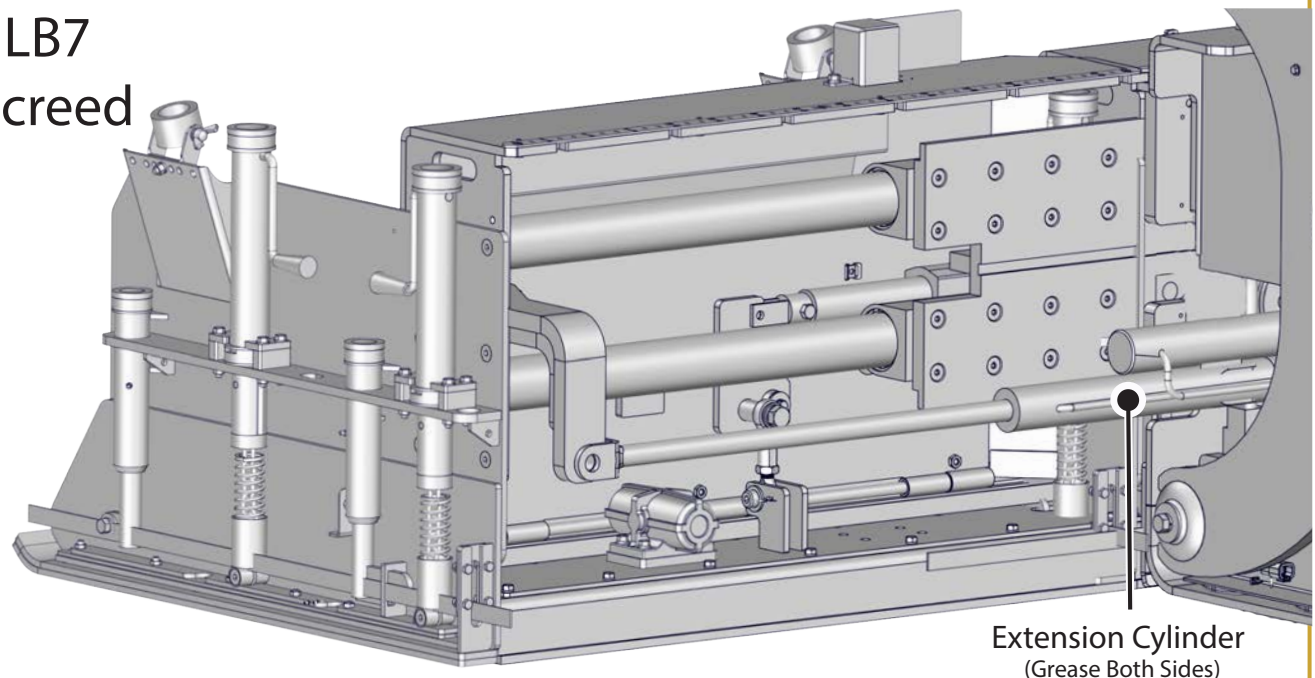
Weekly Lubrication

LB5 Screed



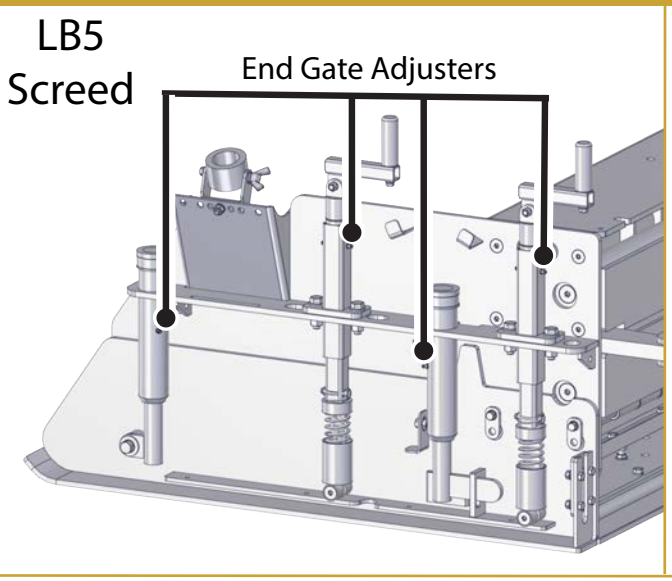
Extension Cylinder
(Grease Both Sides)

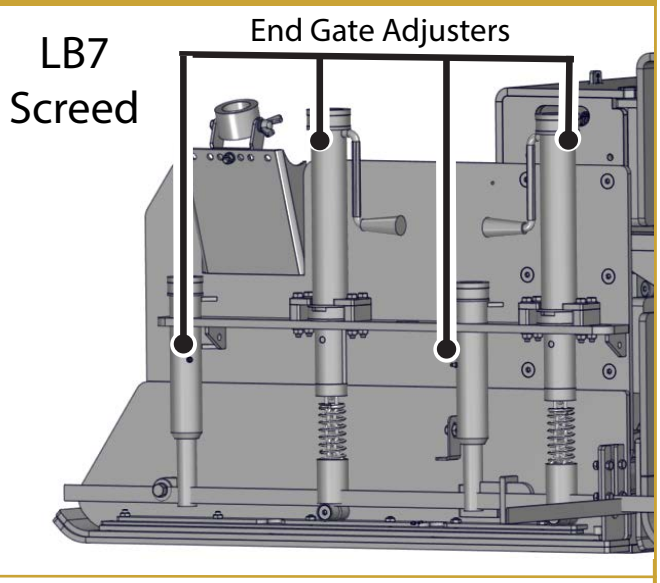
LB7 Screed



Extension Cylinder
(Grease Both Sides)

Weekly Lubrication





Torque Hub Lubrication

Check the torque hub oil level at the rear of the tracks weekly to ensure optimum operation. To check the oil level:

1. Move paver so that one of the hub plugs is in the 12 o'clock position and the other is in the 3 or 9 o'clock position as shown in **Figure 5-3.**
2. Remove the plug in the 3 (or 9) o'clock position.
3. If oil comes out, no oil is needed. Reinsert the plug and tighten.
4. If oil does not come out, remove the top plug and fill with SAE 75 W 90 synthetic gear oil until oil starts to appear at the plug.
5. Replace plug and repeat process on the other hub.

5

MAINTENANCE ADJUSTMENTS

Certain components of the paver need adjustments for optimum performance, operation and general maintenance. Keep chains clean, lubricated and properly adjusted.

⚠WARNING **Entanglement hazard! DO NOT run the engine while checking and servicing conveyor components. NEVER perform any adjustments while the engine is running.**

Conveyor Flight Chains

The conveyor flight chains must be adjusted every 100 hours. If irregular movement of the conveyor occurs, an adjustment is needed regardless of the time interval. ***(See Page 5-?? for slack tension adjustment.)***

⚠CAUTION **DO NOT adjust the conveyor flight chains too tight! Clearance (slack) is required for material to pass under and between chain rollers and the front idler sprocket. There must be *mild* tension--just enough to take up slack and prevent the chains from "knuckling."**

This chain operates most efficiently with a slight amount of slack on the return side. The chains should not:

- Be pulled too tight.
- Hang too loose.

The chains should be slightly slack with constant tension. A chain(s) "popping and cracking" indicates the chain is climbing the sprocket, which can be caused by**:**

- Excessive run-out of the front sprockets:
    - Check to verify sprocket run-out pin on the shaft. Run-out should not exceed .030 inches.
    - If excessive, replace chain or sprocket.
- Idler bearing housing brackets are not parallel to each other and the conveyor bed:
    - Check the idler bracket. If it appears twisted or misaligned, slide shaft 1-1/2 inches through the bearing inside diameter. If the centers are too far off (more than 1/4 inch), this is why the chain is climbing the sprocket.
    - If excessive, replace chain or sprocket.

Use the following procedure to make this adjustment on both sides of the conveyor:

1. Move front of paver onto a ramp so you can get underneath the front of the paver.
2. Loosen the four (4) bolts on each conveyor adjustment plate. **(Figure 5-3)**

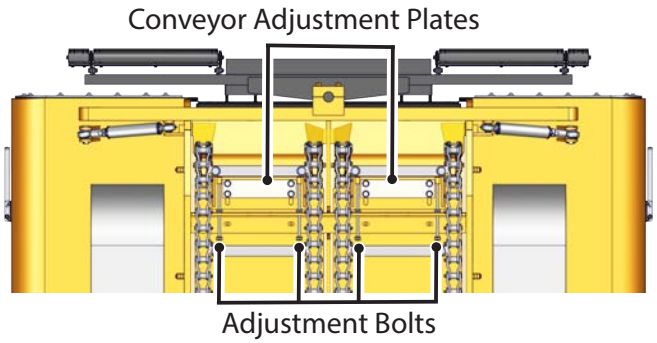


Figure 5-3. Conveyor Flight Chain Adjustment Bolts

3. Loosen the jam nuts in front of the adjustment bolts.
4. Turn each adjustment bolt alternately the same amount. For example, turn one bolt one half turn, then the other bolt one half turn, etc.
5. Continue alternating tightening until the flight chains are tight. (The pressure on the chain will be noticeable as the bolts are tightened.)
6. After the conveyor chain tension is set, retighten locknuts and bolts holding the flight chain assembly.
7. Repeat steps 1 through 4 for the opposite side.

Conveyor Drive Chains

Maintaining proper conveyor drive chain alignment is essential for conveyor performance and maximizing chain and sprocket life. Inspect the drive chain tension on each side of the paver regularly. There should only be 1/4-inch of slack on each chain. If you notice any slippage on either conveyor, the drive chain(s) may be loose. If the chains are loose, adjustment is necessary. ***(See Page 5-51 for slack tension adjustment.)***

Use the following procedure to adjust the conveyor drive chains:

1. Shut off engine.
2. Open the engine access door and locate the conveyor drive chain adjustment bolts just inside (toward rear of paver). **(Figure 5-4)**
3. Loosen the jam nuts on the chain adjuster.

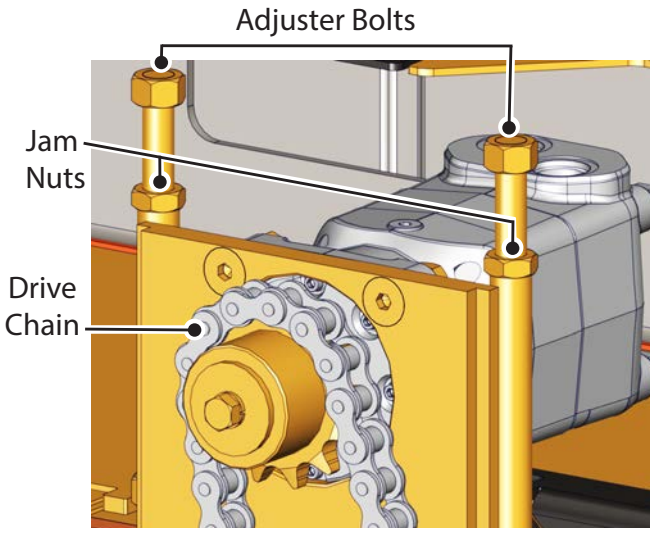


Figure 5-4. Conveyor Drive Chain Bolts

4. Turn the chain adjuster counterclockwise to loosen the chain, or clockwise to tighten the chain, leaving approximately 1/4-inch of slack. DO NOT overtighten.
5. Retighten jam nuts.
6. Repeat steps on the other side if needed.

Auger Drive Chain

The auger drive chains must also be properly aligned for optimum performance. The chains should be snug, not loose. ***(See Page 5-51 for slack tension adjustment.)*** If the chains need to be tightened, follow this procedure:

1. Loosen the lower adjuster jam nut on the right and left auger chains at the base of each auger motor. **(Figure 5-5)**
2. Turn the tension adjusters bolts counterclockwise to tighten chains.
3. Once adjusted, retighten jam nuts.

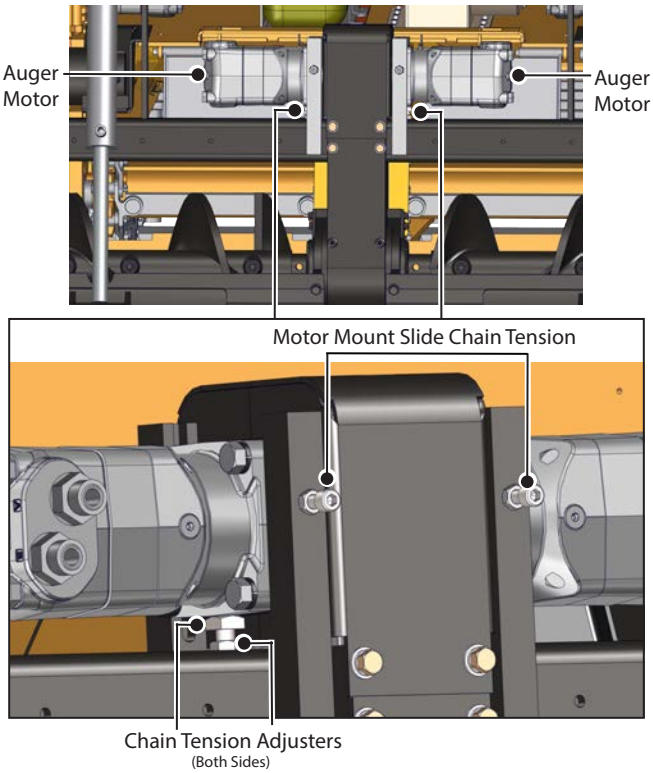


Figure 5-5. Auger Drive Chain Bolts

5

ENGINE MAINTENANCE

The LeeBoy 8680 Commercial Paver features the latest in emissions compliance and electronically-controlled engine technology. **(Figure 5-6)** Advanced emission-control devices and low-sulfur fuel requirements--only Ultra-Low Sulfur Diesel (ULSD) can be used in this engine--combine with after-treatment methods such as Exhaust Gas Recirculation (EGR), Diesel Particulate Filters (DPF), and Diesel Oxidation Catalyst (DOC). This is described in more detail on the following pages.

A common rail system optimizes combustion for a durable, quiet and fuel-efficient, turbo-charged engine. Preventive maintenance on the engine (and other paver components) is important for longevity and avoiding costly, unnecessary repairs.

Even a dirty air filter element can allow debris and dirt into the engine that can cause major problems and costly repairs in time, so be sure to follow the maintenance schedule on ***Page 5-5.*** Change oil and filters according to the maintenance schedule, keep fluids full, adjust and replace fan belt (when needed), and inspect/repair hoses to reduce wear and tear on the engine.

You can access the engine and other components via the access panels on each side of the paver or by raising the hood. ***(Page 5-6)*** The maintenance procedures on the following pages are most often required for the engine. Refer to the Engine Operator's Manual included with your paver for additional information and procedures.

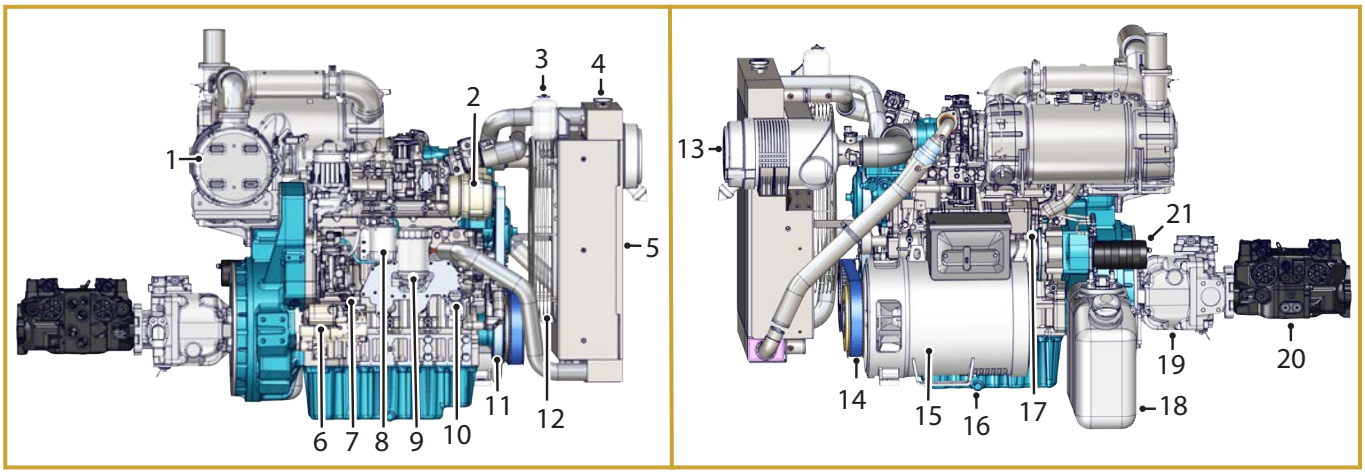


Figure 5-6. Engine and Related Components

1 - SCR (Selective Catalytic Reduction)

2 - Alternator

3 - Coolant Overflow Filler Cap

4 - Radiator Cap

5 - Radiator

6 - Starter

7 - Engine Oil Dipstick

8 - Fuel Filter

9 - Engine Oil Filter

10 - Engine Oil Filler Cap

11 - Engine Belt

12 - Cooling Fan and Safety Shield

13 - Air Filter

14 - Generator Belt

15 - Generator

16 - Engine Oil Drain Plug

17 - Auxiliary Gear Pump

18 - DEF (Diesel Exhaust Fluid) Tank/Cap

19 - Load Sense Pump

20 - Tandem Drive Pump

21 - Tandem Gear Pump

Check Engine Lubrication Oil

The engine lubrication oil must be kept at a level above the ADD mark, but not above the FULL mark, on the engine oil dipstick. **(Figure 5-7)**

⚠WARNING DO NOT check the oil while the engine is running. Hot oil can spew and cause serious personal injury.

Engine oil should be API classification CJ-4. Change the type of oil according to the ambient temperature:

Table 5-1. Ambient Temperature Oil Chart

Temperature	Engine Oil
Above 77°F (25°C)	SAE 10W-30 SAE 15W-40 SAE 30
14°F to 77°F (-10°C to 25°C)	SAE 10W-30 SAE 15W-40
Below 14°F (-10°C)	SAE 10W-30

To check the engine lubrication oil level:

1. Park the machine on a level surface and turn off the engine.
2. Wait five minutes to allow the oil to return to the oil pan so you can obtain an accurate measurement of the engine oil level.
3. Open the front hopper access panel to reach the dipstick.
4. Clean the area around the engine oil dipstick before removing it from the engine. **(Figure 5-8)**
5. Remove the dipstick from the engine and checking the oil level. The oil level should be above the ADD mark--but not above the FULL mark--on the dipstick.
6. Add oil if needed.

NOTICE DO NOT overfill. Oil can drain through the air intake system if overfilled and cause oil leaks from breather piping.

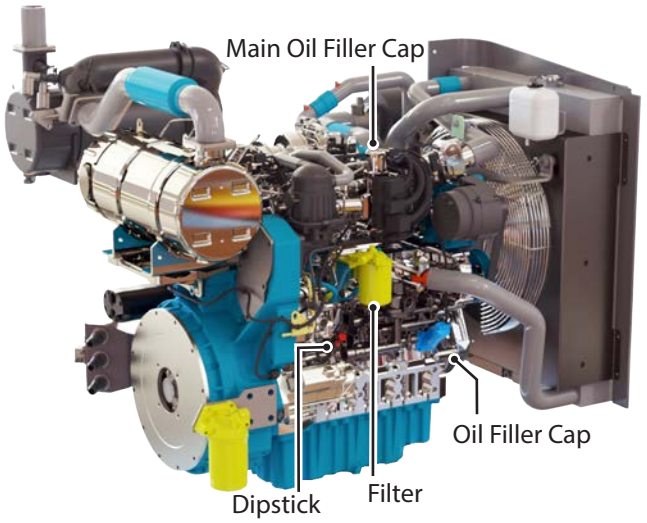


Figure 5-7. Engine Oil Dipstick

Change Engine Oil and Filter

The engine lubrication oil and oil filter must be changed at the same time. **(Figure 5-7)**

NOTE: The engine oil and filter cartridge should be changed after the initial 50 hours of operation and every 250 hour thereafter.

⚠WARNING Change the engine lubrication oil while the engine is WARM, not hot. Hot oil can cause serious personal injury.

NOTICE DO NOT change the engine lubrication oil filter while the engine is running. Serious engine damage will occur.

To change the engine oil and filter:

1. Park the paver on a level surface and turn off the engine. Allow the engine to cool down sufficiently as oil can be hot enough to cause burns.
2. Locate oil drain hose on right side of the paver.
3. Place a container large enough to hold at least five (5) gallons of oil underneath the drain hose.
4. Remove cap on end of the hose and drain all engine oil into the container.
5. Reinstall cap back onto the drain hose and reinsert hose back into place.

6. Wipe the area around the oil filter and its mounting base with a clean cloth.
7. Place the empty container under the oil filter.

NOTE: Before replacing the oil filter cartridge, loosen the cartridge bolt on top of the filter to allow air into the cartridge for complete drainage and less spillage. (Figure 5-8)

8. Use a filter removal wrench to remove the filter, turning counterclockwise.
9. Wipe inside the oil filter head using a clean, lint-free cloth.
10. Apply a very thin layer of oil on the rubber gasket area of the new filter element.
11. Install the new filter element onto the filter head by hand. When the gasket contacts the seal surface, fully tighten the filter (by hand only). Do not overtighten!
12. Fill the engine with five (5) gallons of oil at the fill cap.
13. Check oil level using the oil dipstick.
14. Dispose of used oil and oil filter (including rubber gasket) in accordance with local and federal environmental laws.

Change Air Filter

NOTICE **Never operate the engine without an air filter properly installed. Severe damage to the machine will occur.**

The air filter should be cleaned or changed every 250 hours of operation. With extensive use, the air filter may need to be changed more frequently. **(Figure 5-8)**

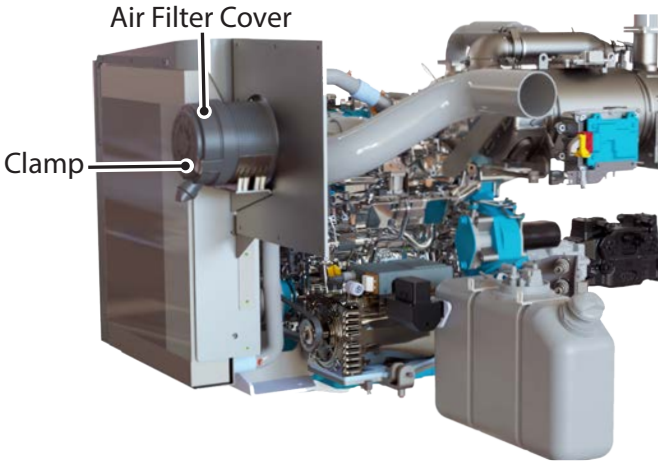


Figure 5-8. Air Filter Cover

To change the air filter elements:

1. Raise the engine hood. ***(Page 5-6)***
2. Unclamp and remove air filter cover.
3. Remove the primary and secondary air filter elements. Discard.
4. Clean the inside of the air cleaner body with a clean cloth.
5. Install new air filter elements inside the canister.
6. Reclamp cap onto the air filter cover.
7. Check clamps to ensure filter elements are tight and filter elements are making proper contact.
8. Start engine. If engine does not run smoothly, recheck filter element contact.

NOTE: The air filter elements should be replaced every 250 service hours of normal paver operation, but more often if the paver is operated in severe conditions.

Engine Drive Belt

This engine is equipped with an auto-tensioner. **(Figure 5-9)** Inspect the engine drive belt every 250 service hours and replace it every 500 hours. Regardless of the maintenance interval, the engine fan belt should be replaced if it appears frayed or cracked.

NOTE: If the fan belt is loose or damaged, and/or the fan is damaged, it can result in overheating or insufficient charging. Replace the fan belt and any other damaged components immediately.

⚠WARNING Engine must be turned off and the key removed before checking, adjusting or replacing the fan belt to avoid personal injury or even death.

To replace the belt:

1. Turn off engine and remove the key.
2. Raise the engine hood. ***(Page 5-6)***
3. Remove the safety shield from the engine cooling fan between the engine and radiator.
4. Using a 17mm socket and a ratched on the auto-tensioner pulley bolt, push counterclockwise to take tension off the belt to remove it.
5. Install new belt.
6. Reinstall cooling fan safety shield.

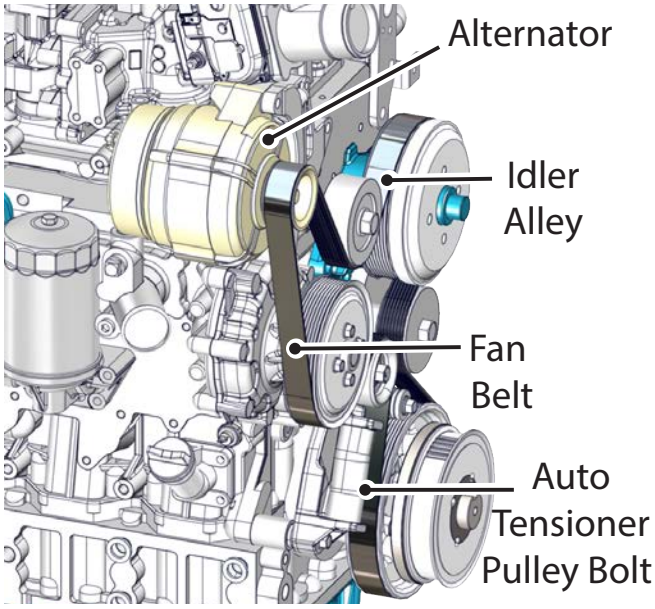


Figure 5-9. Engine Belt Tension Bolt

Change Oil Separator Element

The oil separator captures oil mist carried over from the lubrication process. The oil separator element must be changed every 1500 hours of operation. **(Figure 5-10)**

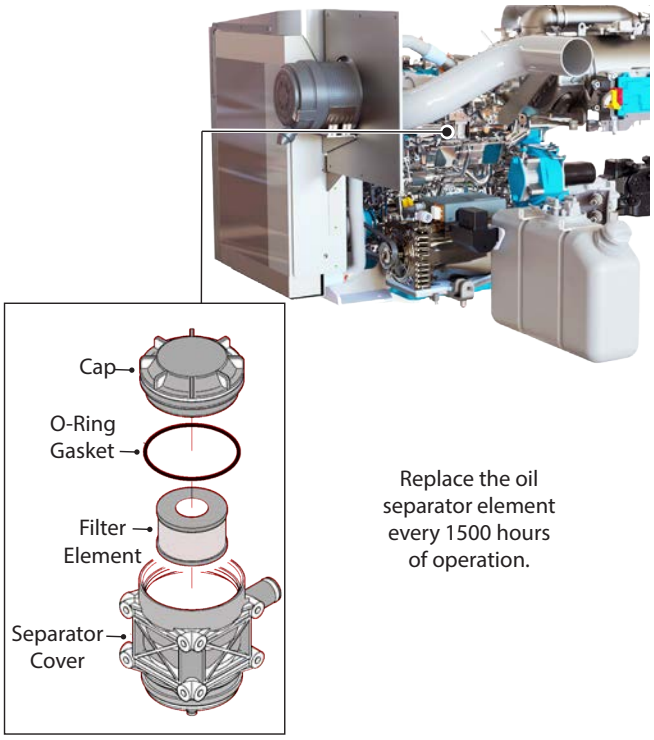


Figure 5-10. Oil Separator

To change the oil separator element:

1. Unscrew the oil separator cover cap and remove from the body.
2. Remove the used gasket and filter element. Dispose of in accordance with environmental laws and requirements.
3. Wipe oil and grease off the oil separator body and cover.
4. Insert a new filter element and gasket.
5. Reattach the cover cap to the oil separator body.

5

Selective Catalytic Reduction (SCR)

The SCR system is an efficient, effective means of filtering contaminates for optimum engine performance while reducing air pollution from exhaust emissions. The SCR system precisely injects Diesel Exhaust Fluid (DEF) into the engine's exhaust system to transform those toxic exhaust gases into harmless water vapor and nitrogen. SCR technology is also cost-effective and fuel-efficient. **(Figure 5-11)**

NOTICE **NEVER tamper with or remove any emission-related component under penalty of law.**

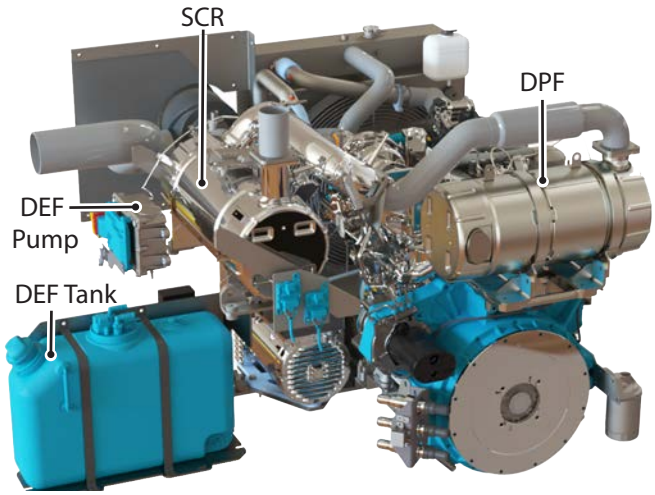


Figure 5-11. SCR System

A sensor on the Diesel Particulate Filter (DEF) tank header monitors the fluid level, temperature and quality of DEF in the tank. The header is mounted on top of the DEF tank and provides filtered DEF supply from the tank to the DEF injector through the DEF pump.

NOTE: Check the fittings around the DEF tank header for leaks or any signs of damage regularly. DEF leaks will leave a white deposit around the fittings.

⚠CAUTION **DO NOT pressure wash or steam clean this unit to avoid damaging the components. Use compressed air to remove any loose debris.**

DEF (Diesel Exhaust Fluid)

DEF is a mixture of pure urea and de-ionized water that is injected into the exhaust stream by the SCR system. Temperature and humidity affect DEF consumption, with hot and dry climates resulting in greater DEF consumption. Typically, machines will consume an amount equal to 2 - 10 percent of the fuel you burn.

The DEF tank is located just inside the right engine access door. **(Figure 5-12)** DEF tanks always have a blue cap with a smaller fill neck to prevent erroneously filling with the wrong fluid.

Figure 5-12. DEF Tank (Blue) Fill Cap

DEF fluid is available at most truck stops, diesel engine dealerships, auto supply stores and large gas stations. To find DEF locations in the U.S., go to http://www.finddef.com/def_map.php.

⚠WARNING **DEF contains urea, therefore, wear eye protection, gloves and appropriate protective clothing when filling the tank and handling DEF. Avoid skin contact. Use only with adequate ventilation and avoid breathing vapor or mist. DO NOT get the substance in your eyes. In case of contact, immediately flush eyes for a minimum of 15 minutes. ALWAYS wash hands thoroughly after handling DEF.**

NOTICE **DO NOT use DEF after its expiration date. Always dispose of DEF in compliance with federal and local environmental laws.**

It is critical that you prevent contamination entering into the SCR system. Even trace amounts of contaminants introduced into the SCR system can cause damage.

- **Always** clean around the fill neck and dispensing nozzle when refilling the DEF tank. If you're refilling the DEF tank often, the opportunity for airborne dust and dirt ingestion is high.

- Be especially careful handling DEF at the job site. While it's a good idea to have DEF on hand, especially in rural areas, the risk of contamination is higher. Be sure the ambient temperature is within the correct storage range if bringing DEF to the work site.

DEF Storage

DEF is a vital element in engine performance. Proper storage and use will help to insure maximum engine performance and a long, trouble-free experience with the installed engine.

DEF is a mixture of pure urea and de-ionized water that is injected into the exhaust stream by the SCR system. Temperature and humidity affect DEF consumption, with hot and dry climates resulting in greater DEF consumption. Temperatures also affect the quality of DEF and can impeded peak performance.

- **DEF freezes at 12°F.** In winter months it should be stored above this temperature. If DEF does freeze in the machine tank during overnight storage between jobs, onboard systems enter into Thaw Control Mode to heat the DEF back to usable temperatures. Engine maximum torque may be limited while DEF is frozen.
- **Temperatures above 86°F causes DEF to break down**, which releases ammonia into the onboard DEF tank or storage container. This breakdown of the DEF affects the quality of the DEF, which could yield warning messages or reduce engine performance of the machine, and may increase pressures within the storage containers. During DEF fill, or after any prolonged storage of DEF, if the smell of ammonia is present, the DEF fluid has started to break down and LeeBoy recommends it be changed or discarded.

If DEF is stored or used above 86°F, the DEF life will be reduced. DEF storage recommendations include:

- Store DEF in a clean, dry, well-sealed container in a well-ventilated location away from dust and direct sunlight.
- Keep DEF away from any container of gasoline and diesel fuel or other lubricants.
- Store DEF in a dedicated container that has not been used with other fluids, including funnels and hoses.
- If using pumps to fill DEF tanks, ensure the seals, O-rings and fittings on the dispensers are DEF-compatible.
- The best practice is to pour directly from the DEF container into DEF tank.
- Use a container size that allows DEF consumption in 2 months.

Other best practices for the use of DEF fluid:

- Fill DEF tank with every fuel fill to maintain fresh high quality DEF within the system.
- Avoid prolonged storage of DEF within the parked machine.
- Drain DEF fluid during prolonged storage.
- Drain DEF fluid during winter months when DEF may freeze for prolonged periods.

NOTICE **At 80°F, DEF has a storage life of one (1) year.**

DPF Service

The DPF is designed to burn trapped particulate matter automatically or via operator-induced regeneration. The longer the engine runs, the more ash (burnt residue) is collected on the filter. Too much residue adversely affects DPF performance, therefore, see your authorized dealer for professional cleaning (or replacement) if needed.

Do not be alarmed if you notice white exhaust from the exhaust pipe at start-up or when speeding up. White exhaust or water may come out during cold weather or after long-hour idling due to condensation.

If DPF regeneration occurs frequently, i.e., five hours or shorter, on a routine basis, change the oil. If this condition continues after an oil change, contact your authorized dealer for service. (Refer to your Engine Operator's Manual for service information.)

NOTICE **Too much idling can cause DEF to crystallize in the injectors if the machine isn't hot enough. Don't idle machine any longer than necessary as too much idling can cause soot to build up in the Diesel Particulate Filters (DPF), requiring more frequent regeneration.**

NOTICE **To reduce the amount of soot and maximize DPF efficiency, use API-CJ4 (or ACEA-E9 European equivalent) engine oil. Running the engine at 1800 RPMs will also provide more efficient operation and allow automatic regeneration more efficiently while paving.**

COOLING SYSTEM

Inspect the radiator and oil cooler daily. Use a low-pressure water hose to clean as needed once the engine has cooled down.

Inspect the coolant system daily. Wash out the radiator with a low-pressure water hose ONLY when the engine is cool.

Check radiator hoses and clamps every six months for leaks or damage, ensuring hose clamps are securely tightened. If radiator hoses are swollen, hardened or cracked, replace them immediately.

⚠CAUTION Inspect the radiator and oil cooler daily. Wash out as needed with low-pressure water hose after the engine has cooled down. Plugged or dirty radiators and oil coolers cause engine or hydraulic overheating.

⚠CAUTION Replace radiator hoses and clamps every two years, or sooner if needed. Damaged radiators hoses or coolant leaks also cause overheating.

If the coolant temperature warning icon illuminates, or steam emits from the radiator or overflow pipe, turn off the load but **idle the engine at least five minutes** to cool down gradually and check the following items:

- Inspect for any coolant leaks, obstacles or debris around the air inlet or outlet, dirt or dust between radiator fins and tube, or clogged water pipe.
- Check the fan belt as described in previous section.
- Check that radiator cap is properly tightened. A gap between the cap and seal accelerates coolant loss.
- Check for any internal blockage in the radiator hose. If scale forms on the hose, clean with scala inhibitor (or equivalent).
- Check engine cooling fan for damaged or bent blades. Replace if damaged.

Replace Radiator Hoses

Radiator hoses should be replaced if worn or frayed. Access the radiator from the top access panel.

NOTE: Inspect the radiator and oil coolers daily. If needed, wash out using a low-pressure water hose once the engine has cooled down.

⚠CAUTION Dirty or plugged radiators and oil coolers will cause engine or hydraulic overheat conditions.

⚠WARNING This procedure should only be performed when the machine is not running.

⚠WARNING DO NOT remove the radiator cap while the engine is still hot. Verify there is no pressure in the radiator BEFORE removing the radiator cap. Serious burn injuries can occur if cap is removed while hot and under pressure.

1. Turn off ignition, remove key, and allow the engine to cool down.
2. Raise the engine hood. ***(Page 5-6)***
3. Carefully remove the radiator coolant fill cap. **(Figure 5-13)**

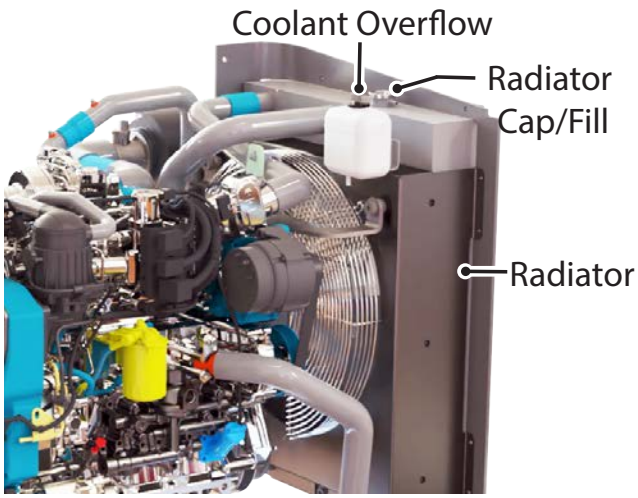


Figure 5-13. Radiator and Coolant Fluid Caps

4. Locate the radiator drain hose (with petcock) inside the left engine panel. Drain radiator fluid into a container large enough to hold seven (7) gallons.
5. Remove the upper and lower hoses by loosening the hose clamps on each.
6. Disconnect hoses and inspect for damage.
   - If clamps are not damaged, retain for reinstallment (or replace if damaged).
7. Install new hoses and reconnect with hose clamps.

FUEL SYSTEM

Fuel Tank

The fuel level is indicated on the digital display fuel gauge. Use only low-sulfer diesel fuel and always fill the fuel tank at the end of the work day to prevent condensation. The fuel tank has a green cap and is located inside the left access panel of the paver. **(Figure 5-14)**

NOTICE **Fill up with low-sulfer diesel fuel at the end of the day to reduce condensation in the tank.**

⚠WARNING **Diesel fuel is highly FLAMMABLE. DO NOT smoke while filling the fuel tank. Fill the fuel tank in a designated area that provides proper ventilation with a fire extinguisher available.**

⚠WARNING **Explosion Hazard! Never fill the fuel tank near an open flame, or near equipment that can create sparks. Never check fuel level or fuel system components near an open flame.**

⚠WARNING **Explosion Hazard! Stop the engine immediately if any fuel leakage is noted. DO NOT start the engine until the leak has been corrected.**

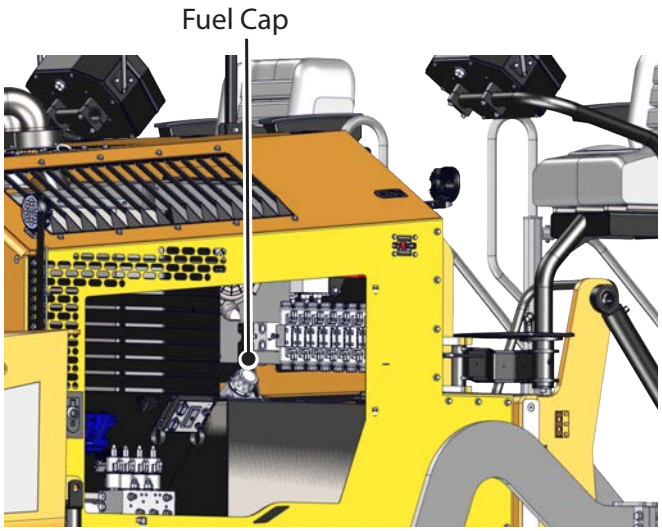


Figure 5-14. Fuel Cap

To fill the fuel tank:

1. Stop engine.
2. Remove fuel cap from fuel tank.
3. Fill with diesel fuel until full.
4. Replace fuel cap.

Change Fuel Filter

The fuel filter, water separator and fuel pump with bowl filter are conveniently located together on the engine access door in the hopper. **(Figure 5-15)**

⚠WARNING **DO NOT change the fuel filter while the engine is running. DO NOT smoke or change fuel filter near open flames as diesel fuel is highly combustible. Be careful not to spill fuel.**

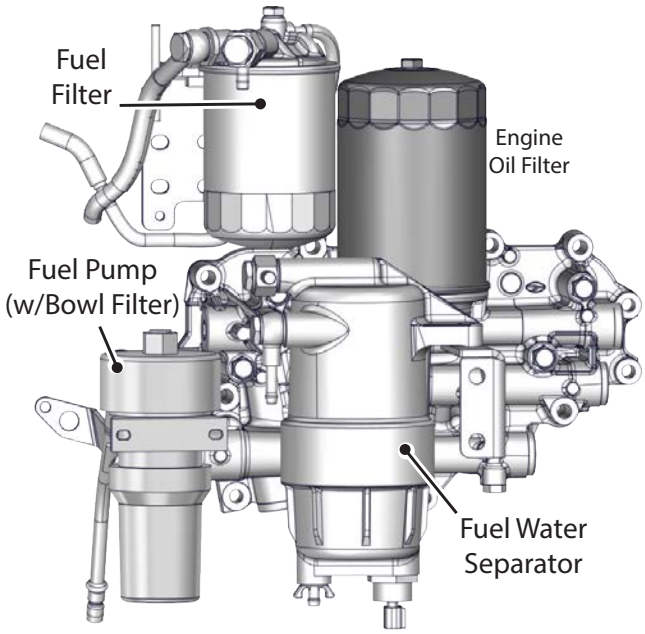


Figure 5-15. Fuel Filters

Replace the fuel filter using the following general procedure:

1. Stop the engine and allow it to cool. Put a container under the filter before removing it.
2. Use a filter removal wrench to loosen and remove the filter by turning it clockwise.
3. Drain and discard the removed filter in compliance with local and federal environmental laws..
4. Wipe a little fuel on the rubber gasket. Install the new fuel filter element onto the filter head, **carefully tightening the element by hand. DO NOT overtighten.**
5. Turn the key to the ON position and allow the fuel pump to fill the filter, assisting with the primer bulb.
6. Start the engine and check for any fuel leaks. **Stop the engine immediately if any leakage is noted. DO NOT start the engine again until the leak has been corrected.**

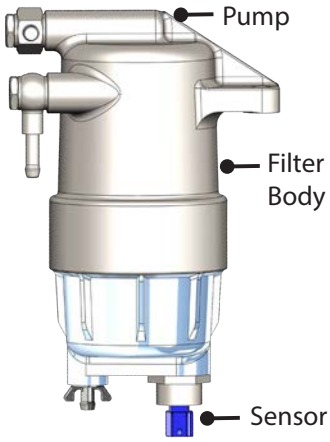
Fuel Water Separator

Water and dirt that gets into fuel settles into the water separator. **(Figure 5-16)** An error code will appear on the DM430 digital display when it needs to be emptied. It is important to drain it immediately when this occurs. The fuel water separator is located on the access door in the hopper.

NOTICE **If you run out of fuel, it is important to pump the water separator to prime fuel injection lines. Failure to do so can adversely affect performance.**

To drain the water separator:

1. Pull retainer clip from plug to disconnect the harness.
2. Unplug the harness, then unscrew the plug.
3. Drain and reinsert components.



To clean the water separator:

1. Follow the previous steps for draining the water separator.
2. Loosen and remove the filter body, then clean it with light oil.
3. Reinsert and tighten.

Figure 5-16. Fuel Water Separator

HYDRAULIC SYSTEM

Check Hydraulic Oil

The hydraulic motors and hydraulic cylinders use the same hydraulic oil tank and hydraulic oil supply.

NOTICE **Take care when working with the hydraulic system to ensure it is completely clean.**

NOTICE **Never allow the hydraulic oil level to fall too low. Pump damage will occur.**

⚠WARNING **DO NOT use your hands on any hydraulic hoses, fittings or system component when checking for leaks. Serious personal injury can result from an oil leak under high pressure. Oil can be injected under the skin by high pressure. Protect the eyes by wearing safety glasses.**

⚠WARNING **DO NOT loosen or remove the hydraulic oil strainer fill cap when the hydraulic oil is hot. Always loosen the cap slowly to relieve any pressure in the hydraulic oil reservoir.**

Check the hydraulic tank oil level daily using the sight gauge just inside the right engine access panel. **(Figure 5-17)** Add hydraulic oil if needed (the hydraulic oil cap is just above the sight gauge):

- Fill to the black line on decal to the right of the sight gauge if the hydraulic oil is cold.
- Fill to the fill line if the hydraulic oil is at normal operating temperature (preferred method for an accurate reading), shown at **Figure 5-17**.

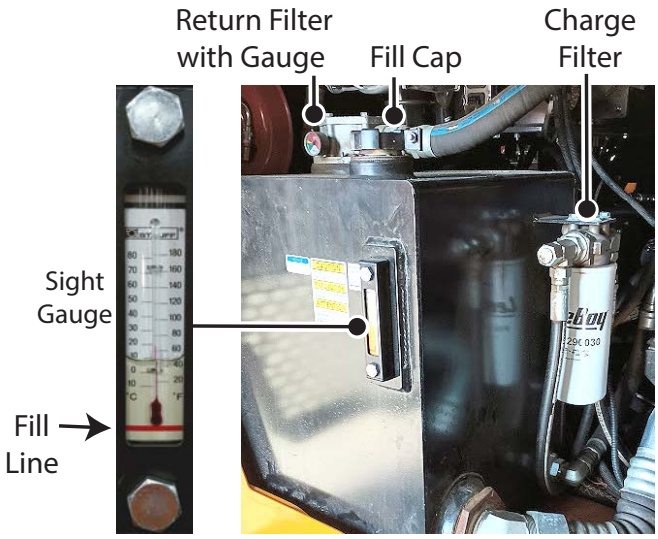


Figure 5-17. Hydraulic Oil Tank, Cap, Filters and Sight Gauge

- Slowly unscrew the hydraulic oil strainer fill cap (above sight gauge) and add hydraulic oil.

NOTE: DO NOT use unfiltered hydraulic oil.

- Check the hydraulic oil level using the sight gauge.
- Keep the hydraulic oil filled to the correct level.

NOTICE **DO NOT overfill the hydraulic oil reservoir.**

NOTICE **Leeboy uses a leak-detection dye that may turn the hydraulic oil a slight yellow tint.**

Change Hydraulic Oil and Filters

Changing the hydraulic oil removes the accumulation of dirt, water and mechanical wear particles from the hydraulic oil system. Clean, filtered oil is critical to optimize hydraulic system operation. The hydraulic oil system holds approximately 37 gallons.

There are two hydraulic filters. The hydraulic charge filter is located inside the right engine access panel on the right side of the hydraulic oil tank. **(Figure 5-18)** The hydraulic return filter is located

NOTICE **Contaminated hydraulic oil can shorten the expected service life of components in the hydraulic system.**

⚠WARNING **DO NOT use your hands on any hydraulic hoses, fittings or system component when checking for leaks. Serious personal injury can result from an oil leak under high pressure. Oil can be injected under the skin by high pressure. Protect the eyes by wearing safety glasses.**

Use the following procedure to change the hydraulic oil and filter:

1. Stop the engine.
2. Allow the hydraulic oil to cool, until it is at a warm temperature only.
3. Slowly remove the hydraulic oil strainer fill cap. Do not remove the strainer portion under the cap.
4. Place a clean, lint-free cloth over the reservoir fill tube opening and secure in place with tape.

⚠WARNING **DO NOT drain the hydraulic oil from the tank when it is hot as the oil can cause serious injury. Drain at a warm temperature only.**

5. Place a container with sufficient capacity to collect the hydraulic oil under the hydraulic tank plug.
6. Carefully remove the plug from the bottom of the hydraulic tank and allow all the hydraulic oil to drain into the container.
7. Reinstall the hydraulic oil drain plug.
8. Change the hydraulic oil filter by turning the filter counterclockwise, then replace with new filter turning clockwise until snug (by hand only). **(Figure 5-18)**
9. Remove the strainer inside the oil filler cap and replace with a new one, aligning the holes on the strainer with the mounting holes on the gasket.
10. Fill the hydraulic oil tank with filtered hydraulic oil until proper level is shown on the sight gauge.

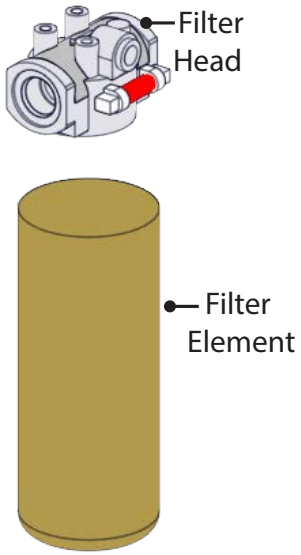


Figure 5-18. Hydraulic Charge Filter

11. Check the oil level in the hydraulic oil tank again. Add oil if needed.
12. Reinstall the hydraulic oil cap onto the reservoir filler neck and tighten securely.
13. Start the engine.
14. Check the hydraulic system for any leaks.

⚠CAUTION **Stop the engine immediately if any hydraulic leak is noted. DO NOT start the engine until any problem noted has been corrected.**

5

ELECTRICAL SYSTEM

The electrical system is a 12-volt negative ground system utilizing two batteries. Raise the hood to access the batteries directly in front of the right operator platform. **(Figure 5-19)**

Figure 5-19. Batteries

Connect the positive (+) cable to the positive (+) post of the battery. Connect the ground cable to the negative (-) post of the battery. It is advisable to disconnect the negative (-) cable first and connect it last. Reversed polarity can damage the electrical system.

⚠WARNING Fire and Explosion Hazard! Always observe battery polarity when connecting a battery charger or jumper cables to the battery: negative (-) to negative (-), positive (+) to positive (+). Failure to do so can produce sparks.

When connecting a booster battery, connect one end of the first jumper cable to the positive (+) terminal of the dead battery and the other end to the positive (+) terminal of the booster battery. Connect one end of the second jumper cable to the negative (-) terminal of the booster battery and the other end to the frame of the paver with the dead battery.

⚠WARNING Fire and Explosion Hazard! Be sure that the battery charger is in the OFF position before connecting it to the battery.

Be sure to keep the battery fully charged during cold weather to keep it from freezing. Freezing weather has little effect on a fully-charged battery.

NOTICE ALWAYS turn the master battery switch off when working on the electrical system or welding anywhere on the machine. Damage to electrical components could result.

NOTICE DO NOT pressure-wash electrical components, especially around the generator. Protect electrical components when washing paver.

Battery Servicing

⚠WARNING Batteries contain sulfuric acid. ALWAYS wear safety goggles and protective clothing when servicing the battery.

⚠WARNING NEVER allow battery fluid to come into contact with clothing, skin or eyes. If battery fluid contacts the eyes or skin, immediately flush with clean water and obtain prompt medical treatment.

⚠WARNING Keep sparks and flames away from the batteries as electrolyte gas is highly flammable.

1. Raise the hood to access the batteries. ***(Page 5-6)***

NOTE: When replacing the battery, discard the old battery properly.

2. Before connecting the batteries, turn off the master switch, located in the right access panel at the front of the operator station. **(Figure 5-20)**

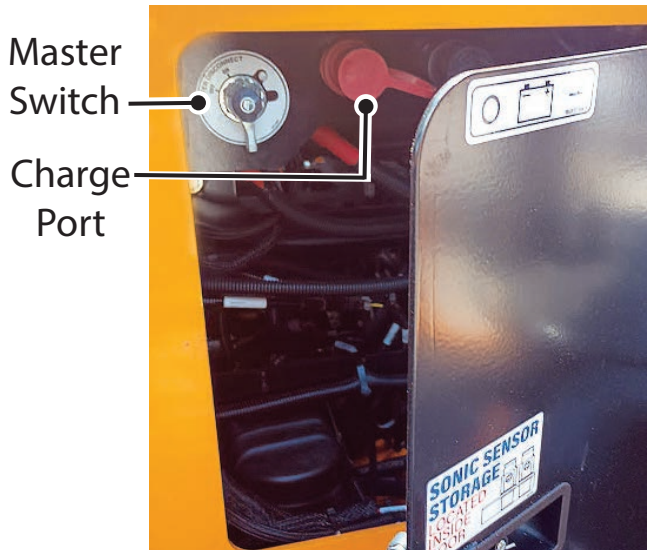


Figure 5-20. Battery Master Switch

3. Be certain that battery terminals and posts are clean and that the battery cable terminals are tight.

NOTICE Dirty or loose battery connections can create high electrical resistance and permit arcing.

4. If corrosion is present around terminal connections, remove and wash with ammonia or a solution of 1/4-pound baking soda added to one quart of warm water. Ensure the vent caps are tight to prevent solution from entering the cells.
5. Pour clean water over the battery and surrounding area to rinse away the solution.
6. Check vent cap breather openings to ensure they are clear.

Alternator

The alternator supplies electrical power to electronic controls and charges the battery. Its built-in regulator controls voltage output. **(Figure 5-21)**

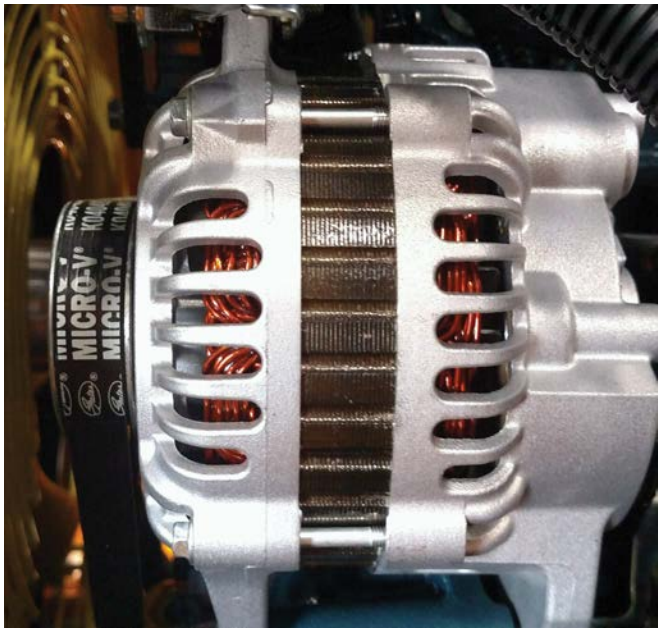


Figure 5-21. Alternator

Use the following precautions to prevent damage to the alternator or regulator:

- Never polarize the alternator.
- Never ground any alternator terminals or circuits.
- Always disconnect the battery before disconnecting or connecting the alternator.
- Never disconnect the alternator while machine is operating.
- Ensure wiring is properly connected before connecting the battery.
- Always connect a booster battery in the proper polarity: Negative (-) to negative (-) and positive (+) to positive (+).

Adjust Generator Belt

Adjusting the generator belt every 250 hours (quarterly) for optimum performance is a simple procedure. You can access the generator from the center engine access panel or by raising the hood. ***(Page 5-6)***

NOTICE **Be sure to maintain the belt alignment during and after adjustment.**

The key to adjusting the generator belt is loosening and/or tightening all four adjustment bolts to align the generator pulley and tension the belt:

1. ***Slightly*** loosen the four bolts on the generator plate. **(Figure 5-22)**
2. ***Slightly*** tighten each mount bolt until you have approximately 1/2 to 3/4 inches deflection with minimal force.
3. Retighten the bolts on the generator plate.

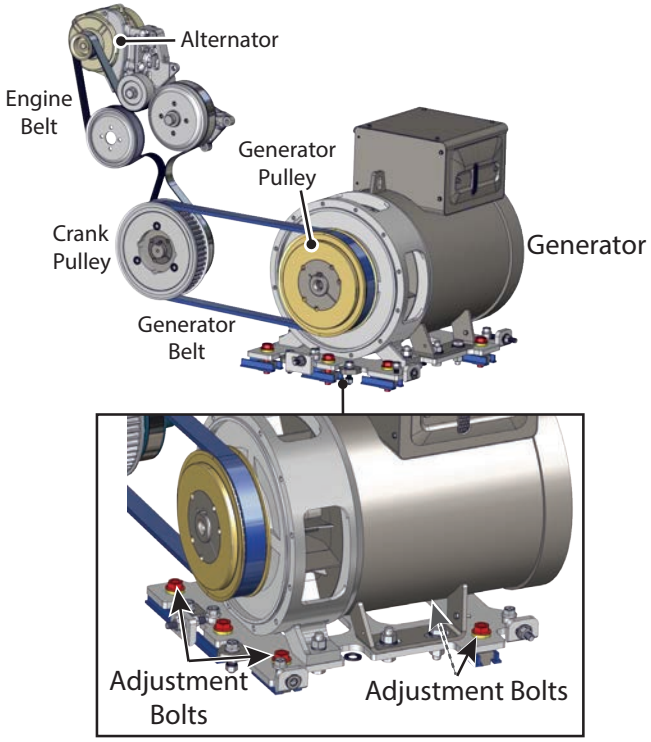


Figure 5-22. Generator Belt Adjustment

5

Fuse Panel

The electrical system is protected from overload damage by fuses. If an electrical component fails to operate, check the fuse box panel first to ensure that a fuse has not "blown." A blown fuse is an indication that there is an overload that needs to be corrected.

The fuse panel is located inside the junction box for convenient access. **(Figure 5-23)**

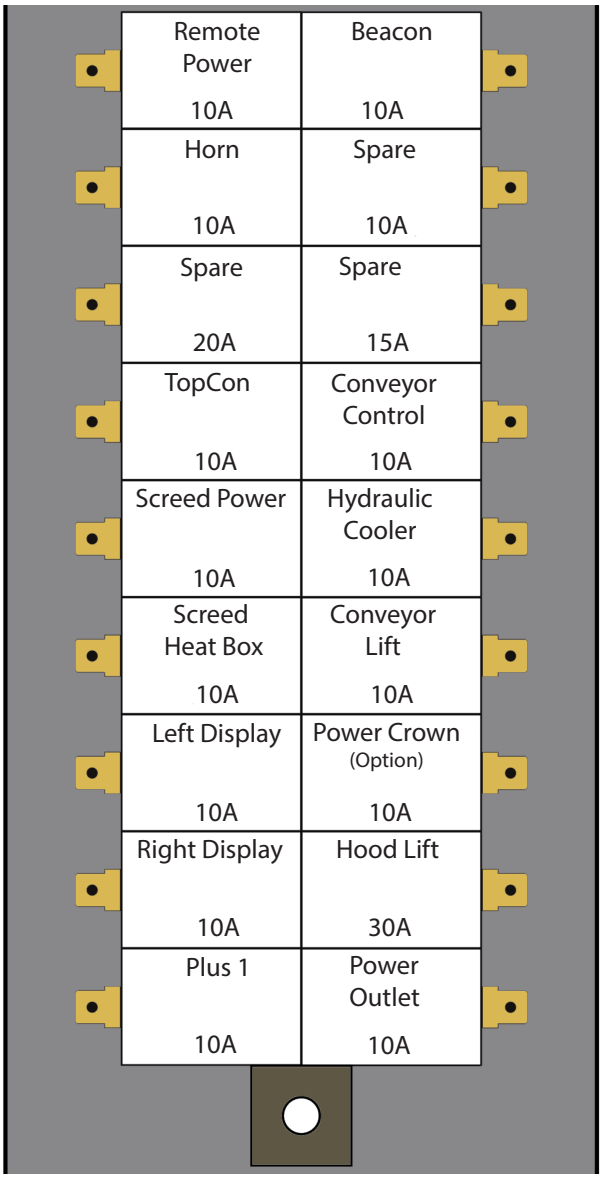


Figure 5-23. Fuse Panel

Electric Heat Control Box

The electric heat control box contains the element breakers and main outputs for the screed heating system on both the standard LB5 and optional LB7 screeds. **(Figures 5-24 and 5-25 *on the following page.)***

The standard LB5 screed is powered at 240 VAC while the optional LB7 screed is powered at 480 VAC. Heating element relays "make" or "break" the circuit to each element to start or stop the heating cycles. When the HEAT ON button is depressed, 12VDC is momentarily applied to the main timer relay to start the timing cycle.

NOTE: Control boxes are manufactured to fit all screed and paver combinations. If not all plugs are connected to wires, it may be normal. Be sure to keep unused plugs covered with mating protective caps.

DANGER The heat box is high voltage ! ALWAYS be careful when working with electrical components to avoid electrical shock. Take extra precautions to avoid electrical shock that could cause serious injury or even death.

- Each element output consists of two wires--one connects to the L-1 circuit and the other to the L-2 circuit.
- Each breaker has two terminals--one is connected to the main input and the other terminal to an element output wire.
- The L-1 circuit is the left bank of element breakers.
- The L-2 circuit is the right bank of element breakers.

Heating element relays "make" or "break" the circuit to each element to start or stop the heating cycles. When the HEAT ON button is depressed, 12VDC is momentarily applied to the main timer relay to start the timing cycle.

LB 5 Screed Heat Box

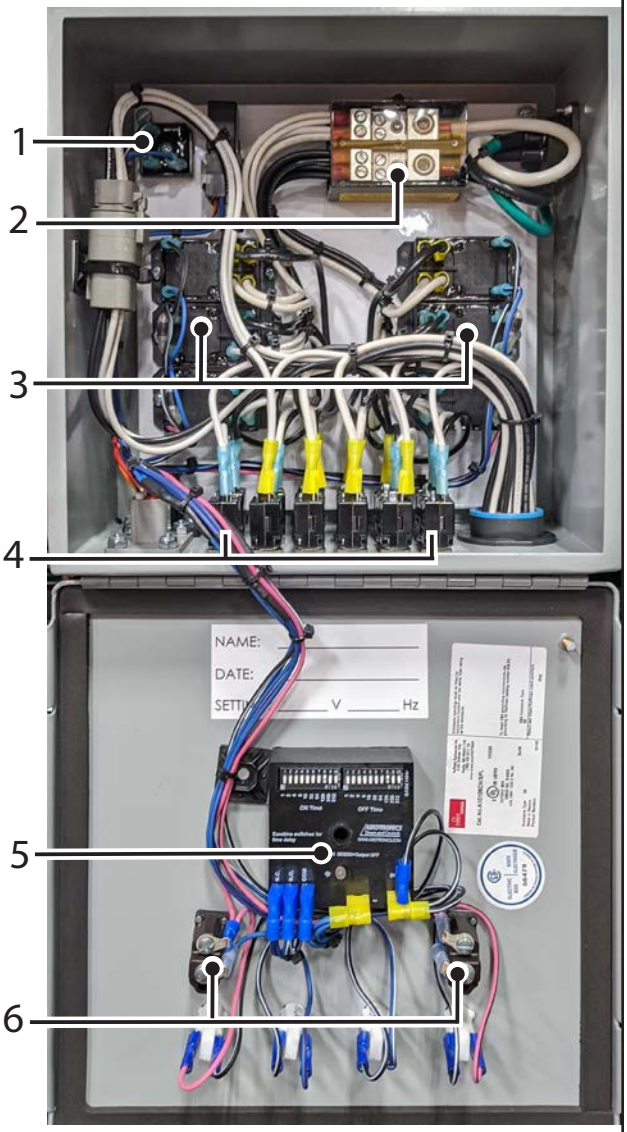


Figure 5-24. Electric Heat Elements

1 - Rectifier

2 - Terminal Block

3 - Heating Element Relays

4 - Element Circuit Breakers

5 - Cycling Timer

6 - Toggle for Left/Right End Gates

LB 7 Screed Heat Box

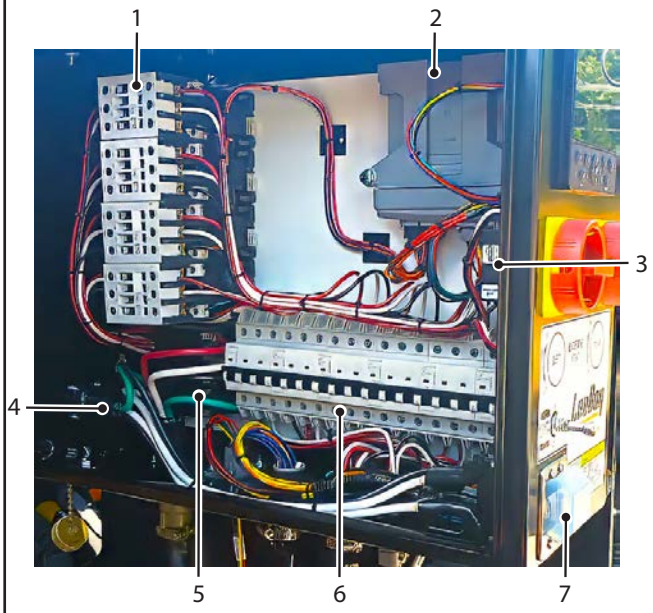


Figure 5-25. LB7 Screed Electric Heat Elements

1 - Heating Element Contactors

2 - OX Module

3 - Switches for Left/Right End Gates

4 - 110 VAC Connector (OPTION)

5 - Terminal Block

6 - Element Circuit Breakers

7 - 110 VAC Circuit Breaker (OPTION)

5

Heating Elements

Each heating element is sized to fit properly and provide sufficient heat to the screed plate. Elevated temperature prevents mix from sticking to the lower surface of the screed plate.

⚠ DANGER DO NOT operate an element with a known electrical shortage. Replace faulty elements and wiring immediately.

An element assembly consists of four main components:

- The element.
- The wire protector adapter.
- The wire protector.
- The four-pin wire plug at the end of the element protector.

Each element is covered by a thin strip of insulation to keep heat from escaping. A support bar and shield protects the element assembly.

Each element is clamped onto the screed plate to provide a positive and efficient connection. Enough pressure should be applied to the element assembly to sufficiently hold it securely against the screed plate surface.

To remove an element:

1. Remove top cover from the extension screed plate.
2. Loosen all the clamping studs over the element.
3. Remove and replace element.
4. After tightening the clamping stud, lock the clamp by tightening the stud jam nut.

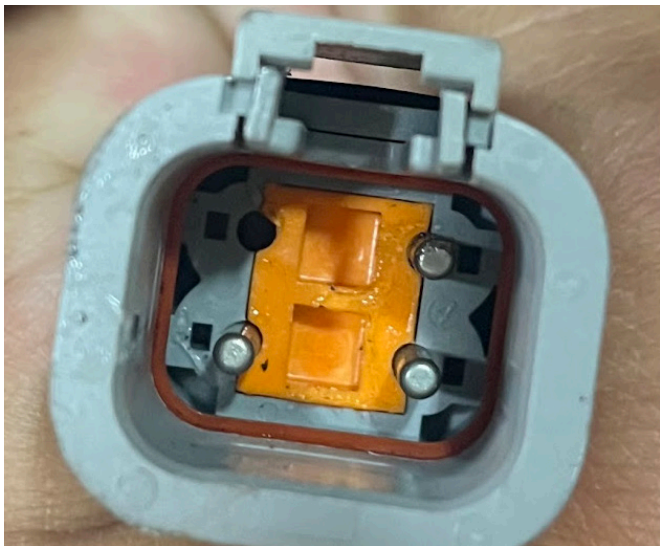


Figure 5-26. 4-Pin Plug

Element Resistance Testing

The breakers are wired into each leg of each element. If an element has a fault, either in the wiring, or in the element itself, the breaker will trip and power will no longer be applied to that leg of the element.

When a breaker in the control box has “tripped,” there may be a problem in the wiring or an element in the circuit.

The breakers can be manually reset by depressing the trip button back into place when they are extended. If the breaker still does not reset, you need to test or possibly replace an element.

If the element is functioning correctly, you should read a resistance between the connector pins when testing. A faulty element will show high resistance, indicating a bad element. ***(Refer to* Tables 5-2 and 5-3 *on the following page.)***

⚠ DANGER HIGH VOLTAGE. This procedure must be performed by a qualified electrical technician due to the danger of electrical shock, injury or even death. See your authorized dealer.

To test the elements:

1. Disconnect elements one at a time from the connection point in the screed, keeping track of connector placement.
2. Using an ohm meter, test the resistance through the element between every combination of ins (i.e., 1 to 2, 2 to 3, 1 to 3) in the plug. **(Figure 5-26)**

NOTE: You do not have to test the plug attached to the lower side of the control box.

3. Before replugging the element, check each wire pin with an ohm meter test lead, placing the lead on a steel section of the screed frame. If there is any continuity through the element to the frame, the element must be replaced.

Table 5-2. LB5 Screed Element Resistance Table

LB5 SCREED (Standard) 220 VAC	
Generator Size (kW)	14.4 kW
Breaker Size	60 Amp
Main Screed Elements (2)	
Wattage Per Element	3500 W
Total Wattage	7000 W
Resistance per Element	15 - 16.8 Ω
Screed Extension Elements (2)	
Wattage Per Element	2000 W
Total Wattage	4000 W
Resistance per Element	26 - 32 Ω
Heated End Gate Elements (2) **	
Wattage Per Element	500
Total Wattage	1000
Resistance per Element	88 - 108 Ω

**** Option item.**

Table 5-3. LB7 Screed Element Resistance Table

LB5 SCREED (Option) 480 VAC	
Generator Size (kW)	34 kW (3-Phase)
Breaker Size	63 Amp
Main Screed Elements (4)	
Wattage Per Element	3500 W
Total Wattage	14,000 W
Resistance per Element	94 - 101 Ω
Screed Extension Elements (2)	
Wattage Per Element	3000 W
Total Wattage	6000 W
Resistance per Element	108 - 118 Ω
Heated End Gate Elements (2) **	
Wattage Per Element	1000
Total Wattage	2000
Resistance per Element	325 - 335 Ω

**** Option item.**

Generator Speed Check

NOTICE **The generator should never be operated at a speed of 2100 RPMs or greater.**

A voltmeter set to read frequency (Hz) can be used to test generator speed. When the engine is running at 2000 RPMs, the generator should operate at 60 Hz. Make certain that the paver engine is running at 2000 RPMs and the hydraulic temperature is at normal operating levels.

Generator Voltage Testing

The generator is belt-driven (from the engine). While the generators are different, the procedure is the same for the standard LB5 or optional LB7 screeds. **(Figure 5-23 or 5-24 on *Page 5-28*)**

DANGER **HIGH VOLTAGE. This procedure must be performed by a qualified electrical technician due to the danger of electrical shock, injury or even death. See your authorized dealer.**

Test the generator voltage at the electric heat control box:

1. Use a volt meter to measure between two of the main input wires L-1 (black wire), L-2 (white wire) or L-3 (red wire) at the terminal block, which should measure 480 ± 5 VAC. If you measure from L-1 to the frame of the paver, or ground, the voltage should be approximately 277 VAC.
2. If the voltage is lower, make certain the generator is turning the correct speed by testing with a Hz meter (valve should be 60 - 63 Hz.
3. If the voltage is still low, contact your authorized LeeBoy dealer for generator speed tuning (see next section).

5

Generator Capacitor Replacement

The two generator capacitors control and regulate voltage while the machine operates. **(Figures 5-27 and 5-28)** If the capacitor fails, the voltage will drop below normal or no output at all. It is located under the generator cover.

If experiencing a drop in voltage, replacing the capacitor will help determine if the capacitor is at fault.

To perform this procedure:

1. Shut down the machine.
2. Detach the wires on top of the capacitor.
3. Remove the capacitor and replace with new one.
4. Reattach wires.

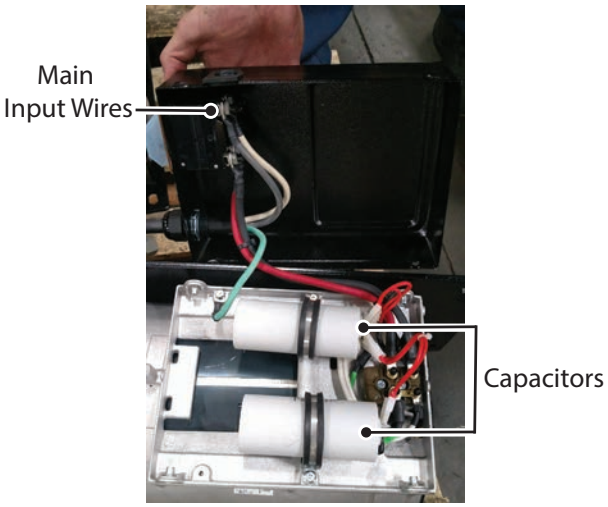


Figure 5-27. LB5 Screed Generator Capacitor

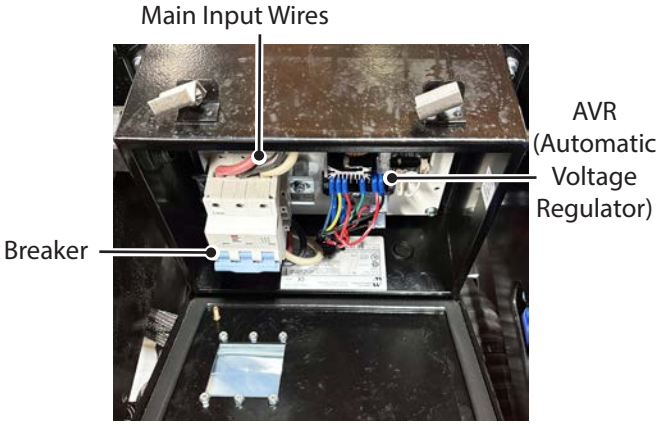


Figure 5-28. LB7 Screed Generator Capacitors

Dielectric Grease

Dielectric grease is a nonconductive, insulating compound applied to electrical connectors. When applied correctly, a high-quality dielectric grease displaces debris and moisture in the connector. The proper application of dielectric grease fills any gaps between pins or in the connector which prevents debris and moisture from creating corrosion of the electrical components.

⚠CAUTION Equipment damage can occur if using low-quality, incompatible dielectric grease.

If not applied or incorrectly applied, corrosion can result that causes rapid increase in resistance, intermittent connections, overheating, and a variety of electrical problems. Numerous electrical issues throughout the machine can be avoided by the proper application and use of dielectric grease.

Figure 5-29. Correct Application of Dielectric Grease

To correctly apply dielectric grease:

- Apply a thin, visible, uniform coat using **ONLY** high-quality dielectric grease (Dow Corning 4 or equivalent) across the **MALE** connector as shown in **Figure 5-29.**
    - Coating too thin may not fill the gaps between pins.
    - Coating too thick may prevent proper connection or force pins or seals out. **(Figure 5-30)**
- Plug in the male connector.

⚠CAUTION DO NOT force connector to lock if too much grease has been applied.

⚠CAUTION Excessive grease application does not allow a proper connection by preventing the plugs from locking into place, and can cause damage to the connector seals.

Figure 5-30. Incorrect Application of Dielectric Grease

DRIVE SYSTEM

The complete drive unit consists of the two-speed motors, sprockets, planetary torque hubs, and high-speed rubber (or the optional poly steel) track systems.

It is important to keep the undercarriage clean and free of asphalt buildup/debris. Undercarriage components should be inspected and maintained periodically, with any worn or faulty component immediately replaced or repaired for operational safety and performance. Maintenance and replacement procedures are described on the following pages.

⚠CAUTION Ensure the paver cannot be started inadvertently by others by blocking the work area before conducting maintenance or replacement procedures on undercarriage components. Use hoisting gear for lifting heavy components, and always wear protective gear.

⚠DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

For performance and durability:

- Check undercarriage components (i.e. drive wheel, rollers and idler) for wear daily. Wear and damage of undercarriage components adversely affects performance and durability.
- Limit driving on sharp, rocky surfaces, manholes and drainage covers.
- Prevent large foreign objects from getting entangled in your undercarriage.
- Remove asphalt buildup on the undercarriage components and tracks daily. Asphalt buildup can cause misalignment between the drive wheel and guide lugs, poor tracking performance, and premature undercarriage and wear.
- Avoid de-tracking episodes by:
    - Moving to a cold area to make turns.
    - Avoiding sharp turns on hot asphalt
    - Limiting movement in reverse.
    - Making gradual turns in a large arc as possible.
    - Repeating zig-zag motions with gradual turns instead of pivoting.
    - Maneuvering at low speed.

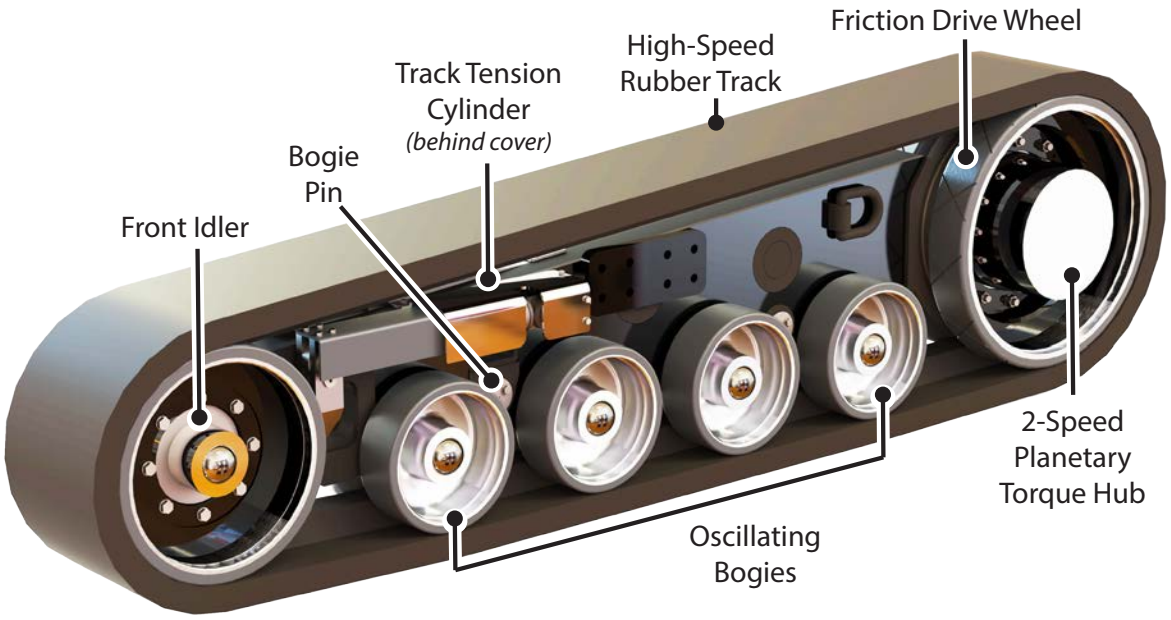


Figure 5-31. Heavy-Duty Rubber Track Components

High-Speed Rubber Tracks

Tracks may need to be replaced due to wear or damage in time, usually caused by reckless driving or load contact between undercarriage parts. Other factors that increase track wear are:

- Tracks resting on uneven rock beds during operation.
- One-sided undercarriage load during turning.
- Driving long distances in reverse.
- Driving over obstructions.
- Improper tension according to operating conditions.
- Excessive dirt and/or debris on the tracks and undercarriage.
- If equipped with the standard rubber tracks, DO NOT drive the paver with the rubber sidewalls pressing against walls, curbs or other objects.

NOTICE **If storing your paver for long periods, keep it indoors in a cool, dry environment away from rain and direct sunlight. If it must be stored outdoors, cover the tracks to reduce exposure. Operate the paver every two weeks for a few minutes to help tracks maintain flexibility and proper shape.**

Rubber track systems require little maintenance, but checking track alignment is key to ensuring longevity. It is essential that you maintain the correct tension with proper track alignment at all times.

Rubber track components are shown at **Figure 5-31.**

DANGER **Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.**

NOTICE **ALWAYS replace BOTH tracks at the same time. Replacing only one may cause alignment issues and damage the undercarriage.**

To replace the high-speed rubber tracks:

1. Park paver on a level surface.
2. Jack up the paver at least two feet off the ground for enough clearance to make it easier to lift the track into place.
3. Find the track tension manifold located inside the left engine access panel on the engine compartment floor. **(Figure 5-32)**

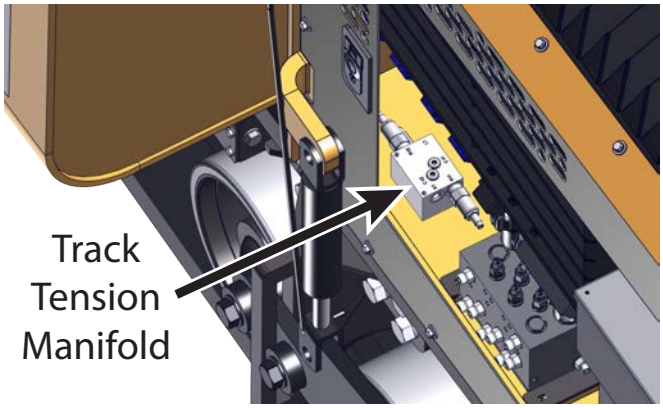


Figure 5-32. Track Tension Manifold Location

4. Loosen the RV1 (left track) or RV2 (right track) relief valve cartridge on the track tension manifold to relieve pressure by turning the cartridge 3 turns counterclockwise. **(Figure 5-33)**

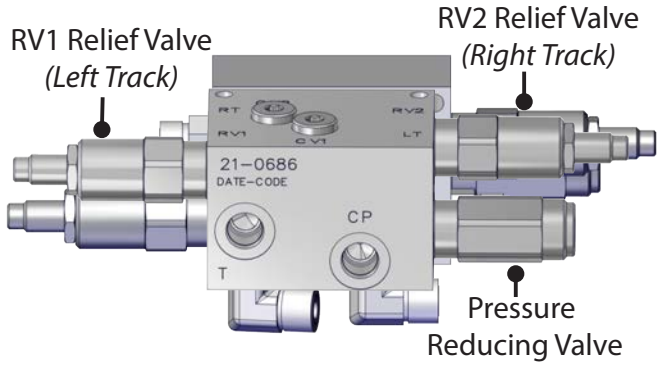


Figure 5-33. Track Tension Main Manifold

5. Remove the outer friction drive wheel (12 nuts) and outer front idler (8 bolts/washers). **(Figure 5-34)**

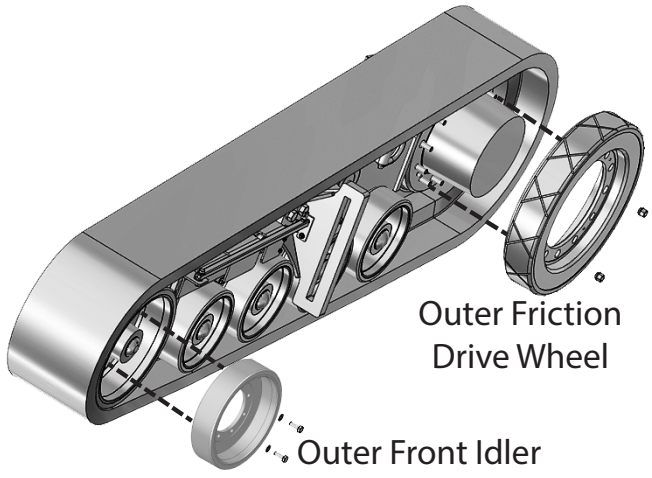


Figure 5-34. Outer Friction Drive Wheel and Outer Front Idler Removal

6. Remove rubber track.
7. Lift the new rubber track into place.
8. Reinstall the outer friction wheel and outer front idler.
9. Screw the track tension manifold relief cartridge back into place and check hydraulic oil level. Add more if needed. **(Figure 5-34)**
10. Run the machine, then re-check the tension after several hours of operation.

Track Tension Cylinder

⚠ DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

1. Follow **Steps 1 - 4** in the previous section. (The track tension cylinders are located behind the front idlers.) **(Figure 5-32)**
2. Remove cylinder covers to access cylinders and disconnect the hydraulic hoses. **(Figure 5-35 *on the following page.)***
3. Remove clevis pins from the cylinders. **(Figure 5-35)**
4. Repack the seal kit or replace cylinder if worn, and reinstall.
5. Reinstall cylinders, pins, hoses and covers.

Oscillating Bogies

⚠ DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

1. Follow the procedure for removing the track on the previous page.
2. Remove the bogie pin. **(Figure 5-31*)***
3. Tap the bogie out of the track frame and replace with new one.
4. Reinstall the bogie pin and track.

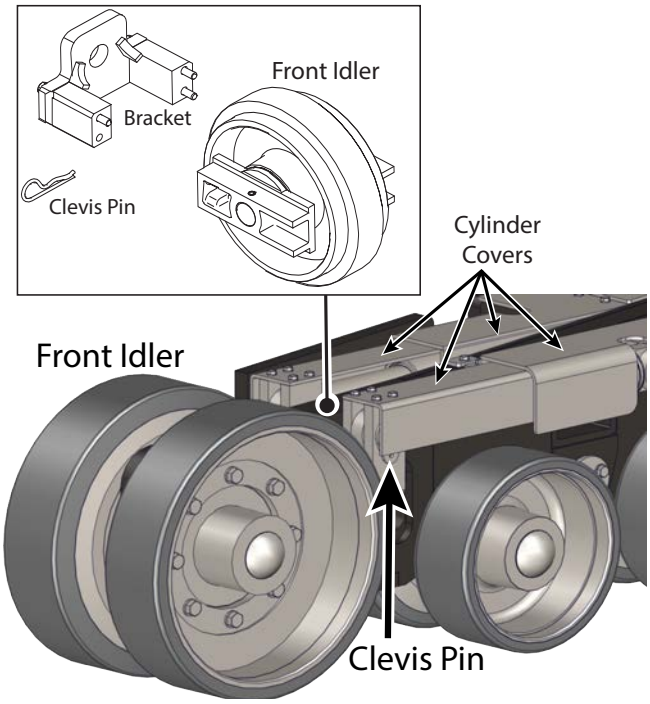


Figure 5-35. Front Idler

Front Idler

DANGER Crush hazard! Always use safety blocks or jack stands when working under the paver to prevent serious injury or even death.

Follow this procedure for replacing the front idler:

1. Follow the procedure for removing the track on ***Pages 5-34 to 5-35.***
2. Remove the front cylinder covers. **(Figure 5-35)**
3. Remove the front idler.
4. Install new idler, ensuring cylinder and clevis pin are aligned properly. Reattach front cylinder covers and track.

Two-Speed Hydraulic Motors

Move paver onto a level ramp where you can access the undercarriage.

DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

1. Disconnect, label and plug fittings on hydraulic hoses to the hydraulic motors. **(Figure 5-36)**

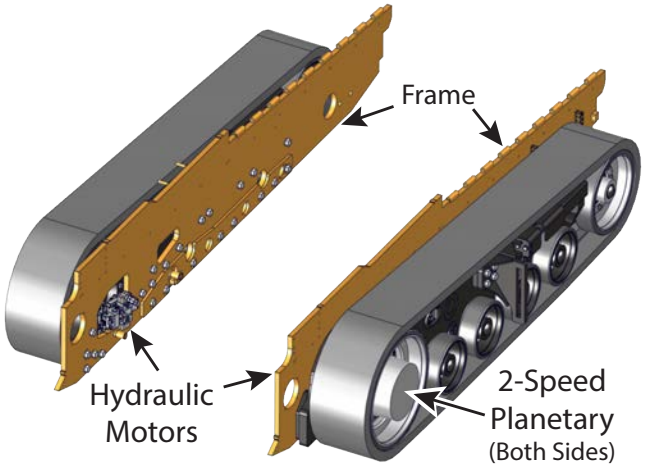


Figure 5-36. Hydraulic Motors and Torque Hubs

2. Place blocks sufficient to support the hydraulic motors underneath them.
3. Remove the hardware attaching the hydraulic motor to the planetary torque hub and carefully separate motor from the planetary. **(Figure 5-37)**

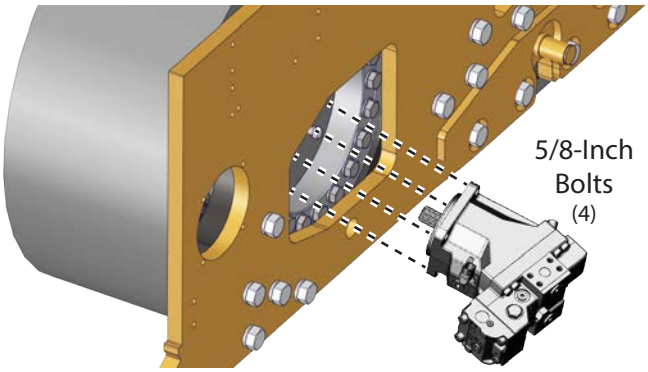


Figure 5-37. Remove Hydraulic Motor

4. Remove the O-Ring and drain hydraulic oil from the motor. Discard or repair the motor as appropriate.
5. Lubricate a new O-ring with hydraulic oil and install onto the planetary.
6. Reattach new or rebuilt hydraulic motor to planetary and refer to ***Torque Tables in* Section 2** to torque bolts. Reconnect hydraulic hoses.
7. Start the engine and let machine run for 10 minutes to work air out of the system. Check for any leaks.

Planetary Torque Hub

If the planetary torque hub has been damaged or worn, use the following procedures for replacement.

⚠ DANGER Crush hazard! Always use safety blocks in addition to jack when working under the paver to prevent serious injury or even death.

1. Remove the track by following **Steps 1 - 6** on ***Page 5-34.***
2. Remove all bolts around the hydraulic motor, then remove the four bolts holding the hydraulic motor and carefully separate motor. **(Figure 5-38)**

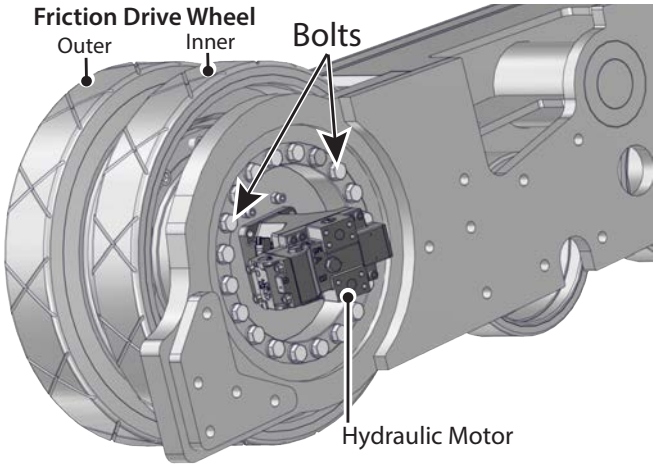


Figure 5-38. Inner Friction Drive Wheel Bolts

3. Inspect the O-ring on the hydraulic motor. Replace if worn, described in the next section.
4. Remove the inner and outer friction drive wheels, then remove the torque hub. **(Figure 5-39)**

NOTE: Mark location of the torque hub to the undercarriage before removing to ensure the planetary is reinstalled in the same position.

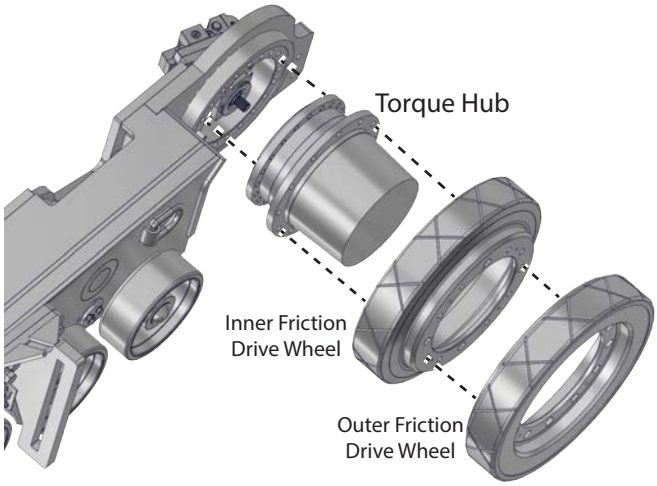


Figure 5-39. Friction Drive Wheel and Torque Hub

5. Install new torque hub into position to align with the hydraulic motor. Place thread-locking adhesive on hub bolts and torque. ***(See Torque Tables in* Section 2**).
6. Reinstall components in reverse order.
7. Lower paver to the ground.
8. Fill hub with oil. ***(Page 5-10)***
9. Start paver and check for any leaks with engine running.

NOTICE DO NOT use your bare hands when checking for hydraulic leaks as serious injury can result from an oil leak under high pressure. Always wear safety glasses and protective clothing.

⚠ WARNING Stop the engine immediately if any hydraulic leak is noted. DO NOT start the engine until the leak has been repaired.

5

REPLACEMENT PROCEDURES

Gear Pumps

Hydraulic fluid is transmitted throughout the machine to motors and cylinders, becoming pressurized according to the resistance present. The fluid is controlled by valves and distributed through hoses and tubes. One load-sense pump, one tandem variable volume drive pump, and two auxiliary pumps play a significant role in producing hydraulic power by supplying hydraulic fluid to system components: **(Figure 5-40)**

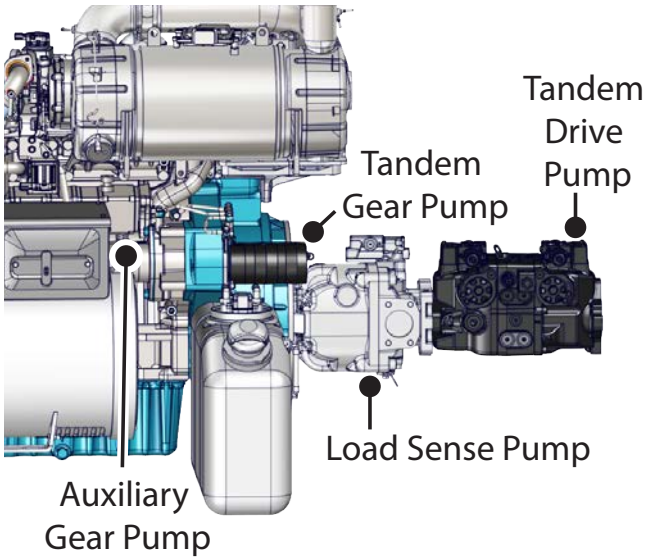


Figure 5-40. Gear Pumps

The tandem drive, load-sense, and rear auxiliary pumps are accessed via the right access panel. The tandem drive pump is attached to the load-sense pump. **(Figure 5-40).** The front auxiliary gear pump is easier to access from the top by raising the hood.

⚠CAUTION **The pump assemblies are very heavy and must be properly supported with a sling before continuing the following steps. DO NOT loosen mounting bolts unless the pump is sufficiently supported.**

1. Identify which pump needs to be replaced.
2. Label and disconnect the hoses to the pump slated for replacement, capping the hoses and fittings on the pump and hoses. Disconnect any wires to pump if applicable.
3. Remove any hardware attaching the pump.
4. Using a sling, carefully lift the pump assembly out of the machine and place on a flat surface.
5. Inspect the O-ring(s) on end of the pump for wear or damage. Replace if needed. Put a small amount of hydraulic oil onto the O-Ring and reinstall.
   - If replacing the load sense pump, check the drive coupling for wear or damage. Remove all bolts and pull out the coupling from the flywheel. Clean the flywheel and threaded bolt holes with ether, then install the new drive coupling into the flywheel if needed.
6. Align the new pump with the mounting holes and reattach pump, covering bolts with thread-locking adhesive. **(Figure 5-41)**

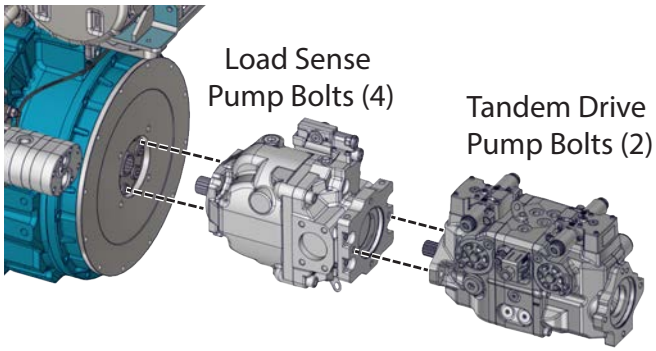


Figure 5-41. Hydraulic Pump Assemblies

7. Refer to ***Torque Tables in*** **Section 2** and torque bolts.
8. Remove plug caps and reconnect hydraulic hoses to pump as previously labeled.
9. Check hydraulic oil level. Add more if needed.
10. Turn the ignition key to Start.
11. Press and hold the Start/Stop button on the main control panel for five (5) seconds and release. Wait another five (5) seconds and repeat several times for one minute.
12. With the engine running, check for leaks.
13. If no leaks are noted, let machine idle approximately 10 minutes to allow the pump and motor cases to fill with hydraulic oil before operating.

Hydraulic Pump/Motor Start-Up

Pre-Start

If the hydraulic system is down as a result of a major component failure:

1. Drain and clean the tank and system components (hoses, valves, fittings and cooler) to ensure it is free from metallic debris and other contamination. Failure to do so may result in damage to the pump(s) and/or other components on start-up.
2. Change all the filters.
3. Change the fluid. On large systems where the cost of changing the fluid may be prohibitive, the fluid should be flushed until a cleanliness level of ISO 4406 18/16/13 or better is achieved.

Installation and Start-Up

When installing hydraulic motors, it is important that the mounting flange of the motor makes full contact with the mounting surface of the application. Mounting hardware of the appropriate grade and size must be used:

- Torque the bolts to the wet torque value defined for the capscrew grade and size. ***(See Section 2 Torque Tables.)***
- Apply Loctite 271 to the bolt threads.

⚠CAUTION **It is CRITICAL to use the correct mounting hardware.**

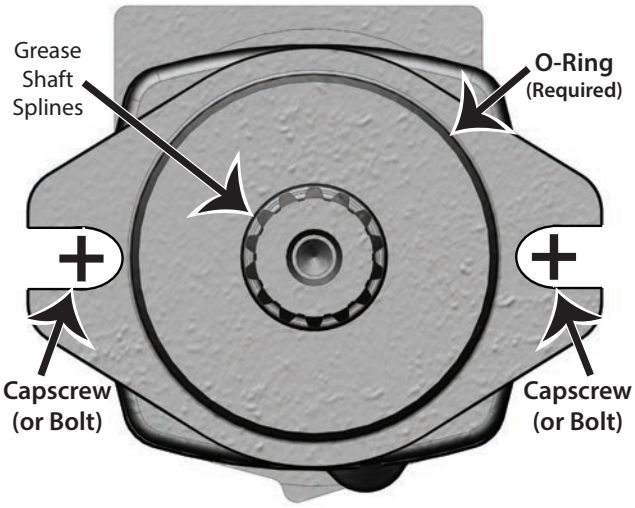


Figure 5-42. Hydraulic Motor Mount

Hubs, pulleys, sprockets and couplings must be properly aligned to avoid inducing excessive thrust or radial loads. Although the output device must fit the shaft tightly, a hammer should never be used to install any type of output device onto the shaft. The port plugs should only be removed from the motor when the system connections are ready to be made.

To avoid contamination:

- Remove all matter from around the ports of the motor and the threads of fittings.
- Fill the case of the motor being used (piston-type, gear motor, gerotor motor, geroller motor or vane motor) with clean hydraulic fluid through the highest case drain port and connect the case drain line. Failure to do so will result in damage to the motor through inadequate lubrication on start-up.
- Units that are mounted vertically with the shaft up require special attention to ensure the fluid level in the case is high enough to lubricate the front shaft bearing(s).

Once all system connections are made, the motor should be run for 15 - 30 minutes at no-load and half-speed to bleed air from the hydraulic system.

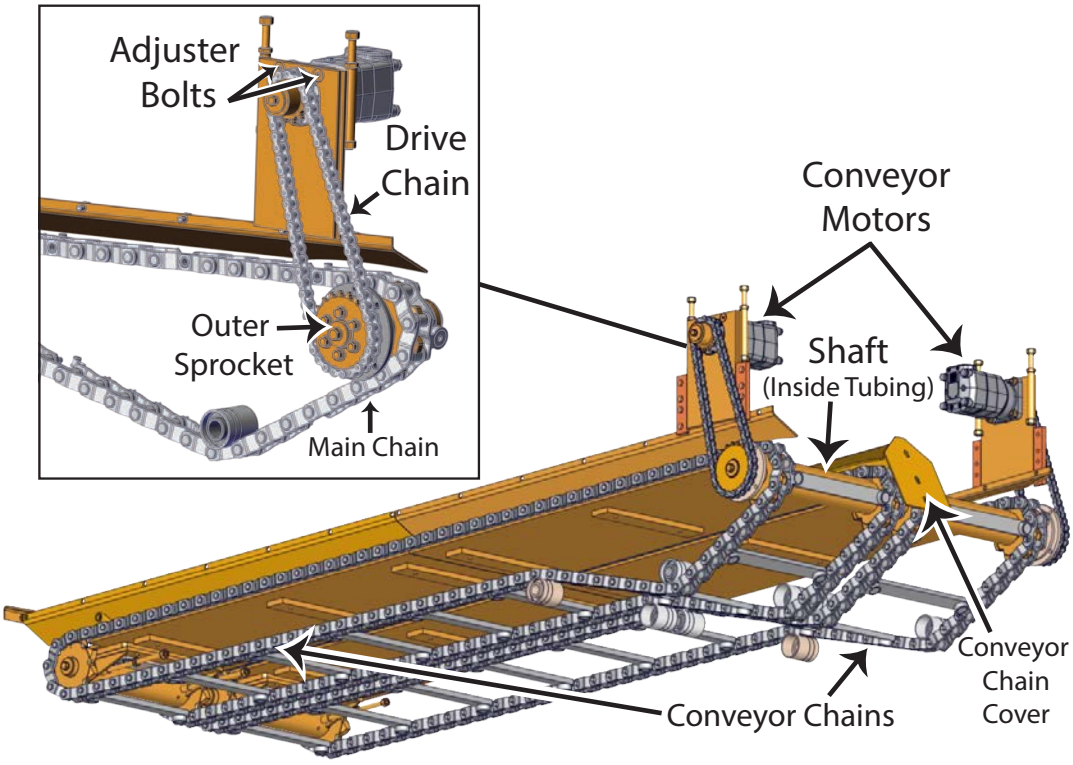


Figure 5-43. Conveyor Components

Rear Conveyor Shaft, Bearings and Sprockets

1. Raise the front lip at front of conveyor.
2. Relieve the chain tension.
3. Remove the rear conveyor chain cover. **(Figure 5-43)**
4. Remove the drive chain.
5. Remove the bolt and washers from the outer sprocket. **(Figures 5-44 and 5-45)**

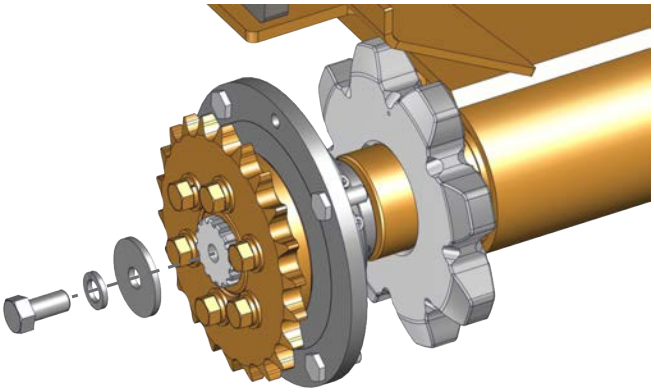


Figure 5-44. Remove Outer Bolt and Washers

6. Remove the remaining six bolts and washers. **(Figure 5-45)**
7. Remove the outer sprocket weldment.
8. Remove the conveyor drive coupling weldment.

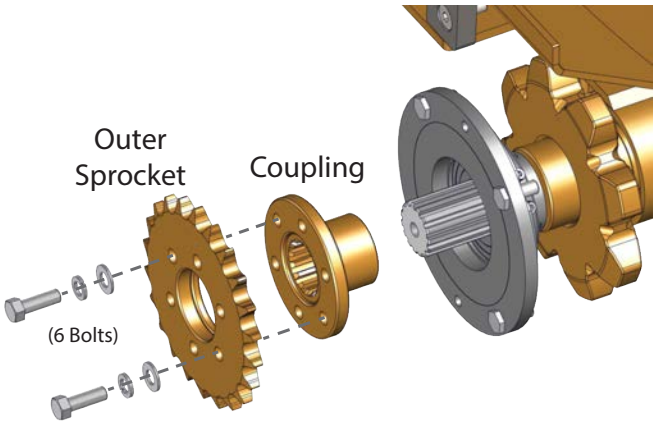


Figure 5-45. Remove Outer Sprocket and Coupling

9. Loosen and remove set screws from the outer drive sprocket bearing, then remove. **(Figure 5-46)**
10. Remove the outer conveyor sprocket bearing and mount plate (attached to the frame).
11. Remove the outer conveyor sprocket weldment.
12. Remove the hardware from the inner side of the shaft, then remove shaft.

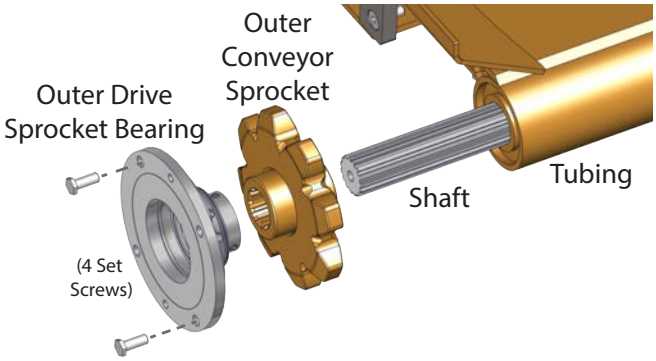


Figure 5-46. Remove Outer Drive Sprocket Weldment

13. Remove the hardware from the inner conveyor sprocket on other side of the shaft, then remove shaft. **(Figure 5-47)**
14. Remove outer conveyor bearings.
15. If replacing inner conveyor bearings and/or sprocket, remove using the same procedure as **Steps 5 - 11.**

NOTE: Rear inside tubing can be removed for changing the inner bearings.

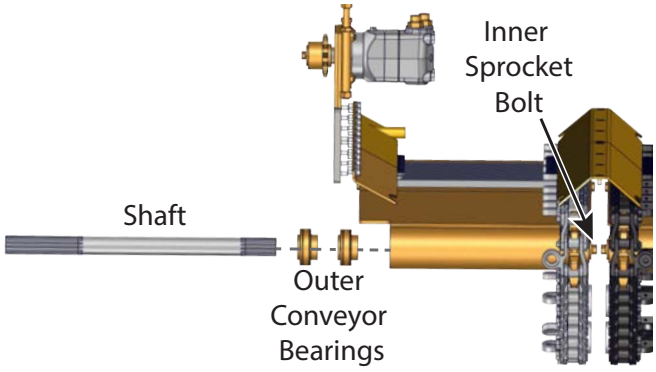


Figure 5-47. Remove Outer Sprocket and Coupling

16. Reinstall components in reverse order, replacing worn or damaged components with new ones, ensuring proper alignment.

NOTE: The outer conveyor sprocket must be aligned with the inner sprocket.

Conveyor Drive Motors

1. Relieve the conveyor chain tension. **(Figure 5-43)**
2. Remove the drive chain.
3. Disconnect hydraulic hoses from conveyor motor.
4. Remove four nuts and cap screws holding hydraulic motor to the slide mount. **(Figure 5-48)**

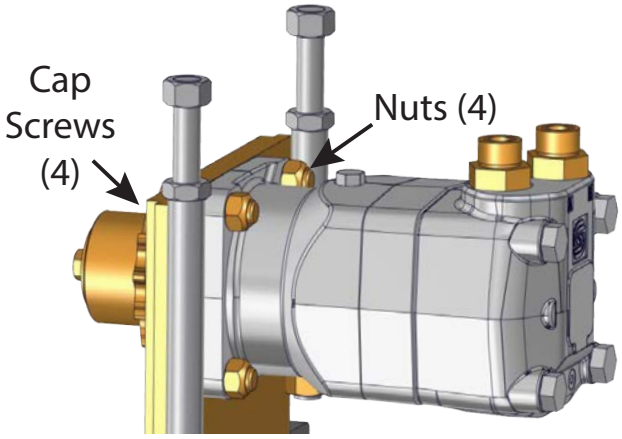


Figure 5-48. Remove Conveyor Drive Motor

5. Carefully slide the conveyor drive motor out of slide mount.
6. Slide new or repaired drive motor into slide mount.
7. Secure with cap screws and nuts.
8. Reconnect hydraulic hoses to motor.
9. Reinstall the drive chain and readjust conveyor chain tension.
10. Check for leaks.

Figure 5-49. Hopper Wing Cylinders

Front Hopper Wing Cylinders

Use the following procedure to replace the front hopper wing cylinder(s). **(Figure 5-49)**

1. Retract hopper wings completely.
2. Secure closed wing so it cannot fall open after cylinder is disconnected.
3. Disconnect hydraulic hose.
4. Remove clevis pin from the top cylinder mount and bolt from the bottom cylinder mount.
5. Slide cylinder off.
6. Slide new cylinder onto the cylinder mount.
7. Reinstall bolt.
8. Clean hydraulic hoses and connect to the cylinder.
9. Extend cylinder rod so that the hole in rod lines up with upper mount.
10. Attach clevis pin.

Rear Hopper Wing Cylinders

Use the following procedure to replace the rear hopper wing cylinder(s). **(Figure 5-49)**

1. Follow **Steps 1 - 2** in the previous section.
2. Raise the hood to access the bolt cover at the rod end of the cylinder.
3. Remove top bolt.
4. Fold out cylinder and disconnect hoses.
5. Disconnect the cylinder mount and slide off cylinder.
6. Slide new cylinder onto the mount.
7. Cover bolts with thread-locking adhesive before attaching new cylinder back onto the mount, then torque bolts ***(see Section 2).***
8. Clean the hydraulic hoses and reattach.
9. Extend the cylinder so that the bolt at the rod end lines up with the mount.
10. Cover the mount bolt with thread-locking adhesive and tighten.
11. Reattach bolt cover and close hood.

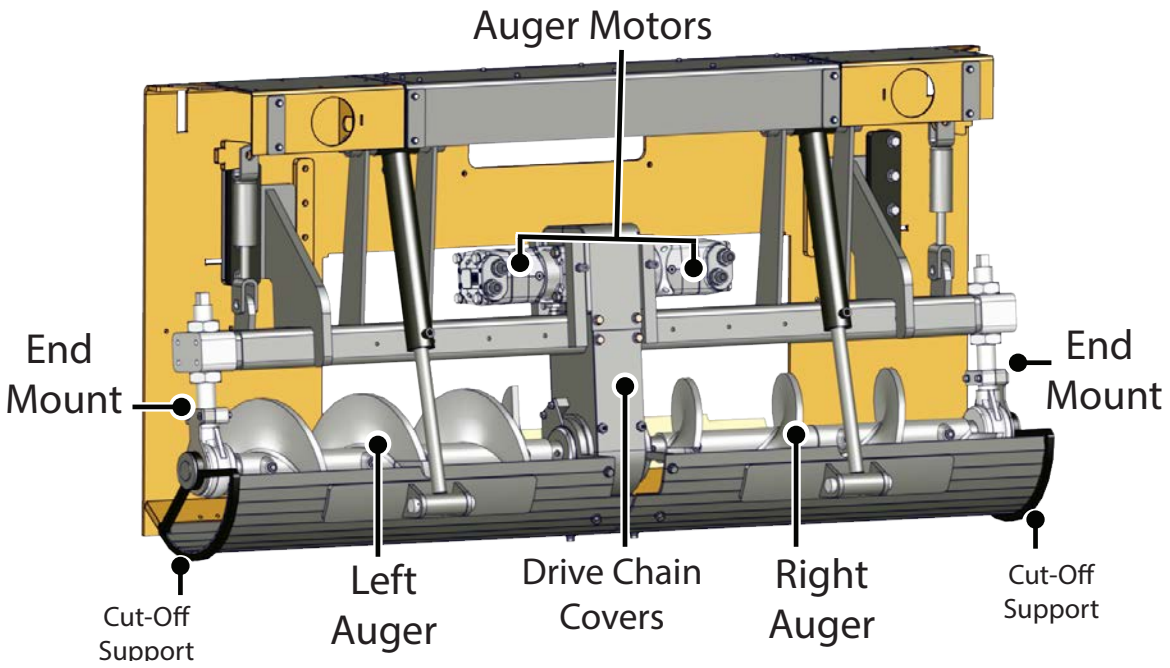


Figure 5-50. Auger Components

Auger Motors

1. Extend screed extension completely.
2. Clean asphalt buildup around the motor(s).
3. Disconnect and label the hydraulic hoses. Plug fittings.
4. Loosen the auger tension adjuster(s). **(Figure 5-51)**

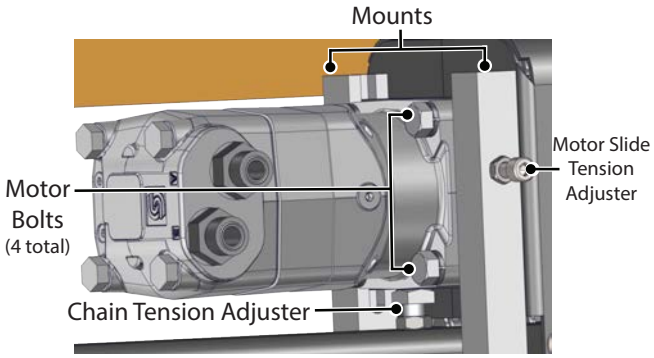


Figure 5-51. Auger Motor Replacement

5. Remove the cap screw on each side of the motor mount and remove mount.
6. Remove the four motor bolts and washers, and carefully slide out the auger motor.
7. Align and reattach the motor mounts.
8. Cover motor bolts with thread-locking adhesive, reinsert and tighten.
9. Align and reattach the motor mounts.
10. Retighten the tension adjuster to achieve 1/4 inches of auger chain slack.
11. Lubricate with chain lube.
12. Reconnect the hydraulic hoses and check for leaks.

Auger End Mounts

1. Extend the screed extensions completely and open the Under Auger Material Cutoffs.
2. Remove the auger cover(s) bolts (4) and washers attaching auger cover to the auger chain cover. **(Figure 5-52)**

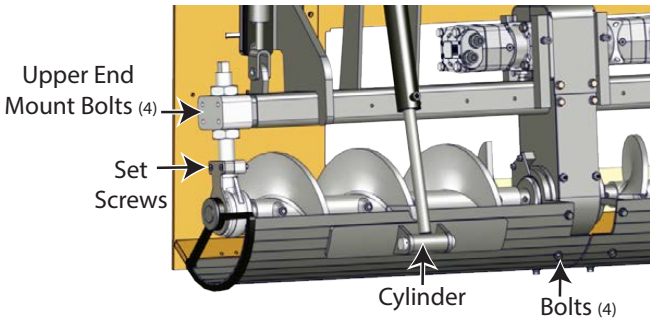


Figure 5-52. Replace Auger End Mounts

3. Remove the clevis pins holding the cylinder onto the cover, and carefully pull the cover outward away from the frame.
4. Remove the four (4) upper end mount bolts from the auger support beam.
5. Loosen the two set screws on the auger end mount collar, then slide the end mount and collar off the auger shaft.
6. Replace with new end mount, ensuring the end mount assembly is properly aligned with the auger support beam and auger shaft.
7. Retighten the auger shaft set screws and reinstall upper end mount bolts.
8. Align the cover with the auger shaft and reattach with the four bolts and torque. ***(See Section 2 for torque specifications.)***
9. Align and reinsert the clevis pins holding the cylinder in place.

Auger Inner Bearings

In time, the auger and inner bearings may become worn from use and need replacement. **(Figure 5-53)**

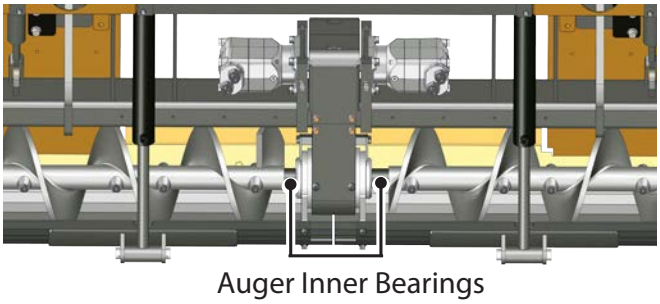


Figure 5-53. Auger Inner Bearings

To replace the auger inner bearings:

1. Extend the screed extensions completely and open the Under Auger Material Cutoffs. You will need to swing out the operator platforms for space to work.
2. Loosen the four (4) bolts on each auger motor mounting bracket. **(Figure 5-54)**

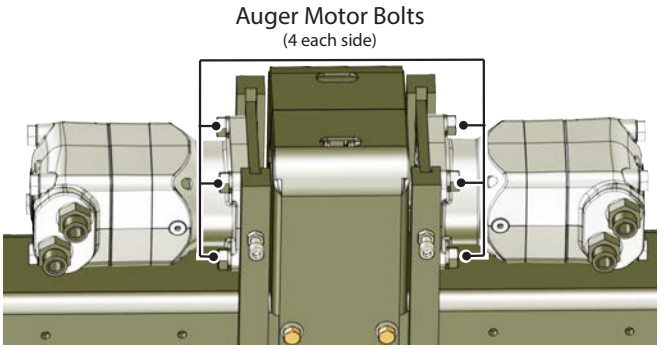


Figure 5-54. Auger Motor Bolts

3. Loosen the two (2) drive chain tensioners. **(Figure 5-51 *on the previous page.)*** The auger motors should slide down as you loosen the tensioners (tap with a rubber mallet if needed).
4. Remove the four (4) auger end mount bolts on each side of the auger, then remove end mount. **(Figure 5-52)**
5. Remove the auger inner shaft bolts. **(Figure 5-55)**

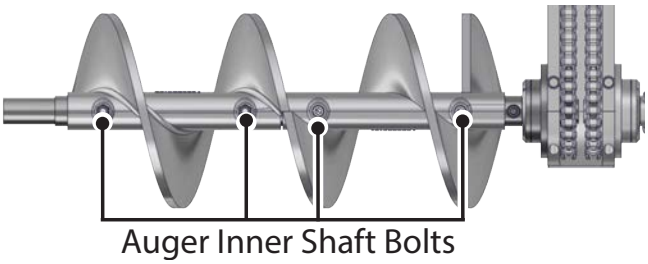


Figure 5-55. Auger Inner Shaft Bolts

6. Carefully remove the auger.
7. Remove the four (4) bolts from the inner bearing, then remove bearing and replace with new one. **(Figure 5-56)**

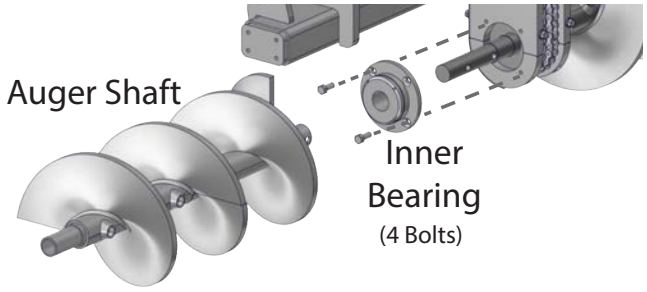


Figure 5-56. Remove Auger Inner Bearing

8. Reach in between the front of the auger and paver frame to remove the auger motor cover. Then remove the six (6) bolts holding the auger support cover in place and remove. **(Figure 5-50 *on Page 5-43.)***
   - You may need to get under the paver if you can't reach the lower two bolts on the auger support cover.
9. Remove the auger inner bearing clamps holding inner bearings into place.
10. Align new augers the same as removed augers, ensuring they are on correct side to auger material outward.
11. Reinstall auger components in reverse order.
    - Feed auger chains back onto the sprocket, ensuring chains have approximately 1/4" of slack.
    - Adjust the auger motors up by tightening the drive chain tensioners, then tighten the motor mount bolts and torque. ***(See Torque Specifications in Section 2.)***
    - Lubricate chains.
12. Operate augers to ensure proper function.

NOTE: Auger chains should be lubricated each day by spraying oil or chain lube through the slots where the auger motor is adjusted.

Main Screed Wear Plate

The screed wear plates can wear out over time and must be replaced if the plate has worn to less than 1/4-inch thick. Poor maintenance, paver cleaning, asphalt mixes containing hard aggregate and improper material flow can contribute to excessive wear on the main screed and extension wear plates. Correct screed heating also helps prevent excessive material build-up. An uneven, worn screed plate will produce an inferior mat

1. Retract screed extensions completely.
2. Set the hydraulic tow points and crown/valley to 0 (center).
3. Lower screed onto set-up blocks and null with flight screws.
4. Remove quick-release latch and then bolts holding rear walk boards (both sides). **(Figure 5-57)**

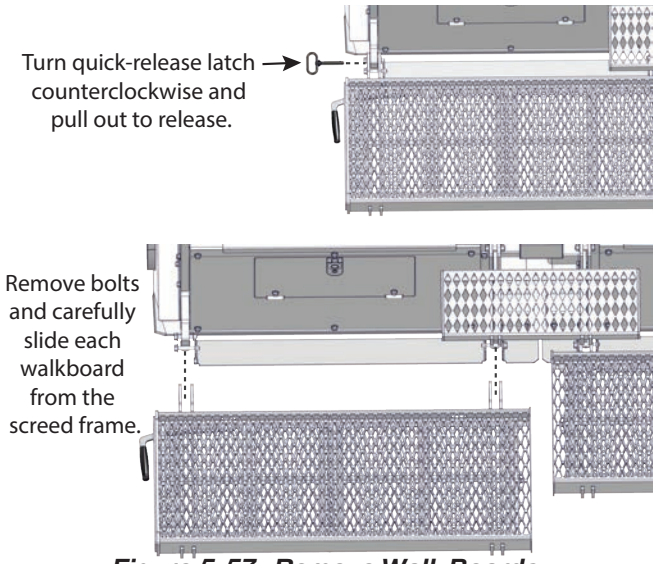


Figure 5-57. Remove Walk Boards

5

5. Remove the four (bolts) bolts holding the first screed stair and remove stair. **(Figure 5-58)**

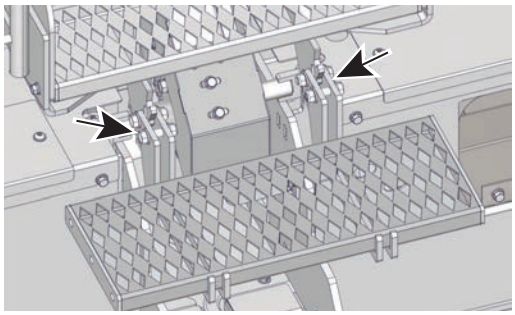


Figure 5-58. Remove First Stair

6. Remove bolts, nuts and washers on each screed cover. Remove main screed covers and disconnect wiring. **(Figure 5-59)**

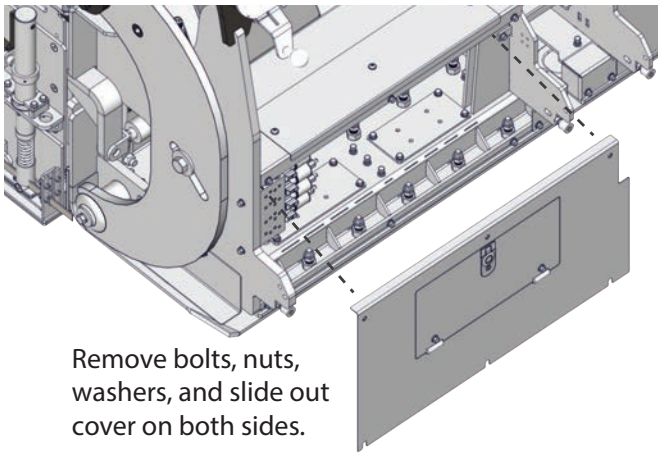


Figure 5-59. Remove Main Screed Covers

7. Install the screed lock tool (provided with your paver) between the screed frames to hold them together. **Clamp the center portion of the screed frame to hold the frame in place before lifting it from the worn wear plate. (Figure 5-60)**

Figure 5-60. Screed Lock Tool

8. Remove both rows of nuts holding the wear plate to the bottom of each side of the screed. **(Figure 5-61)**

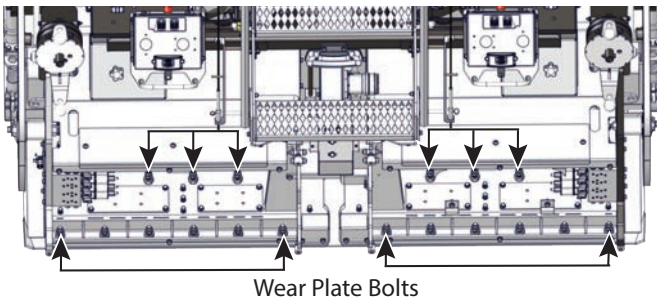


Figure 5-61. Wear Plate Bolts

9. Raise the screed completely.
10. Slide the worn wear plate from under the screed.
11. Clean all material buildup on the screed frame.
12. Set new wear plate level on three blocks, placing one block in the center and one at each end.
13. Lower the screed frame down onto the new wear plate.
14. Reinstall lock nuts and washers, but do not tighten nuts until all have been installed.
15. Adjust screed so the front bullnose of the wear plate is flat.
16. Adjust crown so the rear of the wear plate is flat. Use rear adjusters to ensure rear of wear plate is flat.
17. Reinstall the screed covers in reverse order, then the walk boards and stair.

Screed Extension Heat Box

1. Extend the screed completely.
2. Remove the sonic auger sensor if connected.
3. Remove the end gate by removing the six (6) cap screws holding the end gate. The end gate will tilt forward and slide off. **(Figure 5-62)**

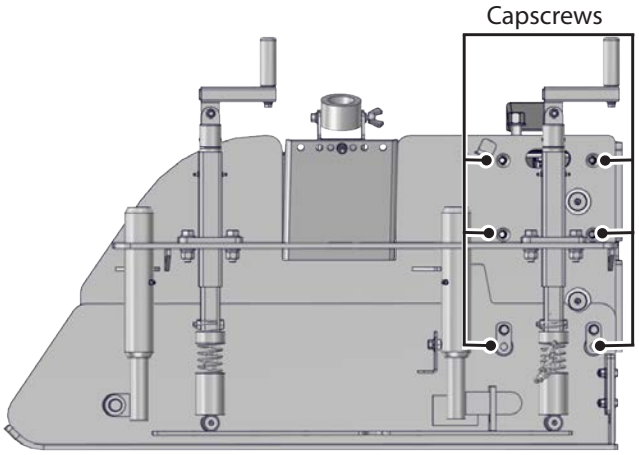


Figure 5-62. Remove End Gate

4. Disconnect the extension AOA adjuster **(Figure 5-63)** on top of the heat box by removing locknut, washer and shoulder bolt. **(Figure 5-63)**
5. Carefully slide out the hinge pin. The heat box assembly will disconnect and fall to the ground.
6. Remove the bolts and nuts holding the heat box cover, then remove.
7. Remove internal components.
8. Hold the new heat box in position and slide the hinge pin back into place.
9. Fasten the AOA adjuster back to the heat box with locknut, washer and shoulder bolt.
10. Reattach wiring.
11. Reinstall the end gate.

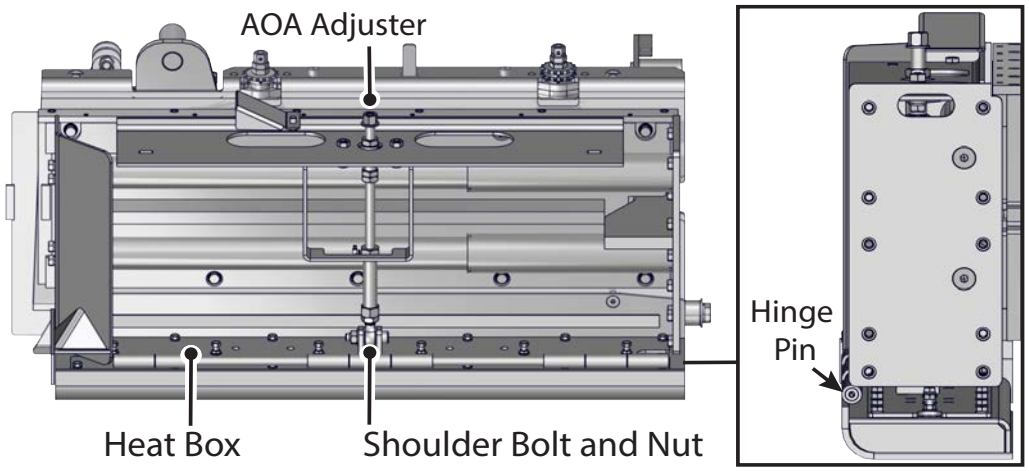


Figure 5-63. Extension Heat Box Replacement

WELDING

If the machine has been damaged or metal parts worn due to extensive use, minor welding may be necessary for repair. Welding is very dangerous. Only individuals who are properly trained should weld in a dedicated area away from others as even looking at ultraviolet light from the flame can cause serious eye injury.

Welding hazards include toxic fumes, harmful dust, light radiation, burns, combustion from sparks and the potential for extensive damage to the eye. There is also a danger of electric shock. If combustible or flammable materials are nearby, the heat and sparks produced by welding can cause fires or explosions.

TURN THE BATTERY SWITCH OFF AND DISCONNECT BATTERY TERMINALS BEFORE WELDING ON THE MACHINE.

Electrical current can ignite fire or cause an explosion, therefore it is critical to turn off power to the machine. It is also important to grind off the paint to ensure proper ground on the area being welded.

DO NOT MAKE MODIFICATIONS TO THE MACHINE. ONLY weld if needed for repair purposes. Weldment modifications to the machine can void your warranty.

Before welding on the machine:

PREPARE:

- Clear the area of any fire hazards, flammable liquids and solids, or any items that could catch on fire.
- Ensure a fire extinguisher is available in the work area.
- Hang welding blanket(s) around and underneath the work area.
- Paint and rust should be removed from the welding area to prevent arcing in other areas or components. An indirect ground can cause arcing that will damage machine components. Only connect the ground to the component being welded. DO NOT connect the ground to other bolted assemblies, wheels, bearings, motors, tracks, torque hubs, etc.

DISCONNECT:

- Turn the master battery switch to the OFF position.
- Disconnect the battery positive and negative terminals.
- Disconnect Plus 1 controller(s).
- Disconnect engine ECU(s) controller (if equipped).
- Disconnect engine DEF controller (if equipped).
- If machine has truck chassis: Disconnect the battery positive and negative terminals and unplug the ECU(s) for the engine, transmission and ABS systems if equipped. Refer to the truck owner manual for any other recommendations.

WINTERIZING FOR STORAGE

Perform the following procedures to prepare the machine for winterizing and storage.

1. Store the machine in a dry, protected area:
   - Cover seat with plastic. Place a "DO NOT OPERATE" tag on the main control panel.
   - Cover all exposed control panels with tarps or plastic.
   - Ensure all cab doors and windows are closed and seal properly.
   - Ensure the tires are properly inflated.
   - Replace worn or damaged decals if needed.
2. Inspect the machine thoroughly and repair worn or damaged parts. Inspect all hoses, couplers, fittings and cylinders. Tighten loose fittings and replace any worn hoses found.
3. Lubricate all grease points:
   - Coat all exposed cylinder shafts with lithium-based grease.
   - Ensure all exposed threads are lubricated.
   - Ensure cables are lubricated and move freely.
   - If the machine has high- and low-speed chains, be sure they are oiled.
4. Check air filter. Clean or replace air filter elements if needed.
5. Check engine oil level and fill if needed.
   - Drain dirty engine oil, fill with new oil and run the engine five minutes to allow the oil to penetrate components.

⚠WARNING To avoid the danger of exhaust fume poisoning, DO NOT operate the engine in a closed building without proper ventilation. Check hydraulic oil sight gauge and fill if needed.

NOTICE **Hydraulic oil that is contaminated must be drained, the filter elements replaced and the hydraulic tank refilled. Failure to perform this maintenance procedure if needed could result in premature failure of the pumps or motors.**

6. Ensure the fuel tank is full to prevent condensation in the fuel tank. Add cold-weather additive when appropriate.
7. If machine is equipped with Diesel Exhaust Fluid (DEF), the DEF may stored in the machine's tank up to four months. If storing any longer, drain the DEF fluid into an approved container.

NOTICE **Failure to drain the DEF tank if storing for extended periods may result in fluid deterioration that can cause DEF sensor damage. Replace the DEF fluid before operating to prevent SCR system malfunction and/or damage.**

8. If machine is equipped with a citrus tank, ensure the tank is full to prevent rust.
9. If machine is equipped with water tank(s), the tank should be drained. The pump and filter must also be purged to prevent freezing.
10. If machine is equipped with windshield washer bottle, ensure the bottle is full of window washer fluid with a cold-weather blend.
11. Perform specific gravity test on engine coolant. Drain and replace or fill with antifreeze mixture to prevent freeze damage if needed.

KUBOTA ENGINE NOTICE: If not using anti-freeze, open the cock at the bottom of radiator and remove the pressure cap to drain water completely. Leave the cock open. Place a "No Water" sign on the pressure cap.

12. Disconnect battery, adjust the electrolyte level, and recharge it. Store battery inside in a cool, dry place. Ensure breakers are in the OFF position.
13. Remove steering control boxes and electrical cords. Store inside in a cool, dry place.
    - If equipped with any plugged-in remote sensors (i.e., sonic auger sensor), disconnect, remove and store inside in a cool dry place.
14. Cover the exhaust and intake filter. Be sure to seal the exhaust cover with tape to prevent moisture from entering, especially if storing outside.
15. If the machine is equipped with a propane system, remove the propane gas bottle and store in a safe area away from the machine. Purge the gas system and seal connection(s).
16. Lower machine components and attachments (i.e., screed, broom, blade, etc.) to the lowest position, resting on blocks to prevent ground contact or pavement damage.
17. Clean the machine, including the undercarriage. Paint chipped or rusty areas to prevent further rusting. Lock all lockable panels, cab doors and spare tire. Remove ignition key and store keys in a safe place.
18. If storing machine for an extended time, run the engine five minutes every two to three months (under no load) to keep engine free from rust caused by moisture in the air. If you do not run the engine for six months or longer, apply engine oil to the valve guide and stem seal before starting.

TROUBLESHOOTING

Hydraulic Pressure

The hydraulic pressure screen on the DM430 digital controller provides an easy and convenient means of checking pressure for the five main functions of the paver: Forward propel (left and right sides), conveyors, cylinders, and augers. **(Figure 5-61)**

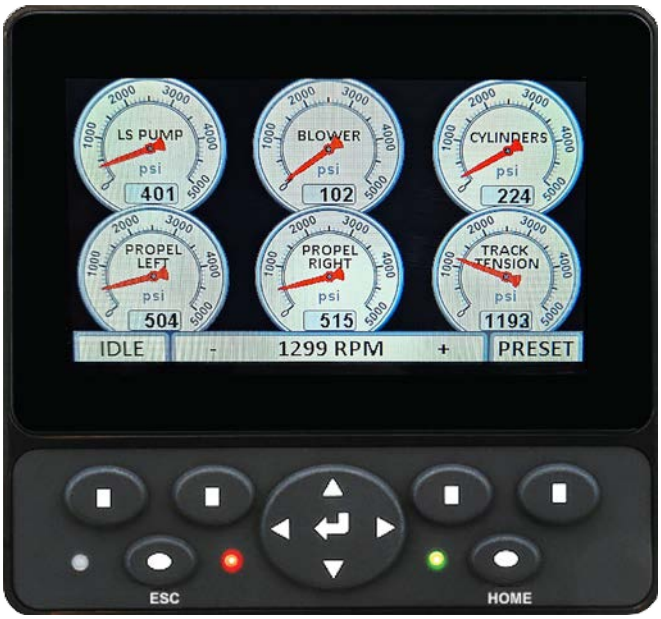


Figure 5-64. Pressure Gauge Screen

Relief pressure PSI for the paver functions are shown in the following table:

Table 5-4. Relief PSI Chart

ITEM	SPECIFICATION
Drive	5510 psi
Conveyors/Augers (Load Sense)	3480 psi
Cylinders	3000 psi
Track Tension Pressure	1200 psi
Track Tension Relieving	3600 psi
Standby Pressure	350 psi
Charge Pressure	500 psi

Track Tension Pressure Relief

The hydraulic adjustment cylinders are automatic and provide even tension on the tracks to prevent excessive wear to the paver's undercarriage. The track tension main manifold is mounted on the side of the lower manifold on the left side of the paver. There are two relief cartridge vales that function independently with ports for the left track (LT) and right track (RT). **(Figure 5-62)**

The track hydraulic pressure is set at 900 PSI at the track tension main manifold.

To check the relief pressure, perform the following procedure:

1. Connect a 3000 PSI gauge in-line with either Port LT or RT on the track tension main manifold.
2. Place a Porta Power between the front idler and frame to release pressure on the idler.
3. Increase the Porta Power pressure until the front idler cylinder compresses.
4. Pressure should read 3600 PSI.
5. If the pressure is not correct, remove cap on the relief cartridge valve. Using an allen wrench, adjust the relief cartridge valve IN for more pressure or OUT for less pressure until the PSI gauge reads 3600 PSI.
6. When the pressure gauge reads 3600 PSI, disconnect Porta Power and gauge.

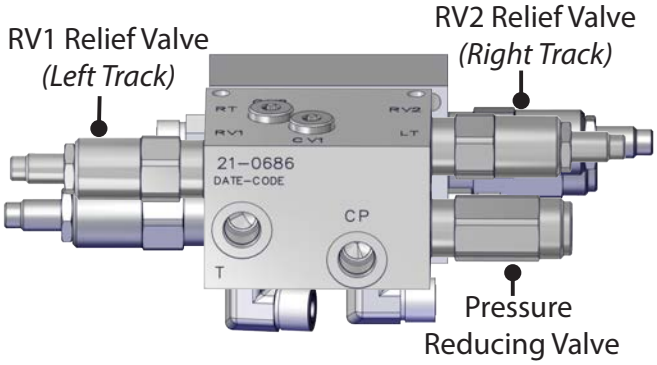


Figure 5-65. Track Tension Main Manifold

Spray Down Pump

The following troubleshooting information is provided in the event of any issues with the spray down pump (located inside the left engine access door). **(Figure 5-63)**

Figure 5-66. Spray Down Pump Location

An electronic pressure sensor displays solid or flashing LED lights on top of the pressure switch for pump status notifications. **(Figure 5-62)**

Green:

- Solid green means the pump is spraying.
- Flashing green means the pump is in a stand-by mode, ready to spray.

Yellow:

- Solid yellow means the pump is priming.

Yellow/Blue:

- Flashing yellow and blue means the pump has incorrect voltage.

Red/Blue:

- Flashing red and blue means the pump has timed-out without fluid. Cycle pump power to reset.

Red:

- Solid or flashing red light means there is an error. Consult owner's manual for troubleshooting.

NOTE: If an overload occurs, the pump stops and a yellow-green LED will flash. The electronic system will try to reactivate the pump at preset intervals.

If the overload reaches a dangerous level, the red LED will become solid and you must switch the power off and start the pump again.

NOTE: When starting the spray down pump or emptying the tank, open the bleeder valve slightly to expel air and facilitate priming. (Figure 5-64) As soon as the pump is operating, close the bleeder valve.

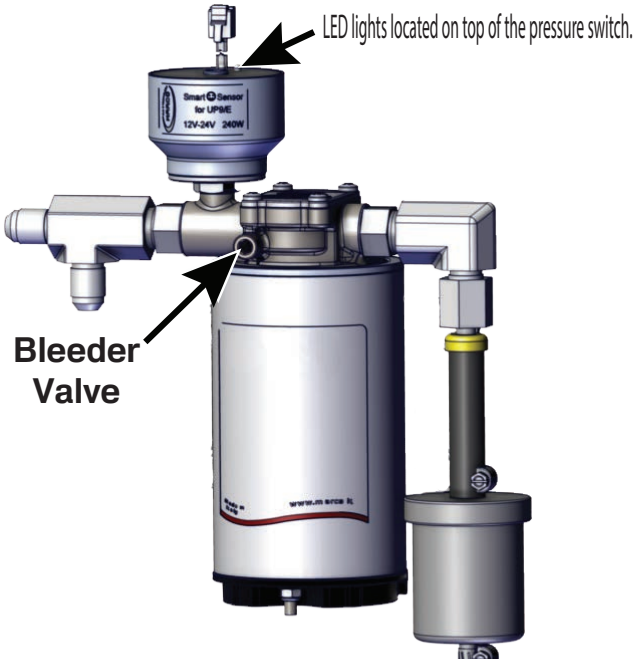


Figure 5-67. Spray Down Pump

If the spray down pump has stopped or will not start:

1. Check the effectiveness of the battery power supply (voltage activity).
2. Check that the fuse has not blown.
3. Check for any foreign material present in the hose.
4. Avoid running the pump dry for more than a few minutes. (Pumps run dry by the absence of fluid are not covered by warranty.)
5. If the pump does not prime itself:
   - Raise the screed completely, ensuring citrus tank on the screed is filled.
   - Loosen the bleeder valve inside of the cut-out until the pump primes.
   - Retighten the bleeder valve.

Conveyor Chain Tension Slack Adjustment

The conveyor chains will need periodic adjustment. Generally, the slack of a roller chain should be on the lower side. **(Figure 5-65)**

Adequate slack (S) is about four (4) percent of the span for normal drives, but the slack should be about two (2) percent of the span for the auger chains:

- Vertical drive or close to vertical drive.
- Center distance between two shafts is greater than three (3) feet.
- Chain is operated under heavy load and high frequency of on and off drive.
- Direction of the drive is often changed.

⚠CAUTION **Conveyor chains must have a slack of 3/8 to 1/2 inch. To measure deflection, exert 30 pounds of force to chain in either direction.**

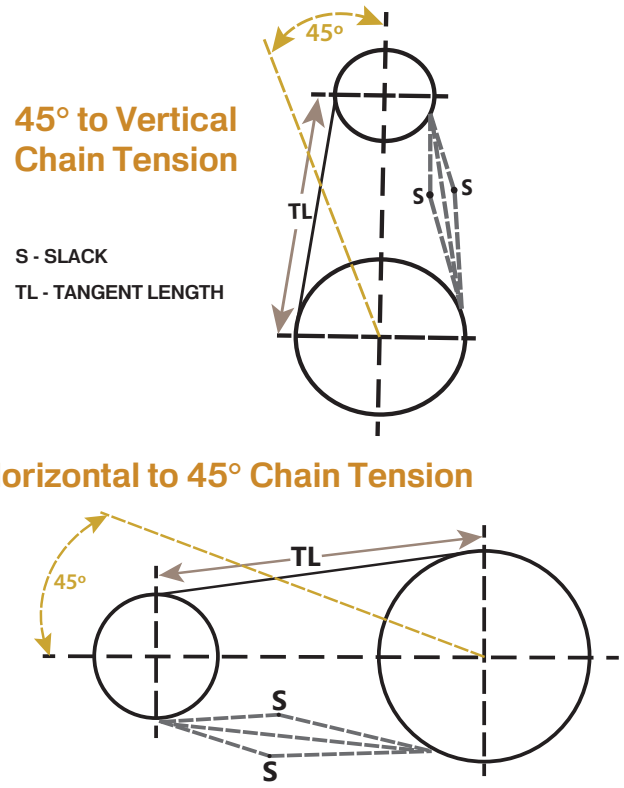


Figure 5-68. Chain Slack Adjustment for Proper Tension

REQUIRED MID-SPAN MOVEMENT

	Tangent Length (TL) Between Sprockets in Inches (Centimeters)					
	10 (25)	20 (51)	30 (76)	50 (127)	70 (178)	100 (254)
	Tangent Length Between Sprockets in Inches (Millimeters)					
Horizontal To 45°	0.4 - 0.6 (10 - 15)	0.8 - 1.2 (20 - 30)	1.2 - 1.8 (30 - 46)	2.0 - 3.0 (51 - 76)	2.8 - 4.2 (71 - 107)	4.0 - 6.0 (102 - 152)
45° To Vertical	0.2 - 0.3 (5 - 8)	0.4 - 0.6 (10 - 15)	0.6 - 0.9 (15 - 23)	1.0 - 1.5 (25 - 38)	1.4 - 2.1 (36 - 53)	2.0 - 3.0 (51 - 76)

TROUBLESHOOTING CHART

The troubleshooting chart below identifies the most common symptoms of mechanical problems. Fault codes are also shown on the DM430 digital display unit on the operator control panel consoles.

Contact your authorized LeeBoy dealer for assistance, service or repair.

Table 5-1. Paver Troubleshooting

SYMPTOM	CAUSE	REMEDY
Engine does not start.	Defective battery or low battery charge.	Replace or charge battery as applicable.
	Master switch is not set at the ON position.	Set switch to the ON position.
	Insufficient fuel supply.	Fill fuel tank.
	Safety switch is faulty.	Replace.
	Solenoid doesn't function properly.	Ensure wires are tight. Clean solenoid plunger. Replace solenoid coil. Replace or rebuild starter or solenoid.
	Switch box unplugged.	Replug switch box.
	Neutral switch defective.	Replace.
	Engine starter relay faulty.	Replace.
	E-Stop engaged.	Disengage E-Stop.
Engine turns over but does not start.	Low fuel.	Add fuel to fuel tank.
Engine is difficult to start.	Fuel is thick and doesn't flow.	Check the fuel tank and remove water, dirt or other impurities. Check the fuel filter cartridge and replace if necessary.
	Engine oil becomes thick in cold weather.	Use oil of different viscosities suitable to ambient temperatures. (Use 10W-30 in winter.)
	Battery.	Charge battery.
Engine suddenly stops.	Lack of fuel.	Refuel. Check the fuel system. Bleed the fuel system if necessary.
	Moving parts overheated.	Contact your authorized dealer.
	E-Stop button is engaged.	Check E-Stop button setting.
	Forced stop due to an engine issue.	The ECU may have forced the engine to a stop. Check the engine warning shown on the DM430 digital display and/or Diagnostic Trouble Code. If you cannot repair the problem, contact your authorized dealer.

SYMPTOM	CAUSE	REMEDY
Engine overheats.	Low oil.	Check oil level. Add oil if needed.
	Fan belt stretched, damaged, or broken.	Adjust belt tension or replace.
	Coolant low.	Add more coolant.
	Excessive concentration of antifreeze.	Add water only. Coolant should always be mixed 50/50 of coolant and water.
	Radiator net or fin clogged with dust and dirt.	Clean or replace defective parts.
	Thermostat defective.	Check thermostat and replace if needed.
	Defective temperature gauge or sensor.	Check temperature with a thermometer. Replace gauge or sensor if needed.
	Overload running.	Reduce load.
	Head gasket leaking.	Replace gasket.
Low battery.	Faulty alternator.	Replace or rebuild.
	Indicator light, bad bulb.	Replace.
Machine will not move.	RUN/STOP switch faulty.	Check RUN/STOP switch.
Machine does not steer straight.	One of the hydraulic drive motors needs adjustment.	Readjust motor(s).
	Steering control is not centered.	Center steering control.
	Travel pump defective.	Replace or rebuild travel pump.
Machine does not change speed when the 2-Speed switch is engaged.	Defective relay.	Replace relay.
	Defective solenoid.	Replace solenoid.
	Defective drive motor.	Replace drive motor.
Conveyor does not function on one or both sides.	Defective Automatic/Manual switch.	Replace switch.
	Solenoid defective.	Replace.
	Feeder drive chain broken.	Repair or replace chain.
	Defective conveyor motor.	Replace.
	Rear conveyor shaft broken.	Replace.
Tracks don't operate smoothly.	Tracks too loose.	Tighten track pad bolts.
	Track rollers worn.	Replace.
	Track tension pressure.	Check tension pressure and relieve or adjust.
Machine will not pull on one or both sides.	Faulty hydraulic motor.	Replace.
	Pump pressure too low.	Check pump pressure and relieve or adjust.
	Faulty torque hub.	Rebuild or replace.
Engine runs but there is no hydraulic pressure.	Pump drive coupling faulty.	Replace.
	Defective pump.	Replace.

SYMPTOM	CAUSE	REMEDY
Auger not turning properly.	Drive chain is too loose.	Adjust.
	Drive chain broken.	Replace.
	Faulty motor.	Replace.
	Solenoid valve defective.	Replace solenoid.
	Asphalt buildup.	Clean thoroughly and grease.
	Overload of material to augers.	Set conveyor to proper speed in AUTO or use paddles in MANUAL position.
Screed extensions binding.	Asphalt buildup.	Keep clean and greased.
Screed extension loose.	Out of adjustment.	Adjust hold-downs on extensions.
	Screed worn out or damaged.	Replace.
	Extensions set too low.	Adjust extension. Always begin operation with extensions extended up with no down pressure.
	Screed not heated properly.	Check elements and heat control box.
Screed leaving ripples in pavement.	Extension set too low.	Readjust extensions.
	Extensions work up and down.	Adjust top guide.
	Extension rod bushings worn.	Replace bushings.
Flight screw locking up.	Twisting screed too far.	Give screed time to react.
	Screw seized.	Replace screw.
Flight screw bearing damage.	Twisting screed too far.	Give screed time to react.
	Loading and unloading.	Check ramps for easy access.
Hydraulic oil leaking from breather cap.	Hydraulic oil tank overfilled.	Drain to proper fill level.
	Oil overheated.	Slow paver down.
		Check oil cooler and thermostat.
Hydraulic pump cavitation or lost power.	Low level in hydraulic tank.	Refill to proper level.
	Clogged filters.	Replace filters.
	Suction hose loose.	Retighten.
	Charge pump worn.	Rebuild or replace charge pump.
Feeder does not work on one or both sides.	Defective switch.	Adjust or replace switch.
	Solenoid defective.	Replace solenoid.
	Feeder drive chain broken.	Adjust or repair chain.
	Defective conveyor motor.	Replace motor.
	Rear conveyor shaft broken.	Replace conveyor shaft.
Feeder flight bars don't function properly.	Flight chains too loose.	Adjust.
	Feeder drive chain too loose.	Adjust.
Loss of power to drives.	Relief out of adjustment.	Check pressure and adjust if needed.
	Piston groups worn.	Replace.
	No charge pressure.	Check that charge pressure is 280 - 425 PSI.

SYMPTOM	CAUSE	REMEDY
Electric screed heating system will not operate.	Control box power switch not in ON position.	Ensure that the screed operator control box Power switch is ON.
	Breakers are in a “tripped” position.	Ensure all element breakers are in the proper set position.
	Generator malfunction.	Check voltage and frequency. Rebuild or replace if needed.
Only part of the electric screed heats.	Screed section not plugged into bottom of the control box.	Replug.
	Element breakers for screed section in a “tripped” condition.	Ensure the element breakers for that screed section are not “tripped.”
Electric screed heat system doesn’t start, or starts and doesn’t run very long.	Heating system timed out.	Check timing relay. Adjust or replace.
Electric Screed is heating, but never gets hot enough to pave.	Elements improperly clamped.	Ensure elements are clamped properly.
	Generator malfunction.	Ensure generator is “spinning.” Check voltage and frequency. Rebuild or replace if needed.
Elements have been tested, but the breaker still trips.	Faulty element wiring.	Inspect and test element wiring.

DISPLAY DIAGNOSTIC TROUBLE CODE (DTC) CHART

LEGEND

SPN: Suspect Parameter Number
DPF: Diesel Particulate Filter
FMI: Failure Mode Identifier
EGR: Exhaust Gas Recirculation
ECU: Engine Control Unit
CPU: Central Processing Unit
NE-G: Negative Ground
NE: Neutral
G: Ground
SCV: Selective Control Valve
MAF: Mass Air Flow
PM: Particulate Matter
DC: Direct Current
IC: Integrated Circuit
EBC: Electronic Brake Control
PLV: Pressure Limiting Valve
CAN: Controller Area Network
CM: Control Module
DDC: Direct Digital Control
SW: Switch
TSC: Torque Speed Control
ETC: Electronic Throttle Control
CCVS: Current Controlled Voltage Source
MPROP: Manifold Pressure Rail Operating Pressure
EEPROM: Electrically Erasable Programmable Read-Only Memory

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
29	3	Accelerator position sensor 2: High	Battery short circuit out of sensor/ harness. Voltage of accelerator position sensor 2 is 4.8V or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
29	4	Accelerator position sensor 2: Low	Ground short circuit or open circuit of sensor/harness. Voltage of accelerator position sensor 2 is 0.3V or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
91	2	Accelerator position sensor correlation error	Deviation from designed correlation in two sensors. Voltage of accelerator position sensor 1 is 4.8V or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
91	3	Accelerator position sensor 1: High	Battery short circuit or open circuit of sensor/harness. Voltage of accelerator position sensor 1 is 4.8V or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
91	4	Accelerator position sensor 1: Low	Ground short circuit or open circuit of sensor/harness. Voltage of accelerator position sensor 1 is 0.3 or less.	Forced idle. (Accelerator = 0%) Diagnostic counter = zero
100	1	Oil pressure error.	Engine oil pressure.	Key switch turn OFF.
102	3	Boost pressure sensor: High	Failure of sensor. Voltage of boost pressure sensor is 4.9V or above.	Key switch turn OFF. (Default value is set in consideration with high altitude usage. Engine power drops.)
102	4	Boost pressure sensor: Low	Failure of sensor. Voltage of boost pressure sensor is 0.2V or below.	Key switch turn OFF. (Default value is set in consideration with high altitude usage. Engine power drops.)
108	3	Barometric pressure sensor error: High	Barometric voltage 4.4V or more.	Diagnostic counter = zero (Default value is set in consideration with high altitude usage. Engine power drops.)

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
108	4	Barometric pressure sensor error: Low	Barometric voltage 1.6V or less.	Diagnostic counter = zero (Default value is set in consideration with high altitude usage. Engine power drops.)
110	0	Engine overheat.	Overheat of engine temperature greater than 248° F (120° C)	Diagnostic counter = zero Output limitation: approximately 75% of normal condition. EGR stop. Intake throttle 100% open.
110	3	Coolant temperature sensor: High	Open circuit or +B short circuit of sensor/harness. Voltage of coolant temperature sensor is 4.9V or above.	Key switch turn OFF. (White smoke increases at low temperature.)
110	4	Coolant temperature sensor: Low	Ground short circuit of sensor/ harness. Voltage of coolant temperature sensor is 0.1V or less.	Key switch turn OFF. (White smoke increases at low temperature.)
132	1	Intake air volume: Low	Engine inlet air mass flow rate tacking. (Disconnect turbo blower intake hose.)	Key switch turn OFF.
132	3	MAF sensor: High	Engine inlet air mass flow rate less than half of target value. MAF sensor voltage: 4.9V or more at normal operating condition.	Key switch turn OFF. **Engine is not forcibly stopped by ECU. STRONGLY recommend operator stop engine promptly.**
132	4	MAF sensor: Low	Open circuit or ground short circuit of sensor/harness. MAF sensor voltage: 0.1V or less.	Key switch turn OFF. **Engine is not forcibly stopped by ECU. STRONGLY recommend operator stop engine promptly.**
132	15	Boost pressure: Low	Boost pressure sensor output is below target level in high air flow operating conditions.	Key switch turn OFF. EGR stop. Engine power is restricted by boost pressure signal accordingly.
157	0	High rail pressure.	Actual pressure exceeds the command pressure. Actual pressure ≥ 197MPa (28600 psi or 2010 kgf/cm^2	Diagnostic counter = zero EGR stop.
157	3	Rail pressure sensor: High	Open circuit or +B short circuit of sensor/harness. Failure of sensor. Voltage or rail pressure sensor is 4.9V or above.	Key switch turn OFF. Engine running noise and white smoke increases. Engine forcibly stopped.
157	4	Rail pressure sensor: Low	Ground short circuit of sensor/ harness. Failure of sensor. Voltage of rail pressure sensor is 0.7V or less.	Key switch turn OFF. Engine running noise and white smoke increases. Engine forcibly stopped.
168	3	Battery voltage: High	Open circuit, short circuit or damage to harness. Failure of battery. ECU recognition of battery voltage is above 16V in 12V system. Faulty starting.	Key switch turn OFF. EGR stop.
168	4	Battery voltage: Low	Open circuit, short circuit or damage to harness. Failure of battery. ECU recognition of battery voltage is below 8V in 12V system. Faulty starting or engine may stop completely. (Not monitored during cranking.)	Diagnostic counter = zero EGR stop.

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
171	3	Intake air temperature built-in MAF sensor: High	Open circuit or +B short circuit of sensor/harness. Intake air temperature built-in MAF sensor voltage: 4.9V or more.	Diagnostic counter = zero
171	4	Intake air temperature built-in MAF sensor: Low	Ground short circuit of sensor/ harness. Intake air temperature built-in MAF sensor voltage: 0.1V or less.	Diagnostic counter = zero
172	0	Intake air temperature: High	Intake air temperature higher than 140° F (60° C).	Key switch turn OFF.
172	3	Intake air temperature error: High	Open circuit or +B short circuit of sensor/harness. Intake air temperature sensor is 4.9V or above.	Diagnostic counter = zero (White smoke increases at low temperature.)
172	4	Intake air temperature error: Low	Ground short circuit of sensor/ harness. Intake air temperature sensor is 0.05V or less.	Diagnostic counter = zero (White smoke increases at low temperature.)
174	0	Fuel temperature high.	Fuel temperature higher than 194° F (90° C).	Diagnostic counter = zero
174	3	Fuel temperature sensor: High	Open circuit or +B short circuit of sensor/harness. Voltage of temperature sensor in supply pump is 4.9V or above.	Diagnostic counter = zero
174	4	Fuel temperature sensor: Low	Ground short circuit of sensor/ harness. Voltage of temperature sensor in supply pump is 0.1V or less.	Diagnostic counter = zero
190	0	Engine speed.	Engine speed exceeds threshold speed. Engine speed ≥ 3500 min-1 (rpm).	Diagnostic counter = zero
628	2	Program memory.	Check-Sum error.	Key switch turn OFF. Engine stopped without delay.
633	7	Pressure limiter emergency open.	Combination of fuel leak and one of conditions below: 1. Rail pressure exceeds 191MPa (27700 psi or 1950 kgf/cm^2). 2. Within one second after the rail pressure goes below temperature in #1, it increases above.	Key switch turn OFF. EGR stop. (Engine speed may go down due to lack of fuel pressure.)
636	2	Crank position sensor pulse number error.	Open circuit or short circuit of sensor/ harness. Failure of sensor. Pulse rotation is not 56 teeth. Faulty starting. Engine vibration increases slightly.	Diagnostic counter = zero
636	7	NE-G phase shift. NE: Crankshaft position sensor. G: Camshaft position sensor.	Large phase shift between NE and G pulses within +/- 15 degrees. Engine hesitates at start-up.	Diagnostic counter = zero
636	8	No input of NE sensor pulse.	Open circuit or short circuit of sensor/ harness. Failure of sensor. No recognition of NE sensor pulse. Faulty starting. Engine vibration increases slightly.	Diagnostic counter = zero

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
651	3	Open circuit of harness/ coil in 1st cylinder injector.	Open circuit of harness or injector coil. Engine vibration increases. Injectors that have no error are operated.	Key switch turn OFF.
652	3	Open circuit of harness/ coil in 2nd cylinder injector.	Open circuit of harness or injector coil. Engine vibration increases. Injectors that have no error are operated.	Key switch turn OFF.
653	3	Open circuit of harness/ coil in 3rd cylinder injector.	Open circuit of harness or injector coil. Engine vibration increases. Injectors that have no error are operated.	Key switch turn OFF.
654	3	Open circuit of harness/ coil in 4th cylinder injector.	Open circuit of harness or injector coil. Engine vibration increases. Injectors that have no error are operated.	Key switch turn OFF.
676	0	Glow heater relay driving circuit overheat.	Glow relay coil resistance or load is too high. Faulty starting. White smoke increases.	Key switch turn OFF.
676	5	Open circuit of glow relay driving circuit.	Open circuit of harness or relay coil. Faulty starting. White smoke increases.	Key switch turn OFF.
679	7	Pressure limiter not open.	Rail pressure value is sticking or not enough engine power to open PLV forcibly.	Key switch turn OFF.
723	2	G sensor pulse number error.	After fault opening PLV, rail pressure is above 160 MPa.	Diagnostic counter = zero Engine stops.
723	8	No input of G sensor pulse.	Open circuit or short circuit of sensor/ harness. Failure of sensor. No recognition of G sensor pulse. Engine hesitates at start-up.	Diagnostic counter = zero
1077	2	Failure of CPU.	CPU fatal error. Engine stops.	Key switch turn OFF.
1239	1	Fuel leak in high-pressured fuel system.	Fuel consumption is calculated from the difference of fuel pressure of, before and after the injection. Error is detected when excess fuel consumption is found. Engine could stop.	Key switch turn OFF.
1347	3	+B short circuit of SCV.	EGR stops.	Key switch turn OFF.
1347	4	SVC drive system error.	Open circuit or ground short circuit of SCV.	Key switch turn OFF. EGR stops, engine forcibly stops later.
1347	5	Open circuit of SCV (MPROP)	Open circuit short of SVC.	Key switch turn OFF. EGR stops, engine forcibly stops later.
1347	7	SCV stuck in open position.	Actual rail pressure continuously exceeds the command rail pressure. Discharge request of supply pump goes below 0 mm^3/st and the actual rail pressure is 10 MPa for 26 seconds or more. Engine may stop.	Key switch turn OFF.
1485	2	Main relay locked in closed position.	Failure of main relay. Stays active longer than one second without command. Battery goes dead.	Diagnostic counter = zero

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
3242	0	Emergency exhaust gas temperature sensor 1: High	DPF inlet temperature higher than 1319° F (715° C). Engine stops. Inhibit starter relay activation until exhaust temperatures reduces down to 572° F (300° C).	Under 572° F (300° C) and key switch turn OFF.
3242	3	Exhaust gas temperature sensor 1: High	Open circuit or +B short circuit of sensor/harness. DPF inlet temperature sensor voltage 4.92V or more.	Key switch turn OFF.
3242	4	Exhaust gas temperature sensor 1: Low	Ground short circuit of sensor/ harness. DPF inlet temperature sensor voltage 0.08V or less.	Key switch turn OFF.
3246	0	Emergency exhaust gas temperature sensor 2: High	DPF outlet temperature 1508° F (820° C) or more.	Under 572° F (300° C) and key switch turn OFF. Inhibit starter relay activation until exhaust temperature reduces down to 572° F (300° C).
3246	3	Exhaust gas temperature sensor 2: High	Open circuit or +B short circuit of sensor/harness. DPF outlet temperature sensor voltage 4.92V or more.	Key switch turn OFF.
3246	4	Exhaust gas temperature sensor 2: Low	Ground short circuit of sensor/ harness. DPF outlet temperature sensor voltage 0.08V or less.	Key switch turn OFF.
3251	3	Differential pressure sensor 1: High	Open circuit or +B short circuit of sensor/harness. DPF differential pressure sensor voltage 4.7V or more.	Key switch turn OFF.
3251	4	Differential pressure sensor 1: Low	Ground short circuit of sensor/ harness. DPF differential pressure sensor voltage 0.21V or less.	Key switch turn OFF.
3252	0	Emission deterioration.	DOC is heated due to unburned fuel. Output and accelerator limitation 50%. EGR stops.	Key switch turn OFF to minimize emission to DPF.
3509	3	Sensor supply voltage 1: High	Voltage to sensor is above 5.625V. EGR stops. Faulty starting. Engine may stop. Emission related.	Key switch turn OFF.
3509	4	Sensor supply voltage 1: Low	Voltage to sensor is below 4.375V. EGR stops. Faulty starting. Engine may stop. Emission related.	Key switch turn OFF.
3510	3	Sensor supply voltage 2: High	Voltage to sensor is above 5.625V. Faulty starting. Emission related.	Key switch turn OFF.
3510	4	Sensor supply voltage 2: Low	Voltage to sensor is below 4.375V. Faulty starting. Emission related.	Key switch turn OFF.
3511	3	Sensor supply voltage 3: High	Voltage to sensor is above 5.25V. Faulty starting.	Key switch turn OFF.
3511	4	Sensor supply voltage 3: Low	Voltage to sensor is below 4.75V. Faulty starting.	Key switch turn OFF.

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
3701	0	Excessive PM 5.	PM accumulation more than trigger level. Regeneration Level = 5.	Key switch turn OFF. **Engine not forcibly stopped by ECU. STRONGLY recommend operator to stop engine promptly.** Must be reset by Kubota service tool.
3701	15	Excessive PM 3.	PM accumulation more than trigger level. Regeneration Level = 3.	Diagnostic counter = 0
3701	16	Excessive PM 4.	PM accumulation more than trigger level. Regeneration Level = 4.	Diagnostic counter = 0
4765	0	Emergency exhaust gas temperature sensor 0: High	DOC inlet temperature 1292° F (700° C) or more. Engine stops.	Key switch turn OFF. Inhibit starter relay activation until exhaust temperature reduces down to 527° F (300° C).
4765	3	Exhaust gas temperature sensor 0: High	Open circuit or +B short circuit of sensor/harness. DOC inlet temperature sensor voltage 4.92V or more.	Key switch turn OFF.
4765	4	Exhaust gas temperature sensor 0: Low	Open circuit or +B short circuit of sensor/harness. DOC inlet temperature sensor voltage 0.08V or less.	Key switch turn OFF.
523523	2	Injector drive circuit open in Nos. 1 & 4 cylinder simultaneously.	Wiring harness open circuit. Injectors that have no DTC are operated. Engine vibration increases and may stop.	Key switch turn OFF.
523523	3	Nos. 1 & 4 cylinder injector short to +B at power supply side, or all cylinder injectors short to +B.	Wiring harness short to +B. Injectors that have no DTC are operated. Engine vibration increases and may stop.	Key switch turn OFF.
523523	4	Nos. 1 & 4 cylinder injector short to ground at power supply side, or all cylinder injectors short to ground.	Wiring harness short to ground. Injectors that have no DTC are operated. Engine vibration increases and may stop.	Key switch turn OFF.
523524	2	Injector drive circuit open in Nos. 2 & 3 cylinder simultaneously.	Wiring harness open circuit. Injectors that have no DTC are operated. Engine vibration increases and may stop.	Key switch turn OFF.
523524	3	Nos. 2 & 3 cylinder injector short to +B at power supply side, or all cylinder injectors short to +B.	Wiring harness short to +B. Injectors that have no DTC are operated. Engine vibration increases and may stop.	Key switch turn OFF.
523524	4	Nos. 2 & 3 cylinder injector short to ground at power supply side, or all cylinder injectors short to ground.	Wiring harness short to ground. Injectors that have no DTC are operated. Engine vibration increases and may stop.	Key switch turn OFF.
523525	1	Injector charge voltage low.	Injector charge voltage low. Failure of charge circuit of ECU. Engine may stop.	Key switch turn OFF.

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
523527	2	ECU monitoring error.	Failure of monitoring IC of CPU. Engine stops.	Key switch turn OFF.
523535	0	Injector charge voltage: High	EGR stops. Engine forcibly stops 60 seconds later.	Key switch turn OFF.
523536	2	EGR feedback error.	DC motor feedback control error. During wrong duty feedback, activation of restoration will repeat 30 times. If the restoration was not successfully completed, it will be judged as a feedback error. EGR stops.	Diagnostic counter = zero
523537	2	EGR DC motor temperature error.	Estimated temperature is 113° F (45° C) or above. EGR stops.	Diagnostic counter = zero
523538	2	QR data error.	QR data read error from EEPROM. Nozzle correction is not executed.	Key switch turn OFF.
523539	2	Pump seizing 1.	Rail pressure of 230 MPa (33400 psi or 2350 kgf/cm^2) or more continues one second under the condition of above 800 min-1 (rpm). Under 800 min-1 (rpm) condition, rail pressure of 220 MPa (31900 psi or 2250 kgf/cm^2) or more continues one second under the condition of below 800 min-1 (rpm). [Threshold changes depending upon the engine speed. 700 min-1 (rpm) should be used as a reference.	Key switch turn OFF to minimize PM emission to DPF and avoid extremely high pressure in injection system.
523540	2	Pump seizing 2.	Rail pressure of 197MPa (28600 psi or 2010 kgf/cm^2) or more continues one second under the condition of above 800 min-1 (rpm). Under 800 min-1 (rpm) condition, rail pressure of 220 MPa (31900 psi or 2250 kgf/cm^2) or more continues one second under the condition of below 800 min-1 (rpm). [Threshold changes depending upon the engine speed. 700 min-1 (rpm) should be used as a reference.	Key switch turn OFF.
523541	3	EGR lift sensor: High	+B short circuit of sensor/harness. Failure of sensor. Voltage of ECR lift sensor is 4.5V or above. EGR stops.	Key switch turn OFF.
523541	4	EGR lift sensor: Low	Open circuit or short circuit of sensor/ harness. Failure of sensor. Voltage of ECR lift sensor is 0.3V or less. EGR stops.	Key switch turn OFF.
523543	2	Accelerator position sensor error (CAN).	Accelerator position sensor signal error (sensor/harness open circuit, ground short circuit, etc.).	Diagnostic counter = zero. CAN signal recovers.
523544	3	+B short of air heater relay driving circuit.	+B short circuit of harness. Faulty starting. White smoke increases.	Key switch turn OFF.

SPN	FMI	DTC	SYMPTOM/CAUSE	RECOVERY FROM ERROR
523544	4	Ground short of air heater relay driving circuit.	Ground short or open circuit of harness.	Key switch turn OFF.
523547	2	CAN2 bus off.	CAN2 bus off.	Key switch turn OFF.
523548	2	CAN-KBT frame error.	CAN-KBT frame open circuit error.	Key switch turn OFF.
523572	4	EGR position sensor failure.	EGR position sensor error signal received via CAN. EGR stops.	Key switch turn OFF.
523574	3	EGR actuator open circuit.	EGR actuator open error signal received via CAN. EGR stops.	Key switch turn OFF.
523575	7	EGR actuator valve stuck.	EGR actuator valve stuck error received via CAN. EGR stops.	Key switch turn OFF.
523576	2	EGR (DC motor) overheat.	EGR (DC motor) temperature error signal received via CAN (exceeds 257° F or 125° C). EGR stops.	Key switch turn OFF.
523577	2	EGR (DC motor) temperature sensor failure.	EGR (DC motor) temperature sensor error signal received via CAN. EGR stops.	Key switch turn OFF.
523578	2	No communication with EGR.	Interruption of CAN. EGR stops.	Key switch turn OFF.
523580	2	Intake throttle feedback error.	Deviation of throttle position is not corrected in 20 times of duty error recovery action.	Key switch turn OFF.
523582	3	Intake throttle lift sensor: High	Intake throttle lift sensor voltage 4.89V or more.	Key switch turn OFF.
523582	4	Intake throttle lift sensor: Low	Intake throttle lift sensor voltage 0.1V or less.	Key switch turn OFF.
523589	17	Low coolant temperature in parked regeneration.	During regeneration mode, engine warm-up condition is not satisfied because coolant temperature is too low (below 149° F or 65° C).	Diagnostic counter = zero Leaving from Parked active regeneration status.
523590	16	Parked regeneration timed out.	Regeneration incomplete due to low temperature of DPF.	Diagnostic counter = zero Leaving from Parked active regeneration status.
523591	2	CAN CCVS (parking and vehicle speed) frame error.	CAN CCVS frame time-out error. Parking SW = off. Vehicle speed = 0	Key switch turn OFF.
523592	2	CAN CM1 (Regen SW) frame error.	CAN CM1 frame time-out error. Regeneration inhibit = on. Parked regeneration SW = off.	Key switch turn OFF.
523593	2	CAN DDC1 (transmission) frame error.	CAN DDC1 frame time-out error.	Key switch turn OFF.
523594	2	CAN ETC2 (neutral SW) frame error.	CAN ETC2 frame time-out error	Key switch turn OFF.
523595	2	CAN ETC5 (neutral SW) frame error.	CAN ETC5 frame time-out error.	Key switch turn OFF.
523596	2	CAN TSC1 frame error.	No request to “TSC1 buffer” continues three times after over-ride control request (other than 0x00).	Diagnostic counter = zero

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